Supporting Information: *From bioavailability scarcity to energy barriers limitations of anaerobic microbial reductive defluorination*

Table S1 Molecular mechanics parameters for selected ligands for molecular dynamics simulation

Tetrachloroethene	Bonds	Distance r0 (nm)	k (kJ/mol/nm^2)	
	C1-C2	0.133430	4.764739E+05	
	C1-Cl3	0.173080	2.688638E+05	
	C1-Cl4	0.173080	2.688638E+05	
	C2-Cl5	0.173080	2.688638E+05	
	C2-Cl6	0.173080	2.688638E+05	
	Angles	Angle a0 (Deg.)	(kJ/mol/rad^2)	
	C2-C1-Cl3	123.110	4.853440E+02	
	C2-C1-Cl4	123.110	4.853440E+02	
	Cl3-C1-Cl4	114.340	4.610768E+02	
	C1-C2-Cl5	123.110	4.853440E+02	
	C1-C2-Cl6	123.110	4.853440E+02	
	Cl5-C2-Cl6	114.340	4.610768E+02	
	Dihedrals	Angle a0 (Deg.)	kd (kJ/mol)	Multiplicity
	Cl3-C1-C2-Cl5	180.000	27.82360	2
	Cl3-C1-C2-Cl6	180.000	27.82360	2
	Cl4-C1-C2-Cl5	180.000	27.82360	2
	Cl4-C1-C2-Cl6	180.000	27.82360	2
	Improper Dihedrals	Angle a0 (Deg.)	kd (kJ/mol)	Multiplicity
	C2-Cl3-C1-Cl4	180.000	4.60240	2
	C1-Cl5-C2-Cl6	180.000	4.60240	2
Trichlorofluoroethene	Bonds	Distance r0 (nm)	k (kJ/mol/nm^2)	
	C1-C2	0.133430	4.764739E+05	
	C1-Cl3	0.173080	2.688638E+05	
	C1-Cl4	0.173080	2.688638E+05	
	C2-F5	0.133850	3.101181E+05	
	C2-Cl6	0.173080	2.688638E+05	
	Angles	Angle a0 (Deg.)	(kJ/mol/rad^2)	
	C2-C1-Cl3	123.110	4.853440E+02	
	C2-C1-Cl4	123.110	4.853440E+02	
	Cl3-C1-Cl4	114.340	4.610768E+02	
	C1-C2-F5	122.870	5.681872E+02	
	C1-C2-Cl6	123.110	4.853440E+02	
	F5-C2-Cl6	113.162	5.000000E+02	
	Dihedrals	Angle a0 (Deg.)	kd (kJ/mol)	Multiplicity
	Cl3-C1-C2-F5	180.000	27.82360	2
	Cl3-C1-C2-Cl6	180.000	27.82360	2
	Cl4-C1-C2-F5	180.000	27.82360	2
	Cl4-C1-C2-Cl6	180.000	27.82360	2
	Improper Dihedrals	Angle a0 (Deg.)	kd (kJ/mol)	Multiplicity

	C2-Cl3-C1-Cl4	180.000	4.60240	2
	C1-F5-C2-Cl6	180.000	4.60240	2
Dichlorodifluoroethene	Bonds	Distance r0 (nm)	kd (kJ/mol)	
	C1-C2	0.133430	4.764739E+05	
	C1-F3	0.133850	3.101181E+05	
	C1-Cl4	0.173080	2.688638E+05	
	C2-F5	0.133850	3.101181E+05	
	C2-Cl6	0.173080	2.688638E+05	
	Angles	Angle a0 (Deg.)	(kJ/mol/rad^2)	
	C2-C1-F3	122.870	5.681872E+02	
	C2-C1-Cl4	123.110	4.853440E+02	
	F3-C1-Cl4	114.661	5.000000E+02	
	C1-C2-F5	122.870	5.681872E+02	
	C1-C2-Cl6	123.110	4.853440E+02	
	F5-C2-Cl6	114.661	5.000000E+02	
	Dihedrals	Angle a0 (Deg.)	kd (kJ/mol)	Multiplicity
	F3-C1-C2-F5	180.000	27.82360	2
	F3-C1-C2-Cl6	180.000	27.82360	2
	Cl4-C1-C2-F5	180.000	27.82360	2
	Cl4-C1-C2-Cl6	180.000	27.82360	2
	Improper Dihedrals	Angle a0 (Deg.)	kd (kJ/mol)	Multiplicity
	C2-F3-C1-Cl4	180.000	4.60240	2
	C1-F5-C2-Cl6	180.000	4.60240	2
Chlorotrifluoroethene	Bonds	Distance r0 (nm)	k (kJ/mol/nm^2)	
	C1-C2	0.133430	4.764739E+05	
	C1-F3	0.133850	3.101181E+05	
	C1-F4	0.133850	3.101181E+05	
	C2-F5	0.133850	3.101181E+05	
	C2-Cl6	0.173080	2.688638E+05	
	Angles	Angle a0 (Deg.)	(kJ/mol/rad^2)	
	C2-C1-F3	122.870	5.681872E+02	
	C2-C1-F4	122.870	5.681872E+02	
	F3-C1-F4	111.640	5.865968E+02	
	C1-C2-F5	122.870	5.681872E+02	
	C1-C2-Cl6	115.602	5.000000E+02	
	Dihedrals	Angle a0 (Deg.)	kd (kJ/mol)	Multiplicity
	F3-C1-C2-F5	180.000	27.82360	2
	F3-C1-C2-Cl6	180.000	27.82360	2
	F4-C1-C2-F5	180.000	27.82360	2
	F4-C1-C2-Cl6	180.000	27.82360	2
	Improper Dihedrals	Angle a0 (Deg.)	kd (kJ/mol)	Multiplicity
	C2-F3-C1-F4	180.000	4.60240	2
	C1-F5-C2-Cl6	180.000	4.60240	2
Tetrafluoroethene	Bonds	Distance r0 (nm)	k (kJ/mol/nm^2)	
	C1-C2	0.133430	4.764739E+05	
	C1-F3	0.133850	3.101181E+05	
	C1-F4	0.133850	3.101181E+05	
	C2-F5	0.133850	3.101181E+05	
	C2-F6	0.133850	3.101181E+05	

	Angles	Angle a0 (Deg.)	(kJ/mol/rad^2)	
	C2-C1-F3	122.870	5.681872E+02	
	C2-C1-F4	122.870	5.681872E+02	
	F3-C1-F4	111.640	5.865968E+02	
	C1-C2-F5	122.870	5.681872E+02	
	C1-C2-F6	122.870	5.681872E+02	
	F5-C2-F6	111.640	5.865968E+02	
	Dihedrals	Angle a0 (Deg.)	kd (kJ/mol)	Multiplicity
	F3-C1-C2-F5	180.000	27.82360	2
	F3-C1-C2-F6	180.000	27.82360	2
	F4-C1-C2-F5	180.000	27.82360	2
	F4-C1-C2-F6	180.000	27.82360	2
	Improper Dihedrals	Angle a0 (Deg.)	kd (kJ/mol)	Multiplicity
	C2-F3-C1-Cl4	180.000	4.60240	2
	C1-F5-C2-Cl6	180.000	4.60240	2

Table S2 Molecular mechanics parameters of Cobalamin cofactor are calculated at B3LYP/ 6-311G(d,p) level of theory using Gaussian 09.

Cobalamin	Bonds	Distance r0 (nm)	k (kJ/mol/nm^2)
	P1-O10	0.168432	1.435555E+05
	P1-O14	0.162309	2.074604E+05
	P1-O18	0.15065	4.547930E+05
	P1-O22	0.149647	4.822287E+05
	C2-C3	0.152066	1.880194E+05
	C2-N84	0.146062	2.060285E+05
	C2-H90	0.109302	2.902079E+05
	C2-H91	0.108998	2.981623E+05
	C3-O14	0.142911	1.872643E+05
	C3-H92	0.109327	2.930088E+05
	C3-H93	0.109451	2.901647E+05
	C4-C7	0.159338	1.078546E+05
	C4-C44	0.1573	1.352165E+05
	C4-C45	0.153607	1.606226E+05
	C4-N46	0.148781	1.163437E+05
	C5-N6	0.148077	1.659440E+05
	C5-C9	0.154819	1.609055E+05
	C5-O25	0.140729	1.785645E+05
	C5-H94	0.109131	2.964871E+05
	N6-C8	0.136454	2.816376E+05
	N6-C31	0.13937	2.428004E+05
	C7-C11	0.158468	1.182475E+05
	C7-C50	0.15393	1.577023E+05
	C7-C51	0.155797	1.402655E+05
	C8-N13	0.131467	3.865079E+05
	C8-H95	0.107518	3.347700E+05

C9-C12	0.154772	1.552299E+05
C9-O29	0.139124	2.516765E+05
C9-H96	0.110123	2.694010E+05
O10-C12	0.14213	1.990860E+05
C11-C15	0.151184	1.625195E+05
C11-C55	0.155709	1.373060E+05
C11-H97	0.109134	2.896392E+05
C12-C16	0.153032	1.733606E+05
C12-H98	0.109367	2.895782E+05
N13-C34	0.138106	2.628651E+05
C15-C19	0.145377	2.076909E+05
C15-N46	0.129292	4.568943E+05
C16-C21	0.152365	1.938888E+05
C16-O25	0.14601	1.304497E+05
C16-H99	0.109167	2.938909E+05
N17-C20	0.13261	3.426758E+05
N17-C34	0.134272	3.162095E+05
C19-C23	0.136694	3.731029E+05
C19-C60	0.151465	1.874472E+05
C20-N24	0.134451	2.858693E+05
C20-H100	0.108662	3.022795E+05
C21-O32	0.14234	2.061561E+05
C21-H101	0.109784	2.780806E+05
C21-H102	0.109848	2.782120E+05
C23-C26	0.154198	1.427932E+05
C23-N47	0.140395	1.983828E+05
N24-C27	0.133832	3.007531E+05
C26-C30	0.156792	1.297937E+05
C26-C61	0.153536	1.665003E+05
C26-C62	0.156969	1.318175E+05
C27-N28	0.136993	3.274588E+05
C27-C31	0.140835	2.665630E+05
N28-H103	0.101107	4.129947E+05
N28-H168	0.10217	3.769363E+05
O29-H104	0.100778	3.104257E+05
C30-C33	0.150901	1.715259E+05
C30-C66	0.154975	1.538879E+05
C30-H105	0.109448	2.838883E+05
C31-C34	0.140518	2.637235E+05
O32-H106	0.097091	4.502024E+05
C33-C35	0.139173	3.050013E+05
C33-N47	0.133938	3.156975E+05
C35-C36	0.138883	3.115421E+05
C35-H107	0.108103	3.180181E+05
C36-C37	0.152145	1.666174E+05
C36-N48	0.134982	2.934090E+05
C37-C38	0.155287	1.439999E+05
C37-C71	0.155089	1.453897E+05

C37-C72	0.153364	1.641019E+05
C38-C39	0.151953	1.614514E+05
C38-C73	0.155763	1.348903E+05
C38-H108	0.109134	2.894835E+05
C39-C40	0.137533	3.389891E+05
C39-N48	0.138711	2.271734E+05
C40-C41	0.144538	2.314316E+05
C40-C78	0.151826	1.839459E+05
C41-C42	0.153541	1.530316E+05
C41-N49	0.130544	4.130161E+05
C42-C43	0.157424	1.240947E+05
C42-C79	0.154664	1.533576E+05
C42-C80	0.154748	1.493486E+05
C43-C44	0.154577	1.512605E+05
C43-C85	0.154537	1.627429E+05
C43-H109	0.109041	2.903877E+05
C44-N49	0.149445	1.173889E+05
C44-H110	0.109406	2.837580E+05
C45-H111	0.108831	2.989695E+05
C45-H112	0.108892	2.999802E+05
C45-H113	0.109175	2.895167E+05
N46-Co89	0.186555	1.167639E+05
N47-Co89	0.190632	1.005997E+05
N48-Co89	0.190951	9.779004E+04
N49-Co89	0.188518	1.039842E+05
C50-H114	0.109024	2.944928E+05
C50-H115	0.108737	3.004873E+05
C50-H116	0.108722	3.033149E+05
C51-C52	0.152764	1.694560E+05
C51-H117	0.10941	2.866621E+05
C51-H118	0.108665	3.013635E+05
C52-O53	0.123319	5.761214E+05
C52-N54	0.134646	3.573648E+05
N54-H119	0.101889	3.944099E+05
N54-H164	0.100907	4.226893E+05
C55-C56	0.154076	1.600660E+05
C55-H120	0.109083	2.928350E+05
C55-H121	0.109242	2.888968E+05
C56-C57	0.152807	1.658924E+05
C56-H122	0.10901	2.945689E+05
C56-H123	0.10929	2.913081E+05
C57-N58	0.135765	3.343252E+05
C57-O59	0.122458	6.098995E+05
N58-H124	0.101453	4.085304E+05
N58-H170	0.100915	4.210133E+05
C60-H125	0.109112	2.934085E+05
C60-H126	0.108533	3.055404E+05
C60-H127	0.109327	2.887728E+05

C61-H128	0.108801	3.000477E+05
C61-H129	0.109472	2.846938E+05
C61-H130	0.109025	2.972506E+05
C62-C63	0.152495	1.667847E+05
C62-H131	0.109489	2.840276E+05
C62-H132	0.109169	2.931784E+05
C63-O64	0.121692	6.394249E+05
C63-N65	0.136727	3.143421E+05
N65-H133	0.100558	4.363162E+05
N65-H163	0.100881	4.246775E+05
C66-C67	0.154108	1.687397E+05
C66-H134	0.10926	2.884981E+05
C66-H135	0.109344	2.893949E+05
C67-C68	0.152258	1.718023E+05
C67-H136	0.109085	2.943031E+05
C67-H137	0.109527	2.827370E+05
C68-O69	0.122451	6.093408E+05
C68-N70	0.135609	3.380116E+05
N70-H138	0.100535	4.376008E+05
N70-H166	0.100884	4.250134E+05
C71-H139	0.109291	2.910531E+05
C71-H140	0.10937	2.888253E+05
C71-H141	0.109272	2.909100E+05
C72-H142	0.109286	2.908454E+05
C72-H143	0.109005	2.963766E+05
C72-H144	0.109312	2.881649E+05
C73-C74	0.153547	1.766088E+05
C73-H145	0.109094	2.929820E+05
C73-H146	0.109391	2.861840E+05
C74-C75	0.154089	1.541286E+05
C74-H147	0.109242	2.880551E+05
C74-H148	0.109461	2.832516E+05
C75-O76	0.123483	5.630791E+05
C75-N77	0.133816	3.672961E+05
N77-H149	0.101097	4.191717E+05
N77-H167	0.103433	3.405608E+05
C78-H150	0.108832	2.977166E+05
C78-H151	0.109227	2.884495E+05
C78-H152	0.108941	2.958494E+05
C79-H153	0.109175	2.920247E+05
C79-H154	0.109094	2.935363E+05
C79-H155	0.1093	2.897381E+05
C80-C81	0.153969	1.690472E+05
C80-H156	0.109278	2.857072E+05
C80-H157	0.10897	2.939795E+05
C81-C82	0.152931	1.695460E+05
C81-H158	0.108923	2.983662E+05
C81-H159	0.109558	2.813309E+05

C82-O83	0.123678	5.517634E+05
C82-N84	0.134311	3.465324E+05
N84-H169	0.101772	3.992591E+05
C85-C86	0.153187	1.653691E+05
C85-H160	0.109053	2.940828E+05
C85-H161	0.109366	2.857929E+05
C86-N87	0.133943	3.678772E+05
C86-O88	0.123614	5.637013E+05
N87-H162	0.101998	3.893749E+05
N87-H165	0.101986	3.902548E+05
Angles	Angle a0 (Deg.)	(kJ/mol/rad^2)
O10-P1-O14	98.125	7.930648E+02
O10-P1-O18	105.602	7.033504E+02
O10-P1-O22	108.162	9.804159E+02
O14-P1-O18	112.129	8.038566E+02
O14-P1-O22	107.043	7.638910E+02
O18-P1-O22	122.875	6.885093E+02
C3-C2-N84	112.252	9.125102E+02
C3-C2-H90	109.314	4.323415E+02
C3-C2-H91	110.19	4.222340E+02
N84-C2-H90	108.197	5.472160E+02
N84-C2-H91	107.729	5.637321E+02
H90-C2-H91	109.087	3.191664E+02
C2-C3-O14	106.168	8.362003E+02
C2-C3-H92	110.453	5.500660E+02
C2-C3-H93	110.538	5.589935E+02
O14-C3-H92	110.53	4.489004E+02
O14-C3-H93	110.296	3.834149E+02
H92-C3-H93	108.844	3.222491E+02
C7-C4-C44	121.037	8.998268E+02
C7-C4-C45	112.297	8.945332E+02
C7-C4-N46	103.098	9.249266E+02
C44-C4-C45	109.682	9.405335E+02
C44-C4-N46	101.667	1.140683E+03
C45-C4-N46	107.433	7.887602E+02
N6-C5-C9	115.299	7.084665E+02
N6-C5-O25	110.334	8.497937E+02
N6-C5-H94	105.918	3.573845E+02
C9-C5-O25	104.961	9.724897E+02
C9-C5-H94	111.643	5.263007E+02
O25-C5-H94	108.598	5.567847E+02
C5-N6-C8	125.134	7.948584E+02
C5-N6-C31	127.714	7.943109E+02
C8-N6-C31	105.077	7.006528E+02
C4-C7-C11	102.567	1.050551E+03
C4-C7-C50	113.111	9.399099E+02
C4-C7-C51	110.566	1.026338E+03

C11-C7-C50	113.081	1.011017E+03
C11-C7-C51	108.23	8.874564E+02
C50-C7-C51	109.076	9.806722E+02
N6-C8-N13	114.692	5.761719E+02
N6-C8-H95	120.439	2.530760E+02
N13-C8-H95	124.868	2.419767E+02
C5-C9-C12	102.53	1.124260E+03
C5-C9-O29	114.014	7.789825E+02
C5-C9-H96	104.775	5.449149E+02
C12-C9-O29	118.994	9.906738E+02
C12-C9-H96	106.108	5.217512E+02
O29-C9-H96	109.232	4.994065E+02
P1-O10-C12	116.142	1.273370E+03
C7-C11-C15	103.167	1.044804E+03
C7-C11-C55	118.016	9.506584E+02
C7-C11-H97	107.912	4.435066E+02
C15-C11-C55	111.066	8.780770E+02
C15-C11-H97	109.359	4.210697E+02
C55-C11-H97	107.091	3.208137E+02
C9-C12-O10	113.025	7.431318E+02
C9-C12-C16	102.149	8.425811E+02
C9-C12-H98	111.731	4.942070E+02
O10-C12-C16	108.536	8.998817E+02
O10-C12-H98	109.882	4.243982E+02
C16-C12-H98	111.273	5.452839E+02
C8-N13-C34	104.493	1.211073E+03
P1-O14-C3	122.755	1.047600E+03
C11-C15-C19	124.416	6.994654E+02
C11-C15-N46	112.686	6.122804E+02
C19-C15-N46	122.853	6.163368E+02
C12-C16-C21	116.194	8.558839E+02
C12-C16-O25	106.695	8.381648E+02
C12-C16-H99	108.792	5.282352E+02
C21-C16-O25	107.672	8.423599E+02
C21-C16-H99	108.562	4.321402E+02
O25-C16-H99	108.714	4.768982E+02
C20-N17-C34	112.88	1.304564E+03
C15-C19-C23	119.925	7.514234E+02
C15-C19-C60	116.428	5.946835E+02
C23-C19-C60	123.641	5.572766E+02
N17-C20-N24	128.607	5.216433E+02
N17-C20-H100	116.342	2.204435E+02
N24-C20-H100	115.041	2.244276E+02
C16-C21-O32	108.798	9.686978E+02
C16-C21-H101	107.719	5.422979E+02
C16-C21-H102	110.429	5.709494E+02
O32-C21-H101	111.084	4.642298E+02
O32-C21-H102	111.225	4.622712E+02

H101-C21-H102	107.528	3.532424E+02
C19-C23-C26	128.141	5.669662E+02
C19-C23-N47	124.315	6.161168E+02
C26-C23-N47	107.535	5.730397E+02
C20-N24-C27	118.799	1.293839E+03
C5-O25-C16	111.253	1.515921E+03
C23-C26-C30	99.511	8.575939E+02
C23-C26-C61	117.404	1.009033E+03
C23-C26-C62	110.653	7.690387E+02
C30-C26-C61	112.594	9.845015E+02
C30-C26-C62	105.766	1.034863E+03
C61-C26-C62	109.967	7.586322E+02
N24-C27-N28	118.954	7.274780E+02
N24-C27-C31	117.578	6.218381E+02
N28-C27-C31	123.457	6.563137E+02
C27-N28-H103	112.095	4.148181E+02
C27-N28-H168	114.594	3.959062E+02
H103-N28-H168	114.864	2.602798E+02
C9-O29-H104	112.483	6.526681E+02
C26-C30-C33	100.215	7.987473E+02
C26-C30-C66	114.437	7.080236E+02
C26-C30-H105	110.639	4.956791E+02
C33-C30-C66	110.403	8.903194E+02
C33-C30-H105	111.818	4.708883E+02
C66-C30-H105	109.138	3.281931E+02
N6-C31-C27	135.991	6.778088E+02
N6-C31-C34	105.808	6.321304E+02
C27-C31-C34	118.182	7.445557E+02
C21-O32-H106	106.483	6.277396E+02
C30-C33-C35	124.46	5.999254E+02
C30-C33-N47	110.393	5.300112E+02
C35-C33-N47	125.137	6.467822E+02
N13-C34-N17	126.314	6.083842E+02
N13-C34-C31	109.92	6.561433E+02
N17-C34-C31	123.753	6.894041E+02
C33-C35-C36	125.226	8.084263E+02
C33-C35-H107	117.272	2.666141E+02
C36-C35-H107	117.456	2.617429E+02
C35-C36-C37	123.635	6.897572E+02
C35-C36-N48	124.601	6.502075E+02
C37-C36-N48	111.665	6.008452E+02
C36-C37-C38	100.631	8.354323E+02
C36-C37-C71	107.63	8.439696E+02
C36-C37-C72	114.677	1.000061E+03
C38-C37-C71	109.142	9.124331E+02
C38-C37-C72	115.36	1.028577E+03
C71-C37-C72	108.912	7.490609E+02
C37-C38-C39	102.467	9.017524E+02

C37-C38-C73	117.309	7.383461E+02
C37-C38-H108	109.301	4.717844E+02
C39-C38-C73	109.128	8.424808E+02
C39-C38-H108	111.988	4.380483E+02
C73-C38-H108	106.731	3.167759E+02
C38-C39-C40	124.137	6.673703E+02
C38-C39-N48	109.083	6.450610E+02
C40-C39-N48	126.379	6.239872E+02
C39-C40-C41	121.307	8.112673E+02
C39-C40-C78	118.133	5.668228E+02
C41-C40-C78	120.424	6.418858E+02
C40-C41-C42	126.024	7.361702E+02
C40-C41-N49	122.701	6.720836E+02
C42-C41-N49	111.225	6.312541E+02
C41-C42-C43	100.917	7.628905E+02
C41-C42-C79	108.659	8.053736E+02
C41-C42-C80	115.721	1.087488E+03
C43-C42-C79	111.975	8.946805E+02
C43-C42-C80	108.917	1.064572E+03
C79-C42-C80	110.374	8.204323E+02
C42-C43-C44	102.987	9.242793E+02
C42-C43-C85	113.048	8.465647E+02
C42-C43-H109	105.769	3.978320E+02
C44-C43-C85	119.585	9.964196E+02
C44-C43-H109	107.253	4.769605E+02
C85-C43-H109	107.304	3.873230E+02
C4-C44-C43	123.332	9.106347E+02
C4-C44-N49	104.475	1.065523E+03
C4-C44-H110	107.663	4.666938E+02
C43-C44-N49	102.486	8.682388E+02
C43-C44-H110	110.458	3.990496E+02
N49-C44-H110	107.123	3.929285E+02
C4-C45-H111	110.775	3.730143E+02
C4-C45-H112	111.302	3.946714E+02
C4-C45-H113	110.636	3.687671E+02
H111-C45-H112	107.958	2.301331E+02
H111-C45-H113	108.322	3.276610E+02
H112-C45-H113	107.723	2.969043E+02
C4-N46-C15	113.911	7.872910E+02
C4-N46-Co89	114.196	6.649788E+02
C15-N46-Co89	131.444	6.059782E+02
C23-N47-C33	110.408	8.413863E+02
C23-N47-Co89	125.038	7.224761E+02
C33-N47-Co89	124.339	6.860782E+02
C36-N48-C39	110.096	8.445101E+02
C36-N48-Co89	124.545	7.004622E+02
C39-N48-Co89	125.089	7.326023E+02
C41-N49-C44	112.824	7.901352E+02

C41-N49-Co89	131.362	6.523989E+02
C44-N49-Co89	115.74	6.578457E+02
C7-C50-H114	110.281	4.626413E+02
C7-C50-H115	113.005	4.163972E+02
C7-C50-H116	111.526	4.130193E+02
H114-C50-H115	107.272	2.786366E+02
H114-C50-H116	106.262	2.725392E+02
H115-C50-H116	108.18	3.051526E+02
C7-C51-C52	113.135	8.681520E+02
C7-C51-H117	110.365	4.062121E+02
C7-C51-H118	109.442	4.132621E+02
C52-C51-H117	107.113	4.331677E+02
C52-C51-H118	108.094	5.274265E+02
H117-C51-H118	108.558	2.998336E+02
C51-C52-O53	122.008	4.972214E+02
C51-C52-N54	114.754	6.223406E+02
O53-C52-N54	123.236	5.818226E+02
C52-N54-H119	120.443	3.449566E+02
C52-N54-H164	117.769	3.467546E+02
H119-N54-H164	119.584	2.130291E+02
C11-C55-C56	113.888	9.142743E+02
C11-C55-H120	110.407	4.187521E+02
C11-C55-H121	109.549	4.677927E+02
C56-C55-H120	109.557	3.752307E+02
C56-C55-H121	107.154	3.606391E+02
H120-C55-H121	105.935	3.182121E+02
C55-C56-C57	111.408	1.069468E+03
C55-C56-H122	111.372	3.572945E+02
C55-C56-H123	108.721	3.866658E+02
C57-C56-H122	111.141	5.061981E+02
C57-C56-H123	106.185	5.017308E+02
H122-C56-H123	107.78	2.969424E+02
C56-C57-N58	115.812	6.175420E+02
C56-C57-O59	121.021	4.782882E+02
N58-C57-O59	123.167	5.701091E+02
C57-N58-H124	121.117	3.514522E+02
C57-N58-H170	116.745	3.526002E+02
H124-N58-H170	118.162	2.128804E+02
C19-C60-H125	109.509	4.616257E+02
C19-C60-H126	111.605	4.213901E+02
C19-C60-H127	113.982	3.990707E+02
H125-C60-H126	106.007	2.887618E+02
H125-C60-H127	106.726	2.583418E+02
H126-C60-H127	108.599	2.768339E+02
C26-C61-H128	112.167	4.360127E+02
C26-C61-H129	109.351	4.892936E+02
C26-C61-H130	112.387	3.693263E+02
H128-C61-H129	107.153	2.822453E+02

H128-C61-H130	107.465	3.147577E+02
H129-C61-H130	108.104	2.527828E+02
C26-C62-C63	115.542	8.737360E+02
C26-C62-H131	107.498	3.562327E+02
C26-C62-H132	109.231	4.074954E+02
C63-C62-H131	111.315	5.182966E+02
C63-C62-H132	105.342	4.326983E+02
H131-C62-H132	107.648	3.061862E+02
C62-C63-O64	121.664	4.739955E+02
C62-C63-N65	115.981	5.980666E+02
O64-C63-N65	122.348	5.629960E+02
C63-N65-H133	122.443	3.252481E+02
C63-N65-H163	117.916	3.174387E+02
H133-N65-H163	118.5	1.947276E+02
C30-C66-C67	113.539	1.184738E+03
C30-C66-H134	109.152	4.177106E+02
C30-C66-H135	109.438	4.914680E+02
C67-C66-H134	108.639	4.130074E+02
C67-C66-H135	108.807	4.237318E+02
H134-C66-H135	107.061	2.693186E+02
C66-C67-C68	110.941	8.216443E+02
C66-C67-H136	110.208	4.133746E+02
C66-C67-H137	110.134	3.236166E+02
C68-C67-H136	106.316	4.578156E+02
C68-C67-H137	111.149	5.600419E+02
H136-C67-H137	107.982	3.063992E+02
C67-C68-O69	121.946	4.759962E+02
C67-C68-N70	116.402	6.009582E+02
O69-C68-N70	121.63	5.553310E+02
C68-N70-H138	122.563	3.212505E+02
C68-N70-H166	118.426	3.117460E+02
H138-N70-H166	118.939	1.944965E+02
C37-C71-H139	111.448	4.914337E+02
C37-C71-H140	109.852	3.568353E+02
C37-C71-H141	111.007	4.739558E+02
H139-C71-H140	108.195	2.648470E+02
H139-C71-H141	108.074	2.428399E+02
H140-C71-H141	108.156	3.098036E+02
C37-C72-H142	110.076	4.510235E+02
C37-C72-H143	112.544	4.037501E+02
C37-C72-H144	110.87	4.667523E+02
H142-C72-H143	108.071	2.485219E+02
H142-C72-H144	106.958	2.873480E+02
H143-C72-H144	108.116	2.761284E+02
C38-C73-C74	117.299	8.558335E+02
C38-C73-H145	108.019	4.519373E+02
C38-C73-H146	108.643	3.872270E+02
C74-C73-H145	106.713	4.124409E+02

C74-C73-H146	109.781	3.559946E+02
H145-C73-H146	105.763	3.180639E+02
C73-C74-C75	112.086	1.353913E+03
C73-C74-H147	112.234	3.678724E+02
C73-C74-H148	110.357	4.029828E+02
C75-C74-H147	110.054	5.601038E+02
C75-C74-H148	105.452	4.209255E+02
H147-C74-H148	106.283	2.396052E+02
C74-C75-O76	120.001	4.223831E+02
C74-C75-N77	115.058	4.929713E+02
O76-C75-N77	124.931	5.551938E+02
C75-N77-H149	119.376	3.634727E+02
C75-N77-H167	122.699	3.845976E+02
H149-N77-H167	115.587	2.293581E+02
C40-C78-H150	111.308	4.425176E+02
C40-C78-H151	113.457	3.793885E+02
C40-C78-H152	110.527	4.351667E+02
H150-C78-H151	107.55	2.359329E+02
H150-C78-H152	106.302	2.657326E+02
H151-C78-H152	107.353	2.975927E+02
C42-C79-H153	110.625	4.587489E+02
C42-C79-H154	111.316	4.083190E+02
C42-C79-H155	111.901	4.232018E+02
H153-C79-H154	107.241	2.893766E+02
H153-C79-H155	108.107	2.493993E+02
H154-C79-H155	107.455	2.881014E+02
C42-C80-C81	115.726	9.273527E+02
C42-C80-H156	106.67	4.850288E+02
C42-C80-H157	110.47	3.485172E+02
C81-C80-H156	107.699	3.479596E+02
C81-C80-H157	110.094	4.991634E+02
H156-C80-H157	105.595	2.389441E+02
C80-C81-C82	112.427	1.139240E+03
C80-C81-H158	112.794	4.131821E+02
C80-C81-H159	111.112	4.386046E+02
C82-C81-H158	109.668	4.765557E+02
C82-C81-H159	105.653	4.188782E+02
H158-C81-H159	104.674	2.162847E+02
C81-C82-O83	120.876	5.555998E+02
C81-C82-N84	115.547	6.955717E+02
O83-C82-N84	123.574	5.379132E+02
C2-N84-C82	122.399	7.241501E+02
C2-N84-H169	117.912	3.037043E+02
C82-N84-H169	118.618	3.199349E+02
C43-C85-C86	115.097	1.020867E+03
C43-C85-H160	112.087	3.030852E+02
C43-C85-H161	108.017	3.561455E+02
C86-C85-H160	107.81	5.465926E+02

C86-C85-H161	106.498	5.034253E+02
H160-C85-H161	106.89	2.899652E+02
C85-C86-N87	115.52	6.783558E+02
C85-C86-O88	120.716	5.174412E+02
N87-C86-O88	123.75	5.661122E+02
C86-N87-H162	122.255	3.467472E+02
C86-N87-H165	121.558	3.411342E+02
H162-N87-H165	116.185	2.165290E+02
N46-Co89-N47	89.705	6.900396E+02
N46-Co89-N48	169.669	9.406187E+02
N46-Co89-N49	83.331	7.560495E+02
N47-Co89-N48	95.306	7.416455E+02
N47-Co89-N49	172.473	7.663071E+02
N48-Co89-N49	92.024	7.403389E+02
Dihedrals	Angle a0 (Deg.)	k (kJ/mol/rad^2)
P1-O10-C12-C9	91.342	8.244285E+02
P1-O10-C12-C16	-156.095	7.978024E+02
P1-O10-C12-H98	-34.213	3.921597E+02
P1-O14-C3-C2	-165.371	6.858432E+02
P1-O14-C3-H92	-45.565	4.212973E+02
P1-O14-C3-H93	74.848	4.030740E+02
C2-N84-C82-C81	-176.348	7.091876E+02
C2-N84-C82-O83	3.083	4.855882E+02
C3-C2-N84-C82	101.908	5.795064E+02
C3-C2-N84-H169	-66.084	1.473763E+02
C3-O14-P1-O10	61.216	6.933571E+02
C3-O14-P1-O18	-49.353	6.422580E+02
C3-O14-P1-O22	173.123	6.587319E+02
C4-C7-C11-C15	-19.353	1.232727E+03
C4-C7-C11-C55	103.509	9.208501E+02
C4-C7-C11-H97	-135.036	5.180777E+02
C4-C7-C50-H114	-175.378	4.697651E+02
C4-C7-C50-H115	-55.327	4.594416E+02
C4-C7-C50-H116	66.803	4.277601E+02
C4-C7-C51-C52	-175.241	1.138305E+03
C4-C7-C51-H117	64.764	4.401069E+02
C4-C7-C51-H118	-54.645	4.922830E+02
C4-C44-C43-C42	146.03	1.072576E+03
C4-C44-C43-C85	-87.603	8.921094E+02
C4-C44-C43-H109	34.704	4.052133E+02
C4-C44-N49-C41	-149.028	6.446664E+02
C4-C44-N49-Co89	28.204	8.737069E+02
C4-N46-C15-C11	1.837	8.652356E+02
C4-N46-C15-C19	-175.803	7.122644E+02
C4-N46-Co89-N47	156.567	8.129394E+02
C4-N46-Co89-N48	37.397	8.180320E+01
C4-N46-Co89-N49	-26.296	7.331810E+02

C5-N6-C8-N13	164.545	6.079050E+02
C5-N6-C8-H95	-15.148	2.479610E+02
C5-N6-C31-C27	18.319	3.530201E+02
C5-N6-C31-C34	-163.402	5.769681E+02
C5-C9-C12-O10	83.244	9.752677E+02
C5-C9-C12-C16	-33.169	1.316190E+03
C5-C9-C12-H98	-152.202	6.035327E+02
C5-C9-O29-H104	-117.996	6.157982E+01
C5-O25-C16-C12	-1.194	8.489115E+02
C5-O25-C16-C21	124.202	8.171286E+02
C5-O25-C16-H99	-118.367	3.993602E+02
N6-C5-C9-C12	-88.062	1.175968E+03
N6-C5-C9-O29	41.925	1.200958E+03
N6-C5-C9-H96	161.294	6.647699E+02
N6-C5-O25-C16	104.296	6.821553E+02
N6-C8-N13-C34	-0.572	6.219322E+02
N6-C31-C27-N24	-176.904	3.975939E+02
N6-C31-C27-N28	1.916	4.544899E+02
N6-C31-C34-N13	-1.078	6.076961E+02
N6-C31-C34-N17	177.644	4.837449E+02
C7-C4-C44-C43	87.2	9.108664E+02
C7-C4-C44-N49	-156.885	9.713069E+02
C7-C4-C44-H110	-43.225	4.783947E+02
C7-C4-C45-H111	-68.28	4.588073E+02
C7-C4-C45-H112	51.843	5.387885E+02
C7-C4-C45-H113	171.577	4.615891E+02
C7-C4-N46-C15	-14.734	6.416680E+02
C7-C4-N46-Co89	172.053	8.634333E+02
C7-C11-C15-C19	-170.402	8.655146E+02
C7-C11-C15-N46	12.001	6.103883E+02
C7-C11-C55-C56	91.593	9.668439E+02
C7-C11-C55-H120	-32.144	5.235230E+02
C7-C11-C55-H121	-148.436	5.070942E+02
C7-C51-C52-O53	-72.608	5.583118E+02
C7-C51-C52-N54	107.917	6.700495E+02
C8-N6-C5-C9	109.934	6.752582E+02
C8-N6-C5-O25	-8.732	6.170336E+02
C8-N6-C5-H94	-126.078	4.167820E+02
C8-N6-C31-C27	-177.584	4.052959E+02
C8-N6-C31-C34	0.695	7.310845E+02
C8-N13-C34-N17	-177.672	4.584286E+02
C8-N13-C34-C31	1.009	6.359900E+02
C9-C5-N6-C31	-88.942	6.808036E+02
C9-C5-O25-C16	-20.507	7.204845E+02
C9-C12-C16-C21	-97.937	1.103379E+03
C9-C12-C16-O25	22.112	7.632885E+02
C9-C12-C16-H99	139.233	5.359179E+02
O10-C12-C9-O29	-43.611	6.107947E+02

O10-C12-C9-H96	-167.116	5.464453E+02
O10-C12-C16-C21	142.447	8.774038E+02
O10-C12-C16-O25	-97.503	6.478615E+02
O10-C12-C16-H99	19.618	4.763319E+02
C11-C7-C4-C44	132.78	1.118517E+03
C11-C7-C4-C45	-95.03	1.198105E+03
C11-C7-C4-N46	20.302	1.250714E+03
C11-C7-C50-H114	-59.338	4.603459E+02
C11-C7-C50-H115	60.713	4.504279E+02
C11-C7-C50-H116	-177.157	4.199360E+02
C11-C7-C51-C52	73.152	1.071850E+03
C11-C7-C51-H117	-46.843	4.298038E+02
C11-C7-C51-H118	-166.252	4.794278E+02
C11-C15-C19-C23	-157.095	7.407113E+02
C11-C15-C19-C60	23.725	8.811360E+02
C11-C15-N46-Co89	173.568	6.400593E+02
C11-C55-C56-C57	75.07	1.037869E+03
C11-C55-C56-H122	-49.641	4.323987E+02
C11-C55-C56-H123	-168.251	4.650431E+02
C12-C9-C5-O25	33.549	1.037407E+03
C12-C9-C5-H94	151.013	5.498309E+02
C12-C9-O29-H104	3.232	6.175544E+01
C12-O10-P1-O14	171.007	8.180716E+02
C12-O10-P1-O18	-73.211	7.203659E+02
C12-O10-P1-O22	60.004	7.499719E+02
C12-C16-C21-O32	-80.135	7.496442E+02
C12-C16-C21-H101	159.356	6.121674E+02
C12-C16-C21-H102	42.202	6.019000E+02
N13-C8-N6-C31	-0.085	5.459720E+02
N13-C34-N17-C20	178.471	5.132810E+02
N13-C34-C31-C27	177.566	5.520197E+02
O14-C3-C2-N84	58.027	9.933982E+02
O14-C3-C2-H90	-62.034	5.568892E+02
O14-C3-C2-H91	178.094	4.506472E+02
C15-C11-C7-C50	-141.502	9.555475E+02
C15-C11-C7-C51	97.541	1.103899E+03
C15-C11-C55-C56	-149.626	9.536542E+02
C15-C11-C55-H120	86.636	5.196315E+02
C15-C11-C55-H121	-29.656	5.034422E+02
C15-C19-C23-C26	170.105	8.997389E+02
C15-C19-C23-N47	-8.666	5.967293E+02
C15-C19-C60-H125	33.743	4.946729E+02
C15-C19-C60-H126	150.806	4.105838E+02
C15-C19-C60-H127	-85.715	4.829229E+02
C15-N46-C4-C44	-140.79	6.122117E+02
C15-N46-C4-C45	104.04	6.217329E+02
C15-N46-Co89-N47	-15.146	4.473485E+02
C15-N46-Co89-N48	-134.316	7.558721E+01

C15-N46-Co89-N49	161.991	4.220817E+02
C16-C12-C9-O29	-160.024	7.290647E+02
C16-C12-C9-H96	76.471	6.392152E+02
C16-C21-O32-H106	179.673	9.316803E+01
C16-O25-C5-H94	-140.027	4.285187E+02
N17-C20-N24-C27	-1.398	4.414519E+02
N17-C34-C31-C27	-3.713	4.477929E+02
C19-C15-C11-C55	62.22	9.152085E+02
C19-C15-C11-H97	-55.758	4.135211E+02
C19-C15-N46-Co89	-4.072	5.523111E+02
C19-C23-C26-C30	-150.773	5.558826E+02
C19-C23-C26-C61	-29.061	5.448715E+02
C19-C23-C26-C62	98.275	5.332793E+02
C19-C23-N47-C33	167.525	4.752719E+02
C19-C23-N47-Co89	-17.621	5.185842E+02
C20-N17-C34-C31	-0.038	5.642004E+02
C20-N24-C27-N28	178.434	6.132255E+02
C20-N24-C27-C31	-2.691	4.991433E+02
C21-C16-C12-H98	21.419	4.944083E+02
C23-C19-C15-N46	20.265	4.365838E+02
C23-C19-C60-H125	-145.403	4.194791E+02
C23-C19-C60-H126	-28.339	3.574074E+02
C23-C19-C60-H127	95.139	4.109992E+02
C23-C26-C30-C33	-32.46	1.143350E+03
C23-C26-C30-C66	85.638	1.117040E+03
C23-C26-C30-H105	-150.586	4.913570E+02
C23-C26-C61-H128	79.773	3.823232E+02
C23-C26-C61-H129	-161.502	4.381016E+02
C23-C26-C61-H130	-41.446	3.990530E+02
C23-C26-C62-C63	-54.321	1.249543E+03
C23-C26-C62-H131	-179.285	4.861292E+02
C23-C26-C62-H132	64.186	4.891933E+02
C23-N47-C33-C30	-11.931	7.589330E+02
C23-N47-C33-C35	166.935	5.050910E+02
C23-N47-Co89-N46	24.958	6.922708E+02
C23-N47-Co89-N48	-164.091	6.517946E+02
C23-N47-Co89-N49	2.705	3.602600E+01
N24-C20-N17-C34	2.772	4.727721E+02
N24-C27-N28-H103	-9.885	3.286237E+02
N24-C27-N28-H168	-143.111	3.618243E+02
N24-C27-C31-C34	4.975	6.019748E+02
O25-C5-N6-C31	152.392	6.216606E+02
O25-C5-C9-O29	163.536	1.056806E+03
O25-C5-C9-H96	-77.094	6.181011E+02
O25-C16-C12-H98	141.468	4.121275E+02
O25-C16-C21-O32	160.344	7.508757E+02
O25-C16-C21-H101	39.835	6.129885E+02
O25-C16-C21-H102	-77.32	6.026937E+02

C26-C23-C19-C60	-10.778	7.858485E+02
C26-C23-N47-C33	-11.461	8.173155E+02
C26-C23-N47-Co89	163.392	9.543934E+02
C26-C30-C33-C35	-149.719	6.251648E+02
C26-C30-C33-N47	29.157	7.271412E+02
C26-C30-C66-C67	177.752	8.919481E+02
C26-C30-C66-H134	56.398	4.909045E+02
C26-C30-C66-H135	-60.47	4.881396E+02
C26-C62-C63-O64	78.825	5.882032E+02
C26-C62-C63-N65	-100.232	6.567624E+02
C27-N24-C20-H100	179.771	3.216320E+02
N28-C27-C31-C34	-176.205	7.427550E+02
O29-C9-C5-H94	-79.001	5.552328E+02
O29-C9-C12-H98	80.943	4.407689E+02
C30-C26-C23-N47	28.163	7.512640E+02
C30-C26-C61-H128	-165.56	4.133208E+02
C30-C26-C61-H129	-46.834	4.792907E+02
C30-C26-C61-H130	73.221	4.329430E+02
C30-C26-C62-C63	-161.174	1.209296E+03
C30-C26-C62-H131	73.862	4.799152E+02
C30-C26-C62-H132	-42.667	4.829013E+02
C30-C33-C35-C36	178.738	5.323644E+02
C30-C33-C35-H107	1.265	2.276038E+02
C30-C33-N47-Co89	173.172	8.467477E+02
C30-C66-C67-C68	172.584	1.050397E+03
C30-C66-C67-H136	55.107	4.676766E+02
C30-C66-C67-H137	-63.928	4.583614E+02
C31-N6-C5-H94	35.046	4.188880E+02
C31-N6-C8-H95	-179.778	2.369952E+02
C31-C27-N28-H103	171.31	3.164713E+02
C31-C27-N28-H168	38.084	3.471472E+02
O32-C21-C16-H99	42.815	5.061572E+02
C33-C30-C26-C61	-157.573	1.086868E+03
C33-C30-C26-C62	82.306	1.140153E+03
C33-C30-C66-C67	-70.11	9.884316E+02
C33-C30-C66-H134	168.536	5.187749E+02
C33-C30-C66-H135	51.669	5.156881E+02
C33-C35-C36-C37	-173.279	5.217942E+02
C33-C35-C36-N48	2.8	4.261467E+02
C33-N47-Co89-N46	-160.887	5.292053E+02
C33-N47-Co89-N48	10.065	5.052214E+02
C33-N47-Co89-N49	176.861	3.545743E+01
C34-N13-C8-H95	179.106	2.249173E+02
C34-N17-C20-H100	-178.41	3.215728E+02
C35-C33-C30-C66	89.249	5.702844E+02
C35-C33-C30-H105	-32.465	3.748876E+02
C35-C33-N47-Co89	-7.961	5.425373E+02
C35-C36-C37-C38	-164.022	6.559801E+02

C35-C36-C37-C71	81.79	6.148643E+02
C35-C36-C37-C72	-39.562	6.290117E+02
C35-C36-N48-C39	177.025	4.413515E+02
C35-C36-N48-Co89	2.747	4.349741E+02
C36-C35-C33-N47	0.027	4.021647E+02
C36-C37-C38-C39	-23.245	1.192061E+03
C36-C37-C38-C73	96.217	1.151142E+03
C36-C37-C38-H108	-142.174	4.749645E+02
C36-C37-C71-H139	56.804	4.260584E+02
C36-C37-C71-H140	176.707	4.002619E+02
C36-C37-C71-H141	-63.721	4.450657E+02
C36-C37-C72-H142	-173.911	4.917057E+02
C36-C37-C72-H143	-53.293	4.715243E+02
C36-C37-C72-H144	67.944	4.635487E+02
C36-N48-C39-C38	-10.232	7.113545E+02
C36-N48-C39-C40	176.823	4.976488E+02
C36-N48-Co89-N46	111.115	6.945626E+01
C36-N48-Co89-N47	-7.612	5.549182E+02
C36-N48-Co89-N49	174.104	5.755338E+02
C37-C36-C35-H107	4.19	2.043661E+02
C37-C36-N48-C39	-6.487	6.480293E+02
C37-C36-N48-Co89	179.235	6.343729E+02
C37-C38-C39-C40	-165.112	8.104304E+02
C37-C38-C39-N48	21.75	8.729995E+02
C37-C38-C73-C74	-67.506	1.105231E+03
C37-C38-C73-H145	171.951	4.710773E+02
C37-C38-C73-H146	57.669	4.978835E+02
C38-C37-C36-N48	19.45	7.515230E+02
C38-C37-C71-H139	-51.569	4.450262E+02
C38-C37-C71-H140	68.333	4.169574E+02
C38-C37-C71-H141	-172.095	4.658049E+02
C38-C37-C72-H142	-57.65	5.005958E+02
C38-C37-C72-H143	62.968	4.796936E+02
C38-C37-C72-H144	-175.795	4.714416E+02
C38-C39-C40-C41	-157.991	8.007546E+02
C38-C39-C40-C78	17.78	7.278609E+02
C38-C39-N48-Co89	164.007	7.316565E+02
C38-C73-C74-C75	-157.31	1.072165E+03
C38-C73-C74-H147	78.243	5.110422E+02
C38-C73-C74-H148	-40.101	5.081000E+02
C39-C38-C37-C71	89.794	1.260903E+03
C39-C38-C37-C72	-147.236	1.067543E+03
C39-C38-C73-C74	48.358	1.080108E+03
C39-C38-C73-H145	-72.185	4.664530E+02
C39-C38-C73-H146	173.533	4.927208E+02
C39-C40-C41-C42	167.412	7.024860E+02
C39-C40-C41-N49	-9.771	4.483800E+02
C39-C40-C78-H150	146.633	3.900686E+02

C39-C40-C78-H151	-91.896	4.058608E+02
C39-C40-C78-H152	28.744	4.136931E+02
C39-N48-Co89-N46	-62.314	6.984614E+01
C39-N48-Co89-N47	178.959	5.808209E+02
C39-N48-Co89-N49	0.675	6.034453E+02
C40-C39-C38-C73	69.858	7.131189E+02
C40-C39-C38-H108	-48.089	3.832616E+02
C40-C39-N48-Co89	-8.937	5.075003E+02
C40-C41-C42-C43	-158.919	6.691710E+02
C40-C41-C42-C79	83.218	6.392138E+02
C40-C41-C42-C80	-41.571	6.563080E+02
C40-C41-N49-C44	177.779	5.985620E+02
C40-C41-N49-Co89	1.102	4.317336E+02
C41-C40-C39-N48	13.946	5.684336E+02
C41-C40-C78-H150	-37.557	4.086349E+02
C41-C40-C78-H151	83.914	4.259996E+02
C41-C40-C78-H152	-155.446	4.346368E+02
C41-C42-C43-C44	-28.657	1.326553E+03
C41-C42-C43-C85	-159.104	1.228724E+03
C41-C42-C43-H109	83.764	4.585217E+02
C41-C42-C79-H153	179.93	4.522798E+02
C41-C42-C79-H154	-60.94	5.191855E+02
C41-C42-C79-H155	59.318	4.779958E+02
C41-C42-C80-C81	-46.086	7.723500E+02
C41-C42-C80-H156	-165.858	4.348290E+02
C41-C42-C80-H157	79.854	4.589768E+02
C41-N49-C44-C43	-19.355	6.135361E+02
C41-N49-C44-H110	96.928	3.526332E+02
C41-N49-Co89-N46	173.647	4.791204E+02
C41-N49-Co89-N47	-163.941	2.399092E+01
C41-N49-Co89-N48	2.904	4.508190E+02
C42-C41-C40-C78	-8.263	7.228186E+02
C42-C41-N49-C44	0.223	9.065327E+02
C42-C41-N49-Co89	-176.454	5.718610E+02
C42-C43-C44-N49	29.154	1.113361E+03
C42-C43-C44-H110	-84.703	4.857694E+02
C42-C43-C85-C86	-150.249	1.113374E+03
C42-C43-C85-H160	86.07	5.695878E+02
C42-C43-C85-H161	-31.427	4.664834E+02
C42-C80-C81-C82	-128.327	1.040177E+03
C42-C80-C81-H158	107.034	4.724905E+02
C42-C80-C81-H159	-10.147	4.384112E+02
C43-C42-C41-N49	18.538	6.467508E+02
C43-C42-C79-H153	69.332	4.541523E+02
C43-C42-C79-H154	-171.538	5.216545E+02
C43-C42-C79-H155	-51.28	4.800878E+02
C43-C42-C80-C81	66.701	9.049447E+02
C43-C42-C80-H156	-53.072	4.739236E+02

C43-C42-C80-H157	-167.359	5.027527E+02
C43-C44-C4-C45	-46.077	9.950591E+02
C43-C44-C4-N46	-159.577	9.447279E+02
C43-C44-N49-Co89	157.877	8.174911E+02
C43-C85-C86-N87	62.017	6.196909E+02
C43-C85-C86-O88	-116.657	5.818814E+02
C44-C4-C7-C50	-105.091	9.869405E+02
C44-C4-C7-C51	17.569	9.247484E+02
C44-C4-C45-H111	69.325	4.616833E+02
C44-C4-C45-H112	-170.552	5.427591E+02
C44-C4-C45-H113	-50.817	4.645003E+02
C44-C4-N46-Co89	45.997	8.109310E+02
C44-C43-C42-C79	86.759	1.073995E+03
C44-C43-C42-C80	-150.887	1.252169E+03
C44-C43-C85-C86	88.258	1.262833E+03
C44-C43-C85-H160	-35.423	6.062978E+02
C44-C43-C85-H161	-152.92	4.908221E+02
C44-N49-Co89-N46	-2.954	8.925437E+02
C44-N49-Co89-N47	19.459	2.456057E+01
C44-N49-Co89-N48	-173.697	7.990920E+02
C45-C4-C7-C50	27.099	1.048391E+03
C45-C4-C7-C51	149.759	9.784873E+02
C45-C4-C44-N49	69.838	1.067634E+03
C45-C4-C44-H110	-176.501	5.006424E+02
C45-C4-N46-Co89	-69.172	8.277212E+02
N46-C4-C7-C50	142.431	1.088453E+03
N46-C4-C7-C51	-94.909	1.013297E+03
N46-C4-C44-N49	-43.662	1.009907E+03
N46-C4-C44-H110	69.998	4.875732E+02
N46-C4-C45-H111	179.046	4.335175E+02
N46-C4-C45-H112	-60.831	5.042449E+02
N46-C4-C45-H113	58.903	4.360004E+02
N46-C15-C11-C55	-115.377	6.346924E+02
N46-C15-C11-H97	126.645	3.446879E+02
N46-C15-C19-C60	-158.914	4.818451E+02
N47-C23-C19-C60	170.451	5.444019E+02
N47-C23-C26-C61	149.875	7.312912E+02
N47-C23-C26-C62	-82.789	7.105608E+02
N47-C33-C30-C66	-91.875	6.539445E+02
N47-C33-C30-H105	146.41	4.093098E+02
N47-C33-C35-H107	-177.447	1.999308E+02
N48-C36-C35-H107	-179.731	1.878525E+02
N48-C36-C37-C71	-94.737	6.980462E+02
N48-C36-C37-C72	143.911	7.163373E+02
N48-C39-C38-C73	-103.28	7.611193E+02
N48-C39-C38-H108	138.773	3.967077E+02
N48-C39-C40-C78	-170.282	5.307047E+02
N49-C41-C40-C78	174.554	4.565776E+02

N49-C41-C42-C79	-99.325	6.187254E+02
N49-C41-C42-C80	135.886	6.347275E+02
N49-C44-C43-C85	155.521	9.201456E+02
N49-C44-C43-H109	-82.171	4.109001E+02
C50-C7-C11-C55	-18.64	7.568512E+02
C50-C7-C11-H97	102.815	4.617822E+02
C50-C7-C51-C52	-50.258	1.157257E+03
C50-C7-C51-H117	-170.253	4.429113E+02
C50-C7-C51-H118	70.338	4.957944E+02
C51-C7-C11-C55	-139.597	8.470102E+02
C51-C7-C11-H97	-18.142	4.938558E+02
C51-C7-C50-H114	61.138	4.611762E+02
C51-C7-C50-H115	-178.811	4.512228E+02
C51-C7-C50-H116	-56.681	4.206269E+02
C51-C52-N54-H119	-12.117	2.308170E+02
C51-C52-N54-H164	-175.182	2.221597E+02
O53-C52-C51-H117	49.23	4.067101E+02
O53-C52-C51-H118	166.033	3.959360E+02
O53-C52-N54-H119	168.416	1.932526E+02
O53-C52-N54-H164	5.35	1.871465E+02
N54-C52-C51-H117	-130.245	4.629486E+02
N54-C52-C51-H118	-13.442	4.490398E+02
C55-C56-C57-N58	-112.922	7.391544E+02
C55-C56-C57-O59	67.029	6.032584E+02
C56-C55-C11-H97	-30.282	4.358910E+02
C56-C57-N58-H124	17.794	2.239131E+02
C56-C57-N58-H170	174.942	2.454025E+02
C57-C56-C55-H120	-160.735	4.680459E+02
C57-C56-C55-H121	-46.243	4.592946E+02
N58-C57-C56-H122	11.918	4.406161E+02
N58-C57-C56-H123	128.864	4.457797E+02
O59-C57-C56-H122	-168.131	3.884527E+02
O59-C57-C56-H123	-51.185	3.924604E+02
O59-C57-N58-H124	-162.157	1.953366E+02
O59-C57-N58-H170	-5.008	2.114931E+02
C61-C26-C30-C66	-39.476	1.063066E+03
C61-C26-C30-H105	84.301	4.806232E+02
C61-C26-C62-C63	77.001	1.035580E+03
C61-C26-C62-H131	-47.963	4.499606E+02
C61-C26-C62-H132	-164.492	4.525846E+02
C62-C26-C30-C66	-159.596	1.113988E+03
C62-C26-C30-H105	-35.82	4.907657E+02
C62-C26-C61-H128	-47.894	4.271649E+02
C62-C26-C61-H129	70.832	4.980070E+02
C62-C26-C61-H130	-169.113	4.481570E+02
C62-C63-N65-H133	-8.642	8.776407E+01
C62-C63-N65-H163	-176.241	1.640800E+02
O64-C63-C62-H131	-158.207	3.646442E+02

O64-C63-C62-H132	-41.816	3.833805E+02
O64-C63-N65-H133	172.308	8.151756E+01
O64-C63-N65-H163	4.709	1.435194E+02
N65-C63-C62-H131	22.735	3.898746E+02
N65-C63-C62-H132	139.127	4.113698E+02
C66-C67-C68-O69	-77.937	6.224350E+02
C66-C67-C68-N70	100.358	5.724632E+02
C67-C66-C30-H105	53.176	4.399420E+02
C67-C68-N70-H138	3.792	2.236709E+01
C67-C68-N70-H166	-179.34	8.618700E+01
C68-C67-C66-H134	-65.774	5.490783E+02
C68-C67-C66-H135	50.455	5.853887E+02
O69-C68-C67-H136	41.891	3.966945E+02
O69-C68-C67-H137	159.16	3.846539E+02
O69-C68-N70-H138	-177.907	2.196792E+01
O69-C68-N70-H166	-1.039	8.054731E+01
N70-C68-C67-H136	-139.813	3.757880E+02
N70-C68-C67-H137	-22.545	3.649657E+02
C71-C37-C38-C73	-150.744	1.215211E+03
C71-C37-C38-H108	-29.135	4.855264E+02
C71-C37-C72-H142	65.441	4.824553E+02
C71-C37-C72-H143	-173.941	4.630111E+02
C71-C37-C72-H144	-52.703	4.553185E+02
C72-C37-C38-C73	-27.774	1.034607E+03
C72-C37-C38-H108	93.835	4.538711E+02
C72-C37-C71-H139	-178.309	4.222562E+02
C72-C37-C71-H140	-58.406	3.969044E+02
C72-C37-C71-H141	61.166	4.409184E+02
C73-C74-C75-O76	35.739	6.101451E+02
C73-C74-C75-N77	-145.341	6.490585E+02
C74-C73-C38-H108	169.562	4.693811E+02
C74-C75-N77-H149	-12.098	1.968104E+02
C74-C75-N77-H167	-173.961	2.581407E+02
C75-C74-C73-H145	-36.083	5.350453E+02
C75-C74-C73-H146	78.085	5.976670E+02
O76-C75-C74-H147	161.389	3.954426E+02
O76-C75-C74-H148	-84.37	3.935128E+02
O76-C75-N77-H149	166.76	1.700192E+02
O76-C75-N77-H167	4.897	2.139261E+02
N77-C75-C74-H147	-19.692	4.114293E+02
N77-C75-C74-H148	94.549	4.093408E+02
C79-C42-C43-C85	-43.689	1.008958E+03
C79-C42-C43-H109	-160.82	4.240538E+02
C79-C42-C80-C81	-169.985	9.552753E+02
C79-C42-C80-H156	70.243	4.873713E+02
C79-C42-C80-H157	-44.045	5.179125E+02
C80-C42-C43-C85	78.666	1.164642E+03
C80-C42-C43-H109	-38.466	4.492963E+02

C80-C42-C79-H153	-52.188	4.295082E+02
C80-C42-C79-H154	66.943	4.894003E+02
C80-C42-C79-H155	-172.799	4.526338E+02
C80-C81-C82-O83	28.897	5.926962E+02
C80-C81-C82-N84	-151.656	6.622152E+02
C81-C82-N84-H169	-8.436	1.867646E+02
C82-C81-C80-H156	-9.118	5.410892E+02
C82-C81-C80-H157	105.542	4.655487E+02
C82-N84-C2-H90	-137.383	3.504223E+02
C82-N84-C2-H91	-19.578	3.641865E+02
O83-C82-C81-H158	155.24	3.669091E+02
O83-C82-C81-H159	-92.456	3.645826E+02
O83-C82-N84-H169	170.994	1.665660E+02
N84-C2-C3-H92	-61.829	6.193306E+02
N84-C2-C3-H93	177.65	6.017217E+02
N84-C82-C81-H158	-25.313	3.924110E+02
N84-C82-C81-H159	86.991	3.897510E+02
C85-C43-C44-H110	41.664	4.449995E+02
C85-C86-N87-H162	9.282	1.326036E+02
C85-C86-N87-H165	-170.174	1.273515E+02
C86-C85-C43-H109	-34.025	5.911566E+02
N87-C86-C85-H160	-172.068	3.124440E+02
N87-C86-C85-H161	-57.649	3.975229E+02
O88-C86-C85-H160	9.258	3.025326E+02
O88-C86-C85-H161	123.677	3.816162E+02
O88-C86-N87-H162	-172.09	1.195817E+02
O88-C86-N87-H165	8.455	1.152938E+02
Co89-N49-C44-H110	-85.84	4.116638E+02
H90-C2-C3-H92	178.109	4.160270E+02
H90-C2-C3-H93	57.589	4.080065E+02
H90-C2-N84-H169	54.624	1.263672E+02
H91-C2-C3-H92	58.237	3.537279E+02
H91-C2-C3-H93	-62.283	3.479129E+02
H91-C2-N84-H169	172.43	1.281133E+02
H94-C5-C9-H96	40.369	4.044232E+02
H96-C9-C12-H98	-42.562	4.062463E+02
H96-C9-O29-H104	125.188	5.786703E+01
H97-C11-C55-H120	-154.019	3.154580E+02
H97-C11-C55-H121	89.689	3.094176E+02
H98-C12-C16-H99	-101.41	3.353150E+02
H99-C16-C21-H101	-77.694	4.395132E+02
H99-C16-C21-H102	165.152	4.341954E+02
H101-C21-O32-H106	-61.917	8.607397E+01
H102-C21-O32-H106	57.819	8.567141E+01
H105-C30-C66-H134	-68.179	3.135837E+02
H105-C30-C66-H135	174.954	3.124531E+02
H108-C38-C73-H145	49.018	2.986438E+02
H108-C38-C73-H146	-65.264	3.091975E+02

H109-C43-C44-H110	163.972	2.782345E+02
H109-C43-C85-H160	-157.705	3.922978E+02
H109-C43-C85-H161	84.798	3.404688E+02
H120-C55-C56-H122	74.554	2.868856E+02
H120-C55-C56-H123	-44.056	3.008995E+02
H121-C55-C56-H122	-170.954	2.835738E+02
H121-C55-C56-H123	70.436	2.972583E+02
H134-C66-C67-H136	176.748	3.325084E+02
H134-C66-C67-H137	57.713	3.277723E+02
H135-C66-C67-H136	-67.022	3.454857E+02
H135-C66-C67-H137	173.943	3.403756E+02
H145-C73-C74-H147	-160.531	3.456507E+02
H145-C73-C74-H148	81.125	3.443022E+02
H146-C73-C74-H147	-46.363	3.707458E+02
H146-C73-C74-H148	-164.706	3.691948E+02
H156-C80-C81-H158	-133.757	3.329788E+02
H156-C80-C81-H159	109.062	3.156851E+02
H157-C80-C81-H158	-19.097	3.027484E+02
H157-C80-C81-H159	-136.279	2.883845E+02

Table S3 Atom types of the selected ligands, with charges, $\mathcal{E}$ and σ.

Ligand	Atom	Atom type	Charge	$\mathcal{E}$ (kJ/mol)	σ (nm)
Tetrachloroethene	C1	c2	0.0244283873	3.598240E-01	3.399670E-01
	C2	c2	0.0231595789	1.108760E+00	3.470941E-01
	Cl3	cl	-0.0118464596	1.108760E+00	3.470941E-01
	Cl4	cl	-0.0123700344	1.108760E+00	3.470941E-01
	Cl5	cl	-0.0115306264	1.108760E+00	3.470941E-01
	Cl6	cl	-0.0118408458	1.108760E+00	3.470941E-01
Trichlorofluoroethene	C1	c2	-0.1222272710	3.598240E-01	3.399670E-01
	C2	c2	0.2778824262	1.108760E+00	3.470941E-01
	Cl3	cl	-0.0076027402	1.108760E+00	3.470941E-01
	Cl4	cl	0.0237807706	1.108760E+00	3.470941E-01
	F5	f	-0.1420374753	2.552240E-01	3.118146E-01
	Cl6	cl	-0.0297957103	1.108760E+00	3.470941E-01
Dichlorodifluoroethene	C1	c2	0.1288906884	3.598240E-01	3.399670E-01
	C2	c2	0.1303143661	3.598240E-01	3.399670E-01
	F3	f	-0.1304644761	2.552240E-01	3.118146E-01
	Cl4	cl	0.0010970493	1.108760E+00	3.470941E-01
	F5	f	-0.1310184142	2.552240E-01	3.118146E-01
	Cl6	cl	0.0011807865	1.108760E+00	3.470941E-01
Chlorotrifluoroethene	C1	c2	0.4132618799	3.598240E-01	3.399670E-01
	C2	c2	-0.0491462395	3.598240E-01	3.399670E-01
	F3	f	-0.1631267707	2.552240E-01	3.118146E-01
	F4	f	-0.1311873332	2.552240E-01	3.118146E-01
	F5	f	-0.0805543561	2.552240E-01	3.118146E-01
	Cl6	cl	0.0107528196	1.108760E+00	3.470941E-01
Tetrafluoroethene	C1	c2	0.2335531635	3.598240E-01	3.399670E-01

C2	c2	0.2298709233	3.598240E-01	3.399670E-01
F3	f	-0.1164880499	2.552240E-01	3.118146E-01
F4	f	-0.1161371019	2.552240E-01	3.118146E-01
F5	f	-0.1154224463	2.552240E-01	3.118146E-01
F6	f	-0.1153764887	2.552240E-01	3.118146E-01

Table S4 Atom types of the cobalamin cofactor, with charges, $\mathcal{E}$ and σ.

Cobalamin	Atom	Atom type	Charge	$\mathcal{E}$ (kJ/mol)	σ (nm)
	P1	p5	1.21476	8.368000E-01	3.741775E-01
	C2	c3	-0.00804	4.577296E-01	3.399670E-01
	C3	c3	0.161564	4.577296E-01	3.399670E-01
	C4	c3	-0.22799	4.577296E-01	3.399670E-01
	C5	c3	0.28114	4.577296E-01	3.399670E-01
	N6	na	0.131089	7.112800E-01	3.249999E-01
	C7	c3	0.515853	4.577296E-01	3.399670E-01
	C8	cc	0.193878	3.598240E-01	3.399670E-01
	C9	c3	0.208463	4.577296E-01	3.399670E-01
	O10	os	-0.55322	7.112800E-01	3.000012E-01
	C11	c3	-0.05206	4.577296E-01	3.399670E-01
	C12	c3	0.091287	4.577296E-01	3.399670E-01
	N13	n2	-0.59793	7.112800E-01	3.249999E-01
	O14	os	-0.41738	7.112800E-01	3.000012E-01
	C15	ce	-0.05706	3.598240E-01	3.399670E-01
	C16	c3	0.118575	4.577296E-01	3.399670E-01
	N17	nb	-0.77442	7.112800E-01	3.249999E-01
	O18	o	-0.72034	8.786400E-01	2.959922E-01
	C19	ce	0.213632	3.598240E-01	3.399670E-01
	C20	ca	0.614429	3.598240E-01	3.399670E-01
	C21	c3	0.221986	4.577296E-01	3.399670E-01
	O22	o	-0.75828	8.786400E-01	2.959922E-01
	C23	c2	-0.28322	3.598240E-01	3.399670E-01
	N24	nb	-0.79927	7.112800E-01	3.249999E-01
	O25	os	-0.52375	7.112800E-01	3.000012E-01
	C26	c3	0.258056	4.577296E-01	3.399670E-01
	C27	ca	0.896048	3.598240E-01	3.399670E-01
	N28	nh	-0.88742	7.112800E-01	3.249999E-01
	O29	oh	-0.66389	8.803136E-01	3.066473E-01
	C30	c3	-0.14015	4.577296E-01	3.399670E-01
	C31	ca	-0.64573	3.598240E-01	3.399670E-01
	O32	oh	-0.70218	8.803136E-01	3.066473E-01
	C33	c2	0.261148	3.598240E-01	3.399670E-01
	C34	ca	0.849704	3.598240E-01	3.399670E-01
	C35	c2	-0.51267	3.598240E-01	3.399670E-01
	C36	c2	0.228708	3.598240E-01	3.399670E-01
	C37	c3	0.414693	4.577296E-01	3.399670E-01
	C38	c3	-0.00956	4.577296E-01	3.399670E-01

C39	c2	-0.02623	3.598240E-01	3.399670E-01
C40	c2	0.034337	3.598240E-01	3.399670E-01
C41	c2	0.095434	3.598240E-01	3.399670E-01
C42	c3	0.239788	4.577296E-01	3.399670E-01
C43	c3	0.106086	4.577296E-01	3.399670E-01
C44	c3	-0.18099	4.577296E-01	3.399670E-01
C45	c3	-0.57678	4.577296E-01	3.399670E-01
N46	nh	0.084831	7.112800E-01	3.249999E-01
N47	nh	0.027701	7.112800E-01	3.249999E-01
N48	nh	-0.21885	7.112800E-01	3.249999E-01
N49	nh	-0.11441	7.112800E-01	3.249999E-01
C50	c3	-0.41691	4.577296E-01	3.399670E-01
C51	c3	-0.31575	4.577296E-01	3.399670E-01
C52	c	0.727595	3.598240E-01	3.399670E-01
O53	o	-0.60029	8.786400E-01	2.959922E-01
N54	n	-0.8647	7.112800E-01	3.249999E-01
C55	c3	-0.1678	4.577296E-01	3.399670E-01
C56	c3	-0.2363	4.577296E-01	3.399670E-01
C57	c	0.695444	3.598240E-01	3.399670E-01
N58	n	-0.85899	7.112800E-01	3.249999E-01
O59	o	-0.56814	8.786400E-01	2.959922E-01
C60	c3	-0.32213	4.577296E-01	3.399670E-01
C61	c3	-0.44391	4.577296E-01	3.399670E-01
C62	c3	-0.35606	4.577296E-01	3.399670E-01
C63	c	0.753321	3.598240E-01	3.399670E-01
O64	o	-0.53067	8.786400E-01	2.959922E-01
N65	n	-0.93334	7.112800E-01	3.249999E-01
C66	c3	0.062233	4.577296E-01	3.399670E-01
C67	c3	-0.34481	4.577296E-01	3.399670E-01
C68	c	0.754389	3.598240E-01	3.399670E-01
O69	o	-0.55782	8.786400E-01	2.959922E-01
N70	n	-0.92679	7.112800E-01	3.249999E-01
C71	c3	-0.41825	4.577296E-01	3.399670E-01
C72	c3	-0.50896	4.577296E-01	3.399670E-01
C73	c3	-0.15453	4.577296E-01	3.399670E-01
C74	c3	-0.07009	4.577296E-01	3.399670E-01
C75	c	0.653557	3.598240E-01	3.399670E-01
O76	o	-0.56462	8.786400E-01	2.959922E-01
N77	n	-0.86671	7.112800E-01	3.249999E-01
C78	c3	-0.47213	4.577296E-01	3.399670E-01
C79	c3	-0.46951	4.577296E-01	3.399670E-01
C80	c3	-0.02033	4.577296E-01	3.399670E-01
C81	c3	-0.19111	4.577296E-01	3.399670E-01
C82	c	0.531898	3.598240E-01	3.399670E-01
O83	o	-0.52126	8.786400E-01	2.959922E-01
N84	n	-0.41353	7.112800E-01	3.249999E-01
C85	c3	-0.54719	4.577296E-01	3.399670E-01
C86	c	0.794095	3.598240E-01	3.399670E-01
N87	n	-0.77769	8.786400E-01	2.959922E-01
O88	o	-0.6138	7.112800E-01	3.249999E-01

Co89	UF_Co	0.236163	5.857600E-02	2.558661E-01
H90	h1	0.071874	6.568880E-02	2.471353E-01
H91	h1	0.071874	6.568880E-02	2.471353E-01
H92	h1	0.041143	6.568880E-02	2.471353E-01
H93	h1	0.041143	6.568880E-02	2.471353E-01
H94	h2	0.074346	6.568880E-02	2.293173E-01
H95	h5	0.119438	6.276000E-02	2.421463E-01
H96	h1	0.033136	6.568880E-02	2.471353E-01
H97	hc	0.066511	6.568880E-02	2.649533E-01
H98	h1	0.113745	6.568880E-02	2.471353E-01
H99	h1	0.074742	6.568880E-02	2.471353E-01
H100	h5	0.015122	6.276000E-02	2.421463E-01
H101	h1	0.007655	6.568880E-02	2.471353E-01
H102	h1	0.007655	6.568880E-02	2.471353E-01
H103	hn	0.369101	6.568880E-02	1.069078E-01
H104	ho	0.454745	0.000000E+00	0.000000E+00
H105	hc	0.089523	6.568880E-02	2.649533E-01
H106	ho	0.43676	0.000000E+00	0.000000E+00
H107	ha	0.168701	6.276000E-02	2.599642E-01
H108	hc	0.053206	6.568880E-02	2.649533E-01
H109	hc	0.049828	6.568880E-02	2.649533E-01
H110	h1	0.15277	6.568880E-02	2.471353E-01
H111	hc	0.210751	6.568880E-02	2.649533E-01
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H118	hc	0.071256	6.568880E-02	2.649533E-01
H119	hn	0.412731	6.568880E-02	1.069078E-01
H120	hc	0.090195	6.568880E-02	2.649533E-01
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H124	hn	0.3809	6.568880E-02	1.069078E-01
H125	hc	0.10889	6.568880E-02	2.649533E-01
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H127	hc	0.10889	6.568880E-02	2.649533E-01
H128	hc	0.128921	6.568880E-02	2.649533E-01
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H132	hc	0.097691	6.568880E-02	2.649533E-01
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H143	hc	0.130749	6.568880E-02	2.649533E-01
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Charges and atom names of selected ligands for molecular dynamics simulations

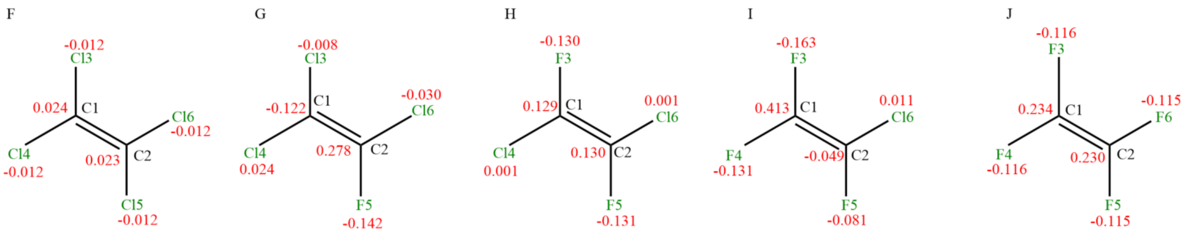

Figure **S1** The atom names and charges of halogenated ethene

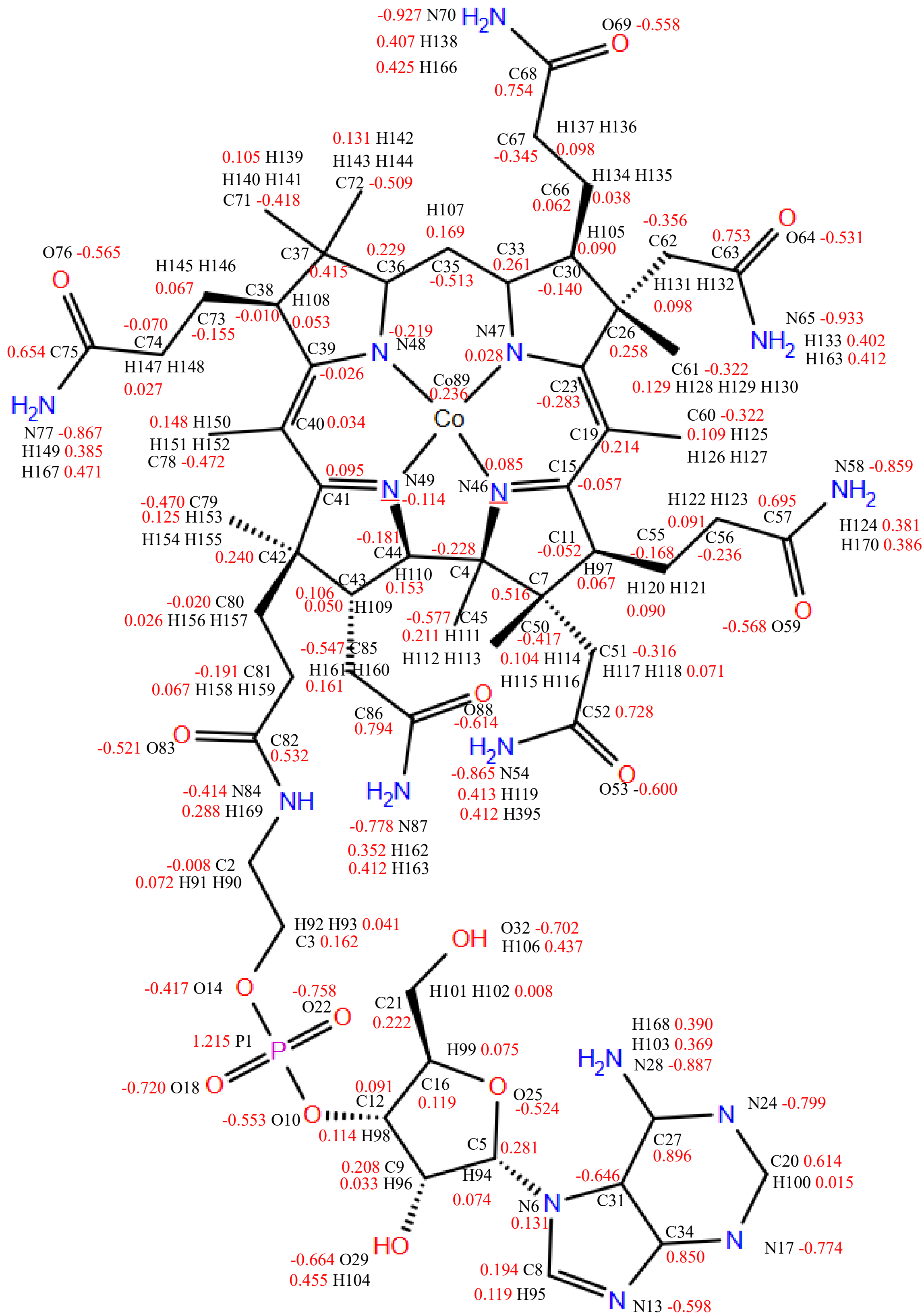

Figure S2 The atom names and charges of cobalamin cofactor

Validation of cobalamin cofactor

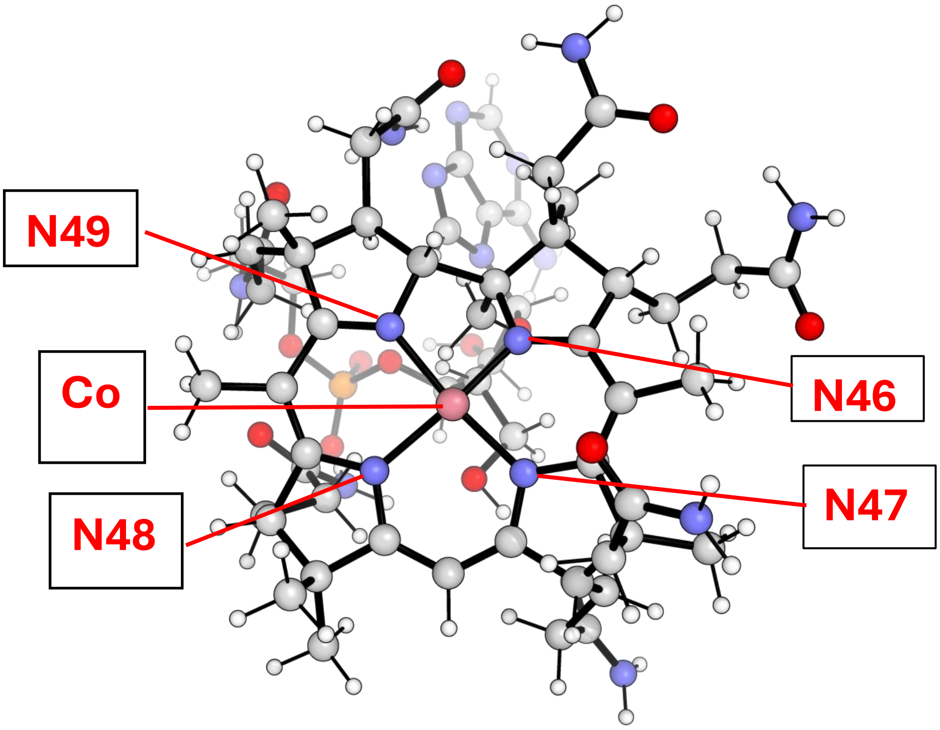

Figure S3 The selected view of the cobalamin cofactor and atoms that are used to validate the structure in MD simulation.

Table S5 Optimized structure of cobalamin cofactor, calculated at B3LYP/ 6-311G(d,p) level of theory using Gaussian 09. The bonds and angles used to validate the geometry in MD simulation.

Optimized structure of Cobalt and nitrogen atom	
N46_Co_bond_distance (Å)	1.86555
N47_Co_bond_distance (Å)	1.90632
N48_Co_bond_distance (Å)	1.90951
N49_Co_bond_distance (Å)	1.88518
angle_N46_Co_N47 (Degree °)	89.70536
angle_N47_Co_N48 (Degree °)	95.30555
angle_N48_Co_N49 (Degree °)	92.02366
angle_N49_Co_N46 (Degree °)	83.33081

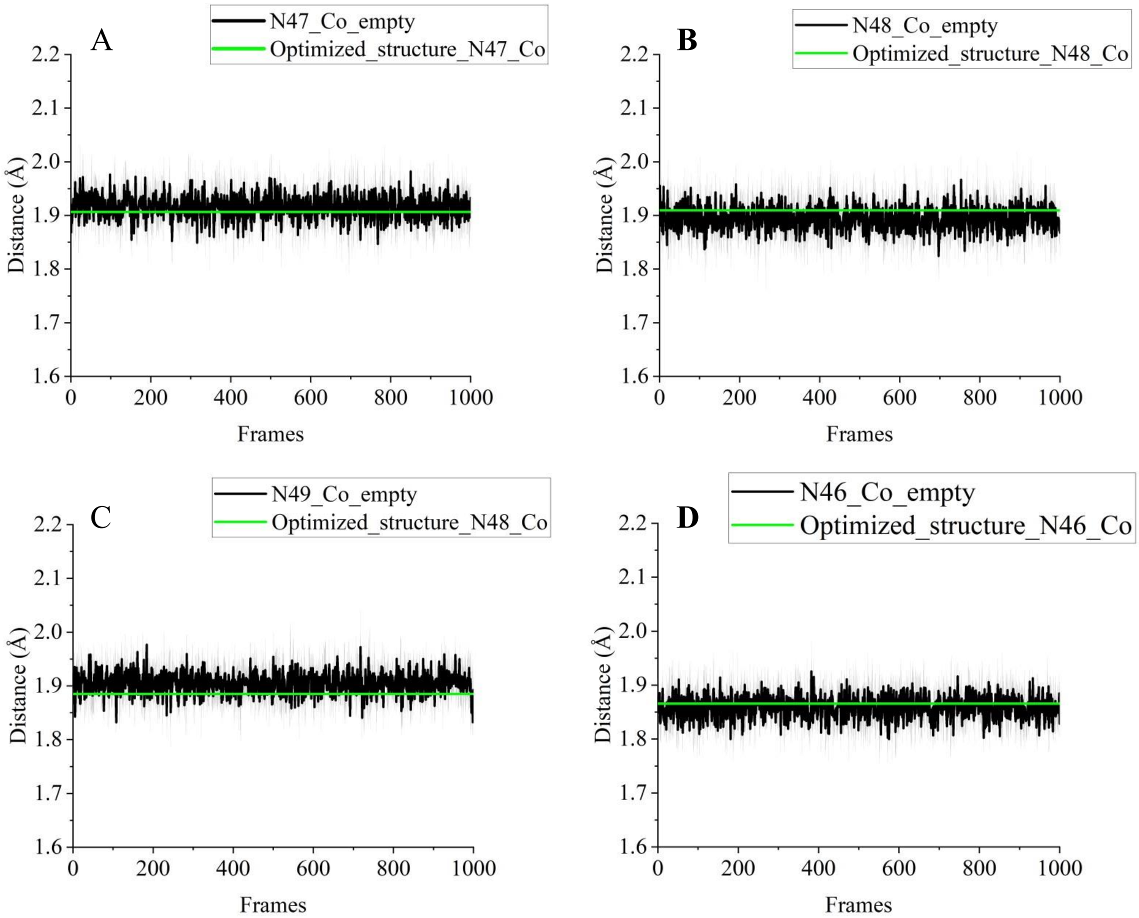

Figure S4 The comparison between the selected bond distances during the MD simulation with the geometry optimized bond distances.

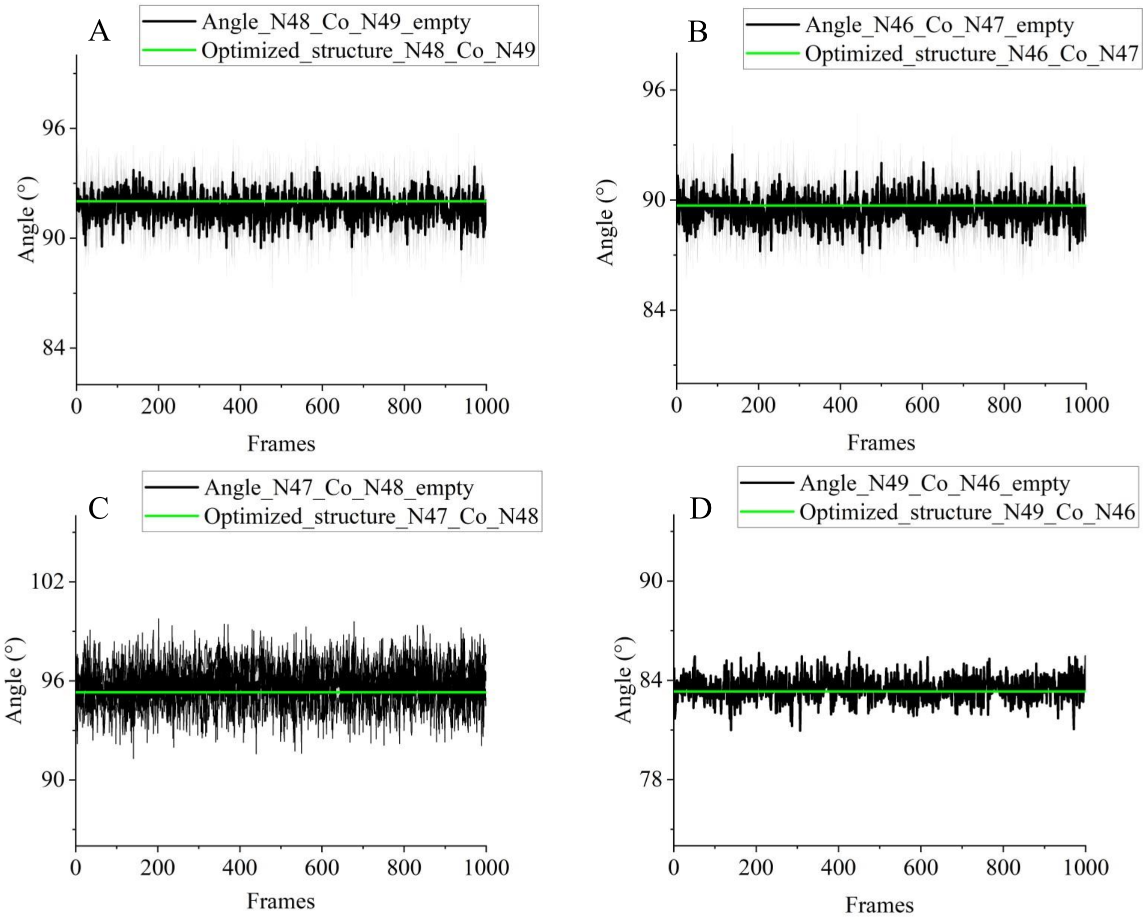

Figure S5 The comparison between the selected angles during the MD simulation with the geometry optimized angles.

Validation of molecular dynamic simulation

The entire PceA chain-B of PDB id: 4UR0 was simulated in presence of the substrate trichloroethene (TCE), which is present in the crystal structure of PceA PDB id: 4UR0. Three independent simulations were conducted to assess the validity and reproducibility of the model. Each independent simulation was run 10 ns, subsequently, the representative structure was generated by using gmx cluster tool. Then, with the obtained representative structures from the three parallel molecular dynamics simulations of PceA, the RMSDs of all non-hydrogen atoms alignments with the PceA crystal structure chain-B are shown in Table 1. For the PceA-TCE complex, the RMSDs of all non-hydrogen atoms alignments are 1.524 Å, 1.427 Å, and 1.332 Å. All RMSD values are below the resolution of the crystal structure 1.65 Å, indicating that the simulated structures are in good agreement with the atomic positions in the crystallographic coordinates, and the results are reproducible.

Table S6. The RMSD (Å) results of alignment between the crystal structure and simulated structures PceA_trichloroethene. The active site includes residues Trp 56, Trp 96, Thr 242, Tyr 246, Arg 305, Trp376 and Tyr382.

System	Replication 1	Replication 2	Replication 3
PceA_trichloroethene	1.524	1.427	1.332
PceA_trichloroethene active site	0.953	0.581	0.9

The RMSDs of the all non-hydrogen atoms alignments between the active sites of the crystal structure and PceA-TCE, the values are reduced to 0.953 Å, 0.581 Å, and 0.900 Å. As shown in the Supplementary Information Figure S6, the alignments between the molecular dynamics simulated structures and the atomic positions of the PceA crystal structure demonstrate good agreement. These findings indicate that the molecular dynamics simulations reliably generate the influence of ligands association in the binding pocket between key residues, as compared to the crystal structure.

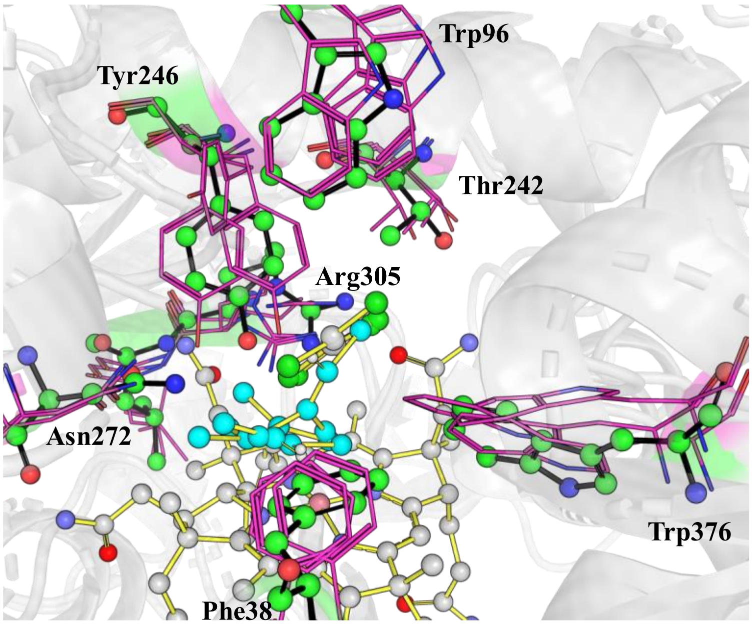

Figure S6 The alignments for between the PceA crystal structure (shown in sticks, with green sphere) chain-B, with trichloroethene associated, and PceA_trichloroethene MD simulation structures (shown in line with purple color), and the trichloroethene ligands (shown in blue sphere).

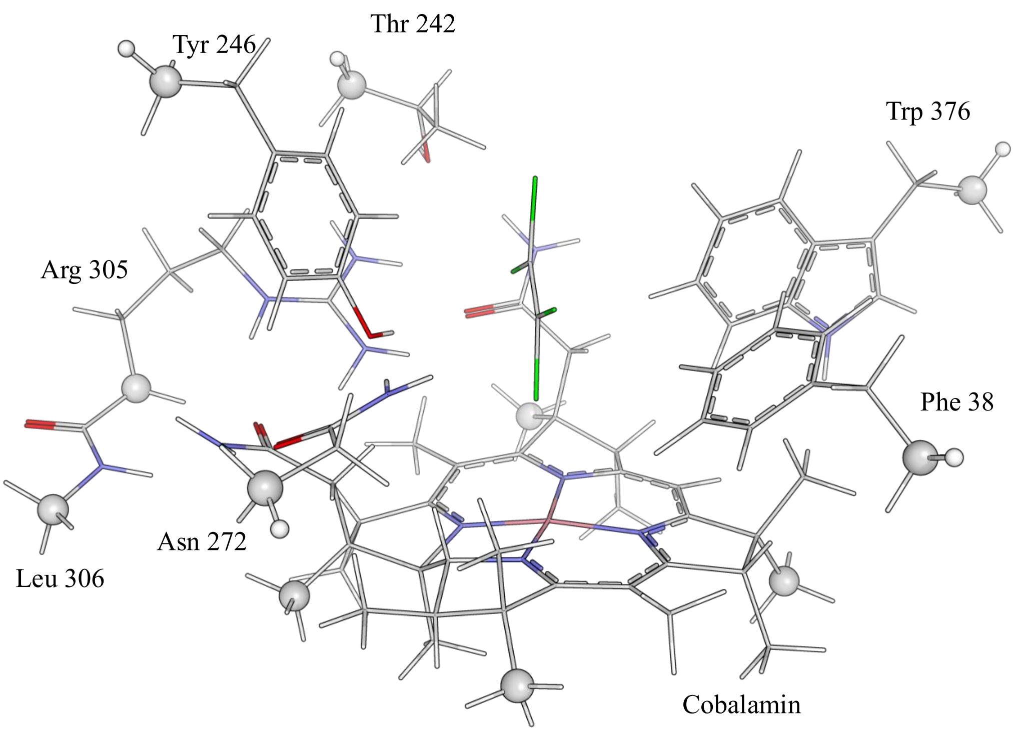

Figure S7 The cluster model of PceA for QM calculation, the representation is the ligand Cl4 binding at the active site centre of PceA cluster, the atoms shown in sphere are the atoms fixed during the optimization.

Cartesian coordinates for all DFT optimized structures

The electronic energies obtained from geometry optimization (B3LYP-D3/lanl08(f) 6-31g(d,p)) are given below

Cl3F1_React, E_{opt}= -5737.8591267

Center number	Atomic Symbol	Coordinates (Angstroms)		
		X	Y	Z
1	C	7.457154	-4.410071	3.946189
2	C	6.143829	-3.785111	4.464369
3	C	5.443029	-2.817426	3.536007
4	C	5.929643	-1.512263	3.361823
5	C	4.260061	-3.175547	2.876046
6	C	5.248592	-0.588683	2.568858
7	C	3.571906	-2.256951	2.079213
8	C	4.062712	-0.959391	1.927538
9	C	-3.498039	7.18852	1.544855
10	C	-2.172047	6.524874	1.138591
11	O	-2.068431	6.429921	-0.309114
12	C	-2.035915	5.108359	1.684867
13	C	-6.711634	2.927653	5.387697
14	C	-5.274601	2.53833	5.793327
15	C	-4.468793	1.918927	4.671405
16	C	-4.883943	0.699138	4.1173
17	C	-3.286524	2.480735	4.172678
18	C	-4.153223	0.0574	3.122741
19	C	-2.533522	1.845823	3.183872
20	C	-2.936288	0.602382	2.643061
21	O	-2.234218	-0.014572	1.693314
22	C	-4.045077	-5.537865	3.752354
23	C	-2.902804	-4.5265	3.77603
24	C	-3.127546	-3.355338	2.820135
25	O	-4.160802	-3.275012	2.13545
26	N	-2.112205	-2.483627	2.719765
27	C	-7.761339	1.441571	-2.167686
28	C	-7.30769	1.814428	-0.743948
29	C	-6.587303	3.165818	-0.67793
30	C	-5.710544	3.296228	0.576908
31	N	-4.557386	2.39605	0.548279
32	C	-3.490995	2.5485	-0.250907
33	N	-3.401295	3.637565	-1.062136

34	N	-2.533462	1.627165	-0.29376
35	C	10.035993	5.629865	0.674265
36	C	8.793907	4.951208	1.258806
37	C	7.914392	4.284394	0.242524
38	C	8.135991	4.136573	-1.102833
39	C	6.636213	3.674763	0.510524
40	N	7.069342	3.469132	-1.690649
41	C	6.13084	3.17895	-0.723583
42	C	5.87114	3.509466	1.676912
43	C	4.890273	2.542205	-0.813203
44	C	4.636575	2.874186	1.592447
45	C	4.152311	2.396102	0.357532
46	Co	1.088363	-1.608765	-2.007652
47	C	-1.50534	-2.716095	-2.331527
48	C	-2.992453	-2.210627	-2.314001
49	C	-2.876764	-0.882562	-3.162153
50	C	-1.423824	-0.484847	-2.980962
51	C	-0.88008	0.821668	-3.31979
52	C	0.474653	1.058169	-3.279277
53	C	1.256642	2.355142	-3.658897
54	C	2.619571	1.722754	-4.071371
55	C	2.658104	0.498063	-3.199911
56	C	3.837838	-0.139414	-2.8615
57	C	3.964337	-1.342943	-2.178546
58	C	5.320732	-1.969381	-1.933889
59	C	4.930744	-3.337097	-1.279541
60	C	3.409326	-3.269684	-1.16099
61	C	2.635337	-4.247464	-0.570565
62	C	1.192833	-4.20795	-0.623107
63	C	0.214406	-5.264704	-0.054759
64	C	-1.101115	-4.873925	-0.806604
65	C	-0.925345	-3.37133	-1.054606
66	C	-1.164243	-3.57453	-3.568587
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72	C	-3.421337	-1.841574	-0.871121
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74	O	-5.324727	-0.582182	-1.688614
75	N	-5.531905	-1.64105	0.306353
76	C	-3.326907	-0.969528	-4.631228
77	C	-1.875483	1.879845	-3.752967
78	C	0.714344	3.251588	-4.783712
79	C	1.56821	3.239489	-2.398954

80	C	0.387364	3.969629	-1.786769
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83	C	2.720306	1.285223	-5.5473
84	C	6.17157	-1.11162	-0.975539
85	C	6.050349	-2.077723	-3.285463
86	C	5.418041	-4.577498	-2.051865
87	C	3.309151	-5.379312	0.181695
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187	H	-2.593352	-4.720333	0.785479
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192	H	-4.386656	-1.231113	-4.667469
193	H	3.647892	0.727857	-5.704168
194	H	-8.282034	2.28247	-2.634916
195	H	-4.823066	-5.289566	4.478545
196	H	-3.68999	-6.548531	3.953547
197	H	-7.386276	2.87896	6.242535
198	H	-6.756284	3.93093	4.952143
199	H	-3.466315	8.279663	1.47448
200	H	-3.768957	6.897644	2.559848
201	H	7.288746	-5.041177	3.068354
202	H	7.850183	-5.019362	4.760364
203	H	10.695047	4.903986	0.186287
204	H	10.600177	6.120617	1.467196
205	H	6.994157	3.248909	-2.669404
206	H	1.281924	5.82639	-1.917185
207	H	-5.110685	-2.227457	1.031929
208	C	-8.747085	0.283666	-2.124469
209	O	-9.958974	0.438329	-2.234685
210	N	-8.156674	-0.916501	-1.863578
211	H	-7.141716	-0.952285	-1.908697
212	C	-8.930981	-2.131264	-1.784921
213	H	-8.452164	-2.836955	-1.098248
214	H	-9.929797	-1.888608	-1.41709
215	H	-9.044862	-2.62228	-2.761197

Cl3F1_Int, E_{opt}= -5738.4370552

Center number	Atomic Symbol	Coordinates (Angstroms)		
		X	Y	Z
1	C	7.598088	-4.295696	4.045093
2	C	6.792231	-3.124157	4.64335
3	C	5.929053	-2.346801	3.67053
4	C	6.110177	-0.972502	3.465279
5	C	4.876467	-2.982896	2.994055
6	C	5.260978	-0.246375	2.624311
7	C	4.0263	-2.266517	2.153474
8	C	4.212346	-0.89363	1.96939
9	C	-3.523069	7.104487	1.465946
10	C	-2.024585	6.892695	1.199603
11	O	-1.783879	6.620917	-0.204373
12	C	-1.458272	5.704039	1.958916
13	C	-7.310021	2.999127	5.404997
14	C	-5.862617	2.612864	5.776019
15	C	-5.089203	2.021052	4.615394
16	C	-5.287405	0.674037	4.271219
17	C	-4.174761	2.759622	3.855304
18	C	-4.589421	0.07214	3.229524
19	C	-3.45363	2.171583	2.810544
20	C	-3.656495	0.823594	2.509841
21	O	-2.928018	0.164668	1.543203
22	C	-3.891481	-5.564297	4.014358
23	C	-2.760205	-4.544511	3.897137
24	C	-3.118111	-3.35922	3.004171
25	O	-4.148806	-3.345941	2.317094
26	N	-2.214351	-2.35732	2.959202
27	C	-7.758643	1.211634	-2.044129
28	C	-8.064733	2.140055	-0.846623
29	C	-7.247955	3.446387	-0.772051
30	C	-6.005637	3.375252	0.124367
31	N	-4.980972	2.458155	-0.373338
32	C	-3.704734	2.745381	-0.624507
33	N	-3.195466	3.976633	-0.552506
34	N	-2.853969	1.716795	-0.893385
35	C	10.017434	5.683936	0.480244
36	C	8.666956	5.243401	1.072415
37	C	7.813098	4.363587	0.200718
38	C	8.116897	3.826756	-1.024687
39	C	6.490754	3.88248	0.531639

40	N	7.063727	3.044867	-1.479772
41	C	6.053247	3.055569	-0.541774
42	C	5.638065	4.064082	1.635215
43	C	4.807686	2.417995	-0.536655
44	C	4.396147	3.435141	1.646401
45	C	3.987819	2.61962	0.569787
46	Co	1.133638	-1.779272	-2.016566
47	C	-1.419927	-2.921211	-2.28556
48	C	-2.903813	-2.406911	-2.250062
49	C	-2.824081	-1.178314	-3.221011
50	C	-1.376813	-0.734758	-3.051336
51	C	-0.872207	0.556216	-3.426741
52	C	0.476035	0.846548	-3.306324
53	C	1.211285	2.171524	-3.668247
54	C	2.606397	1.591559	-4.041427
55	C	2.666495	0.384225	-3.144422
56	C	3.858275	-0.220329	-2.792363
57	C	4.004345	-1.421149	-2.112275
58	C	5.371722	-2.03169	-1.870875
59	C	4.998696	-3.386665	-1.186442
60	C	3.474587	-3.336	-1.081001
61	C	2.723271	-4.315159	-0.455523
62	C	1.290246	-4.306889	-0.521057
63	C	0.324332	-5.336816	0.109042
64	C	-0.99278	-5.00459	-0.661523
65	C	-0.81694	-3.519748	-0.99549
66	C	-1.116937	-3.861065	-3.471698
67	N	-0.639267	-1.669104	-2.483266
68	N	1.427791	-0.072716	-2.851337
69	N	2.974553	-2.187075	-1.687931
70	N	0.639234	-3.337764	-1.128194
71	C	-3.94191	-3.441074	-2.666734
72	C	-3.224228	-1.922058	-0.798414
73	C	-4.577882	-1.281315	-0.587872
74	O	-4.913411	-0.234124	-1.196461
75	N	-5.411155	-1.883998	0.268028
76	C	-3.24023	-1.422055	-4.686735
77	C	-1.859354	1.542994	-4.031262
78	C	0.664535	3.051902	-4.805276
79	C	1.467723	3.031059	-2.379911
80	C	0.334779	3.875967	-1.84002
81	O	-0.843021	3.476084	-1.804784
82	N	0.655376	5.089791	-1.351368
83	C	2.75184	1.143207	-5.50893
84	C	6.244867	-1.160677	-0.946842
85	C	6.080918	-2.172441	-3.228142

86	C	5.507566	-4.639667	-1.923759
87	C	3.422206	-5.400826	0.343238
88	C	0.185867	-5.056688	1.622478
89	C	0.647787	-6.825498	-0.11847
90	C	-2.279817	-5.398456	0.062203
91	Cl	1.62731	1.213752	3.303454
92	C	0.666164	1.23901	1.874549
93	C	0.305694	0.136359	1.20066
94	Cl	0.146822	2.79536	1.308517
95	F	-0.439952	0.232101	0.094331
96	H	7.48029	-2.437203	5.149015
97	H	6.138098	-3.526	5.429272
98	H	6.921096	-0.461723	3.979015
99	H	4.712219	-4.048175	3.140804
100	H	5.416044	0.818301	2.473609
101	H	3.227411	-2.777745	1.629437
102	H	3.557411	-0.337473	1.306315
103	H	8.413832	-3.943202	3.407559
104	H	-1.469638	7.796893	1.487293
105	H	-0.400352	5.565815	1.721246
106	H	-1.999032	4.793638	1.689107
107	H	-1.547423	5.857116	3.037722
108	H	-2.01478	7.408432	-0.715226
109	H	-4.107669	6.43991	0.820309
110	H	-5.333003	3.483258	6.179069
111	H	-5.887604	1.871412	6.583066
112	H	-3.995733	3.804437	4.098392
113	H	-5.997131	0.080814	4.842795
114	H	-2.70853	2.738771	2.260629
115	H	-4.737674	-0.973004	2.979226
116	H	-2.568555	0.790297	0.886626
117	H	-7.715948	2.287877	4.676829
118	H	-2.439868	-4.172919	4.878041
119	H	-1.873246	-5.005023	3.449081
120	H	-2.373022	-1.54911	2.364626
121	H	-1.38935	-2.369582	3.53738
122	H	-4.434657	-5.620485	3.067803
123	H	-9.131101	2.375298	-0.900569
124	H	-7.932241	1.575677	0.088122
125	H	-7.879531	4.239759	-0.355135
126	H	-6.954293	3.776511	-1.77595
127	H	-5.571994	4.372121	0.23561
128	H	-6.28118	3.049978	1.135248
129	H	-5.226726	1.469438	-0.459894
130	H	-1.9622	1.964304	-1.323592
131	H	-3.318694	0.87886	-1.242568

132	H	-3.776416	4.784213	-0.415565
133	H	-2.252307	4.121806	-0.932185
134	H	-6.709886	0.913421	-2.07295
135	H	8.090638	6.136777	1.351708
136	H	8.852412	4.71544	2.018928
137	H	9.010328	3.938352	-1.62186
138	H	5.949277	4.685839	2.470104
139	H	3.732564	3.560867	2.497026
140	H	4.495172	1.766306	-1.344763
141	H	3.022734	2.123581	0.610141
142	H	9.88664	6.351791	-0.376669
143	H	-3.507481	-0.410831	-2.857965
144	H	3.404573	2.308918	-3.816569
145	H	4.764281	0.271653	-3.128903
146	H	5.420972	-3.39153	-0.174084
147	H	-0.939836	-5.571038	-1.59679
148	H	-1.122191	-2.9106	-0.140143
149	H	-1.303234	-3.366339	-4.42339
150	H	-1.711236	-4.776911	-3.435308
151	H	-0.057258	-4.124053	-3.444624
152	H	-4.945618	-3.00037	-2.672271
153	H	-3.760908	-3.834267	-3.666928
154	H	-3.966426	-4.285657	-1.975676
155	H	-2.473498	-1.187263	-0.49761
156	H	-3.143647	-2.761006	-0.108254
157	H	-6.29021	-1.420403	0.461056
158	H	-3.119768	-0.504297	-5.266148
159	H	-2.649433	-2.195192	-5.17906
160	H	-1.688468	2.555689	-3.678515
161	H	-1.809579	1.552637	-5.125822
162	H	-2.887622	1.291875	-3.767531
163	H	1.446432	3.751248	-5.12167
164	H	-0.198468	3.6443	-4.501283
165	H	0.376906	2.455952	-5.67247
166	H	2.326885	3.683329	-2.568962
167	H	1.770134	2.365791	-1.562183
168	H	-0.072288	5.661953	-0.924608
169	H	2.736433	1.989701	-6.201001
170	H	1.949524	0.451058	-5.783813
171	H	5.774751	-1.010219	0.02486
172	H	6.430332	-0.174483	-1.385842
173	H	7.215646	-1.643154	-0.78632
174	H	5.478364	-2.722773	-3.955393
175	H	6.285683	-1.185668	-3.652693
176	H	7.044507	-2.68026	-3.123585
177	H	6.594893	-4.606627	-2.035059

178	H	5.267703	-5.552972	-1.380498
179	H	4.416948	-5.084718	0.65979
180	H	2.873148	-5.632867	1.257676
181	H	3.529976	-6.34159	-0.207435
182	H	-0.278377	-4.085071	1.812084
183	H	-0.432665	-5.827145	2.093268
184	H	1.157283	-5.067771	2.122505
185	H	-0.245875	-7.425942	0.080676
186	H	1.436479	-7.197843	0.537405
187	H	-3.144321	-5.344757	-0.602043
188	H	-2.490732	-4.756194	0.917863
189	Cl	0.702689	-1.46627	1.633001
190	H	5.064563	-4.723264	-2.919651
191	H	0.952017	-7.007699	-1.153995
192	H	-2.226755	-6.429773	0.425161
193	H	-4.293824	-1.710289	-4.745318
194	H	3.702535	0.618107	-5.636203
195	H	-7.988672	1.745365	-2.972752
196	H	-4.611912	-5.278437	4.785362
197	H	-3.521878	-6.564754	4.238197
198	H	-7.974584	2.964846	6.26869
199	H	-7.362207	3.994608	4.953654
200	H	-3.847068	8.130191	1.268832
201	H	-3.779738	6.837987	2.491387
202	H	6.961271	-4.956878	3.44849
203	H	8.007073	-4.87856	4.87059
204	H	10.607573	4.820334	0.155266
205	H	10.58498	6.201897	1.253183
206	H	7.061317	2.515575	-2.3356
207	H	1.61164	5.40827	-1.325837
208	H	-5.047532	-2.554589	0.950489
209	C	-8.688711	0.02236	-1.943256
210	O	-9.906155	0.140718	-2.008545
211	N	-8.075396	-1.164405	-1.658058
212	H	-7.080765	-1.237566	-1.818173
213	C	-8.880391	-2.363474	-1.553798
214	H	-8.305313	-3.14444	-1.04952
215	H	-9.776647	-2.138599	-0.973442
216	H	-9.202941	-2.73609	-2.533885

Cl3F1_Int, E_{opt}= -5738.3883937

1	C	7.598096	-4.295702	4.045102
2	C	6.882495	-3.060525	4.624636
3	C	6.027776	-2.275159	3.650619
4	C	6.20938	-0.8989	3.460498
5	C	4.974513	-2.902602	2.967454
6	C	5.352709	-0.160641	2.637269
7	C	4.119648	-2.174396	2.143147
8	C	4.297519	-0.797261	1.983495
9	C	-3.523154	7.104518	1.466011
10	C	-2.082079	6.62979	1.219991
11	O	-1.764284	6.566597	-0.19189
12	C	-1.835301	5.231105	1.771454
13	C	-6.641516	2.909952	5.456653
14	C	-5.287146	2.25416	5.782324
15	C	-4.503694	1.810046	4.564211
16	C	-5.052574	0.845529	3.703863
17	C	-3.212101	2.269698	4.287084
18	C	-4.330269	0.335309	2.633114
19	C	-2.468382	1.771768	3.210069
20	C	-3.028472	0.786791	2.398507
21	O	-2.352564	0.189453	1.347668
22	C	-3.891475	-5.564374	4.014419
23	C	-2.786119	-4.523397	3.896476
24	C	-3.158935	-3.406239	2.926896
25	O	-4.166975	-3.472264	2.211303
26	N	-2.285662	-2.386391	2.844018
27	C	-7.758689	1.211683	-2.044213
28	C	-8.090271	2.155821	-0.867141
29	C	-7.168963	3.377778	-0.690529
30	C	-5.946632	3.138676	0.211206
31	N	-4.887365	2.311122	-0.363791
32	C	-3.708488	2.76596	-0.805065
33	N	-3.529373	4.061258	-1.112719
34	N	-2.68524	1.913253	-0.927797
35	C	10.017452	5.683942	0.480231
36	C	8.642595	5.287327	1.030035
37	C	7.818426	4.42621	0.117612
38	C	8.150415	3.93603	-1.119764
39	C	6.502417	3.916171	0.413377
40	N	7.116611	3.152279	-1.614835
41	C	6.094677	3.115769	-0.689788
42	C	5.636948	4.053315	1.511917
43	C	4.865869	2.450274	-0.710552
44	C	4.410991	3.396177	1.495123

45	C	4.034793	2.595711	0.39595
46	Co	1.097461	-1.66697	-1.805298
47	C	-1.438497	-2.880522	-2.198975
48	C	-2.932629	-2.397259	-2.191125
49	C	-2.851568	-1.125491	-3.115584
50	C	-1.402811	-0.686299	-2.942353
51	C	-0.889462	0.609758	-3.342896
52	C	0.452546	0.905016	-3.226193
53	C	1.201671	2.208862	-3.640794
54	C	2.586881	1.594407	-4.006815
55	C	2.638102	0.401318	-3.091913
56	C	3.833069	-0.226817	-2.779321
57	C	3.986905	-1.415732	-2.081054
58	C	5.355588	-2.041351	-1.872527
59	C	4.977471	-3.397316	-1.18956
60	C	3.461783	-3.309312	-1.007405
61	C	2.710112	-4.285723	-0.381975
62	C	1.270069	-4.27736	-0.441318
63	C	0.309648	-5.344	0.141819
64	C	-1.002204	-5.000082	-0.63354
65	C	-0.849803	-3.499482	-0.907878
66	C	-1.085759	-3.772744	-3.408449
67	N	-0.683587	-1.607543	-2.361514
68	N	1.405945	-0.00588	-2.743681
69	N	2.967521	-2.141373	-1.577847
70	N	0.606863	-3.298364	-0.999297
71	C	-3.94186	-3.441056	-2.66671
72	C	-3.315243	-1.973646	-0.743698
73	C	-4.710664	-1.407237	-0.591593
74	O	-5.082796	-0.390059	-1.212576
75	N	-5.538742	-2.061913	0.241315
76	C	-3.28786	-1.324859	-4.580464
77	C	-1.877198	1.570197	-3.981883
78	C	0.664518	3.05187	-4.805315
79	C	1.471603	3.121316	-2.399339
80	C	0.343278	3.997102	-1.87553
81	O	-0.859539	3.736957	-2.05095
82	N	0.709697	5.084833	-1.17135
83	C	2.72423	1.112825	-5.465161
84	C	6.226157	-1.172635	-0.940066
85	C	6.081007	-2.172475	-3.228244
86	C	5.401168	-4.65259	-1.975183
87	C	3.408779	-5.386753	0.394126
88	C	0.155561	-5.093439	1.657907
89	C	0.647817	-6.825526	-0.118475
90	C	-2.292135	-5.448106	0.049991

91	Cl	1.827784	1.518535	3.26344
92	C	0.879182	1.060039	1.869152
93	C	0.446787	-0.17912	1.543988
94	Cl	0.365462	2.410002	0.878765
95	F	0.544615	-0.664363	-0.02655
96	H	7.62451	-2.394369	5.079181
97	H	6.235355	-3.393517	5.447874
98	H	7.0223	-0.394809	3.977763
99	H	4.805757	-3.968309	3.104393
100	H	5.503066	0.907171	2.505449
101	H	3.306199	-2.67322	1.631953
102	H	3.622774	-0.230367	1.350156
103	H	8.407799	-4.012353	3.366235
104	H	-1.386104	7.329571	1.707422
105	H	-0.815624	4.903647	1.556515
106	H	-2.528036	4.510969	1.328691
107	H	-1.974258	5.217677	2.856017
108	H	-1.876557	7.449215	-0.569906
109	H	-4.204778	6.566807	0.798183
110	H	-4.680139	2.933299	6.390929
111	H	-5.466939	1.369246	6.406816
112	H	-2.760551	3.016482	4.935322
113	H	-6.047793	0.455452	3.901132
114	H	-1.461918	2.128079	3.023006
115	H	-4.737785	-0.451548	2.009684
116	H	-1.355475	0.186782	1.458935
117	H	-7.152502	2.378058	4.647192
118	H	-2.520087	-4.087737	4.866841
119	H	-1.865494	-4.971841	3.508361
120	H	-2.433976	-1.635976	2.179882
121	H	-1.463704	-2.339605	3.424628
122	H	-4.434016	-5.629339	3.068293
123	H	-9.123791	2.485903	-1.00427
124	H	-8.085784	1.576043	0.06701
125	H	-7.743827	4.184515	-0.219124
126	H	-6.84297	3.759725	-1.66752
127	H	-5.49778	4.091685	0.50851
128	H	-6.262562	2.649568	1.140363
129	H	-5.025578	1.301435	-0.433604
130	H	-1.82005	2.263234	-1.330899
131	H	-2.631397	1.12801	-0.283398
132	H	-4.318052	4.650245	-1.321341
133	H	-2.592019	4.389204	-1.342384
134	H	-6.714429	0.902573	-2.030629
135	H	8.07597	6.194725	1.282878
136	H	8.776348	4.756015	1.983109

137	H	9.050663	4.080326	-1.699077
138	H	5.927497	4.660603	2.364629
139	H	3.736599	3.482561	2.340368
140	H	4.57499	1.816137	-1.538442
141	H	3.09005	2.064931	0.420683
142	H	9.930077	6.33966	-0.391906
143	H	-3.504149	-0.357647	-2.696712
144	H	3.39911	2.298462	-3.7943
145	H	4.731884	0.235814	-3.168682
146	H	5.447437	-3.433502	-0.199546
147	H	-0.922882	-5.526292	-1.590551
148	H	-1.179325	-2.914694	-0.044815
149	H	-1.284585	-3.253775	-4.345347
150	H	-1.64895	-4.707939	-3.407558
151	H	-0.018166	-4.005433	-3.380912
152	H	-4.949689	-3.013992	-2.693338
153	H	-3.726418	-3.813243	-3.668631
154	H	-3.97316	-4.297638	-1.991
155	H	-2.621396	-1.208443	-0.393767
156	H	-3.22561	-2.82731	-0.072801
157	H	-6.454116	-1.656086	0.392107
158	H	-3.174204	-0.399778	-5.14701
159	H	-2.714198	-2.096502	-5.09727
160	H	-1.66736	2.59789	-3.709628
161	H	-1.862113	1.493766	-5.074983
162	H	-2.894752	1.362856	-3.652243
163	H	1.458686	3.723196	-5.15013
164	H	-0.181961	3.671362	-4.514644
165	H	0.35886	2.433085	-5.650455
166	H	2.31426	3.778383	-2.642557
167	H	1.80905	2.506846	-1.557898
168	H	-0.026077	5.65942	-0.757384
169	H	2.719283	1.94766	-6.169981
170	H	1.910298	0.429622	-5.72827
171	H	5.756977	-1.030044	0.032774
172	H	6.409782	-0.183719	-1.372455
173	H	7.195494	-1.65879	-0.78502
174	H	5.488449	-2.713888	-3.970132
175	H	6.300652	-1.182675	-3.637138
176	H	7.038016	-2.687254	-3.107568
177	H	6.481091	-4.656103	-2.143426
178	H	5.156615	-5.566077	-1.434606
179	H	4.413106	-5.084353	0.693097
180	H	2.872032	-5.612793	1.317
181	H	3.492682	-6.325602	-0.163184
182	H	-0.270916	-4.108086	1.859434

183	H	-0.495966	-5.855953	2.09446
184	H	1.116663	-5.149906	2.173813
185	H	-0.245723	-7.432953	0.053944
186	H	1.427413	-7.207739	0.542357
187	H	-3.148376	-5.364821	-0.621549
188	H	-2.521509	-4.859344	0.936833
189	Cl	0.870418	-1.53981	2.537386
190	H	4.903317	-4.701341	-2.94758
191	H	0.96786	-6.983995	-1.153352
192	H	-2.231104	-6.498321	0.351641
193	H	-4.343953	-1.603817	-4.623729
194	H	3.66694	0.572226	-5.586542
195	H	-7.949395	1.738203	-2.986138
196	H	-4.62033	-5.289823	4.782022
197	H	-3.521873	-6.564868	4.238163
198	H	-7.30608	2.875391	6.320363
199	H	-6.520367	3.950423	5.140272
200	H	-3.65073	8.180516	1.318368
201	H	-3.779675	6.837967	2.491373
202	H	6.906095	-4.941119	3.494579
203	H	8.007068	-4.878563	4.870595
204	H	10.59249	4.799657	0.184686
205	H	10.585004	6.201911	1.253197
206	H	7.133951	2.6563	-2.489996
207	H	1.67671	5.266301	-0.947653
208	H	-5.161035	-2.723053	0.925406
209	C	-8.701385	0.029674	-1.96068
210	O	-9.917315	0.151193	-2.048541
211	N	-8.09113	-1.15641	-1.665166
212	H	-7.090523	-1.210665	-1.794809
213	C	-8.880366	-2.363481	-1.553779
214	H	-8.319024	-3.117133	-0.994426
215	H	-9.80699	-2.131654	-1.025895
216	H	-9.150727	-2.780485	-2.532281

Cl2F2_React, E_{opt}= -5377.4889161

Center number	Atomic Symbol	Coordinates (Angstroms)		
		X	Y	Z
1	C	7.59812	-4.295627	4.045177
2	C	6.759727	-3.151006	4.649111
3	C	5.862238	-2.409233	3.682132
4	C	6.064992	-1.056614	3.382194
5	C	4.754755	-3.054757	3.109524
6	C	5.178203	-0.356809	2.559412
7	C	3.864152	-2.362673	2.289999
8	C	4.067363	-1.006402	2.018846
9	C	-3.523151	7.104529	1.466007
10	C	-2.041629	6.796238	1.20993
11	O	-1.765849	6.662571	-0.208899
12	C	-1.601676	5.485496	1.84237
13	C	-7.310112	2.999209	5.404992
14	C	-5.862016	2.572149	5.73302
15	C	-5.097141	1.980126	4.559932
16	C	-5.439783	0.70475	4.079193
17	C	-4.00212	2.610891	3.952997
18	C	-4.713738	0.075355	3.073785
19	C	-3.256492	1.989145	2.945401
20	C	-3.574065	0.686563	2.491848
21	O	-2.840768	0.045162	1.58228
22	C	-3.891428	-5.564269	4.014509
23	C	-2.78698	-4.505096	3.967886
24	C	-3.108941	-3.391123	2.968256
25	O	-3.977098	-3.580082	2.097967
26	N	-2.366388	-2.276254	3.011999
27	C	-7.758642	1.211684	-2.044178
28	C	-8.095725	2.006889	-0.760963
29	C	-7.206205	3.231035	-0.462159
30	C	-5.998911	2.946581	0.444437
31	N	-4.952699	2.120325	-0.156726
32	C	-3.724113	2.543617	-0.480421
33	N	-3.519846	3.845599	-0.786826
34	N	-2.714344	1.671972	-0.479469
35	C	10.017475	5.68385	0.480303
36	C	8.687283	5.178436	1.05686
37	C	7.834001	4.416189	0.082948
38	C	8.110721	4.147528	-1.233381
39	C	6.553145	3.806788	0.357751
40	N	7.080265	3.41005	-1.798699
41	C	6.113072	3.183336	-0.844338
42	C	5.736675	3.721288	1.499847

43	C	4.905434	2.485455	-0.922635
44	C	4.531161	3.027259	1.426824
45	C	4.125581	2.407636	0.226418
46	Co	1.110432	-1.740498	-1.945868
47	C	-1.464874	-2.888351	-2.235282
48	C	-2.948331	-2.391949	-2.172251
49	C	-2.881935	-1.085065	-3.050093
50	C	-1.426053	-0.668508	-2.917951
51	C	-0.905307	0.634203	-3.297613
52	C	0.442355	0.902601	-3.237081
53	C	1.198836	2.20616	-3.638287
54	C	2.582903	1.597191	-4.020159
55	C	2.637038	0.381608	-3.139751
56	C	3.830165	-0.24895	-2.82688
57	C	3.982877	-1.436644	-2.127151
58	C	5.349578	-2.039278	-1.876707
59	C	4.979742	-3.392107	-1.180649
60	C	3.460153	-3.332526	-1.039456
61	C	2.703614	-4.30039	-0.409158
62	C	1.263679	-4.295174	-0.483719
63	C	0.29436	-5.346174	0.109025
64	C	-1.003813	-5.028811	-0.699449
65	C	-0.862476	-3.527432	-0.967365
66	C	-1.14841	-3.761744	-3.467861
67	N	-0.700391	-1.619788	-2.398541
68	N	1.403776	-0.033517	-2.805759
69	N	2.966441	-2.185502	-1.655134
70	N	0.59896	-3.332681	-1.069882
71	C	-3.941932	-3.441125	-2.666767
72	C	-3.316304	-2.006162	-0.714016
73	C	-4.729001	-1.479891	-0.542995
74	O	-5.173737	-0.541426	-1.237425
75	N	-5.489287	-2.094119	0.373773
76	C	-3.372274	-1.21686	-4.504378
77	C	-1.911405	1.645125	-3.812496
78	C	0.664539	3.051894	-4.80536
79	C	1.467558	3.112015	-2.384851
80	C	0.31463	3.934832	-1.821585
81	O	-0.827807	3.473834	-1.70165
82	N	0.602937	5.183992	-1.40431
83	C	2.723565	1.142894	-5.48857
84	C	6.187441	-1.145767	-0.938429
85	C	6.080936	-2.17245	-3.228207
86	C	5.452241	-4.65177	-1.930504
87	C	3.394788	-5.382832	0.398477
88	C	0.106687	-5.065864	1.617482

89	C	0.647791	-6.825528	-0.118474
90	C	-2.298116	-5.50823	-0.04637
91	F	1.523564	1.103128	2.979443
92	C	0.858245	1.239602	1.836708
93	C	0.369801	0.185585	1.180618
94	Cl	0.668935	2.858301	1.297404
95	F	-0.239297	0.321375	-0.002045
96	H	7.42844	-2.439573	5.1464
97	H	6.126051	-3.57218	5.440664
98	H	6.91755	-0.538277	3.813932
99	H	4.574355	-4.103611	3.335274
100	H	5.351101	0.692121	2.340879
101	H	3.006761	-2.875122	1.869743
102	H	3.371161	-0.463521	1.387176
103	H	8.413818	-3.919789	3.421153
104	H	-1.422716	7.612158	1.613047
105	H	-0.550697	5.295082	1.613496
106	H	-2.202839	4.653393	1.466427
107	H	-1.708356	5.527462	2.929771
108	H	-2.028886	7.482535	-0.647633
109	H	-4.133654	6.475161	0.809296
110	H	-5.308439	3.423022	6.147218
111	H	-5.893273	1.819417	6.531478
112	H	-3.703166	3.600708	4.296127
113	H	-6.284276	0.18242	4.526832
114	H	-2.382137	2.479656	2.523694
115	H	-4.982935	-0.922535	2.742802
116	H	-7.741213	2.316835	4.663216
117	H	-2.585781	-4.07363	4.955199
118	H	-1.840286	-4.947535	3.636659
119	H	-2.558688	-1.465934	2.378787
120	H	-1.750232	-2.118765	3.795035
121	H	-4.384049	-5.613882	3.041183
122	H	-9.140778	2.320441	-0.844345
123	H	-8.055996	1.322978	0.098713
124	H	-7.80816	3.987702	0.056267
125	H	-6.868622	3.694622	-1.399645
126	H	-5.536984	3.882033	0.773945
127	H	-6.325408	2.432721	1.356128
128	H	-5.077385	1.111837	-0.174863
129	H	-1.826533	2.015341	-0.827528
130	H	-2.71287	0.941003	0.306893
131	H	-4.291511	4.397699	-1.124765
132	H	-2.577898	4.14106	-1.028679
133	H	-6.707034	0.92415	-2.06365
134	H	8.113891	6.030445	1.448478

135	H	8.889724	4.54009	1.928305
136	H	8.971159	4.42325	-1.82502
137	H	6.04602	4.190014	2.429654
138	H	3.894301	2.944664	2.30261
139	H	4.593242	2.002071	-1.838712
140	H	3.194813	1.849318	0.200169
141	H	9.853072	6.38047	-0.348423
142	H	-3.510152	-0.331356	-2.571962
143	H	3.394963	2.296199	-3.794776
144	H	4.73266	0.225852	-3.189119
145	H	5.420797	-3.400149	-0.17702
146	H	-0.886527	-5.549774	-1.656474
147	H	-1.187327	-2.952851	-0.096265
148	H	-1.394289	-3.235374	-4.389876
149	H	-1.699967	-4.70317	-3.453403
150	H	-0.078793	-3.992023	-3.489439
151	H	-4.960174	-3.041165	-2.643468
152	H	-3.750485	-3.764697	-3.690458
153	H	-3.929309	-4.323323	-2.025904
154	H	-2.656238	-1.222355	-0.336552
155	H	-3.203593	-2.863298	-0.053134
156	H	-6.381493	-1.668042	0.58734
157	H	-3.299343	-0.263742	-5.03018
158	H	-2.811926	-1.955906	-5.081896
159	H	-1.646038	2.651984	-3.513927
160	H	-1.996686	1.610227	-4.904165
161	H	-2.899911	1.45374	-3.395189
162	H	1.456722	3.723974	-5.150833
163	H	-0.182846	3.672747	-4.513994
164	H	0.350213	2.432759	-5.647048
165	H	2.296233	3.783413	-2.633449
166	H	1.823435	2.490138	-1.556193
167	H	-0.136246	5.735397	-0.961188
168	H	2.722515	1.991354	-6.176212
169	H	1.909849	0.466664	-5.769181
170	H	5.699388	-1.013113	0.027579
171	H	6.353288	-0.15424	-1.370026
172	H	7.164256	-1.610466	-0.767398
173	H	5.495178	-2.723814	-3.968266
174	H	6.296303	-1.184061	-3.641788
175	H	7.04058	-2.680324	-3.100804
176	H	6.53711	-4.634887	-2.060931
177	H	5.207347	-5.560912	-1.382856
178	H	4.38885	-5.066	0.715103
179	H	2.839875	-5.598831	1.312703
180	H	3.495556	-6.327584	-0.144916

181	H	-0.364468	-4.096755	1.801099
182	H	-0.533141	-5.836797	2.054875
183	H	1.058753	-5.086693	2.153402
184	H	-0.245225	-7.435839	0.042067
185	H	1.412819	-7.194295	0.566605
186	H	-3.132546	-5.466318	-0.747533
187	H	-2.582915	-4.919397	0.824693
188	Cl	0.524934	-1.428727	1.724062
189	H	4.990357	-4.725367	-2.918743
190	H	0.992449	-6.999611	-1.142888
191	H	-2.208119	-6.552872	0.266603
192	H	-4.424614	-1.510039	-4.507704
193	H	3.666504	0.604599	-5.618147
194	H	-7.968569	1.835819	-2.918919
195	H	-4.656164	-5.308978	4.753413
196	H	-3.521877	-6.564876	4.238075
197	H	-7.974626	2.964887	6.268767
198	H	-7.345803	4.004687	4.972988
199	H	-3.775033	8.153924	1.286702
200	H	-3.779674	6.837955	2.491371
201	H	6.979869	-4.962955	3.435271
202	H	8.007086	-4.878635	4.870554
203	H	10.628664	4.853732	0.110651
204	H	10.584976	6.201994	1.253151
205	H	7.058816	3.080297	-2.748995
206	H	1.533771	5.56494	-1.469492
207	H	-5.045246	-2.70896	1.064531
208	C	-8.683395	0.00947	-2.085395
209	O	-9.878004	0.106291	-2.3467
210	N	-8.098334	-1.153279	-1.681552
211	H	-7.088312	-1.170223	-1.610586
212	C	-8.880379	-2.36347	-1.553785
213	H	-8.369685	-3.058286	-0.880875
214	H	-9.861917	-2.113147	-1.145749
215	H	-9.040294	-2.86408	-2.517742

Cl2F2_Int, E_{opt}= -5378.0732168

		Coordinates (Angstroms)		

Center number	Atomic Symbol	X	Y	Z
1	C	7.598088	-4.295696	4.045093
2	C	6.766804	-3.150158	4.657182
3	C	5.912578	-2.361134	3.688416
4	C	6.100793	-0.986205	3.499409
5	C	4.865302	-2.987792	2.995523
6	C	5.26099	-0.248949	2.659714
7	C	4.024093	-2.259642	2.156397
8	C	4.214072	-0.884671	1.991698
9	C	-3.523069	7.104487	1.465946
10	C	-2.027973	6.884135	1.18742
11	O	-1.800829	6.631682	-0.222733
12	C	-1.457857	5.682441	1.924256
13	C	-7.310021	2.999127	5.404997
14	C	-5.858549	2.632947	5.779774
15	C	-5.086211	2.031629	4.624114
16	C	-5.285243	0.681235	4.293623
17	C	-4.178591	2.765388	3.851826
18	C	-4.595474	0.071537	3.251575
19	C	-3.465378	2.169334	2.806159
20	C	-3.670076	0.818884	2.518096
21	O	-2.952021	0.153784	1.549897
22	C	-3.891481	-5.564297	4.014358
23	C	-2.760658	-4.544187	3.895881
24	C	-3.121552	-3.359121	3.003829
25	O	-4.163488	-3.338987	2.334022
26	N	-2.210362	-2.36488	2.941559
27	C	-7.758643	1.211634	-2.044129
28	C	-8.078748	2.147547	-0.856328
29	C	-7.262539	3.453704	-0.780279
30	C	-6.021942	3.381359	0.118161
31	N	-4.998817	2.460448	-0.375571
32	C	-3.721909	2.744451	-0.627617
33	N	-3.212676	3.976551	-0.566331
34	N	-2.872502	1.713601	-0.886983
35	C	10.017434	5.683936	0.480244
36	C	8.671393	5.223913	1.063107
37	C	7.811303	4.410848	0.136338
38	C	8.093063	4.02336	-1.149206
39	C	6.512984	3.857407	0.447448
40	N	7.04732	3.269547	-1.662442
41	C	6.067223	3.144209	-0.701612
42	C	5.689791	3.882742	1.587909
43	C	4.844106	2.468164	-0.734369

44	C	4.470864	3.209821	1.562827
45	C	4.057495	2.507146	0.411841
46	Co	1.128284	-1.775788	-2.009975
47	C	-1.42367	-2.916909	-2.277686
48	C	-2.90835	-2.40532	-2.244015
49	C	-2.829337	-1.173705	-3.211579
50	C	-1.382603	-0.729188	-3.040865
51	C	-0.877354	0.560823	-3.418363
52	C	0.471847	0.848413	-3.302873
53	C	1.20929	2.170426	-3.670194
54	C	2.601403	1.586974	-4.047825
55	C	2.661726	0.379201	-3.151624
56	C	3.852384	-0.230754	-2.808043
57	C	3.998008	-1.426998	-2.120901
58	C	5.365339	-2.034129	-1.874705
59	C	4.993189	-3.389791	-1.191998
60	C	3.469842	-3.336512	-1.080669
61	C	2.719676	-4.315065	-0.453752
62	C	1.286816	-4.305734	-0.517882
63	C	0.322646	-5.336932	0.111577
64	C	-0.996242	-5.003012	-0.655759
65	C	-0.820832	-3.517316	-0.987887
66	C	-1.117738	-3.85402	-3.465424
67	N	-0.644729	-1.66337	-2.472521
68	N	1.422851	-0.071717	-2.849492
69	N	2.968678	-2.188548	-1.688507
70	N	0.635098	-3.335018	-1.12133
71	C	-3.94191	-3.441074	-2.666734
72	C	-3.233019	-1.926896	-0.791359
73	C	-4.58689	-1.286929	-0.579715
74	O	-4.92294	-0.239132	-1.187258
75	N	-5.419577	-1.889828	0.276448
76	C	-3.244415	-1.414359	-4.678138
77	C	-1.863636	1.550546	-4.01941
78	C	0.664535	3.051902	-4.805276
79	C	1.466626	3.035086	-2.385779
80	C	0.328782	3.877261	-1.851823
81	O	-0.836052	3.449541	-1.763253
82	N	0.630381	5.11898	-1.424334
83	C	2.74157	1.139123	-5.516042
84	C	6.228271	-1.15803	-0.945512
85	C	6.080918	-2.172441	-3.228142
86	C	5.494151	-4.642387	-1.935637
87	C	3.419943	-5.40099	0.343388
88	C	0.187941	-5.056756	1.625276
89	C	0.647787	-6.825498	-0.11847

90	C	-2.281758	-5.396571	0.070862
91	F	1.739053	1.175338	2.703524
92	C	0.847191	1.245714	1.720849
93	C	0.347832	0.15134	1.141657
94	Cl	0.388999	2.844741	1.28071
95	F	-0.519356	0.249391	0.126072
96	H	7.437609	-2.467365	5.190921
97	H	6.105192	-3.579846	5.421765
98	H	6.909467	-0.484095	4.025137
99	H	4.698733	-4.054603	3.128334
100	H	5.424122	0.81546	2.519763
101	H	3.229382	-2.762967	1.61867
102	H	3.566992	-0.31995	1.32918
103	H	8.414231	-3.918672	3.422409
104	H	-1.465415	7.780958	1.483482
105	H	-0.398886	5.55736	1.683403
106	H	-1.991445	4.773402	1.636
107	H	-1.547484	5.814871	3.005732
108	H	-2.04142	7.425087	-0.719905
109	H	-4.116247	6.446967	0.820736
110	H	-5.336011	3.514185	6.168111
111	H	-5.874396	1.903135	6.597567
112	H	-4.000379	3.812732	4.084362
113	H	-5.990335	0.092656	4.875612
114	H	-2.727134	2.734268	2.243838
115	H	-4.744118	-0.975614	3.009696
116	H	-2.580802	0.776105	0.897616
117	H	-7.707675	2.276815	4.683276
118	H	-2.440574	-4.171298	4.876425
119	H	-1.873163	-5.003914	3.448282
120	H	-2.374524	-1.553998	2.351767
121	H	-1.375568	-2.383361	3.505374
122	H	-4.435799	-5.620999	3.068504
123	H	-9.144316	2.382642	-0.924533
124	H	-7.957572	1.58932	0.083652
125	H	-7.894829	4.246928	-0.364082
126	H	-6.967271	3.783843	-1.783729
127	H	-5.5855	4.377119	0.228853
128	H	-6.300305	3.058549	1.129075
129	H	-5.244683	1.471332	-0.454645
130	H	-1.977121	1.954038	-1.313285
131	H	-3.33759	0.871747	-1.226597
132	H	-3.792605	4.786293	-0.438353
133	H	-2.263762	4.115624	-0.932839
134	H	-6.709744	0.913793	-2.061009
135	H	8.105715	6.104275	1.400132

136	H	8.858719	4.636469	1.97322
137	H	8.966445	4.225755	-1.751873
138	H	6.007369	4.415746	2.479732
139	H	3.831695	3.205093	2.440953
140	H	4.525931	1.903215	-1.60182
141	H	3.115936	1.96846	0.421228
142	H	9.877441	6.363186	-0.36677
143	H	-3.51368	-0.407783	-2.84706
144	H	3.401301	2.303486	-3.825821
145	H	4.758805	0.254449	-3.152192
146	H	5.420454	-3.398701	-0.181812
147	H	-0.946026	-5.568274	-1.591912
148	H	-1.125734	-2.909126	-0.131547
149	H	-1.303694	-3.357479	-4.416292
150	H	-1.710666	-4.770845	-3.431711
151	H	-0.057728	-4.115462	-3.43774
152	H	-4.947682	-3.005004	-2.669157
153	H	-3.759627	-3.82767	-3.669223
154	H	-3.962262	-4.290138	-1.980936
155	H	-2.483188	-1.193452	-0.485759
156	H	-3.153042	-2.768802	-0.104726
157	H	-6.297147	-1.424298	0.47164
158	H	-3.124379	-0.49518	-5.255411
159	H	-2.652405	-2.185773	-5.171771
160	H	-1.692079	2.562074	-3.662782
161	H	-1.813975	1.56488	-5.113909
162	H	-2.892093	1.299529	-3.756454
163	H	1.447101	3.750934	-5.120752
164	H	-0.198269	3.645467	-4.502021
165	H	0.375665	2.45824	-5.673665
166	H	2.325044	3.687368	-2.577514
167	H	1.768378	2.373497	-1.565508
168	H	-0.098527	5.6884	-0.995991
169	H	2.727118	1.985654	-6.208138
170	H	1.936488	0.449411	-5.788942
171	H	5.750161	-1.010489	0.022833
172	H	6.409272	-0.170737	-1.383133
173	H	7.200462	-1.635447	-0.778357
174	H	5.482635	-2.722193	-3.959349
175	H	6.288329	-1.185121	-3.649983
176	H	7.044335	-2.679755	-3.119321
177	H	6.581082	-4.613414	-2.051426
178	H	5.252654	-5.556538	-1.394468
179	H	4.416149	-5.08573	0.65605
180	H	2.873952	-5.631636	1.25998
181	H	3.524806	-6.342257	-0.206937

182	H	-0.268654	-4.081354	1.814676
183	H	-0.434935	-5.823168	2.096893
184	H	1.159924	-5.074124	2.123883
185	H	-0.245246	-7.427103	0.080177
186	H	1.437077	-7.197737	0.536628
187	H	-3.148842	-5.338172	-0.589703
188	H	-2.487029	-4.756486	0.92951
189	Cl	0.72816	-1.438614	1.633198
190	H	5.046586	-4.720694	-2.929957
191	H	0.951629	-7.005596	-1.154461
192	H	-2.229842	-6.429192	0.430267
193	H	-4.297693	-1.703455	-4.738207
194	H	3.690164	0.610845	-5.645704
195	H	-7.979221	1.739373	-2.978548
196	H	-4.611474	-5.278262	4.785678
197	H	-3.521878	-6.564754	4.238197
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199	H	-7.373577	3.98981	4.944661
200	H	-3.839467	8.133465	1.273192
201	H	-3.779738	6.837987	2.491387
202	H	6.97882	-4.959432	3.433221
203	H	8.007073	-4.87856	4.87059
204	H	10.61264	4.830392	0.138872
205	H	10.58498	6.201897	1.253183
206	H	7.038707	2.829937	-2.567607
207	H	1.575954	5.467431	-1.451368
208	H	-5.054513	-2.557697	0.961149
209	C	-8.688843	0.022079	-1.945986
210	O	-9.906046	0.139943	-2.017478
211	N	-8.076288	-1.16388	-1.657085
212	H	-7.080458	-1.23592	-1.809794
213	C	-8.880391	-2.363474	-1.553798
214	H	-8.307188	-3.142283	-1.044075
215	H	-9.780167	-2.137982	-0.979135
216	H	-9.196996	-2.739415	-2.534565

Cl2F2_Int, E_{opt}= -5378.023701

		Coordinates (Angstroms)		

Center number	Atomic Symbol	X	Y	Z
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2	C	6.851427	-3.092383	4.649099
3	C	6.017166	-2.286259	3.678876
4	C	6.220598	-0.91198	3.50687
5	C	4.96944	-2.895436	2.971407
6	C	5.38693	-0.155741	2.677975
7	C	4.136635	-2.148469	2.142384
8	C	4.332955	-0.771296	2.003779
9	C	-3.523138	7.1045	1.466007
10	C	-2.074146	6.656602	1.215215
11	O	-1.769039	6.578368	-0.198939
12	C	-1.792568	5.271908	1.784411
13	C	-6.641526	2.909876	5.456674
14	C	-5.256889	2.317711	5.784233
15	C	-4.477514	1.845872	4.571361
16	C	-4.9753	0.781669	3.801304
17	C	-3.234661	2.379946	4.213268
18	C	-4.248668	0.246377	2.744566
19	C	-2.489433	1.860259	3.146966
20	C	-2.992863	0.775451	2.430109
21	O	-2.302756	0.165112	1.396871
22	C	-3.891403	-5.564288	4.014362
23	C	-2.777779	-4.525674	3.939988
24	C	-3.112296	-3.392129	2.975158
25	O	-4.140149	-3.401016	2.28543
26	N	-2.186877	-2.42075	2.870134
27	C	-7.758676	1.211687	-2.044217
28	C	-8.071731	2.122134	-0.835759
29	C	-7.149262	3.341512	-0.640777
30	C	-5.930474	3.091252	0.263017
31	N	-4.869894	2.267876	-0.316067
32	C	-3.690636	2.726623	-0.751723
33	N	-3.513187	4.025522	-1.048017
34	N	-2.66534	1.876809	-0.879014
35	C	10.017413	5.683935	0.480253
36	C	8.654403	5.242659	1.027631
37	C	7.815377	4.478249	0.044537
38	C	8.115606	4.199186	-1.264864
39	C	6.532014	3.869612	0.303337
40	N	7.094114	3.459951	-1.842286
41	C	6.115581	3.232897	-0.899441
42	C	5.69926	3.787867	1.434049
43	C	4.919653	2.517938	-0.987786

44	C	4.504972	3.076284	1.351337
45	C	4.126948	2.437779	0.151823
46	Co	1.088115	-1.663553	-1.793548
47	C	-1.444931	-2.876076	-2.1805
48	C	-2.94059	-2.396449	-2.17488
49	C	-2.86139	-1.118929	-3.092186
50	C	-1.412233	-0.680883	-2.922594
51	C	-0.898083	0.613529	-3.327593
52	C	0.445282	0.905046	-3.219699
53	C	1.197931	2.203775	-3.644937
54	C	2.57409	1.57884	-4.023197
55	C	2.630045	0.392778	-3.099151
56	C	3.823979	-0.238763	-2.792788
57	C	3.978128	-1.422159	-2.086403
58	C	5.346728	-2.0475	-1.877267
59	C	4.96751	-3.406043	-1.20168
60	C	3.454417	-3.309569	-1.00369
61	C	2.704782	-4.284743	-0.374503
62	C	1.26503	-4.27412	-0.428509
63	C	0.308319	-5.344232	0.15335
64	C	-1.009153	-4.99481	-0.611479
65	C	-0.855733	-3.493962	-0.887736
66	C	-1.089362	-3.767474	-3.389828
67	N	-0.691794	-1.602475	-2.343506
68	N	1.398217	-0.006332	-2.737789
69	N	2.959	-2.141053	-1.571878
70	N	0.600443	-3.291403	-0.979583
71	C	-3.941901	-3.441079	-2.666751
72	C	-3.328972	-1.985859	-0.724365
73	C	-4.726399	-1.426673	-0.566662
74	O	-5.10922	-0.41596	-1.192442
75	N	-5.546543	-2.079519	0.275163
76	C	-3.300447	-1.311138	-4.557264
77	C	-1.885798	1.577767	-3.960695
78	C	0.664526	3.051839	-4.805273
79	C	1.477792	3.120004	-2.409192
80	C	0.349065	3.992399	-1.881023
81	O	-0.851971	3.699886	-2.005409
82	N	0.711402	5.111436	-1.224049
83	C	2.68845	1.083363	-5.479056
84	C	6.209521	-1.179104	-0.936869
85	C	6.080993	-2.172493	-3.228244
86	C	5.368323	-4.657818	-2.005177
87	C	3.407	-5.388919	0.393946
88	C	0.16553	-5.099917	1.671812
89	C	0.647789	-6.825538	-0.118487

90	C	-2.295822	-5.435602	0.083582
91	F	1.938518	1.348711	2.754431
92	C	1.049655	1.091782	1.783365
93	C	0.476346	-0.107274	1.523549
94	Cl	0.56647	2.53272	0.946388
95	F	0.490294	-0.619162	-0.013487
96	H	7.573051	-2.434385	5.146414
97	H	6.187714	-3.463555	5.44221
98	H	7.032521	-0.424487	4.041275
99	H	4.789261	-3.961273	3.092577
100	H	5.558498	0.908951	2.554567
101	H	3.326864	-2.631933	1.611152
102	H	3.676465	-0.190069	1.36571
103	H	8.410613	-3.979471	3.384451
104	H	-1.388225	7.377922	1.684957
105	H	-0.768921	4.963221	1.55968
106	H	-2.477035	4.531346	1.363136
107	H	-1.915468	5.274338	2.871044
108	H	-1.909226	7.451348	-0.589799
109	H	-4.194852	6.552168	0.799965
110	H	-4.663799	3.044807	6.349656
111	H	-5.392165	1.455961	6.450479
112	H	-2.822679	3.205312	4.788689
113	H	-5.93191	0.335539	4.061899
114	H	-1.520845	2.277217	2.89208
115	H	-4.613651	-0.614225	2.195782
116	H	-1.296136	0.204154	1.495435
117	H	-7.132473	2.334486	4.664118
118	H	-2.543169	-4.100633	4.923358
119	H	-1.845761	-4.975075	3.580516
120	H	-2.317415	-1.653641	2.220175
121	H	-1.354092	-2.412888	3.436988
122	H	-4.407471	-5.619482	3.052891
123	H	-9.107829	2.45429	-0.946672
124	H	-8.050618	1.515792	0.081076
125	H	-7.724484	4.143817	-0.162378
126	H	-6.819301	3.734443	-1.612076
127	H	-5.481154	4.040323	0.571817
128	H	-6.251182	2.593086	1.185639
129	H	-5.01372	1.25999	-0.403524
130	H	-1.803197	2.227677	-1.288869
131	H	-2.608511	1.086823	-0.240158
132	H	-4.30337	4.609702	-1.265076
133	H	-2.578033	4.353152	-1.286697
134	H	-6.713646	0.903667	-2.053179
135	H	8.097798	6.124602	1.375362

136	H	8.805709	4.623053	1.9226
137	H	8.987025	4.468886	-1.843129
138	H	5.991269	4.267896	2.363949
139	H	3.859542	2.985006	2.219052
140	H	4.629031	2.014817	-1.899803
141	H	3.212762	1.855562	0.117848
142	H	9.902977	6.363645	-0.370793
143	H	-3.513165	-0.353378	-2.667895
144	H	3.391003	2.283175	-3.831745
145	H	4.722596	0.216156	-3.190277
146	H	5.450089	-3.456092	-0.218499
147	H	-0.941014	-5.522077	-1.56873
148	H	-1.185757	-2.908273	-0.025551
149	H	-1.290712	-3.248851	-4.326569
150	H	-1.649229	-4.704653	-3.388896
151	H	-0.020912	-3.995995	-3.36268
152	H	-4.953067	-3.021638	-2.687622
153	H	-3.722437	-3.797298	-3.673349
154	H	-3.966226	-4.308225	-2.003779
155	H	-2.638773	-1.22073	-0.365352
156	H	-3.235224	-2.843099	-0.058907
157	H	-6.457877	-1.66934	0.43731
158	H	-3.189223	-0.383027	-5.119157
159	H	-2.725452	-2.078793	-5.07876
160	H	-1.668214	2.604805	-3.691846
161	H	-1.88059	1.499571	-5.053773
162	H	-2.901903	1.376604	-3.622193
163	H	1.462885	3.717028	-5.152265
164	H	-0.176615	3.678169	-4.511707
165	H	0.34902	2.437971	-5.650437
166	H	2.320651	3.774573	-2.657935
167	H	1.817741	2.507762	-1.568299
168	H	-0.023707	5.685176	-0.807988
169	H	2.681604	1.911488	-6.191833
170	H	1.86565	0.404491	-5.725164
171	H	5.740279	-1.053314	0.03813
172	H	6.37851	-0.183245	-1.358549
173	H	7.184446	-1.656241	-0.788561
174	H	5.492523	-2.707385	-3.978051
175	H	6.30789	-1.180923	-3.628753
176	H	7.035501	-2.690963	-3.103282
177	H	6.446106	-4.672532	-2.185775
178	H	5.118319	-5.573985	-1.471508
179	H	4.415183	-5.090182	0.68342
180	H	2.878481	-5.614767	1.321584
181	H	3.483217	-6.327244	-0.165293

182	H	-0.243649	-4.108227	1.879413
183	H	-0.49619	-5.853338	2.108977
184	H	1.128372	-5.174558	2.18195
185	H	-0.244783	-7.43469	0.053568
186	H	1.430346	-7.210616	0.536969
187	H	-3.158278	-5.343447	-0.579159
188	H	-2.510399	-4.846696	0.974529
189	Cl	0.932785	-1.449688	2.541525
190	H	4.858901	-4.689936	-2.972333
191	H	0.963554	-6.976054	-1.155793
192	H	-2.239589	-6.487003	0.38218
193	H	-4.356089	-1.591587	-4.600891
194	H	3.624933	0.533531	-5.606822
195	H	-7.964099	1.763551	-2.968032
196	H	-4.640783	-5.297856	4.764466
197	H	-3.521893	-6.564853	4.238191
198	H	-7.30607	2.87542	6.320368
199	H	-6.56463	3.944427	5.108269
200	H	-3.671143	8.177253	1.313727
201	H	-3.779684	6.837978	2.491376
202	H	6.926461	-4.944191	3.473962
203	H	8.006978	-4.878647	4.87058
204	H	10.606069	4.822023	0.148873
205	H	10.585027	6.201912	1.253186
206	H	7.102196	3.093639	-2.779387
207	H	1.680555	5.337155	-1.058983
208	H	-5.160897	-2.729224	0.963901
209	C	-8.700992	0.0278	-1.978678
210	O	-9.915216	0.148063	-2.090849
211	N	-8.094079	-1.155016	-1.667068
212	H	-7.089351	-1.203768	-1.763344
213	C	-8.88037	-2.363481	-1.553777
214	H	-8.329782	-3.105029	-0.968055
215	H	-9.820867	-2.127043	-1.053014
216	H	-9.124495	-2.79898	-2.531181

Cl2F2-iso1_react, E_{opt}= -5377.4917601

		Coordinates (Angstroms)		

Center number	Atomic Symbol	X	Y	Z
1	C	7.59812	-4.295591	4.045124
2	C	6.744343	-3.159828	4.645873
3	C	5.88923	-2.377879	3.669716
4	C	6.124462	-1.020073	3.415852
5	C	4.787189	-2.985293	3.046294
6	C	5.275712	-0.278769	2.588644
7	C	3.933984	-2.252093	2.221213
8	C	4.17091	-0.892648	1.996128
9	C	-3.523077	7.104484	1.466
10	C	-2.077152	6.669613	1.184538
11	O	-1.817152	6.598827	-0.241363
12	C	-1.769749	5.284946	1.736458
13	C	-7.310109	2.999208	5.404991
14	C	-5.861455	2.566248	5.729628
15	C	-5.10551	1.959885	4.557496
16	C	-5.41344	0.656368	4.132042
17	C	-4.058112	2.61032	3.889694
18	C	-4.706127	0.02176	3.116267
19	C	-3.331906	1.985319	2.870179
20	C	-3.621379	0.660591	2.462868
21	O	-2.9118	0.030749	1.527268
22	C	-3.891441	-5.564261	4.014517
23	C	-2.78535	-4.509351	3.914915
24	C	-3.14223	-3.396155	2.924417
25	O	-4.109874	-3.540822	2.159792
26	N	-2.322483	-2.332255	2.854473
27	C	-7.758639	1.211678	-2.04417
28	C	-8.157041	2.076514	-0.825688
29	C	-7.258875	3.293697	-0.535252
30	C	-6.062966	3.002896	0.383874
31	N	-5.029816	2.145262	-0.194093
32	C	-3.807391	2.550418	-0.56194
33	N	-3.602766	3.837003	-0.91929
34	N	-2.80248	1.674249	-0.552111
35	C	10.017421	5.683762	0.480298
36	C	8.695689	5.187809	1.092477
37	C	7.810345	4.377738	0.186042
38	C	8.060343	3.969444	-1.098843
39	C	6.505191	3.856569	0.525344
40	N	6.99186	3.224186	-1.581756
41	C	6.02029	3.141692	-0.60668
42	C	5.701037	3.923165	1.676895
43	C	4.771373	2.510701	-0.611924

44	C	4.457323	3.297269	1.677916
45	C	3.997685	2.597134	0.542713
46	Co	1.126739	-1.748672	-1.967012
47	C	-1.452804	-2.904505	-2.244234
48	C	-2.935614	-2.399495	-2.187913
49	C	-2.863996	-1.110328	-3.089048
50	C	-1.408207	-0.697936	-2.966304
51	C	-0.892244	0.604501	-3.350884
52	C	0.450626	0.886455	-3.265955
53	C	1.194401	2.204734	-3.634156
54	C	2.59768	1.623787	-3.98893
55	C	2.650404	0.399326	-3.120117
56	C	3.84734	-0.214989	-2.785441
57	C	4.00228	-1.415854	-2.108007
58	C	5.370348	-2.021248	-1.866618
59	C	5.004023	-3.368783	-1.157667
60	C	3.48063	-3.328592	-1.04734
61	C	2.723364	-4.301858	-0.425234
62	C	1.282389	-4.296114	-0.495046
63	C	0.314587	-5.339577	0.114161
64	C	-0.99634	-5.012684	-0.669898
65	C	-0.845755	-3.517434	-0.963584
66	C	-1.145264	-3.805899	-3.458607
67	N	-0.682617	-1.64367	-2.437761
68	N	1.416552	-0.03976	-2.821491
69	N	2.9858	-2.179283	-1.658749
70	N	0.616594	-3.332309	-1.077995
71	C	-3.941918	-3.441112	-2.666762
72	C	-3.303786	-1.988766	-0.736852
73	C	-4.71246	-1.444174	-0.580391
74	O	-5.147441	-0.523775	-1.30458
75	N	-5.479396	-2.024375	0.351305
76	C	-3.354802	-1.265639	-4.54122
77	C	-1.900594	1.601647	-3.888109
78	C	0.664593	3.051874	-4.805271
79	C	1.419365	3.08652	-2.352725
80	C	0.253024	3.915502	-1.825654
81	O	-0.903694	3.475457	-1.788481
82	N	0.543749	5.147834	-1.367975
83	C	2.781034	1.189319	-5.458474
84	C	6.230945	-1.126593	-0.951158
85	C	6.080956	-2.172429	-3.228238
86	C	5.525348	-4.630505	-1.870089
87	C	3.41434	-5.386577	0.379971
88	C	0.159163	-5.043887	1.62336
89	C	0.647803	-6.82551	-0.118465

90	C	-2.28528	-5.460058	0.015445
91	Cl	0.885704	1.968467	3.125907
92	C	0.217115	1.560704	1.604155
93	C	0.107071	0.334715	1.088911
94	F	-0.250263	2.602001	0.90219
95	F	-0.417535	0.181482	-0.134972
96	H	7.400289	-2.468411	5.186498
97	H	6.078789	-3.596439	5.402177
98	H	6.972161	-0.530987	3.889386
99	H	4.580646	-4.036437	3.236364
100	H	5.467845	0.775243	2.409903
101	H	3.081211	-2.73678	1.757693
102	H	3.50372	-0.316391	1.362722
103	H	8.416703	-3.90837	3.431687
104	H	-1.38328	7.397639	1.632127
105	H	-0.764346	4.965411	1.452029
106	H	-2.484678	4.553788	1.353084
107	H	-1.831572	5.288108	2.828491
108	H	-2.024713	7.458731	-0.630865
109	H	-4.197663	6.549387	0.805299
110	H	-5.301669	3.418014	6.133748
111	H	-5.892963	1.820519	6.534276
112	H	-3.783348	3.621019	4.18968
113	H	-6.218461	0.117344	4.629903
114	H	-2.498796	2.494294	2.393234
115	H	-4.953208	-0.996008	2.831611
116	H	-7.742845	2.31873	4.662286
117	H	-2.54705	-4.068654	4.890172
118	H	-1.852015	-4.956238	3.554747
119	H	-2.567039	-1.500003	2.268064
120	H	-1.6321	-2.195965	3.577659
121	H	-4.421169	-5.622952	3.061259
122	H	-9.188196	2.404298	-0.987447
123	H	-8.18323	1.438791	0.069565
124	H	-7.856622	4.063236	-0.030863
125	H	-6.908551	3.741223	-1.476106
126	H	-5.58445	3.935202	0.699076
127	H	-6.405988	2.513049	1.30234
128	H	-5.156976	1.138197	-0.17462
129	H	-1.91241	2.005868	-0.904928
130	H	-2.805874	0.949199	0.231892
131	H	-4.377717	4.391673	-1.244101
132	H	-2.659216	4.12892	-1.16147
133	H	-6.710307	0.919277	-2.000212
134	H	8.126848	6.052746	1.461666
135	H	8.921037	4.589838	1.987081

136	H	8.925048	4.149479	-1.721018
137	H	6.048242	4.458816	2.555783
138	H	3.82963	3.339886	2.562903
139	H	4.422354	1.953234	-1.473129
140	H	3.027225	2.107536	0.575673
141	H	9.842543	6.372976	-0.351751
142	H	-3.489467	-0.346122	-2.624234
143	H	3.393625	2.331638	-3.734175
144	H	4.74995	0.27977	-3.121649
145	H	5.418204	-3.349102	-0.142417
146	H	-0.909515	-5.548997	-1.621207
147	H	-1.161465	-2.923444	-0.101169
148	H	-1.387054	-3.295556	-4.390681
149	H	-1.705973	-4.741452	-3.425847
150	H	-0.077809	-4.046058	-3.476796
151	H	-4.953528	-3.024912	-2.652684
152	H	-3.753421	-3.785712	-3.684153
153	H	-3.944879	-4.313944	-2.012056
154	H	-2.634625	-1.208739	-0.370642
155	H	-3.202274	-2.838988	-0.063917
156	H	-6.370269	-1.587136	0.547472
157	H	-3.267504	-0.324646	-5.086129
158	H	-2.803808	-2.02379	-5.102707
159	H	-1.651607	2.612849	-3.592069
160	H	-1.967895	1.557148	-4.980733
161	H	-2.893535	1.403628	-3.485122
162	H	1.451695	3.737963	-5.134547
163	H	-0.195703	3.657571	-4.522052
164	H	0.37417	2.431289	-5.654298
165	H	2.265576	3.752589	-2.550902
166	H	1.731547	2.441614	-1.522477
167	H	-0.205035	5.69857	-0.93845
168	H	2.779951	2.046052	-6.135752
169	H	1.986336	0.502181	-5.765672
170	H	5.752755	-0.964432	0.015142
171	H	6.416802	-0.147884	-1.404411
172	H	7.199988	-1.606272	-0.777578
173	H	5.486268	-2.739873	-3.948895
174	H	6.280017	-1.189408	-3.662508
175	H	7.046114	-2.67152	-3.109087
176	H	6.611709	-4.584019	-1.979832
177	H	5.298173	-5.534831	-1.307712
178	H	4.400105	-5.062974	0.715627
179	H	2.84879	-5.621738	1.282199
180	H	3.534283	-6.323027	-0.174187
181	H	-0.283941	-4.059663	1.802408

182	H	-0.490114	-5.796504	2.078109
183	H	1.118813	-5.082438	2.144402
184	H	-0.247355	-7.425217	0.068306
185	H	1.427982	-7.200183	0.545969
186	H	-3.135634	-5.401373	-0.665159
187	H	-2.533852	-4.857889	0.887723
188	Cl	0.633709	-1.125781	1.807315
189	H	5.086022	-4.739794	-2.86512
190	H	0.962078	-7.004638	-1.151673
191	H	-2.21256	-6.50378	0.335772
192	H	-4.411105	-1.543922	-4.5406
193	H	3.736391	0.669717	-5.572633
194	H	-7.920587	1.790583	-2.959963
195	H	-4.629076	-5.294487	4.775294
196	H	-3.521869	-6.564871	4.238065
197	H	-7.974626	2.964882	6.268766
198	H	-7.343636	4.005525	4.974786
199	H	-3.679602	8.176895	1.316271
200	H	-3.779697	6.837971	2.491364
201	H	6.992165	-4.964475	3.424602
202	H	8.007045	-4.878602	4.870564
203	H	10.624815	4.84939	0.113998
204	H	10.584921	6.202005	1.253139
205	H	6.936133	2.824365	-2.503236
206	H	1.488993	5.497269	-1.350642
207	H	-5.05058	-2.631699	1.059483
208	C	-8.686067	0.011868	-2.068011
209	O	-9.884307	0.110917	-2.311152
210	N	-8.097598	-1.152829	-1.673449
211	H	-7.085852	-1.178098	-1.640511
212	C	-8.880382	-2.363466	-1.553788
213	H	-8.364227	-3.067385	-0.894766
214	H	-9.856646	-2.115803	-1.132172
215	H	-9.051814	-2.85085	-2.522567

Cl2F2-iso1_Int, E_{opt}= -5378.0748761

		Coordinates (Angstroms)		

Center number	Atomic Symbol	X	Y	Z
1	C	7.598072	-4.295706	4.045087
2	C	6.709954	-3.189816	4.654533
3	C	5.898599	-2.3602	3.679545
4	C	6.139931	-0.989651	3.511889
5	C	4.836142	-2.936535	2.965374
6	C	5.337595	-0.20889	2.674395
7	C	4.030348	-2.164536	2.127942
8	C	4.274784	-0.795136	1.985943
9	C	-3.523109	7.104383	1.465916
10	C	-2.053608	6.745601	1.191477
11	O	-1.822533	6.548727	-0.225578
12	C	-1.638144	5.447453	1.869553
13	C	-7.31001	2.999119	5.404977
14	C	-5.866392	2.593706	5.773006
15	C	-5.097796	2.004259	4.607522
16	C	-5.314032	0.663899	4.247798
17	C	-4.170192	2.738534	3.858702
18	C	-4.629261	0.065899	3.194631
19	C	-3.461795	2.153718	2.803687
20	C	-3.693491	0.816485	2.478838
21	O	-2.98636	0.171162	1.487347
22	C	-3.891498	-5.564291	4.014358
23	C	-2.761869	-4.541925	3.874311
24	C	-3.128887	-3.366546	2.969654
25	O	-4.187762	-3.341801	2.328741
26	N	-2.203649	-2.385627	2.859041
27	C	-7.758628	1.21163	-2.044101
28	C	-8.120787	2.183214	-0.897918
29	C	-7.292286	3.479842	-0.815706
30	C	-6.064917	3.39138	0.099478
31	N	-5.041288	2.463558	-0.378002
32	C	-3.7835	2.758792	-0.706844
33	N	-3.298431	3.997911	-0.738963
34	N	-2.931119	1.726657	-0.955598
35	C	10.017453	5.68385	0.480297
36	C	8.654952	5.269937	1.066826
37	C	7.80375	4.363986	0.218275
38	C	8.120516	3.76179	-0.973044
39	C	6.468381	3.918323	0.547948
40	N	7.064007	2.972536	-1.407621
41	C	6.037091	3.044432	-0.490512
42	C	5.598935	4.167786	1.624739
43	C	4.782066	2.42489	-0.476853

44	C	4.34742	3.558327	1.643926
45	C	3.945233	2.695109	0.602638
46	Co	1.140339	-1.785286	-2.028035
47	C	-1.415397	-2.929445	-2.287236
48	C	-2.899565	-2.412453	-2.255481
49	C	-2.819092	-1.193108	-3.237027
50	C	-1.370384	-0.753448	-3.078556
51	C	-0.867362	0.53468	-3.46276
52	C	0.477924	0.830301	-3.330907
53	C	1.206414	2.162531	-3.671898
54	C	2.609794	1.597524	-4.036174
55	C	2.670335	0.384724	-3.146299
56	C	3.86325	-0.211966	-2.783351
57	C	4.010794	-1.415424	-2.107817
58	C	5.379042	-2.025446	-1.867903
59	C	5.0084	-3.37853	-1.178589
60	C	3.483302	-3.334733	-1.083573
61	C	2.732301	-4.315058	-0.459931
62	C	1.299247	-4.306162	-0.524231
63	C	0.334477	-5.3333	0.111266
64	C	-0.986466	-4.997321	-0.650318
65	C	-0.809512	-3.514534	-0.991299
66	C	-1.117193	-3.88233	-3.464299
67	N	-0.633013	-1.682188	-2.502102
68	N	1.432147	-0.083524	-2.868827
69	N	2.981849	-2.186669	-1.691112
70	N	0.647195	-3.334716	-1.126926
71	C	-3.941894	-3.441063	-2.666731
72	C	-3.225216	-1.921878	-0.807833
73	C	-4.587104	-1.297634	-0.604742
74	O	-4.929589	-0.251378	-1.210776
75	N	-5.423224	-1.914423	0.23865
76	C	-3.242172	-1.449678	-4.698814
77	C	-1.853237	1.51321	-4.082372
78	C	0.664531	3.051885	-4.805227
79	C	1.441722	2.998008	-2.363636
80	C	0.299604	3.830237	-1.81885
81	O	-0.886521	3.458234	-1.862704
82	N	0.625571	5.007353	-1.25151
83	C	2.772501	1.161587	-5.505463
84	C	6.258691	-1.153589	-0.951237
85	C	6.080946	-2.172489	-3.228243
86	C	5.533735	-4.632028	-1.90326
87	C	3.431201	-5.402118	0.336989
88	C	0.211284	-5.038864	1.622831
89	C	0.647794	-6.825497	-0.118466

90	C	-2.271285	-5.38299	0.080857
91	Cl	0.966703	2.109393	2.93726
92	C	0.142321	1.630821	1.516618
93	C	0.08804	0.399625	0.997894
94	F	-0.531562	2.620581	0.90174
95	F	-0.610595	0.214844	-0.128845
96	H	7.338175	-2.523666	5.256863
97	H	6.013314	-3.663962	5.358992
98	H	6.958914	-0.525436	4.05626
99	H	4.628856	-3.998191	3.081991
100	H	5.536862	0.852719	2.555304
101	H	3.224355	-2.631737	1.571311
102	H	3.655918	-0.193493	1.328476
103	H	8.417754	-3.877431	3.453916
104	H	-1.405655	7.559291	1.547907
105	H	-0.624097	5.159965	1.582042
106	H	-2.312818	4.640317	1.578099
107	H	-1.671875	5.557801	2.957015
108	H	-1.984803	7.384059	-0.684424
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110	H	-5.328008	3.454723	6.18448
111	H	-5.900128	1.845216	6.573239
112	H	-3.97287	3.776316	4.117143
113	H	-6.029568	0.073152	4.814671
114	H	-2.703171	2.709684	2.26244
115	H	-4.790603	-0.974436	2.932067
116	H	-2.669062	0.805822	0.816893
117	H	-7.72359	2.298376	4.670969
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119	H	-1.875172	-5.004926	3.428785
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121	H	-1.363583	-2.391303	3.415756
122	H	-4.449017	-5.625017	3.076396
123	H	-9.179144	2.428174	-1.020448
124	H	-8.049966	1.649594	0.06128
125	H	-7.921542	4.280551	-0.408969
126	H	-6.980075	3.804525	-1.815928
127	H	-5.618071	4.380715	0.227557
128	H	-6.361492	3.063857	1.103431
129	H	-5.27548	1.470958	-0.420547
130	H	-2.04119	1.965919	-1.392584
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132	H	-3.859929	4.807311	-0.54502
133	H	-2.339126	4.13261	-1.079777
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136	H	8.824837	4.773789	2.033306
137	H	9.025651	3.830392	-1.558936
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139	H	3.670519	3.73883	2.474088
140	H	4.475135	1.737974	-1.258056
141	H	2.971307	2.214881	0.651746
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143	H	-3.497446	-0.419005	-2.878138
144	H	3.399693	2.31981	-3.798164
145	H	4.768893	0.287352	-3.110394
146	H	5.422006	-3.374194	-0.162514
147	H	-0.941861	-5.566599	-1.584077
148	H	-1.112911	-2.897895	-0.139948
149	H	-1.303794	-3.39529	-4.419969
150	H	-1.714279	-4.795908	-3.418999
151	H	-0.058182	-4.147999	-3.436931
152	H	-4.941834	-2.99244	-2.682882
153	H	-3.759666	-3.847615	-3.661357
154	H	-3.978328	-4.278597	-1.967121
155	H	-2.484435	-1.177437	-0.512027
156	H	-3.135813	-2.756265	-0.113202
157	H	-6.308386	-1.459623	0.425294
158	H	-3.113318	-0.540592	-5.289736
159	H	-2.660236	-2.234792	-5.182716
160	H	-1.689234	2.529089	-3.736969
161	H	-1.794543	1.51344	-5.176579
162	H	-2.882631	1.260828	-3.824877
163	H	1.445868	3.758075	-5.107645
164	H	-0.203749	3.636548	-4.502374
165	H	0.388647	2.462245	-5.680463
166	H	2.299663	3.658828	-2.527651
167	H	1.742207	2.314912	-1.559572
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169	H	2.75675	2.013341	-6.191039
170	H	1.97813	0.465495	-5.793131
171	H	5.794475	-0.998711	0.022699
172	H	6.44471	-0.169181	-1.394495
173	H	7.229475	-1.63725	-0.794403
174	H	5.474717	-2.727585	-3.948776
175	H	6.280628	-1.187591	-3.659675
176	H	7.046079	-2.678068	-3.127178
177	H	6.621285	-4.588386	-2.008877
178	H	5.301024	-5.543681	-1.354403
179	H	4.423115	-5.084166	0.660805
180	H	2.878257	-5.642189	1.246625
181	H	3.545996	-6.339293	-0.218612

182	H	-0.210802	-4.045129	1.801582
183	H	-0.433857	-5.781474	2.102363
184	H	1.183206	-5.082053	2.11971
185	H	-0.246432	-7.420888	0.092672
186	H	1.443167	-7.201205	0.527378
187	H	-3.139981	-5.320719	-0.577301
188	H	-2.46844	-4.739015	0.937814
189	Cl	0.802905	-1.009885	1.641267
190	H	5.097256	-4.728197	-2.900847
191	H	0.937823	-7.009191	-1.157747
192	H	-2.22429	-6.414876	0.443
193	H	-4.298708	-1.727763	-4.751972
194	H	3.72845	0.644939	-5.628049
195	H	-7.947435	1.710331	-3.001545
196	H	-4.600989	-5.27323	4.793412
197	H	-3.521868	-6.564747	4.238194
198	H	-7.974582	2.964822	6.268679
199	H	-7.351036	3.998884	4.962109
200	H	-3.74219	8.162217	1.297269
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203	H	8.007053	-4.878539	4.870612
204	H	10.598688	4.807592	0.173504
205	H	10.584947	6.201993	1.25315
206	H	7.067667	2.402466	-2.236902
207	H	1.586068	5.304762	-1.175985
208	H	-5.060699	-2.582216	0.924071
209	C	-8.689476	0.022757	-1.939904
210	O	-9.906978	0.139681	-2.006126
211	N	-8.075552	-1.163934	-1.654299
212	H	-7.082506	-1.237268	-1.82442
213	C	-8.880394	-2.36347	-1.553804
214	H	-8.301941	-3.148228	-1.059373
215	H	-9.77262	-2.142986	-0.965583
216	H	-9.210047	-2.728666	-2.534316

Cl2F2-iso1_TS, E_{opt}= -5378.0275228

		Coordinates (Angstroms)		

Center number	Atomic Symbol	X	Y	Z
1	C	7.598264	-4.295557	4.045088
2	C	6.717195	-3.185017	4.652591
3	C	5.889978	-2.38141	3.671785
4	C	6.165772	-1.031802	3.41882
5	C	4.779828	-2.961104	3.036542
6	C	5.348006	-0.271031	2.578336
7	C	3.956715	-2.207807	2.199869
8	C	4.234618	-0.856145	1.974182
9	C	-3.523176	7.104528	1.465979
10	C	-2.11691	6.558351	1.169525
11	O	-1.835991	6.547082	-0.250496
12	C	-1.945487	5.121985	1.651435
13	C	-6.641539	2.909824	5.45666
14	C	-5.236831	2.376674	5.800604
15	C	-4.496719	1.842496	4.59301
16	C	-4.947167	0.65845	3.987081
17	C	-3.368335	2.466497	4.051954
18	C	-4.283378	0.098391	2.903146
19	C	-2.6898	1.924338	2.954568
20	C	-3.143252	0.729287	2.399724
21	O	-2.505582	0.106909	1.338693
22	C	-3.89141	-5.564317	4.014358
23	C	-2.762962	-4.535739	3.9428
24	C	-3.070449	-3.397125	2.974774
25	O	-4.11778	-3.363145	2.316545
26	N	-2.104868	-2.465885	2.826586
27	C	-7.758716	1.21168	-2.044228
28	C	-8.150913	2.235626	-0.957725
29	C	-7.211616	3.444439	-0.798139
30	C	-6.015283	3.207591	0.139144
31	N	-4.957425	2.345952	-0.385808
32	C	-3.773647	2.773248	-0.843877
33	N	-3.577574	4.055781	-1.188862
34	N	-2.760183	1.907537	-0.937548
35	C	10.017424	5.683863	0.480266
36	C	8.701784	5.164227	1.069881
37	C	7.828514	4.412732	0.107826
38	C	8.070267	4.116083	-1.209364
39	C	6.539554	3.846369	0.418015
40	N	7.004641	3.401087	-1.740111
41	C	6.050599	3.218951	-0.761492
42	C	5.757764	3.80588	1.584313
43	C	4.815459	2.565037	-0.797434

44	C	4.529033	3.154941	1.554201
45	C	4.063927	2.538277	0.37368
46	Co	1.105128	-1.684333	-1.837063
47	C	-1.432996	-2.895687	-2.216157
48	C	-2.925479	-2.405276	-2.200911
49	C	-2.847723	-1.139723	-3.13149
50	C	-1.397239	-0.705235	-2.969331
51	C	-0.887476	0.590372	-3.368913
52	C	0.453206	0.888607	-3.247514
53	C	1.191889	2.203967	-3.63778
54	C	2.588722	1.609977	-3.991535
55	C	2.643311	0.409299	-3.085818
56	C	3.839619	-0.208847	-2.759891
57	C	3.99444	-1.406161	-2.075257
58	C	5.364735	-2.029643	-1.867755
59	C	4.991448	-3.381955	-1.174761
60	C	3.472597	-3.308173	-1.014692
61	C	2.721439	-4.289925	-0.396906
62	C	1.281968	-4.283544	-0.459699
63	C	0.32181	-5.340709	0.139235
64	C	-0.998088	-4.995464	-0.622577
65	C	-0.837874	-3.500926	-0.920641
66	C	-1.093102	-3.804252	-3.41647
67	N	-0.674325	-1.628648	-2.395617
68	N	1.409441	-0.017366	-2.764234
69	N	2.975267	-2.142754	-1.587968
70	N	0.618587	-3.307159	-1.023184
71	C	-3.941917	-3.441057	-2.666743
72	C	-3.290902	-1.975836	-0.752478
73	C	-4.674946	-1.389277	-0.57456
74	O	-5.0424	-0.363012	-1.182778
75	N	-5.502285	-2.041214	0.261899
76	C	-3.288814	-1.346138	-4.594162
77	C	-1.876117	1.54965	-4.008151
78	C	0.664543	3.05186	-4.805299
79	C	1.423057	3.09444	-2.368993
80	C	0.28659	3.980764	-1.881282
81	O	-0.912256	3.712673	-2.068951
82	N	0.641071	5.088096	-1.203615
83	C	2.749263	1.139796	-5.451553
84	C	6.239634	-1.154636	-0.946387
85	C	6.080987	-2.172495	-3.228225
86	C	5.448908	-4.640095	-1.936336
87	C	3.420528	-5.39002	0.380293
88	C	0.191781	-5.074947	1.655336
89	C	0.647777	-6.825496	-0.118486

90	C	-2.283144	-5.418804	0.086081
91	Cl	1.142686	2.55709	2.530182
92	C	0.496441	1.48364	1.322375
93	C	0.320856	0.15312	1.356109
94	F	0.017665	2.182248	0.263756
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96	H	7.352327	-2.505581	5.232454
97	H	6.033723	-3.649459	5.375738
98	H	7.021141	-0.565337	3.901821
99	H	4.54478	-4.006694	3.225425
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101	H	3.09125	-2.662555	1.731241
102	H	3.589584	-0.268945	1.328354
103	H	8.417173	-3.883904	3.448173
104	H	-1.370297	7.192676	1.671513
105	H	-0.992589	4.702448	1.315956
106	H	-2.749094	4.487829	1.271109
107	H	-1.969467	5.086678	2.744205
108	H	-1.935578	7.447551	-0.588076
109	H	-4.247947	6.622288	0.8012
110	H	-4.646835	3.156684	6.293915
111	H	-5.334719	1.562129	6.528915
112	H	-2.999043	3.386051	4.498982
113	H	-5.818502	0.150927	4.39383
114	H	-1.813113	2.416391	2.549185
115	H	-4.603274	-0.842857	2.471064
116	H	-1.517235	0.253303	1.344402
117	H	-7.108223	2.290517	4.682835
118	H	-2.533528	-4.109921	4.92739
119	H	-1.832332	-4.998527	3.596702
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121	H	-1.275643	-2.457569	3.399196
122	H	-4.406409	-5.615312	3.052136
123	H	-9.164381	2.576666	-1.186206
124	H	-8.22455	1.721151	0.011019
125	H	-7.781626	4.276262	-0.365851
126	H	-6.85482	3.788311	-1.778642
127	H	-5.556735	4.160248	0.423283
128	H	-6.365356	2.749898	1.07179
129	H	-5.101072	1.335269	-0.414278
130	H	-1.877864	2.229597	-1.326155
131	H	-2.766294	1.090705	-0.335792
132	H	-4.358245	4.654436	-1.398038
133	H	-2.634475	4.36285	-1.427558
134	H	-6.718447	0.902704	-1.954361
135	H	8.13208	6.008584	1.483354

136	H	8.922264	4.516076	1.92998
137	H	8.92506	4.357706	-1.824076
138	H	6.112824	4.274238	2.497961
139	H	3.922756	3.105073	2.451817
140	H	4.461404	2.068644	-1.691636
141	H	3.111702	2.016668	0.381603
142	H	9.836684	6.38423	-0.341888
143	H	-3.497545	-0.368564	-2.714051
144	H	3.388159	2.325474	-3.769108
145	H	4.74086	0.268126	-3.125725
146	H	5.444785	-3.399401	-0.176107
147	H	-0.938597	-5.53706	-1.572363
148	H	-1.157755	-2.902039	-0.064215
149	H	-1.294449	-3.29522	-4.35837
150	H	-1.662124	-4.735953	-3.401135
151	H	-0.026676	-4.042414	-3.391934
152	H	-4.948147	-3.00932	-2.683862
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156	H	-3.200914	-2.827675	-0.079484
157	H	-6.40292	-1.614656	0.440271
158	H	-3.173114	-0.424282	-5.165558
159	H	-2.717463	-2.121683	-5.107736
160	H	-1.669582	2.577058	-3.733479
161	H	-1.859581	1.474212	-5.101433
162	H	-2.894088	1.340597	-3.680548
163	H	1.457509	3.732712	-5.133929
164	H	-0.19203	3.661692	-4.524022
165	H	0.377313	2.434519	-5.65781
166	H	2.291374	3.735415	-2.557279
167	H	1.697495	2.456641	-1.522282
168	H	-0.102184	5.666399	-0.809226
169	H	2.743624	1.978469	-6.151891
170	H	1.94668	0.448764	-5.728523
171	H	5.775326	-1.012578	0.029002
172	H	6.415641	-0.166029	-1.382549
173	H	7.212027	-1.63633	-0.795916
174	H	5.484246	-2.724748	-3.958752
175	H	6.290688	-1.185623	-3.649335
176	H	7.041638	-2.680925	-3.110321
177	H	6.530867	-4.624173	-2.090593
178	H	5.216501	-5.550963	-1.386554
179	H	4.417195	-5.079609	0.695926
180	H	2.875268	-5.629483	1.29433
181	H	3.520414	-6.323738	-0.183265

182	H	-0.203343	-4.074874	1.851358
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184	H	1.156059	-5.158154	2.161163
185	H	-0.245023	-7.428034	0.074821
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187	H	-3.148385	-5.333302	-0.5739
188	H	-2.488858	-4.813575	0.968122
189	Cl	0.869911	-0.801092	2.687903
190	H	4.965162	-4.712641	-2.914258
191	H	0.94629	-6.99016	-1.158723
192	H	-2.230715	-6.465294	0.402193
193	H	-4.345734	-1.622485	-4.634918
194	H	3.699333	0.610164	-5.563777
195	H	-7.889296	1.673564	-3.029988
196	H	-4.637963	-5.29273	4.764944
197	H	-3.52191	-6.564874	4.238211
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199	H	-6.603189	3.934793	5.075498
200	H	-3.591292	8.189145	1.341577
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202	H	7.011953	-4.965556	3.407425
203	H	8.006965	-4.878671	4.870605
204	H	10.633248	4.862806	0.098228
205	H	10.585044	6.201976	1.25319
206	H	6.95382	3.051795	-2.6823
207	H	1.605143	5.285908	-0.983058
208	H	-5.124985	-2.702286	0.942077
209	C	-8.702819	0.032244	-1.943733
210	O	-9.918872	0.15335	-2.032243
211	N	-8.092395	-1.153909	-1.649068
212	H	-7.090856	-1.208297	-1.77132
213	C	-8.880361	-2.363501	-1.553772
214	H	-8.32462	-3.118323	-0.990568
215	H	-9.814009	-2.135864	-1.036702
216	H	-9.136747	-2.775822	-2.538024

Cl2F2-iso2_React, E_{opt}= -5377.5007874

		Coordinates (Angstroms)		

Center number	Atomic Symbol	X	Y	Z
1	C	7.598073	-4.295688	4.045198
2	C	6.146907	-3.92717	4.424398
3	C	5.395427	-2.962689	3.523513
4	C	5.898421	-1.680952	3.247416
5	C	4.118528	-3.284131	3.03877
6	C	5.144676	-0.748491	2.532109
7	C	3.357068	-2.356827	2.320842
8	C	3.867816	-1.083163	2.069239
9	C	-3.523095	7.104548	1.466015
10	C	-2.143482	6.549442	1.083031
11	O	-2.037916	6.408807	-0.362642
12	C	-1.882378	5.172663	1.677915
13	C	-7.310121	2.999201	5.404985
14	C	-5.865963	2.560844	5.741434
15	C	-5.102118	1.940897	4.582201
16	C	-5.435618	0.648665	4.141298
17	C	-4.014686	2.56292	3.953216
18	C	-4.713056	-0.002215	3.146715
19	C	-3.27315	1.92046	2.955353
20	C	-3.590467	0.608235	2.535278
21	O	-2.866593	-0.044967	1.623125
22	C	-3.891456	-5.564227	4.014559
23	C	-2.773202	-4.517851	4.000851
24	C	-3.064652	-3.388492	3.012328
25	O	-3.973391	-3.527317	2.17532
26	N	-2.26178	-2.314641	3.025773
27	C	-7.758635	1.211677	-2.044166
28	C	-8.076122	2.036075	-0.776156
29	C	-7.166617	3.255119	-0.52156
30	C	-5.955267	2.971971	0.378511
31	N	-4.950207	2.078942	-0.196182
32	C	-3.695108	2.423567	-0.517877
33	N	-3.411883	3.681069	-0.912897
34	N	-2.732858	1.49952	-0.475991
35	C	10.017455	5.683871	0.480263
36	C	8.777688	5.02568	1.105066
37	C	7.88213	4.305663	0.136202
38	C	8.076453	4.123151	-1.209029
39	C	6.622194	3.671121	0.450114
40	N	7.014931	3.413576	-1.754486
41	C	6.104572	3.127626	-0.759606
42	C	5.87798	3.514466	1.632731
43	C	4.879927	2.454943	-0.81066

44	C	4.658484	2.84346	1.58777
45	C	4.164414	2.318146	0.37534
46	Co	1.114307	-1.737279	-1.933303
47	C	-1.465681	-2.878004	-2.212405
48	C	-2.954277	-2.390371	-2.150129
49	C	-2.886162	-1.060678	-2.999018
50	C	-1.429949	-0.645206	-2.862885
51	C	-0.899292	0.658436	-3.230286
52	C	0.454864	0.899953	-3.224175
53	C	1.21295	2.195842	-3.647845
54	C	2.588614	1.580624	-4.04463
55	C	2.646556	0.369476	-3.157788
56	C	3.837978	-0.254228	-2.833431
57	C	3.985748	-1.44028	-2.126858
58	C	5.351438	-2.043211	-1.876514
59	C	4.982995	-3.399177	-1.187038
60	C	3.464189	-3.338818	-1.042794
61	C	2.708619	-4.309234	-0.419061
62	C	1.268066	-4.294624	-0.479277
63	C	0.302275	-5.344038	0.118607
64	C	-1.0048	-5.018687	-0.673582
65	C	-0.861293	-3.518129	-0.943856
66	C	-1.146247	-3.743681	-3.449118
67	N	-0.702229	-1.607121	-2.366789
68	N	1.4135	-0.043729	-2.817003
69	N	2.970159	-2.187177	-1.653193
70	N	0.601043	-3.323352	-1.048725
71	C	-3.941905	-3.441128	-2.666742
72	C	-3.336505	-2.040577	-0.688108
73	C	-4.749377	-1.51349	-0.515584
74	O	-5.202774	-0.58902	-1.224324
75	N	-5.500299	-2.109703	0.419173
76	C	-3.368107	-1.15642	-4.457903
77	C	-1.894755	1.722916	-3.651694
78	C	0.664547	3.05191	-4.805341
79	C	1.50606	3.101347	-2.396671
80	C	0.32995	3.823243	-1.75702
81	O	-0.549809	3.207159	-1.142599
82	N	0.290816	5.168825	-1.862111
83	C	2.709704	1.122795	-5.513196
84	C	6.182887	-1.149984	-0.933011
85	C	6.080954	-2.172458	-3.228223
86	C	5.454258	-4.653884	-1.946607
87	C	3.398787	-5.410058	0.36244
88	C	0.139437	-5.056994	1.629282
89	C	0.647832	-6.825511	-0.118478

90	C	-2.293871	-5.486399	-0.001419
91	Cl	0.692428	0.006939	3.880913
92	C	0.455421	0.649447	2.29429
93	C	0.115762	-0.146255	1.28182
94	Cl	0.62118	2.354316	2.059878
95	F	-0.087567	0.268581	0.04142
96	H	6.164857	-3.486759	5.430847
97	H	5.563886	-4.850851	4.518966
98	H	6.879835	-1.400345	3.621067
99	H	3.704082	-4.266423	3.252447
100	H	5.540948	0.244882	2.340553
101	H	2.362284	-2.619514	1.974853
102	H	3.278365	-0.352115	1.523836
103	H	8.223504	-3.407718	3.911796
104	H	-1.357019	7.240888	1.419234
105	H	-0.998557	4.714889	1.225454
106	H	-2.730148	4.503857	1.514237
107	H	-1.718926	5.254761	2.756179
108	H	-2.304772	7.248211	-0.761697
109	H	-4.278749	6.635997	0.82446
110	H	-5.304331	3.412043	6.143899
111	H	-5.907462	1.821389	6.551491
112	H	-3.717168	3.561331	4.271572
113	H	-6.270157	0.130933	4.6122
114	H	-2.40083	2.399732	2.519504
115	H	-4.971204	-1.013485	2.849726
116	H	-7.74255	2.320549	4.660608
117	H	-2.588133	-4.098046	4.996527
118	H	-1.823724	-4.967002	3.686137
119	H	-2.480001	-1.491666	2.416138
120	H	-1.608901	-2.183889	3.784245
121	H	-4.365326	-5.603546	3.031977
122	H	-9.118494	2.359122	-0.855212
123	H	-8.034001	1.37205	0.099024
124	H	-7.749175	4.036399	-0.017599
125	H	-6.829833	3.684587	-1.474932
126	H	-5.458275	3.906982	0.656565
127	H	-6.285246	2.518207	1.320735
128	H	-5.12727	1.07801	-0.187128
129	H	-1.810656	1.818275	-0.752529
130	H	-2.774868	0.805152	0.344252
131	H	-4.138176	4.362977	-1.047678
132	H	-2.44653	3.990055	-0.911294
133	H	-6.709259	0.916197	-2.068891
134	H	8.187919	5.792203	1.627038
135	H	9.09755	4.324172	1.888571

136	H	8.895634	4.445305	-1.835091
137	H	6.249538	3.917909	2.57041
138	H	4.072868	2.721474	2.494017
139	H	4.502476	2.042144	-1.737828
140	H	3.207503	1.803038	0.369341
141	H	9.734493	6.415374	-0.283941
142	H	-3.514403	-0.318255	-2.505245
143	H	3.405494	2.277738	-3.829286
144	H	4.742297	0.222401	-3.188764
145	H	5.42702	-3.415213	-0.184788
146	H	-0.905415	-5.542207	-1.631405
147	H	-1.185075	-2.943379	-0.074355
148	H	-1.396486	-3.212817	-4.367727
149	H	-1.694477	-4.687122	-3.438885
150	H	-0.075893	-3.970306	-3.473791
151	H	-4.964403	-3.054798	-2.622325
152	H	-3.755244	-3.732529	-3.701186
153	H	-3.907587	-4.340898	-2.051394
154	H	-2.676696	-1.270732	-0.285401
155	H	-3.231961	-2.912637	-0.045869
156	H	-6.390756	-1.679482	0.631566
157	H	-3.283309	-0.190522	-4.960453
158	H	-2.80788	-1.885723	-5.04832
159	H	-1.625823	2.697785	-3.253073
160	H	-1.973371	1.806936	-4.740841
161	H	-2.887457	1.501432	-3.263103
162	H	1.455381	3.722864	-5.155909
163	H	-0.17729	3.674619	-4.500698
164	H	0.340308	2.438038	-5.647258
165	H	2.26943	3.828342	-2.693417
166	H	1.944148	2.489376	-1.602464
167	H	-0.434638	5.673447	-1.34817
168	H	2.703013	1.969941	-6.202653
169	H	1.890257	0.448799	-5.782296
170	H	5.685975	-1.020631	0.029226
171	H	6.350202	-0.157368	-1.361304
172	H	7.158855	-1.613755	-0.754548
173	H	5.493576	-2.720531	-3.969347
174	H	6.298093	-1.183456	-3.63923
175	H	7.040237	-2.681961	-3.10425
176	H	6.538402	-4.634432	-2.082284
177	H	5.21391	-5.566646	-1.403461
178	H	4.395612	-5.105334	0.679882
179	H	2.850021	-5.638981	1.276165
180	H	3.487879	-6.343837	-0.201735
181	H	-0.278279	-4.063821	1.81812

182	H	-0.529655	-5.799135	2.072587
183	H	1.095654	-5.122993	2.154213
184	H	-0.247719	-7.431724	0.044168
185	H	1.41458	-7.201848	0.560041
186	H	-3.141682	-5.424546	-0.685136
187	H	-2.551952	-4.901025	0.880396
188	F	0.011143	-1.461515	1.375601
189	H	4.98757	-4.722498	-2.932935
190	H	0.984855	-6.995582	-1.146045
191	H	-2.213358	-6.534968	0.300863
192	H	-4.422546	-1.441943	-4.475186
193	H	3.649178	0.581082	-5.653171
194	H	-7.969275	1.821964	-2.928704
195	H	-4.667645	-5.311124	4.74174
196	H	-3.521855	-6.564891	4.238023
197	H	-7.974618	2.964886	6.268771
198	H	-7.338573	4.006355	4.976656
199	H	-3.590605	8.190117	1.347023
200	H	-3.779708	6.837927	2.491362
201	H	7.641703	-4.887694	3.126153
202	H	8.007082	-4.878592	4.870542
203	H	10.672144	4.940032	0.014252
204	H	10.584932	6.201957	1.253158
205	H	6.925236	3.162815	-2.724613
206	H	1.041247	5.685844	-2.29161
207	H	-5.046763	-2.702242	1.122794
208	C	-8.691345	0.015067	-2.063846
209	O	-9.889984	0.120242	-2.303794
210	N	-8.102755	-1.151026	-1.678217
211	H	-7.092218	-1.166167	-1.605922
212	C	-8.880378	-2.363458	-1.553789
213	H	-8.377703	-3.052157	-0.868428
214	H	-9.869101	-2.114432	-1.162797
215	H	-9.022714	-2.871613	-2.516693

Cl2F2-iso2_Int, E_{opt}= -5378.082173

		Coordinates (Angstroms)

Center number	Atomic Symbol	X	Y	Z
1	C	7.598087	-4.295686	4.045108
2	C	6.234958	-3.742657	4.519576
3	C	5.503305	-2.805851	3.578307
4	C	5.96255	-1.495491	3.368677
5	C	4.31069	-3.194464	2.951757
6	C	5.249753	-0.600476	2.569475
7	C	3.590194	-2.304767	2.149247
8	C	4.059568	-1.005288	1.957031
9	C	-3.523066	7.104475	1.465943
10	C	-2.036504	6.823943	1.190046
11	O	-1.810069	6.588387	-0.222931
12	C	-1.52477	5.583242	1.907361
13	C	-7.31002	2.999126	5.404994
14	C	-5.865961	2.601026	5.77896
15	C	-5.089357	2.008627	4.620539
16	C	-5.277671	0.658823	4.281985
17	C	-4.17881	2.750396	3.858274
18	C	-4.577681	0.057893	3.240349
19	C	-3.455492	2.163132	2.814956
20	C	-3.653097	0.814097	2.515993
21	O	-2.924781	0.164124	1.541362
22	C	-3.89148	-5.564296	4.014357
23	C	-2.752874	-4.549054	3.894178
24	C	-3.101569	-3.362887	2.998538
25	O	-4.161358	-3.317211	2.359582
26	N	-2.163511	-2.39412	2.895948
27	C	-7.75864	1.211636	-2.044128
28	C	-8.091033	2.157934	-0.868202
29	C	-7.269015	3.459867	-0.790177
30	C	-6.033316	3.380365	0.114237
31	N	-5.012805	2.451928	-0.371152
32	C	-3.739973	2.734741	-0.643242
33	N	-3.233735	3.966747	-0.61831
34	N	-2.890445	1.696906	-0.889716
35	C	10.017448	5.683925	0.480245
36	C	8.71655	5.144585	1.098982
37	C	7.830015	4.3507	0.179858
38	C	8.060852	3.998724	-1.125669
39	C	6.542798	3.790719	0.524649
40	N	6.99337	3.260676	-1.618738
41	C	6.047338	3.114525	-0.626183
42	C	5.765103	3.788164	1.696476
43	C	4.812162	2.457731	-0.634224

44	C	4.536725	3.132943	1.696888
45	C	4.065434	2.476654	0.540191
46	Co	1.139088	-1.775646	-2.008893
47	C	-1.416148	-2.920251	-2.27494
48	C	-2.903366	-2.409323	-2.243529
49	C	-2.822686	-1.178623	-3.212342
50	C	-1.375479	-0.734589	-3.044324
51	C	-0.871497	0.555647	-3.421443
52	C	0.477798	0.844547	-3.305978
53	C	1.212032	2.169155	-3.670457
54	C	2.605255	1.589027	-4.049024
55	C	2.668814	0.383758	-3.149504
56	C	3.860209	-0.217591	-2.795491
57	C	4.007751	-1.416079	-2.111168
58	C	5.377278	-2.019401	-1.868385
59	C	5.011874	-3.368263	-1.1675
60	C	3.485313	-3.336951	-1.086664
61	C	2.734643	-4.324476	-0.476207
62	C	1.301401	-4.308855	-0.530038
63	C	0.337959	-5.333935	0.109851
64	C	-0.986375	-4.996072	-0.645292
65	C	-0.810302	-3.512497	-0.981771
66	C	-1.112543	-3.863389	-3.45824
67	N	-0.636496	-1.668279	-2.47661
68	N	1.430346	-0.07291	-2.852558
69	N	2.981106	-2.185072	-1.686793
70	N	0.646536	-3.329593	-1.116808
71	C	-3.941915	-3.441075	-2.66673
72	C	-3.232384	-1.932184	-0.791666
73	C	-4.589392	-1.299233	-0.580531
74	O	-4.929937	-0.252458	-1.187542
75	N	-5.420042	-1.905192	0.27545
76	C	-3.237418	-1.421065	-4.678729
77	C	-1.858408	1.546283	-4.020053
78	C	0.66453	3.051908	-4.805284
79	C	1.473741	3.026255	-2.381265
80	C	0.33383	3.845185	-1.816087
81	O	-0.829288	3.410998	-1.735212
82	N	0.632523	5.07557	-1.355316
83	C	2.744645	1.138478	-5.516344
84	C	6.247701	-1.132325	-0.957797
85	C	6.080913	-2.172451	-3.228148
86	C	5.566932	-4.621214	-1.870049
87	C	3.432327	-5.419892	0.308799
88	C	0.225154	-5.036466	1.621946
89	C	0.647789	-6.825497	-0.118477

90	C	-2.267801	-5.379752	0.09324
91	Cl	1.340181	0.663697	3.388683
92	C	0.545956	0.934992	1.883241
93	C	0.119088	-0.073404	1.120275
94	Cl	0.277504	2.574696	1.378665
95	F	-0.522968	0.101665	-0.020059
96	H	6.39782	-3.213531	5.467811
97	H	5.57999	-4.589211	4.757817
98	H	6.879853	-1.170554	3.854477
99	H	3.933368	-4.202261	3.108437
100	H	5.611896	0.413259	2.420621
101	H	2.669867	-2.622889	1.671435
102	H	3.512485	-0.312846	1.325682
103	H	8.297673	-3.490066	3.802018
104	H	-1.437351	7.689441	1.506985
105	H	-0.475049	5.402854	1.660125
106	H	-2.105177	4.707198	1.609214
107	H	-1.603268	5.704845	2.990976
108	H	-2.0179	7.398529	-0.707795
109	H	-4.142658	6.475514	0.816784
110	H	-5.33263	3.466356	6.188051
111	H	-5.899446	1.856724	6.582983
112	H	-4.004777	3.796567	4.099162
113	H	-5.981981	0.062504	4.857012
114	H	-2.709471	2.730197	2.266285
115	H	-4.720284	-0.988646	2.992596
116	H	-2.624533	0.793316	0.85683
117	H	-7.719335	2.29352	4.673248
118	H	-2.43521	-4.173587	4.874679
119	H	-1.864968	-5.015015	3.453601
120	H	-2.341975	-1.575074	2.322881
121	H	-1.316591	-2.421933	3.441683
122	H	-4.436961	-5.619637	3.069144
123	H	-9.154363	2.397574	-0.95289
124	H	-7.985828	1.606441	0.077659
125	H	-7.898916	4.256794	-0.377388
126	H	-6.967401	3.787475	-1.792596
127	H	-5.589749	4.372681	0.228076
128	H	-6.318729	3.059337	1.123651
129	H	-5.2589	1.462372	-0.43761
130	H	-1.996978	1.935265	-1.322146
131	H	-3.36128	0.860052	-1.236392
132	H	-3.802268	4.780753	-0.468528
133	H	-2.273816	4.096371	-0.959558
134	H	-6.709757	0.913852	-2.048986
135	H	8.139924	5.986946	1.50708

136	H	8.968424	4.521549	1.969152
137	H	8.910268	4.217706	-1.756398
138	H	6.122857	4.28989	2.591264
139	H	3.930834	3.118549	2.598346
140	H	4.453456	1.92585	-1.507571
141	H	3.105435	1.969001	0.567335
142	H	9.813752	6.384753	-0.335648
143	H	-3.50664	-0.411603	-2.849193
144	H	3.403585	2.308181	-3.829398
145	H	4.766374	0.274643	-3.130541
146	H	5.409514	-3.346032	-0.144722
147	H	-0.947488	-5.564177	-1.580163
148	H	-1.115403	-2.902055	-0.128376
149	H	-1.299194	-3.370656	-4.411038
150	H	-1.706755	-4.779214	-3.420015
151	H	-0.052802	-4.125958	-3.430768
152	H	-4.944671	-2.998455	-2.673829
153	H	-3.758628	-3.832235	-3.667308
154	H	-3.969262	-4.28819	-1.978342
155	H	-2.488579	-1.193907	-0.487236
156	H	-3.146972	-2.772818	-0.104241
157	H	-6.299278	-1.442582	0.470391
158	H	-3.113995	-0.503343	-5.257646
159	H	-2.646437	-2.194932	-5.169775
160	H	-1.688326	2.556757	-3.65929
161	H	-1.807922	1.565336	-5.114411
162	H	-2.887045	1.293073	-3.759799
163	H	1.446975	3.750061	-5.122839
164	H	-0.19678	3.646296	-4.499013
165	H	0.373126	2.457608	-5.672297
166	H	2.316909	3.695809	-2.582057
167	H	1.802591	2.361255	-1.573675
168	H	-0.101612	5.639075	-0.927649
169	H	2.726983	1.983583	-6.210143
170	H	1.940918	0.446206	-5.78667
171	H	5.770645	-0.966083	0.007946
172	H	6.435439	-0.154078	-1.412574
173	H	7.216921	-1.613034	-0.782622
174	H	5.479202	-2.739064	-3.94365
175	H	6.273326	-1.189767	-3.667369
176	H	7.050003	-2.669283	-3.122534
177	H	6.653991	-4.555265	-1.968991
178	H	5.351276	-5.53015	-1.310499
179	H	4.413903	-5.096335	0.655479
180	H	2.870214	-5.681198	1.20622
181	H	3.561966	-6.344979	-0.263583

182	H	-0.179932	-4.036534	1.802552
183	H	-0.427272	-5.77048	2.104885
184	H	1.199107	-5.092377	2.113595
185	H	-0.247039	-7.419313	0.095025
186	H	1.443561	-7.202764	0.525515
187	H	-3.141627	-5.310204	-0.557616
188	H	-2.454344	-4.73867	0.954987
189	F	0.243044	-1.347934	1.436591
190	H	5.14013	-4.740481	-2.869393
191	H	0.935117	-7.011531	-1.158145
192	H	-2.222817	-6.413484	0.45027
193	H	-4.291312	-1.707612	-4.739801
194	H	3.694409	0.612346	-5.64613
195	H	-7.969946	1.731095	-2.985399
196	H	-4.608565	-5.2747	4.786761
197	H	-3.521878	-6.564755	4.2382
198	H	-7.974585	2.964847	6.268693
199	H	-7.35471	3.997148	4.958584
200	H	-3.79704	8.147252	1.283731
201	H	-3.779737	6.837996	2.491389
202	H	7.491003	-4.937192	3.165359
203	H	8.007076	-4.878569	4.870583
204	H	10.63806	4.871726	0.087398
205	H	10.58497	6.201914	1.253189
206	H	6.94399	2.855504	-2.538488
207	H	1.575206	5.431615	-1.383627
208	H	-5.052794	-2.569265	0.962133
209	C	-8.688959	0.022127	-1.944753
210	O	-9.906099	0.139753	-2.016957
211	N	-8.076289	-1.163625	-1.654851
212	H	-7.08073	-1.235496	-1.809632
213	C	-8.880392	-2.363475	-1.553798
214	H	-8.306497	-3.143731	-1.047081
215	H	-9.779353	-2.139671	-0.977197
216	H	-9.198462	-2.73663	-2.535142

Cl2F2-iso2_TS, E_{opt}= -5378.0304381

		Coordinates (Angstroms)		

Center number	Atomic Symbol	X	Y	Z
1	C	7.598204	-4.295575	4.045124
2	C	6.211476	-3.799983	4.507281
3	C	5.457626	-2.889741	3.561129
4	C	5.940758	-1.606996	3.259445
5	C	4.220014	-3.271833	3.025883
6	C	5.207063	-0.732982	2.457052
7	C	3.478534	-2.401422	2.222919
8	C	3.972803	-1.130277	1.933346
9	C	-3.523138	7.1045	1.466007
10	C	-2.091349	6.60284	1.216146
11	O	-1.778331	6.527466	-0.196198
12	C	-1.871836	5.201574	1.772909
13	C	-6.641526	2.909876	5.456674
14	C	-5.245942	2.343017	5.787211
15	C	-4.488512	1.838115	4.574556
16	C	-4.968006	0.711222	3.886761
17	C	-3.295512	2.415079	4.125273
18	C	-4.27403	0.164027	2.81454
19	C	-2.585345	1.885293	3.040419
20	C	-3.072629	0.746947	2.398185
21	O	-2.421835	0.143581	1.337349
22	C	-3.891403	-5.564288	4.014362
23	C	-2.763469	-4.531936	3.950049
24	C	-3.074539	-3.386705	2.989995
25	O	-4.137577	-3.338486	2.358041
26	N	-2.103008	-2.466183	2.817621
27	C	-7.758676	1.211687	-2.044217
28	C	-8.099102	2.139652	-0.85694
29	C	-7.176073	3.356599	-0.656198
30	C	-5.957383	3.100171	0.245885
31	N	-4.899975	2.274822	-0.336104
32	C	-3.723751	2.732263	-0.781832
33	N	-3.549868	4.02816	-1.090966
34	N	-2.699465	1.881369	-0.906227
35	C	10.017413	5.683935	0.480253
36	C	8.734483	5.08621	1.069949
37	C	7.850483	4.406584	0.063951
38	C	8.046593	4.30861	-1.290326
39	C	6.606608	3.728734	0.341669
40	N	6.995249	3.618468	-1.876474
41	C	6.100222	3.245351	-0.897156
42	C	5.873805	3.477977	1.515019
43	C	4.904172	2.528607	-0.983293

44	C	4.683406	2.760461	1.434904
45	C	4.206764	2.283542	0.19552
46	Co	1.104886	-1.679074	-1.821714
47	C	-1.437245	-2.89153	-2.200712
48	C	-2.931036	-2.404505	-2.185145
49	C	-2.85399	-1.130764	-3.106571
50	C	-1.402899	-0.698345	-2.949239
51	C	-0.891205	0.596368	-3.35336
52	C	0.451315	0.888297	-3.24308
53	C	1.197074	2.195275	-3.647653
54	C	2.579628	1.583741	-4.024043
55	C	2.640278	0.394159	-3.104639
56	C	3.835684	-0.223304	-2.779852
57	C	3.991587	-1.412969	-2.082242
58	C	5.362919	-2.029534	-1.870034
59	C	4.995301	-3.381753	-1.173986
60	C	3.475593	-3.318286	-1.023915
61	C	2.725328	-4.307237	-0.420231
62	C	1.285876	-4.290979	-0.470384
63	C	0.327912	-5.340555	0.14041
64	C	-1.000202	-4.990158	-0.605342
65	C	-0.837292	-3.496535	-0.907016
66	C	-1.097771	-3.796601	-3.404028
67	N	-0.680038	-1.623037	-2.379819
68	N	1.407258	-0.020307	-2.763666
69	N	2.975743	-2.149583	-1.590586
70	N	0.619554	-3.309773	-1.020483
71	C	-3.941901	-3.441079	-2.666751
72	C	-3.30893	-1.989255	-0.735279
73	C	-4.705893	-1.430335	-0.569786
74	O	-5.095283	-0.425069	-1.199817
75	N	-5.520332	-2.082293	0.27862
76	C	-3.302111	-1.325635	-4.568629
77	C	-1.87997	1.559507	-3.985892
78	C	0.664526	3.051839	-4.805273
79	C	1.461746	3.083686	-2.38626
80	C	0.329034	3.945779	-1.847401
81	O	-0.871803	3.695578	-2.043068
82	N	0.69112	5.010987	-1.106101
83	C	2.703111	1.094319	-5.481275
84	C	6.229219	-1.143503	-0.951951
85	C	6.080993	-2.172493	-3.228244
86	C	5.475135	-4.636979	-1.926738
87	C	3.422641	-5.420445	0.338035
88	C	0.22213	-5.062707	1.656586
89	C	0.647789	-6.825538	-0.118487

90	C	-2.27862	-5.40391	0.120483
91	Cl	1.814722	1.047897	3.412574
92	C	0.898172	0.920678	1.939193
93	C	0.28637	-0.213731	1.491508
94	Cl	0.634045	2.411474	1.073541
95	F	0.536042	-0.681931	-0.070416
96	H	6.342938	-3.268994	5.459733
97	H	5.587333	-4.672022	4.736016
98	H	6.892127	-1.284534	3.676169
99	H	3.822842	-4.256675	3.261538
100	H	5.587457	0.26127	2.240912
101	H	2.513006	-2.700505	1.834293
102	H	3.398806	-0.455393	1.307648
103	H	8.27679	-3.464995	3.829006
104	H	-1.380427	7.290955	1.698467
105	H	-0.867235	4.840777	1.539082
106	H	-2.595341	4.49876	1.352971
107	H	-1.986802	5.202393	2.86035
108	H	-1.879018	7.409358	-0.579165
109	H	-4.216272	6.581044	0.79883
110	H	-4.65085	3.095509	6.316068
111	H	-5.362707	1.504218	6.485075
112	H	-2.897143	3.286463	4.639315
113	H	-5.88468	0.23214	4.222351
114	H	-1.65614	2.337391	2.711109
115	H	-4.619738	-0.73948	2.326103
116	H	-1.396634	0.161327	1.400046
117	H	-7.122632	2.315267	4.67221
118	H	-2.537304	-4.111307	4.93755
119	H	-1.83048	-4.990277	3.604019
120	H	-2.26357	-1.682151	2.194792
121	H	-1.240584	-2.476798	3.338956
122	H	-4.402525	-5.613963	3.050001
123	H	-9.130508	2.474794	-0.998423
124	H	-8.10366	1.546995	0.069037
125	H	-7.750853	4.156949	-0.173868
126	H	-6.845973	3.753597	-1.62591
127	H	-5.504865	4.047267	0.556556
128	H	-6.278951	2.600777	1.167414
129	H	-5.03618	1.26488	-0.405111
130	H	-1.835722	2.2319	-1.311905
131	H	-2.651089	1.095078	-0.259867
132	H	-4.341284	4.615202	-1.294576
133	H	-2.614862	4.360308	-1.321724
134	H	-6.713597	0.904522	-2.027199
135	H	8.161996	5.878839	1.57248

136	H	8.995715	4.370416	1.861901
137	H	8.859853	4.679465	-1.89664
138	H	6.236765	3.837561	2.473815
139	H	4.111702	2.553079	2.333795
140	H	4.540338	2.154355	-1.930762
141	H	3.288302	1.706561	0.161286
142	H	9.787562	6.410629	-0.306577
143	H	-3.499722	-0.361395	-2.679731
144	H	3.389823	2.294471	-3.82866
145	H	4.736218	0.248265	-3.15213
146	H	5.437497	-3.391121	-0.169993
147	H	-0.955786	-5.533654	-1.554977
148	H	-1.149402	-2.89392	-0.050252
149	H	-1.308013	-3.287498	-4.344096
150	H	-1.661757	-4.731175	-3.386678
151	H	-0.030252	-4.030534	-3.386567
152	H	-4.949853	-3.014161	-2.681407
153	H	-3.733267	-3.802964	-3.673759
154	H	-3.968341	-4.306937	-2.001473
155	H	-2.619072	-1.222197	-0.384457
156	H	-3.209246	-2.843099	-0.066316
157	H	-6.426947	-1.665998	0.451402
158	H	-3.189519	-0.4002	-5.134529
159	H	-2.733719	-2.09807	-5.090379
160	H	-1.662937	2.586434	-3.717676
161	H	-1.875115	1.479535	-5.078882
162	H	-2.895469	1.358225	-3.646161
163	H	1.463256	3.720072	-5.145545
164	H	-0.177582	3.67489	-4.508929
165	H	0.353503	2.441108	-5.654202
166	H	2.30615	3.74591	-2.60917
167	H	1.792857	2.451207	-1.555033
168	H	-0.047422	5.595456	-0.710948
169	H	2.692516	1.925336	-6.19053
170	H	1.886144	0.410649	-5.733331
171	H	5.758308	-0.999917	0.019433
172	H	6.400479	-0.156688	-1.392888
173	H	7.202653	-1.620281	-0.792983
174	H	5.488512	-2.729733	-3.958544
175	H	6.288226	-1.186246	-3.651642
176	H	7.043654	-2.676143	-3.107403
177	H	6.557822	-4.605673	-2.072734
178	H	5.252716	-5.549123	-1.37574
179	H	4.416999	-5.115215	0.663149
180	H	2.877955	-5.673081	1.247919
181	H	3.519636	-6.343446	-0.243233

182	H	-0.14094	-4.050122	1.849255
183	H	-0.461912	-5.778322	2.121371
184	H	1.190224	-5.168055	2.150953
185	H	-0.246397	-7.425005	0.078432
186	H	1.439962	-7.212055	0.523974
187	H	-3.152814	-5.308436	-0.526455
188	H	-2.464147	-4.798214	1.006607
189	F	0.509344	-1.345482	2.206075
190	H	5.000375	-4.720168	-2.908179
191	H	0.940355	-6.991302	-1.160227
192	H	-2.230896	-6.4514	0.433841
193	H	-4.359261	-1.600933	-4.60506
194	H	3.644227	0.552072	-5.607518
195	H	-7.943491	1.75002	-2.980432
196	H	-4.641633	-5.294454	4.76177
197	H	-3.521893	-6.564853	4.238191
198	H	-7.30607	2.87542	6.320368
199	H	-6.58232	3.940616	5.093827
200	H	-3.631307	8.183114	1.322795
201	H	-3.779684	6.837978	2.491376
202	H	7.528132	-4.92698	3.154292
203	H	8.006978	-4.878647	4.87058
204	H	10.655279	4.906699	0.046839
205	H	10.585027	6.201912	1.253186
206	H	6.928094	3.386122	-2.853108
207	H	1.658325	5.208111	-0.898214
208	H	-5.129677	-2.719142	0.975791
209	C	-8.699658	0.026387	-1.983937
210	O	-9.912744	0.143763	-2.111411
211	N	-8.094228	-1.154235	-1.661987
212	H	-7.088004	-1.201428	-1.744031
213	C	-8.88037	-2.363481	-1.553777
214	H	-8.333654	-3.103657	-0.96277
215	H	-9.825182	-2.127717	-1.060844
216	H	-9.116066	-2.800072	-2.53278

Cl1F3_React, E_{opt}= -5017.1316806

		Coordinates (Angstroms)		

Center number	Atomic Symbol	X	Y	Z
1	C	7.598139	-4.29563	4.045187
2	C	6.750402	-3.157192	4.648065
3	C	5.904844	-2.369093	3.669688
4	C	6.146367	-1.011321	3.422817
5	C	4.808323	-2.972629	3.033207
6	C	5.308723	-0.265814	2.58835
7	C	3.966384	-2.235335	2.200804
8	C	4.209442	-0.875905	1.982251
9	C	-3.523176	7.104429	1.465986
10	C	-2.075789	6.678851	1.178812
11	O	-1.824697	6.604435	-0.248965
12	C	-1.754523	5.298942	1.734219
13	C	-7.31011	2.999257	5.404987
14	C	-5.861848	2.571422	5.736832
15	C	-5.105078	1.957834	4.570212
16	C	-5.403336	0.646694	4.161743
17	C	-4.070498	2.612257	3.887448
18	C	-4.699371	0.008594	3.146124
19	C	-3.347425	1.983811	2.868099
20	C	-3.628186	0.652595	2.476076
21	O	-2.921847	0.022063	1.538845
22	C	-3.891374	-5.564322	4.014434
23	C	-2.783214	-4.511946	3.911963
24	C	-3.143478	-3.395055	2.927248
25	O	-4.131721	-3.524411	2.18679
26	N	-2.308011	-2.344712	2.836247
27	C	-7.758667	1.211676	-2.044181
28	C	-8.154363	2.077677	-0.825755
29	C	-7.25275	3.292685	-0.536592
30	C	-6.058002	2.998706	0.382914
31	N	-5.028525	2.136043	-0.194092
32	C	-3.80054	2.532626	-0.552973
33	N	-3.584087	3.818724	-0.905527
34	N	-2.802881	1.648207	-0.539419
35	C	10.017438	5.683838	0.480212
36	C	8.693355	5.19132	1.091325
37	C	7.809132	4.380164	0.184433
38	C	8.062294	3.975604	-1.101046
39	C	6.5049	3.853561	0.520782
40	N	6.997143	3.228486	-1.587704
41	C	6.024897	3.139571	-0.614129
42	C	5.697564	3.913128	1.670877
43	C	4.779287	2.502589	-0.623628

44	C	4.45679	3.280839	1.667925
45	C	4.003336	2.581592	0.529768
46	Co	1.12651	-1.744772	-1.958392
47	C	-1.4532	-2.901182	-2.236393
48	C	-2.937446	-2.398516	-2.182109
49	C	-2.864675	-1.105318	-3.078619
50	C	-1.40912	-0.692325	-2.953641
51	C	-0.892149	0.611255	-3.335379
52	C	0.451629	0.890884	-3.25666
53	C	1.197524	2.206119	-3.634442
54	C	2.597194	1.619575	-3.994074
55	C	2.650908	0.398541	-3.120276
56	C	3.847701	-0.216725	-2.787468
57	C	4.002755	-1.416177	-2.107111
58	C	5.370813	-2.02228	-1.867052
59	C	5.004473	-3.370535	-1.159694
60	C	3.481297	-3.329027	-1.046774
61	C	2.72425	-4.30261	-0.425306
62	C	1.28295	-4.295277	-0.491961
63	C	0.316369	-5.33913	0.118218
64	C	-0.99783	-5.008378	-0.659478
65	C	-0.845655	-3.513345	-0.954832
66	C	-1.143845	-3.801647	-3.450955
67	N	-0.683369	-1.639619	-2.428571
68	N	1.417232	-0.036457	-2.814688
69	N	2.986401	-2.178065	-1.655392
70	N	0.61678	-3.329711	-1.071363
71	C	-3.941947	-3.441068	-2.666811
72	C	-3.309776	-1.993737	-0.730515
73	C	-4.717191	-1.445173	-0.57474
74	O	-5.15256	-0.529961	-1.305907
75	N	-5.482011	-2.013771	0.364997
76	C	-3.353231	-1.255566	-4.531928
77	C	-1.900745	1.613401	-3.86298
78	C	0.664562	3.051926	-4.805391
79	C	1.433341	3.095814	-2.360701
80	C	0.263921	3.91167	-1.82145
81	O	-0.875559	3.437797	-1.722986
82	N	0.531171	5.169247	-1.421216
83	C	2.771338	1.178662	-5.462752
84	C	6.233891	-1.128761	-0.952977
85	C	6.08094	-2.172394	-3.228082
86	C	5.523089	-4.631599	-1.875382
87	C	3.416017	-5.389528	0.376284
88	C	0.16792	-5.045532	1.628589
89	C	0.647796	-6.825525	-0.118479

90	C	-2.284682	-5.450095	0.033564
91	F	0.796437	1.844889	2.759258
92	C	0.229092	1.542523	1.610476
93	C	0.12479	0.311073	1.123179
94	F	-0.244149	2.601474	0.968581
95	F	-0.43425	0.11059	-0.080363
96	H	7.40915	-2.471262	5.192223
97	H	6.080006	-3.591867	5.401244
98	H	6.99036	-0.526129	3.906853
99	H	4.597728	-4.023914	3.217927
100	H	5.505886	0.787874	2.413651
101	H	3.117681	-2.715768	1.725748
102	H	3.551152	-0.297408	1.341778
103	H	8.416599	-3.911231	3.429794
104	H	-1.384387	7.413664	1.618835
105	H	-0.746604	4.989053	1.447444
106	H	-2.464333	4.560354	1.355384
107	H	-1.812852	5.305603	2.826413
108	H	-2.045123	7.460396	-0.640132
109	H	-4.195916	6.545363	0.806594
110	H	-5.304202	3.42719	6.135402
111	H	-5.893951	1.831789	6.546968
112	H	-3.803795	3.628769	4.174782
113	H	-6.198684	0.105321	4.67239
114	H	-2.52421	2.497398	2.378706
115	H	-4.939116	-1.014371	2.874072
116	H	-7.739023	2.314573	4.664045
117	H	-2.54028	-4.073341	4.887081
118	H	-1.852128	-4.959878	3.5476
119	H	-2.559562	-1.505347	2.262332
120	H	-1.595753	-2.22154	3.540371
121	H	-4.423606	-5.623192	3.062631
122	H	-9.184853	2.407834	-0.986854
123	H	-8.181525	1.440403	0.069799
124	H	-7.84816	4.064511	-0.032947
125	H	-6.900696	3.738074	-1.477776
126	H	-5.575601	3.929366	0.696934
127	H	-6.403261	2.511968	1.302237
128	H	-5.163737	1.129839	-0.181006
129	H	-1.9092	1.972969	-0.889472
130	H	-2.813566	0.925192	0.249972
131	H	-4.352368	4.382456	-1.230383
132	H	-2.636192	4.105208	-1.135967
133	H	-6.710533	0.918403	-2.002005
134	H	8.125247	6.058011	1.457488
135	H	8.916171	4.594968	1.987594

136	H	8.927522	4.159894	-1.721209
137	H	6.040117	4.447735	2.552187
138	H	3.827014	3.316545	2.551981
139	H	4.434693	1.946194	-1.48715
140	H	3.036071	2.085107	0.558773
141	H	9.845232	6.372262	-0.353042
142	H	-3.491229	-0.343336	-2.611995
143	H	3.396423	2.326235	-3.746249
144	H	4.750137	0.275344	-3.128021
145	H	5.420614	-3.353005	-0.145266
146	H	-0.918385	-5.545752	-1.610822
147	H	-1.159778	-2.9175	-0.092991
148	H	-1.385588	-3.290696	-4.382772
149	H	-1.704099	-4.737508	-3.419279
150	H	-0.076274	-4.041215	-3.468652
151	H	-4.954596	-3.027692	-2.649004
152	H	-3.753135	-3.77907	-3.686375
153	H	-3.940695	-4.317868	-2.017277
154	H	-2.638891	-1.217883	-0.358748
155	H	-3.212385	-2.847402	-0.061173
156	H	-6.371639	-1.572986	0.55887
157	H	-3.266519	-0.312184	-5.072992
158	H	-2.80029	-2.01076	-5.095612
159	H	-1.64691	2.623297	-3.565638
160	H	-1.975498	1.573836	-4.955275
161	H	-2.891694	1.4169	-3.454302
162	H	1.452325	3.734362	-5.140505
163	H	-0.192526	3.661749	-4.52041
164	H	0.367474	2.430235	-5.65132
165	H	2.269313	3.769441	-2.575988
166	H	1.765108	2.457631	-1.532984
167	H	-0.217292	5.712686	-0.981976
168	H	2.76922	2.032598	-6.14359
169	H	1.97295	0.492635	-5.762796
170	H	5.759142	-0.967675	0.014815
171	H	6.418174	-0.149628	-1.405972
172	H	7.20346	-1.608733	-0.783198
173	H	5.486091	-2.738128	-3.949997
174	H	6.281611	-1.189166	-3.661179
175	H	7.045534	-2.672695	-3.109508
176	H	6.609337	-4.586433	-1.986655
177	H	5.295596	-5.536701	-1.314425
178	H	4.403185	-5.067827	0.709671
179	H	2.852659	-5.625572	1.279577
180	H	3.53339	-6.325158	-0.179798
181	H	-0.271596	-4.060221	1.810368

182	H	-0.480788	-5.797719	2.084858
183	H	1.129628	-5.086811	2.145538
184	H	-0.247541	-7.424474	0.070117
185	H	1.429667	-7.201929	0.542899
186	H	-3.140016	-5.383633	-0.640125
187	H	-2.521793	-4.848216	0.909159
188	Cl	0.704213	-1.100117	1.900731
189	H	5.082098	-4.738434	-2.869938
190	H	0.958449	-7.002985	-1.153071
191	H	-2.21624	-6.494991	0.350995
192	H	-4.409178	-1.535219	-4.533769
193	H	3.724492	0.655722	-5.57988
194	H	-7.921397	1.790352	-2.959994
195	H	-4.627022	-5.292641	4.776366
196	H	-3.521909	-6.564876	4.238194
197	H	-7.974626	2.964846	6.268762
198	H	-7.345861	4.004189	4.971859
199	H	-3.686592	8.175702	1.315234
200	H	-3.779658	6.838051	2.491406
201	H	6.98863	-4.962983	3.426612
202	H	8.007106	-4.87866	4.870536
203	H	10.623503	4.847569	0.116094
204	H	10.584956	6.201936	1.253172
205	H	6.946093	2.826899	-2.508692
206	H	1.463538	5.550112	-1.45807
207	H	-5.053501	-2.616616	1.077801
208	C	-8.686969	0.012453	-2.066861
209	O	-9.885418	0.112602	-2.308669
210	N	-8.098376	-1.152421	-1.673376
211	H	-7.086527	-1.177345	-1.64015
212	C	-8.880368	-2.363465	-1.553797
213	H	-8.367176	-3.064699	-0.889585
214	H	-9.859074	-2.115203	-1.138296
215	H	-9.046058	-2.854386	-2.521824

Cl1F3_Int, E_{opt}= -5017.7146293

		Coordinates (Angstroms)		

Center number	Atomic Symbol	X	Y	Z
1	C	7.598088	-4.295696	4.045093
2	C	6.71207	-3.190184	4.657023
3	C	5.899042	-2.366172	3.680211
4	C	6.143998	-0.998311	3.499035
5	C	4.834963	-2.947904	2.973185
6	C	5.344211	-0.224687	2.652868
7	C	4.031714	-2.183121	2.127393
8	C	4.280892	-0.816387	1.970068
9	C	-3.523069	7.104487	1.465946
10	C	-2.054161	6.753597	1.180537
11	O	-1.834403	6.570354	-0.240695
12	C	-1.625705	5.452609	1.844804
13	C	-7.310021	2.999127	5.404997
14	C	-5.862038	2.611838	5.774705
15	C	-5.093568	2.015045	4.613189
16	C	-5.3029	0.669319	4.269576
17	C	-4.176127	2.747889	3.850794
18	C	-4.620209	0.06418	3.21949
19	C	-3.469625	2.155705	2.798502
20	C	-3.692691	0.812852	2.491
21	O	-2.985566	0.16099	1.505262
22	C	-3.891481	-5.564297	4.014358
23	C	-2.760324	-4.543189	3.87728
24	C	-3.125128	-3.36594	2.974419
25	O	-4.189808	-3.333382	2.34318
26	N	-2.19399	-2.391853	2.856347
27	C	-7.758643	1.211634	-2.044129
28	C	-8.130754	2.19203	-0.909083
29	C	-7.298448	3.485908	-0.826446
30	C	-6.075559	3.39577	0.094548
31	N	-5.049714	2.466904	-0.376845
32	C	-3.790724	2.762146	-0.700071
33	N	-3.308962	4.002795	-0.739762
34	N	-2.933503	1.730452	-0.934382
35	C	10.017434	5.683936	0.480244
36	C	8.666112	5.249013	1.07773
37	C	7.802572	4.367137	0.216788
38	C	8.094066	3.822524	-1.008124
39	C	6.481092	3.891482	0.561217
40	N	7.034514	3.041859	-1.4506
41	C	6.031806	3.060349	-0.504437
42	C	5.638348	4.07997	1.671465
43	C	4.784476	2.425762	-0.485968

44	C	4.395201	3.4536	1.696627
45	C	3.975157	2.634895	0.627121
46	Co	1.138157	-1.777745	-2.014573
47	C	-1.416491	-2.923756	-2.277852
48	C	-2.902546	-2.410705	-2.247956
49	C	-2.821684	-1.185955	-3.223553
50	C	-1.373403	-0.744543	-3.061592
51	C	-0.86973	0.543974	-3.444873
52	C	0.476319	0.839012	-3.316446
53	C	1.205805	2.169227	-3.66641
54	C	2.608824	1.601677	-4.027946
55	C	2.66904	0.390817	-3.135307
56	C	3.86209	-0.208857	-2.777705
57	C	4.010365	-1.413502	-2.104416
58	C	5.378859	-2.024934	-1.867807
59	C	5.007817	-3.377909	-1.178326
60	C	3.482725	-3.333316	-1.081383
61	C	2.731902	-4.31447	-0.458944
62	C	1.298613	-4.304467	-0.520387
63	C	0.334731	-5.333529	0.113756
64	C	-0.987216	-4.995701	-0.645526
65	C	-0.811121	-3.511846	-0.982704
66	C	-1.114272	-3.87188	-3.457766
67	N	-0.635707	-1.674118	-2.486799
68	N	1.430477	-0.074389	-2.853862
69	N	2.981193	-2.183279	-1.685366
70	N	0.645597	-3.33054	-1.117707
71	C	-3.94191	-3.441074	-2.666734
72	C	-3.232357	-1.928908	-0.798293
73	C	-4.591662	-1.300096	-0.592127
74	O	-4.930496	-0.250178	-1.194241
75	N	-5.428612	-1.915007	0.251767
76	C	-3.242147	-1.437019	-4.687038
77	C	-1.855861	1.524298	-4.061376
78	C	0.664535	3.051902	-4.805276
79	C	1.440194	3.017825	-2.365925
80	C	0.294244	3.853239	-1.834723
81	O	-0.882621	3.450531	-1.825505
82	N	0.604254	5.064721	-1.335063
83	C	2.772151	1.161199	-5.495937
84	C	6.261259	-1.15336	-0.953414
85	C	6.080918	-2.172441	-3.228142
86	C	5.532142	-4.631468	-1.903633
87	C	3.431247	-5.403	0.335559
88	C	0.213547	-5.0419	1.626041
89	C	0.647787	-6.825498	-0.11847

90	C	-2.271115	-5.382857	0.086636
91	F	1.088195	1.96934	2.441565
92	C	0.310649	1.637718	1.430807
93	C	0.136541	0.398489	0.979644
94	F	-0.31238	2.682543	0.891167
95	F	-0.654713	0.202064	-0.085008
96	H	7.341785	-2.522151	5.255681
97	H	6.018064	-3.663642	5.364509
98	H	6.964176	-0.53077	4.03873
99	H	4.625374	-4.007769	3.101554
100	H	5.546081	0.834915	2.522235
101	H	3.223329	-2.652689	1.576535
102	H	3.664555	-0.22229	1.303756
103	H	8.417193	-3.87779	3.452912
104	H	-1.407277	7.567509	1.538263
105	H	-0.609106	5.178797	1.552429
106	H	-2.294089	4.642191	1.547089
107	H	-1.658214	5.552904	2.933293
108	H	-2.013738	7.407109	-0.690611
109	H	-4.175907	6.512386	0.814488
110	H	-5.330085	3.482659	6.173648
111	H	-5.886407	1.87288	6.584033
112	H	-3.985914	3.790228	4.095864
113	H	-6.011781	0.080561	4.846764
114	H	-2.718999	2.712116	2.24525
115	H	-4.776695	-0.979597	2.967737
116	H	-2.66746	0.790174	0.830609
117	H	-7.716721	2.288476	4.676714
118	H	-2.437465	-4.157436	4.852084
119	H	-1.873132	-5.006738	3.433344
120	H	-2.379246	-1.575716	2.280181
121	H	-1.346811	-2.406238	3.40207
122	H	-4.44738	-5.624275	3.075413
123	H	-9.186766	2.439855	-1.045281
124	H	-8.072809	1.665194	0.05467
125	H	-7.926953	4.289521	-0.424236
126	H	-6.980919	3.807297	-1.826095
127	H	-5.628082	4.384445	0.225833
128	H	-6.377705	3.068258	1.096828
129	H	-5.281871	1.473488	-0.414476
130	H	-2.043207	1.967415	-1.37232
131	H	-3.397125	0.878533	-1.255083
132	H	-3.87643	4.812044	-0.563074
133	H	-2.346889	4.136881	-1.072586
134	H	-6.710954	0.913829	-2.007914
135	H	8.094207	6.145771	1.355253

136	H	8.853766	4.725422	2.026238
137	H	8.983085	3.927742	-1.613027
138	H	5.958244	4.704285	2.501108
139	H	3.739118	3.583587	2.552587
140	H	4.464813	1.770038	-1.28844
141	H	3.010714	2.138221	0.682535
142	H	9.887241	6.350631	-0.37759
143	H	-3.501649	-0.414411	-2.862165
144	H	3.399193	2.32425	-3.792047
145	H	4.767668	0.288747	-3.107528
146	H	5.422134	-3.373782	-0.162547
147	H	-0.943915	-5.562802	-1.580701
148	H	-1.115713	-2.897727	-0.129975
149	H	-1.300044	-3.3817	-4.412038
150	H	-1.709956	-4.786594	-3.416816
151	H	-0.054902	-4.135939	-3.429522
152	H	-4.943401	-2.995778	-2.679351
153	H	-3.758129	-3.840073	-3.664113
154	H	-3.974635	-4.283473	-1.97279
155	H	-2.490319	-1.189165	-0.494819
156	H	-3.146789	-2.768283	-0.109225
157	H	-6.310176	-1.455167	0.442974
158	H	-3.11503	-0.525105	-5.273959
159	H	-2.657396	-2.218493	-5.173498
160	H	-1.690482	2.539826	-3.714954
161	H	-1.79968	1.526504	-5.155704
162	H	-2.884964	1.272578	-3.801921
163	H	1.445557	3.756702	-5.11158
164	H	-0.204456	3.638324	-4.50711
165	H	0.388688	2.457204	-5.677096
166	H	2.298718	3.676337	-2.535677
167	H	1.737249	2.342893	-1.553643
168	H	-0.133747	5.634368	-0.921903
169	H	2.757349	2.010571	-6.184501
170	H	1.977741	0.464393	-5.781729
171	H	5.799711	-0.997579	0.021345
172	H	6.447305	-0.169545	-1.397816
173	H	7.231751	-1.638297	-0.798592
174	H	5.475309	-2.728165	-3.948791
175	H	6.280373	-1.187577	-3.659806
176	H	7.046187	-2.677472	-3.126315
177	H	6.619521	-4.587647	-2.01079
178	H	5.300546	-5.543166	-1.354446
179	H	4.422912	-5.085092	0.660161
180	H	2.878499	-5.644933	1.244773
181	H	3.546309	-6.339042	-0.221864

182	H	-0.206482	-4.047785	1.806837
183	H	-0.432244	-5.78449	2.104752
184	H	1.185906	-5.087518	2.121809
185	H	-0.246245	-7.421106	0.092903
186	H	1.443863	-7.202062	0.525958
187	H	-3.141246	-5.317184	-0.569387
188	H	-2.465328	-4.741716	0.946436
189	Cl	0.831431	-1.00165	1.66668
190	H	5.094312	-4.72777	-2.900628
191	H	0.9366	-7.007925	-1.158331
192	H	-2.224492	-6.416084	0.444954
193	H	-4.297854	-1.71777	-4.742844
194	H	3.727853	0.64358	-5.616078
195	H	-7.938038	1.703242	-3.007107
196	H	-4.602136	-5.273389	4.792352
197	H	-3.521878	-6.564754	4.238197
198	H	-7.974584	2.964846	6.26869
199	H	-7.361807	3.994809	4.954099
200	H	-3.746615	8.161754	1.298776
201	H	-3.779738	6.837987	2.491387
202	H	7.01714	-4.965701	3.40301
203	H	8.007073	-4.87856	4.87059
204	H	10.604888	4.817982	0.156852
205	H	10.58498	6.201897	1.253183
206	H	7.024126	2.506498	-2.302625
207	H	1.557487	5.391579	-1.305698
208	H	-5.065657	-2.582302	0.937649
209	C	-8.690016	0.023032	-1.940736
210	O	-9.90726	0.13987	-2.012143
211	N	-8.076794	-1.163004	-1.651833
212	H	-7.082462	-1.235349	-1.814397
213	C	-8.880391	-2.363474	-1.553798
214	H	-8.304523	-3.1458	-1.052552
215	H	-9.777423	-2.14255	-0.973109
216	H	-9.201859	-2.732225	-2.535711

Cl1F3_TS, E_{opt}= -5017.6671066

		Coordinates (Angstroms)		

Center number	Atomic Symbol	X	Y	Z
1	C	7.598162	-4.295662	4.045056
2	C	6.793777	-3.123176	4.64125
3	C	5.957923	-2.322235	3.663761
4	C	6.208215	-0.963743	3.43155
5	C	4.858392	-2.912029	3.018971
6	C	5.37465	-0.204017	2.604823
7	C	4.020811	-2.160872	2.195282
8	C	4.271399	-0.799909	1.993322
9	C	-3.523145	7.104524	1.466011
10	C	-2.115008	6.55864	1.172
11	O	-1.836102	6.532496	-0.24843
12	C	-1.93513	5.127969	1.670355
13	C	-6.641524	2.909909	5.456648
14	C	-5.238281	2.366295	5.792299
15	C	-4.481604	1.851047	4.58396
16	C	-4.951547	0.706547	3.918842
17	C	-3.30149	2.438058	4.115093
18	C	-4.260085	0.150103	2.850253
19	C	-2.593025	1.898432	3.034047
20	C	-3.069327	0.74151	2.418404
21	O	-2.412609	0.119697	1.371848
22	C	-3.891519	-5.564334	4.01449
23	C	-2.764273	-4.533445	3.957462
24	C	-3.064328	-3.390622	2.991565
25	O	-4.107222	-3.352692	2.326608
26	N	-2.096838	-2.460433	2.85454
27	C	-7.7587	1.211683	-2.044211
28	C	-8.12423	2.195557	-0.911881
29	C	-7.186469	3.404134	-0.736754
30	C	-5.983247	3.154767	0.188016
31	N	-4.92821	2.3004	-0.355145
32	C	-3.744515	2.734694	-0.806722
33	N	-3.556952	4.019465	-1.148776
34	N	-2.724996	1.875099	-0.899041
35	C	10.01747	5.683955	0.480218
36	C	8.677271	5.21128	1.061494
37	C	7.819437	4.416166	0.119022
38	C	8.108875	4.040016	-1.168039
39	C	6.512366	3.876655	0.412063
40	N	7.060095	3.301542	-1.697625
41	C	6.06792	3.182373	-0.748434
42	C	5.677828	3.907056	1.543663
43	C	4.833304	2.529916	-0.799084

44	C	4.446419	3.258864	1.49926
45	C	4.032765	2.574293	0.337654
46	Co	1.096884	-1.67721	-1.819315
47	C	-1.438501	-2.88716	-2.197124
48	C	-2.932832	-2.402277	-2.185882
49	C	-2.855261	-1.129912	-3.108601
50	C	-1.404837	-0.694783	-2.945563
51	C	-0.89271	0.599516	-3.349025
52	C	0.449847	0.893199	-3.237149
53	C	1.195241	2.201502	-3.64187
54	C	2.583268	1.594934	-4.007082
55	C	2.63749	0.399806	-3.094494
56	C	3.833084	-0.223039	-2.776016
57	C	3.987125	-1.413565	-2.080217
58	C	5.356698	-2.037796	-1.87172
59	C	4.9807	-3.392541	-1.185863
60	C	3.46455	-3.306646	-1.00603
61	C	2.714387	-4.285014	-0.382177
62	C	1.274792	-4.277928	-0.442087
63	C	0.316817	-5.342004	0.148686
64	C	-1.003098	-4.993747	-0.612274
65	C	-0.845532	-3.496685	-0.902126
66	C	-1.0919	-3.789081	-3.400817
67	N	-0.682196	-1.617726	-2.370553
68	N	1.404544	-0.015624	-2.755447
69	N	2.967689	-2.141013	-1.579385
70	N	0.61076	-3.298239	-0.999866
71	C	-3.941832	-3.441064	-2.66671
72	C	-3.309295	-1.985727	-0.73532
73	C	-4.700796	-1.416519	-0.564122
74	O	-5.0798	-0.398064	-1.178883
75	N	-5.522437	-2.072672	0.274643
76	C	-3.296894	-1.328391	-4.572193
77	C	-1.88126	1.563008	-3.982194
78	C	0.6645	3.051882	-4.80533
79	C	1.449453	3.09806	-2.383052
80	C	0.314286	3.97104	-1.871667
81	O	-0.883724	3.668343	-2.001799
82	N	0.665516	5.104906	-1.237105
83	C	2.724302	1.116497	-5.46623
84	C	6.225396	-1.166445	-0.940548
85	C	6.081012	-2.172481	-3.228171
86	C	5.408892	-4.648321	-1.968413
87	C	3.415538	-5.387079	0.390404
88	C	0.183829	-5.087372	1.666448
89	C	0.647818	-6.825543	-0.118513

90	C	-2.2884	-5.422984	0.09273
91	F	1.443521	2.094866	2.320245
92	C	0.750295	1.441308	1.392412
93	C	0.369674	0.15116	1.392912
94	F	0.342034	2.292936	0.441535
95	F	0.492142	-0.619452	-0.056963
96	H	7.480631	-2.450301	5.167137
97	H	6.121313	-3.526803	5.410226
98	H	7.055879	-0.489357	3.920337
99	H	4.641588	-3.964454	3.190234
100	H	5.579463	0.849909	2.439595
101	H	3.164465	-2.624909	1.719179
102	H	3.613696	-0.213645	1.359419
103	H	8.415403	-3.944393	3.408425
104	H	-1.369779	7.201336	1.665271
105	H	-0.980091	4.711582	1.33484
106	H	-2.73517	4.483395	1.300213
107	H	-1.956846	5.105226	2.763539
108	H	-1.949604	7.42708	-0.597164
109	H	-4.246854	6.620298	0.801515
110	H	-4.650225	3.134646	6.305938
111	H	-5.341476	1.537597	6.504115
112	H	-2.912799	3.324773	4.609728
113	H	-5.859584	0.222134	4.269575
114	H	-1.672387	2.35972	2.690476
115	H	-4.59697	-0.765641	2.377834
116	H	-1.412893	0.255854	1.390126
117	H	-7.114611	2.300609	4.678669
118	H	-2.544706	-4.111645	4.946003
119	H	-1.829792	-4.99358	3.618151
120	H	-2.237165	-1.673357	2.229538
121	H	-1.267015	-2.460949	3.426314
122	H	-4.397313	-5.612721	3.047224
123	H	-9.145169	2.53882	-1.100708
124	H	-8.168034	1.646576	0.039634
125	H	-7.754856	4.226985	-0.285613
126	H	-6.838064	3.766549	-1.713592
127	H	-5.52273	4.103795	0.480838
128	H	-6.32599	2.684473	1.117176
129	H	-5.074954	1.290569	-0.400494
130	H	-1.847506	2.204456	-1.294932
131	H	-2.715151	1.068339	-0.282001
132	H	-4.341674	4.61103	-1.363386
133	H	-2.615227	4.336839	-1.376934
134	H	-6.715972	0.902208	-1.989232
135	H	8.109203	6.08329	1.41535

136	H	8.868395	4.607464	1.960024
137	H	8.989188	4.239872	-1.761265
138	H	5.994639	4.428225	2.442719
139	H	3.794022	3.264117	2.366619
140	H	4.516075	1.981758	-1.676789
141	H	3.081446	2.054187	0.33298
142	H	9.870202	6.367884	-0.361857
143	H	-3.505526	-0.361789	-2.686405
144	H	3.390798	2.304734	-3.795284
145	H	4.732992	0.24405	-3.157604
146	H	5.450079	-3.425991	-0.19529
147	H	-0.942208	-5.529773	-1.565108
148	H	-1.170111	-2.902761	-0.043872
149	H	-1.293786	-3.276363	-4.340709
150	H	-1.656591	-4.723443	-3.391063
151	H	-0.024411	-4.022179	-3.375233
152	H	-4.950509	-3.015299	-2.684809
153	H	-3.730197	-3.804575	-3.672491
154	H	-3.969362	-4.305204	-1.999644
155	H	-2.610589	-1.225627	-0.386227
156	H	-3.214541	-2.840853	-0.067434
157	H	-6.428159	-1.655066	0.448475
158	H	-3.182496	-0.403393	-5.138686
159	H	-2.724497	-2.100363	-5.09012
160	H	-1.669203	2.589587	-3.707766
161	H	-1.871509	1.489209	-5.075612
162	H	-2.898199	1.357328	-3.649018
163	H	1.459468	3.726432	-5.141767
164	H	-0.185423	3.668559	-4.516803
165	H	0.363642	2.436792	-5.654771
166	H	2.306487	3.746585	-2.595448
167	H	1.752649	2.465815	-1.542898
168	H	-0.07714	5.675123	-0.830516
169	H	2.718602	1.951809	-6.170646
170	H	1.912839	0.431122	-5.731192
171	H	5.756768	-1.03118	0.033775
172	H	6.400461	-0.174636	-1.369975
173	H	7.19846	-1.646403	-0.788864
174	H	5.487614	-2.716884	-3.967143
175	H	6.298094	-1.183431	-3.640356
176	H	7.039097	-2.685339	-3.107591
177	H	6.488667	-4.64787	-2.137866
178	H	5.168915	-5.561615	-1.42565
179	H	4.41758	-5.081778	0.693995
180	H	2.878335	-5.621428	1.310697
181	H	3.504843	-6.322647	-0.171784

182	H	-0.211388	-4.088567	1.868922
183	H	-0.487933	-5.827031	2.111821
184	H	1.147271	-5.174134	2.173279
185	H	-0.244517	-7.431147	0.067154
186	H	1.437329	-7.210228	0.528946
187	H	-3.153302	-5.333517	-0.567303
188	H	-2.49549	-4.82419	0.979017
189	Cl	0.902989	-0.871584	2.694182
190	H	4.910156	-4.701496	-2.940118
191	H	0.950881	-6.982053	-1.158678
192	H	-2.235245	-6.471671	0.401412
193	H	-4.353503	-1.605645	-4.614281
194	H	3.66851	0.578135	-5.585986
195	H	-7.915785	1.707489	-3.009039
196	H	-4.645701	-5.297057	4.758979
197	H	-3.521889	-6.564865	4.23813
198	H	-7.306063	2.875385	6.320374
199	H	-6.596944	3.937431	5.082838
200	H	-3.591887	8.188806	1.340155
201	H	-3.779676	6.837958	2.491371
202	H	6.959801	-4.954785	3.447244
203	H	8.007001	-4.87858	4.870599
204	H	10.618068	4.837273	0.131176
205	H	10.585016	6.201908	1.253195
206	H	7.044327	2.894228	-2.617502
207	H	1.631829	5.337306	-1.066008
208	H	-5.13813	-2.725009	0.959404
209	C	-8.702264	0.030364	-1.958584
210	O	-9.917364	0.151015	-2.060963
211	N	-8.093821	-1.153747	-1.654184
212	H	-7.090085	-1.204545	-1.759449
213	C	-8.880363	-2.363483	-1.553786
214	H	-8.327778	-3.112498	-0.979705
215	H	-9.81869	-2.133173	-1.046269
216	H	-9.128282	-2.785587	-2.536081

Cl1F3-conf1_React, E_{opt}= -5017.1328071

		Coordinates (Angstroms)

Center number	Atomic Symbol	X	Y	Z
1	C	7.598151	-4.295627	4.045162
2	C	6.150326	-3.918877	4.427171
3	C	5.411568	-2.942151	3.530095
4	C	5.930271	-1.664491	3.264894
5	C	4.135216	-3.247928	3.034686
6	C	5.192568	-0.721225	2.547843
7	C	3.388741	-2.308981	2.31625
8	C	3.915463	-1.039838	2.074687
9	C	-3.523167	7.104286	1.465936
10	C	-2.139333	6.556874	1.08658
11	O	-2.027646	6.418613	-0.358704
12	C	-1.873266	5.180269	1.6799
13	C	-7.310094	2.999281	5.405022
14	C	-5.865691	2.564594	5.74516
15	C	-5.100484	1.940478	4.589542
16	C	-5.418649	0.638372	4.166977
17	C	-4.02762	2.570816	3.944133
18	C	-4.696329	-0.014251	3.173207
19	C	-3.28616	1.926876	2.947583
20	C	-3.588336	0.605444	2.544283
21	O	-2.862953	-0.042997	1.630713
22	C	-3.891379	-5.564328	4.014426
23	C	-2.771305	-4.52061	3.984062
24	C	-3.075279	-3.389093	3.001441
25	O	-4.023056	-3.506528	2.206606
26	N	-2.242016	-2.337034	2.97431
27	C	-7.758666	1.211684	-2.044177
28	C	-8.087201	2.045168	-0.784927
29	C	-7.177823	3.26358	-0.528607
30	C	-5.969681	2.980269	0.375712
31	N	-4.962689	2.086981	-0.195239
32	C	-3.707041	2.432449	-0.51417
33	N	-3.424918	3.690907	-0.907011
34	N	-2.74376	1.510508	-0.468858
35	C	10.017395	5.683859	0.480198
36	C	8.777564	5.020718	1.097845
37	C	7.88534	4.315044	0.115735
38	C	8.080928	4.16389	-1.233322
39	C	6.629438	3.666407	0.415226
40	N	7.023405	3.461599	-1.794954
41	C	6.115635	3.147177	-0.806771
42	C	5.885781	3.478608	1.593625
43	C	4.896346	2.466994	-0.873061

44	C	4.671282	2.799635	1.533197
45	C	4.182268	2.297329	0.309216
46	Co	1.114846	-1.739379	-1.934955
47	C	-1.465885	-2.880687	-2.210239
48	C	-2.954325	-2.390894	-2.149262
49	C	-2.884608	-1.063235	-3.000978
50	C	-1.428361	-0.649527	-2.867008
51	C	-0.899213	0.655496	-3.232347
52	C	0.454232	0.899401	-3.223581
53	C	1.212106	2.195341	-3.648205
54	C	2.586645	1.578716	-4.046653
55	C	2.646049	0.368879	-3.158037
56	C	3.837443	-0.255332	-2.835336
57	C	3.985662	-1.440374	-2.127174
58	C	5.351636	-2.042162	-1.876338
59	C	4.984079	-3.397415	-1.184948
60	C	3.465181	-3.337637	-1.040234
61	C	2.709934	-4.30763	-0.415369
62	C	1.269171	-4.29423	-0.476412
63	C	0.304379	-5.34361	0.123631
64	C	-1.006921	-5.014206	-0.660777
65	C	-0.859998	-3.514946	-0.938576
66	C	-1.148845	-3.753094	-3.442897
67	N	-0.700681	-1.612007	-2.372648
68	N	1.41318	-0.043583	-2.815038
69	N	2.970536	-2.187049	-1.652244
70	N	0.602262	-3.324391	-1.048257
71	C	-3.94193	-3.441042	-2.666813
72	C	-3.33921	-2.03763	-0.688214
73	C	-4.74962	-1.501341	-0.519994
74	O	-5.197991	-0.581056	-1.237666
75	N	-5.50263	-2.0819	0.422126
76	C	-3.366919	-1.161347	-4.459613
77	C	-1.895797	1.718391	-3.655054
78	C	0.664582	3.051921	-4.805359
79	C	1.506001	3.103099	-2.399237
80	C	0.330294	3.827807	-1.761479
81	O	-0.559901	3.213326	-1.160352
82	N	0.303841	5.174622	-1.85196
83	C	2.703834	1.11772	-5.514614
84	C	6.182416	-1.146855	-0.934204
85	C	6.080942	-2.172397	-3.228092
86	C	5.45577	-4.652752	-1.943176
87	C	3.401292	-5.407238	0.366926
88	C	0.148863	-5.061795	1.635829
89	C	0.64779	-6.825517	-0.118478

90	C	-2.293992	-5.471755	0.02226
91	F	1.098394	0.168966	3.320234
92	C	0.636023	0.622616	2.154561
93	C	0.157032	-0.227275	1.253654
94	Cl	0.692345	2.323456	1.927346
95	F	-0.284365	0.117602	0.054721
96	H	6.171205	-3.484575	5.43618
97	H	5.559108	-4.837965	4.515291
98	H	6.911999	-1.396561	3.646904
99	H	3.709436	-4.22723	3.239787
100	H	5.602559	0.26733	2.360727
101	H	2.394228	-2.559296	1.961851
102	H	3.34379	-0.303212	1.518919
103	H	8.227597	-3.410991	3.908654
104	H	-1.357684	7.251797	1.426896
105	H	-0.982436	4.730599	1.232612
106	H	-2.715268	4.50681	1.506674
107	H	-1.718898	5.260411	2.759635
108	H	-2.295139	7.25782	-0.757729
109	H	-4.27464	6.629929	0.823828
110	H	-5.30583	3.418642	6.144131
111	H	-5.907059	1.828983	6.558613
112	H	-3.741859	3.577111	4.248296
113	H	-6.241588	0.114621	4.651481
114	H	-2.423769	2.412221	2.498375
115	H	-4.944626	-1.032224	2.890869
116	H	-7.739884	2.317658	4.661842
117	H	-2.572805	-4.099946	4.976916
118	H	-1.82671	-4.97148	3.65815
119	H	-2.472246	-1.499703	2.38583
120	H	-1.555925	-2.229908	3.70666
121	H	-4.377064	-5.606872	3.037708
122	H	-9.128046	2.369665	-0.876625
123	H	-8.055422	1.387101	0.095234
124	H	-7.76162	4.045551	-0.027115
125	H	-6.837821	3.692091	-1.481337
126	H	-5.473107	3.915057	0.654949
127	H	-6.303397	2.527077	1.316863
128	H	-5.139088	1.08607	-0.187946
129	H	-1.822366	1.826911	-0.75012
130	H	-2.781452	0.813446	0.348552
131	H	-4.153065	4.365695	-1.065226
132	H	-2.459607	3.999604	-0.916889
133	H	-6.709428	0.915923	-2.05904
134	H	8.18754	5.78148	1.627889
135	H	9.095214	4.308946	1.872838

136	H	8.898409	4.504475	-1.851734
137	H	6.254018	3.863838	2.540216
138	H	4.086826	2.652076	2.436394
139	H	4.521928	2.073437	-1.809388
140	H	3.23177	1.771243	0.290472
141	H	9.733998	6.417969	-0.281499
142	H	-3.512163	-0.319342	-2.508781
143	H	3.404142	2.276205	-3.835315
144	H	4.741591	0.220194	-3.192241
145	H	5.428512	-3.411813	-0.182859
146	H	-0.917856	-5.542065	-1.617096
147	H	-1.179105	-2.934445	-0.070509
148	H	-1.399307	-3.226115	-4.363724
149	H	-1.698141	-4.695839	-3.427995
150	H	-0.078779	-3.980923	-3.467514
151	H	-4.964569	-3.055274	-2.620806
152	H	-3.756014	-3.731047	-3.701721
153	H	-3.906584	-4.342136	-2.053091
154	H	-2.676674	-1.270756	-0.284246
155	H	-3.24114	-2.910264	-0.045138
156	H	-6.390834	-1.644602	0.62959
157	H	-3.280491	-0.19662	-4.964102
158	H	-2.807668	-1.892711	-5.048479
159	H	-1.624873	2.694657	-3.261621
160	H	-1.978075	1.797506	-4.744339
161	H	-2.887325	1.498913	-3.262083
162	H	1.456163	3.721882	-5.156119
163	H	-0.17654	3.675694	-4.500718
164	H	0.339042	2.438742	-5.647287
165	H	2.270062	3.828757	-2.697455
166	H	1.944848	2.49315	-1.60408
167	H	-0.422176	5.6801	-1.339623
168	H	2.696482	1.96338	-6.205882
169	H	1.883076	0.443983	-5.780373
170	H	5.685131	-1.016113	0.027599
171	H	6.348631	-0.154815	-1.364193
172	H	7.158656	-1.609653	-0.754748
173	H	5.493715	-2.721498	-3.96859
174	H	6.297669	-1.183704	-3.640007
175	H	7.040474	-2.681346	-3.103739
176	H	6.539778	-4.632455	-2.079787
177	H	5.216803	-5.565033	-1.398626
178	H	4.397238	-5.100463	0.685354
179	H	2.8521	-5.637855	1.27992
180	H	3.49307	-6.340731	-0.197336
181	H	-0.276564	-4.07308	1.828318

182	H	-0.511046	-5.81105	2.080985
183	H	1.108426	-5.120756	2.155293
184	H	-0.247567	-7.430939	0.048673
185	H	1.418402	-7.203208	0.554882
186	H	-3.147506	-5.402862	-0.653706
187	H	-2.538934	-4.88322	0.905577
188	F	0.142803	-1.541678	1.438822
189	H	4.988347	-4.723098	-2.929043
190	H	0.978368	-6.994146	-1.148388
191	H	-2.219385	-6.52049	0.325487
192	H	-4.421843	-1.445051	-4.476392
193	H	3.642404	0.574686	-5.655496
194	H	-7.962262	1.815882	-2.934634
195	H	-4.659321	-5.306044	4.748391
196	H	-3.521905	-6.564874	4.2382
197	H	-7.974645	2.964829	6.268737
198	H	-7.34049	4.005502	4.974668
199	H	-3.596635	8.189238	1.344862
200	H	-3.779655	6.838161	2.491443
201	H	7.636599	-4.889156	3.126882
202	H	8.007088	-4.878653	4.87055
203	H	10.673577	4.94327	0.011136
204	H	10.584966	6.201911	1.253179
205	H	6.9373	3.228311	-2.769779
206	H	1.060446	5.689774	-2.272779
207	H	-5.054101	-2.671863	1.131758
208	C	-8.691505	0.015296	-2.062335
209	O	-9.890504	0.120835	-2.300213
210	N	-8.102673	-1.150953	-1.677213
211	H	-7.091749	-1.16757	-1.612004
212	C	-8.880371	-2.363474	-1.553798
213	H	-8.378599	-3.052028	-0.867659
214	H	-9.869386	-2.114038	-1.163947
215	H	-9.021721	-2.871597	-2.516876

Cl1F3-conf1_Int, E_{opt}= -5017.7147188

		Coordinates (Angstroms)

Center number	Atomic Symbol	X	Y	Z
1	C	7.598077	-4.295721	4.044993
2	C	6.197331	-3.82308	4.49297
3	C	5.459653	-2.879218	3.565519
4	C	5.933857	-1.577112	3.340957
5	C	4.248555	-3.252799	2.967249
6	C	5.218363	-0.676876	2.551129
7	C	3.524683	-2.356707	2.175525
8	C	4.009721	-1.066281	1.965435
9	C	-3.523055	7.10448	1.465948
10	C	-2.039151	6.817625	1.182076
11	O	-1.821363	6.59215	-0.234204
12	C	-1.52848	5.569473	1.887682
13	C	-7.310023	2.999135	5.404998
14	C	-5.862811	2.61878	5.783987
15	C	-5.085681	2.017872	4.631211
16	C	-5.266297	0.662689	4.31041
17	C	-4.187678	2.758224	3.853391
18	C	-4.569521	0.054573	3.271214
19	C	-3.467713	2.16382	2.811963
20	C	-3.655545	0.809297	2.531992
21	O	-2.928084	0.153078	1.562987
22	C	-3.891482	-5.564288	4.014365
23	C	-2.751128	-4.55023	3.89485
24	C	-3.096488	-3.364462	2.997173
25	O	-4.166265	-3.306688	2.376516
26	N	-2.146224	-2.408985	2.874186
27	C	-7.75864	1.211639	-2.044128
28	C	-8.095898	2.165749	-0.876234
29	C	-7.274416	3.468308	-0.805362
30	C	-6.039499	3.394186	0.100381
31	N	-5.020943	2.458991	-0.376071
32	C	-3.745639	2.735026	-0.644559
33	N	-3.235943	3.966097	-0.628345
34	N	-2.899069	1.693523	-0.87909
35	C	10.017493	5.684	0.480232
36	C	8.739898	5.085457	1.089551
37	C	7.84635	4.369096	0.115956
38	C	8.047621	4.191909	-1.229504
39	C	6.587961	3.727057	0.422393
40	N	6.988267	3.488479	-1.783976
41	C	6.080727	3.184428	-0.792334
42	C	5.842666	3.550048	1.602348
43	C	4.870879	2.487526	-0.85228

44	C	4.638027	2.852451	1.549829
45	C	4.160349	2.324958	0.332009
46	Co	1.13116	-1.768296	-1.989992
47	C	-1.421206	-2.913272	-2.259685
48	C	-2.909911	-2.406895	-2.233227
49	C	-2.828844	-1.172088	-3.197351
50	C	-1.382795	-0.726092	-3.025087
51	C	-0.877532	0.562349	-3.40623
52	C	0.472809	0.848069	-3.296333
53	C	1.210529	2.167347	-3.67241
54	C	2.597307	1.577912	-4.05861
55	C	2.663218	0.377879	-3.151565
56	C	3.853498	-0.229906	-2.806065
57	C	4.000018	-1.424714	-2.115676
58	C	5.368605	-2.030337	-1.87287
59	C	5.00088	-3.384565	-1.184838
60	C	3.476564	-3.341071	-1.084966
61	C	2.72733	-4.327619	-0.472716
62	C	1.294558	-4.307467	-0.520723
63	C	0.333675	-5.335765	0.115564
64	C	-0.991963	-4.995355	-0.63654
65	C	-0.817599	-3.509582	-0.966722
66	C	-1.110522	-3.850622	-3.446053
67	N	-0.644318	-1.658454	-2.454122
68	N	1.424219	-0.069606	-2.841585
69	N	2.972837	-2.186909	-1.680682
70	N	0.638995	-3.324361	-1.099924
71	C	-3.941902	-3.441069	-2.666704
72	C	-3.245823	-1.938584	-0.780179
73	C	-4.601224	-1.302635	-0.568087
74	O	-4.939518	-0.254449	-1.173932
75	N	-5.432074	-1.905803	0.28977
76	C	-3.23993	-1.410414	-4.665505
77	C	-1.86271	1.555017	-4.004254
78	C	0.66455	3.051877	-4.805244
79	C	1.479783	3.031865	-2.390609
80	C	0.337293	3.847456	-1.825335
81	O	-0.814238	3.393206	-1.703132
82	N	0.621372	5.097053	-1.408149
83	C	2.720857	1.116077	-5.524006
84	C	6.230491	-1.148774	-0.948809
85	C	6.080921	-2.172495	-3.228209
86	C	5.527735	-4.634796	-1.914161
87	C	3.426386	-5.428053	0.303635
88	C	0.223992	-5.044516	1.628804
89	C	0.647792	-6.825513	-0.118458

90	C	-2.272322	-5.38289	0.101729
91	F	1.6342	0.747703	2.720504
92	C	0.804622	0.958579	1.700287
93	C	0.23483	-0.053476	1.051163
94	Cl	0.519261	2.610706	1.302664
95	F	-0.589294	0.120235	0.030104
96	H	6.305019	-3.326938	5.46697
97	H	5.572528	-4.70554	4.675024
98	H	6.866524	-1.263329	3.804066
99	H	3.861417	-4.254968	3.135407
100	H	5.596292	0.328037	2.384959
101	H	2.591863	-2.66464	1.716876
102	H	3.463757	-0.37305	1.335922
103	H	8.265826	-3.45205	3.846043
104	H	-1.434093	7.677833	1.502241
105	H	-0.482135	5.385262	1.629089
106	H	-2.115764	4.698033	1.589582
107	H	-1.596747	5.686	2.972548
108	H	-2.035962	7.405093	-0.711449
109	H	-4.147998	6.48015	0.817352
110	H	-5.336129	3.493946	6.180529
111	H	-5.889137	1.884914	6.597764
112	H	-4.021922	3.809046	4.079438
113	H	-5.962631	0.068575	4.897297
114	H	-2.73309	2.731684	2.248429
115	H	-4.706823	-0.995365	3.034993
116	H	-2.612541	0.778664	0.883432
117	H	-7.711609	2.283441	4.678885
118	H	-2.437174	-4.172295	4.87569
119	H	-1.861689	-5.018025	3.459627
120	H	-2.328033	-1.582481	2.311855
121	H	-1.293396	-2.447429	3.410065
122	H	-4.436817	-5.619382	3.069056
123	H	-9.159001	2.404315	-0.966508
124	H	-7.993905	1.621242	0.074043
125	H	-7.904686	4.267723	-0.397957
126	H	-6.972015	3.789646	-1.809577
127	H	-5.593797	4.386564	0.205285
128	H	-6.326605	3.083074	1.112446
129	H	-5.269732	1.469754	-0.434501
130	H	-2.002655	1.922922	-1.309495
131	H	-3.370731	0.852747	-1.214724
132	H	-3.802016	4.783768	-0.489774
133	H	-2.27167	4.088918	-0.959163
134	H	-6.709974	0.913473	-2.04307
135	H	8.16541	5.885391	1.578122

136	H	9.015923	4.393483	1.897718
137	H	8.87096	4.515449	-1.849383
138	H	6.206842	3.949865	2.544646
139	H	4.057495	2.697064	2.454643
140	H	4.505764	2.059099	-1.776579
141	H	3.227804	1.769469	0.314783
142	H	9.77959	6.405603	-0.308505
143	H	-3.514887	-0.407359	-2.833313
144	H	3.399679	2.297079	-3.854283
145	H	4.760035	0.255183	-3.148964
146	H	5.414585	-3.38246	-0.168401
147	H	-0.953699	-5.559439	-1.573879
148	H	-1.125555	-2.902826	-0.111361
149	H	-1.29447	-3.353723	-4.397249
150	H	-1.702865	-4.767916	-3.414592
151	H	-0.050336	-4.110927	-3.415782
152	H	-4.947223	-3.004141	-2.671978
153	H	-3.754584	-3.823654	-3.669781
154	H	-3.96543	-4.293738	-1.984927
155	H	-2.502121	-1.204062	-0.466955
156	H	-3.164363	-2.783979	-0.098103
157	H	-6.308475	-1.439112	0.487601
158	H	-3.11689	-0.490691	-5.24127
159	H	-2.646452	-2.18166	-5.157738
160	H	-1.689732	2.565289	-3.643596
161	H	-1.813323	1.574447	-5.098644
162	H	-2.89168	1.304387	-3.742761
163	H	1.45019	3.745028	-5.125918
164	H	-0.192369	3.652082	-4.497182
165	H	0.365862	2.459848	-5.671372
166	H	2.319374	3.702948	-2.601058
167	H	1.81508	2.373136	-1.581292
168	H	-0.11304	5.657174	-0.976555
169	H	2.702044	1.956128	-6.223947
170	H	1.910693	0.426589	-5.782001
171	H	5.752528	-1.006481	0.020228
172	H	6.403237	-0.16016	-1.386039
173	H	7.205423	-1.621342	-0.783148
174	H	5.482255	-2.728315	-3.954571
175	H	6.282963	-1.186347	-3.655204
176	H	7.046701	-2.675218	-3.120142
177	H	6.61458	-4.586297	-2.024144
178	H	5.301576	-5.548891	-1.367115
179	H	4.414646	-5.112074	0.637636
180	H	2.873994	-5.683998	1.208625
181	H	3.541226	-6.354367	-0.269774

182	H	-0.175924	-4.04322	1.812771
183	H	-0.431177	-5.77744	2.109651
184	H	1.198398	-5.107074	2.118786
185	H	-0.246841	-7.422238	0.08806
186	H	1.441	-7.204137	0.527683
187	H	-3.147478	-5.309159	-0.547006
188	H	-2.456257	-4.746272	0.967297
189	F	0.395907	-1.322588	1.380942
190	H	5.087836	-4.729815	-2.910452
191	H	0.940064	-7.006295	-1.157669
192	H	-2.226996	-6.418598	0.452983
193	H	-4.293188	-1.698572	-4.729946
194	H	3.666246	0.583221	-5.658038
195	H	-7.965443	1.725412	-2.989601
196	H	-4.608581	-5.274218	4.786475
197	H	-3.521875	-6.564753	4.238193
198	H	-7.974586	2.964844	6.268692
199	H	-7.364592	3.992991	4.950502
200	H	-3.791934	8.149119	1.286525
201	H	-3.779749	6.837971	2.491382
202	H	7.549611	-4.91816	3.146767
203	H	8.007018	-4.878491	4.870634
204	H	10.654716	4.905476	0.048206
205	H	10.584953	6.201882	1.253241
206	H	6.928414	3.191682	-2.743628
207	H	1.55434	5.472763	-1.474673
208	H	-5.064002	-2.568143	0.977565
209	C	-8.689708	0.022781	-1.943008
210	O	-9.906825	0.141086	-2.015711
211	N	-8.077414	-1.162983	-1.653669
212	H	-7.081083	-1.233673	-1.803552
213	C	-8.880392	-2.363477	-1.553798
214	H	-8.308189	-3.14182	-1.042227
215	H	-9.782597	-2.139303	-0.982432
216	H	-9.192842	-2.739507	-2.535875

Cl1F3-conf1_TS, E_{opt}= -5017.6621585

1	C	7.598223	-4.295635	4.045024

2	C	6.205786	-3.815142	4.504487
3	C	5.461251	-2.886105	3.570576
4	C	5.951705	-1.600306	3.296322
5	C	4.228225	-3.255204	3.016824
6	C	5.227511	-0.709334	2.504512
7	C	3.495915	-2.367777	2.224475
8	C	3.994417	-1.091156	1.966976
9	C	-3.523126	7.104507	1.466
10	C	-2.088611	6.612899	1.210108
11	O	-1.785364	6.533234	-0.204275
12	C	-1.852962	5.21612	1.771659
13	C	-6.641504	2.909907	5.456621
14	C	-5.238124	2.36005	5.784882
15	C	-4.476102	1.851382	4.57519
16	C	-4.940525	0.710445	3.899816
17	C	-3.28765	2.434653	4.120917
18	C	-4.235841	0.154484	2.838683
19	C	-2.566999	1.895957	3.046945
20	C	-3.036585	0.741306	2.41897
21	O	-2.371314	0.130747	1.37405
22	C	-3.89151	-5.56436	4.014486
23	C	-2.757371	-4.535331	3.966888
24	C	-3.048049	-3.384425	3.007115
25	O	-4.111326	-3.317926	2.377266
26	N	-2.062731	-2.477784	2.83369
27	C	-7.758699	1.211684	-2.044212
28	C	-8.088306	2.11956	-0.838414
29	C	-7.164801	3.334949	-0.627542
30	C	-5.947436	3.071653	0.274267
31	N	-4.892157	2.24487	-0.309708
32	C	-3.714439	2.70043	-0.753169
33	N	-3.539671	3.996747	-1.061616
34	N	-2.690285	1.848915	-0.876202
35	C	10.017445	5.683912	0.480218
36	C	8.729073	5.085232	1.060013
37	C	7.853529	4.416577	0.03807
38	C	8.063954	4.357105	-1.316592
39	C	6.615813	3.71292	0.285548
40	N	7.029031	3.670098	-1.932307
41	C	6.132049	3.255699	-0.972707
42	C	5.869651	3.415063	1.440737
43	C	4.953211	2.517763	-1.095446
44	C	4.694363	2.675966	1.324012
45	C	4.246329	2.223012	0.06542
46	Co	1.095184	-1.679143	-1.81784
47	C	-1.443222	-2.889038	-2.193399

48	C	-2.936763	-2.40286	-2.177319
49	C	-2.861375	-1.125552	-3.094179
50	C	-1.409645	-0.6948	-2.939709
51	C	-0.896832	0.599644	-3.34462
52	C	0.446266	0.889422	-3.238475
53	C	1.19582	2.19267	-3.64981
54	C	2.572411	1.573276	-4.033592
55	C	2.633715	0.386629	-3.110421
56	C	3.828171	-0.23585	-2.791869
57	C	3.982664	-1.421685	-2.088374
58	C	5.352842	-2.040453	-1.875028
59	C	4.9818	-3.39745	-1.192139
60	C	3.465411	-3.320384	-1.021287
61	C	2.716555	-4.306496	-0.412091
62	C	1.27771	-4.288776	-0.460582
63	C	0.321918	-5.342148	0.145377
64	C	-1.006239	-4.991088	-0.600561
65	C	-0.844812	-3.495971	-0.899192
66	C	-1.102466	-3.792754	-3.397521
67	N	-0.686405	-1.620873	-2.37297
68	N	1.40104	-0.021102	-2.759969
69	N	2.96599	-2.150999	-1.586567
70	N	0.611534	-3.306157	-1.00931
71	C	-3.94182	-3.441049	-2.666707
72	C	-3.317454	-1.99501	-0.725429
73	C	-4.717212	-1.444558	-0.55767
74	O	-5.116519	-0.445722	-1.192174
75	N	-5.523692	-2.094969	0.299277
76	C	-3.312661	-1.315517	-4.555867
77	C	-1.885577	1.566205	-3.971879
78	C	0.664491	3.051876	-4.805321
79	C	1.47	3.085284	-2.393477
80	C	0.337323	3.943801	-1.849833
81	O	-0.862507	3.660526	-1.998277
82	N	0.696056	5.041048	-1.154355
83	C	2.68259	1.07801	-5.489954
84	C	6.209323	-1.161415	-0.940634
85	C	6.081011	-2.172482	-3.228162
86	C	5.428059	-4.648823	-1.972086
87	C	3.416213	-5.421747	0.341036
88	C	0.214838	-5.071641	1.662499
89	C	0.647816	-6.825541	-0.118515
90	C	-2.284761	-5.408292	0.12339
91	F	1.91283	0.979683	2.855843
92	C	1.039182	0.931187	1.841359
93	C	0.319711	-0.173756	1.477502

94	Cl	0.760362	2.469731	1.100822
95	F	0.473065	-0.654641	-0.064107
96	H	6.326471	-3.304155	5.469244
97	H	5.581859	-4.693472	4.708374
98	H	6.901895	-1.289772	3.724558
99	H	3.827871	-4.244202	3.228112
100	H	5.617715	0.283895	2.302947
101	H	2.536152	-2.659624	1.817439
102	H	3.432552	-0.404803	1.344278
103	H	8.270774	-3.457867	3.838089
104	H	-1.37997	7.30925	1.683851
105	H	-0.847807	4.862739	1.527637
106	H	-2.57567	4.505705	1.363434
107	H	-1.955492	5.222168	2.860338
108	H	-1.905275	7.410659	-0.591981
109	H	-4.213986	6.576621	0.799861
110	H	-4.649076	3.123087	6.305444
111	H	-5.343145	1.526102	6.490395
112	H	-2.900784	3.317903	4.623582
113	H	-5.853803	0.226443	4.237861
114	H	-1.640816	2.353288	2.714103
115	H	-4.571109	-0.759277	2.361785
116	H	-1.333742	0.161239	1.439406
117	H	-7.11784	2.30521	4.676776
118	H	-2.543805	-4.11835	4.958817
119	H	-1.821623	-4.998002	3.634251
120	H	-2.215281	-1.687299	2.215621
121	H	-1.199558	-2.501386	3.353743
122	H	-4.392228	-5.609339	3.044496
123	H	-9.121543	2.45555	-0.964045
124	H	-8.082484	1.511394	0.077425
125	H	-7.739288	4.132556	-0.140375
126	H	-6.833445	3.738502	-1.59413
127	H	-5.491953	4.01621	0.588236
128	H	-6.271039	2.570185	1.193948
129	H	-5.033894	1.236257	-0.386971
130	H	-1.828124	2.198915	-1.286395
131	H	-2.635054	1.067363	-0.22367
132	H	-4.330895	4.581799	-1.271963
133	H	-2.604439	4.327673	-1.293174
134	H	-6.713051	0.905384	-2.041007
135	H	8.155512	5.875399	1.565121
136	H	8.983535	4.361873	1.847087
137	H	8.878427	4.754095	-1.904192
138	H	6.211186	3.754031	2.414799
139	H	4.112047	2.428587	2.20638

140	H	4.611368	2.165802	-2.059168
141	H	3.345504	1.621412	-0.00061
142	H	9.792018	6.412995	-0.305768
143	H	-3.50585	-0.357485	-2.66323
144	H	3.386671	2.281633	-3.847558
145	H	4.728546	0.228741	-3.172536
146	H	5.441851	-3.428471	-0.196909
147	H	-0.960518	-5.532463	-1.551335
148	H	-1.159917	-2.895662	-0.041986
149	H	-1.314166	-3.283023	-4.337034
150	H	-1.664108	-4.728726	-3.380964
151	H	-0.0343	-4.023647	-3.380404
152	H	-4.952277	-3.019869	-2.678768
153	H	-3.730786	-3.794595	-3.676034
154	H	-3.963483	-4.311956	-2.007764
155	H	-2.631102	-1.226453	-0.369106
156	H	-3.214016	-2.850721	-0.059667
157	H	-6.430741	-1.681365	0.475649
158	H	-3.202348	-0.387895	-5.118624
159	H	-2.743899	-2.085155	-5.081487
160	H	-1.663428	2.592449	-3.70477
161	H	-1.887521	1.486597	-5.06491
162	H	-2.900017	1.368569	-3.62647
163	H	1.46537	3.716068	-5.148323
164	H	-0.173984	3.679392	-4.506484
165	H	0.346238	2.443548	-5.653298
166	H	2.312937	3.746558	-2.624243
167	H	1.80687	2.455755	-1.562848
168	H	-0.043683	5.620811	-0.754696
169	H	2.671927	1.906234	-6.202542
170	H	1.86004	0.397932	-5.733428
171	H	5.731779	-1.033602	0.0301
172	H	6.375492	-0.167578	-1.367307
173	H	7.184812	-1.634276	-0.782411
174	H	5.492249	-2.7192	-3.969161
175	H	6.297346	-1.18304	-3.639511
176	H	7.040424	-2.681836	-3.104378
177	H	6.508715	-4.635852	-2.134624
178	H	5.19587	-5.565045	-1.431597
179	H	4.416432	-5.121827	0.652948
180	H	2.880532	-5.669228	1.257876
181	H	3.501518	-6.346802	-0.23869
182	H	-0.146971	-4.059467	1.85874
183	H	-0.470882	-5.788081	2.123648
184	H	1.182527	-5.180813	2.156916
185	H	-0.245857	-7.428402	0.070545

186	H	1.437088	-7.212335	0.527241
187	H	-3.15862	-5.312588	-0.52407
188	H	-2.472013	-4.805041	1.011018
189	F	0.594129	-1.296935	2.19993
190	H	4.936408	-4.707359	-2.947055
191	H	0.946839	-6.98589	-1.15926
192	H	-2.235577	-6.45653	0.434172
193	H	-4.369532	-1.591995	-4.591393
194	H	3.619651	0.529827	-5.620529
195	H	-7.952794	1.76486	-2.969646
196	H	-4.64818	-5.296338	4.755819
197	H	-3.521898	-6.564839	4.238131
198	H	-7.306083	2.875386	6.3204
199	H	-6.594113	3.938589	5.086088
200	H	-3.638572	8.182278	1.321865
201	H	-3.779688	6.837969	2.49138
202	H	7.538503	-4.922217	3.15017
203	H	8.006947	-4.878598	4.870619
204	H	10.655575	4.906879	0.046966
205	H	10.585026	6.201941	1.253195
206	H	6.979974	3.454222	-2.913849
207	H	1.66389	5.275848	-0.995237
208	H	-5.126199	-2.725817	0.997398
209	C	-8.699284	0.025195	-1.994303
210	O	-9.911017	0.141805	-2.135701
211	N	-8.095928	-1.153445	-1.663259
212	H	-7.087973	-1.196741	-1.725942
213	C	-8.880364	-2.363485	-1.553786
214	H	-8.340053	-3.096684	-0.948276
215	H	-9.832596	-2.12553	-1.076305
216	H	-9.101296	-2.810509	-2.53156

Cl1F3-conf2_React, E_{opt}= -5017.1344936

		Coordinates (Angstroms)

Center number	Atomic Symbol	X	Y	Z
1	C	7.598234	-4.295531	4.045147
2	C	6.15368	-3.89971	4.422764
3	C	5.423247	-2.923055	3.51636
4	C	5.934799	-1.638774	3.267225
5	C	4.15752	-3.235051	2.997282
6	C	5.199362	-0.695016	2.547557
7	C	3.41434	-2.296103	2.274205
8	C	3.932028	-1.019735	2.051757
9	C	-3.523156	7.104432	1.465992
10	C	-2.122451	6.561455	1.136595
11	O	-1.921451	6.475906	-0.299746
12	C	-1.90228	5.158787	1.689707
13	C	-7.310113	2.999243	5.40499
14	C	-5.869117	2.547701	5.738749
15	C	-5.099808	1.943129	4.574352
16	C	-5.420675	0.651814	4.121529
17	C	-4.015303	2.578519	3.952955
18	C	-4.689121	0.013786	3.12442
19	C	-3.265415	1.949452	2.953255
20	C	-3.568652	0.63714	2.521244
21	O	-2.837014	0.003104	1.602628
22	C	-3.891375	-5.564323	4.014434
23	C	-2.775367	-4.517586	3.945858
24	C	-3.108635	-3.389219	2.966472
25	O	-4.0963	-3.495178	2.221586
26	N	-2.253469	-2.355038	2.886276
27	C	-7.758672	1.211676	-2.044183
28	C	-8.129129	2.06471	-0.809406
29	C	-7.20977	3.266627	-0.519694
30	C	-6.020754	2.957767	0.40264
31	N	-4.996851	2.082562	-0.166949
32	C	-3.768402	2.472865	-0.536722
33	N	-3.562552	3.743168	-0.938068
34	N	-2.763725	1.595974	-0.496791
35	C	10.017422	5.683847	0.480204
36	C	8.788395	5.015749	1.11718
37	C	7.872606	4.303258	0.161392
38	C	8.050436	4.098186	-1.182723
39	C	6.599925	3.704981	0.494037
40	N	6.96705	3.406553	-1.709337
41	C	6.055705	3.161301	-0.704061
42	C	5.8646	3.583638	1.686053
43	C	4.807696	2.530565	-0.736342

44	C	4.623628	2.952943	1.660423
45	C	4.098571	2.435407	0.457928
46	Co	1.12706	-1.745843	-1.95558
47	C	-1.454455	-2.899005	-2.223053
48	C	-2.940791	-2.399902	-2.167985
49	C	-2.869365	-1.097236	-3.052286
50	C	-1.412596	-0.685681	-2.931602
51	C	-0.893787	0.618033	-3.31109
52	C	0.453135	0.888681	-3.252499
53	C	1.201858	2.200288	-3.640678
54	C	2.593439	1.604612	-4.014086
55	C	2.651811	0.391016	-3.130447
56	C	3.84723	-0.221946	-2.794444
57	C	4.000881	-1.418732	-2.108278
58	C	5.369154	-2.021881	-1.866262
59	C	5.005472	-3.370405	-1.157657
60	C	3.482543	-3.332437	-1.046913
61	C	2.726523	-4.310716	-0.433413
62	C	1.285835	-4.296397	-0.4899
63	C	0.320893	-5.338396	0.124646
64	C	-0.998085	-5.003178	-0.643303
65	C	-0.845137	-3.508391	-0.940013
66	C	-1.144683	-3.800841	-3.436617
67	N	-0.684832	-1.638151	-2.418619
68	N	1.417691	-0.041467	-2.819097
69	N	2.986327	-2.179688	-1.652356
70	N	0.617659	-3.324871	-1.058795
71	C	-3.941948	-3.441071	-2.666816
72	C	-3.323616	-2.01462	-0.713542
73	C	-4.739095	-1.487413	-0.562699
74	O	-5.184377	-0.577859	-1.29453
75	N	-5.49682	-2.063782	0.378614
76	C	-3.356546	-1.23273	-4.507271
77	C	-1.901204	1.636913	-3.808692
78	C	0.664589	3.0519	-4.805326
79	C	1.455095	3.090335	-2.370432
80	C	0.283802	3.885294	-1.809978
81	O	-0.812128	3.357356	-1.581457
82	N	0.49366	5.187826	-1.538679
83	C	2.744838	1.150038	-5.481129
84	C	6.219223	-1.121961	-0.946417
85	C	6.080957	-2.172413	-3.228132
86	C	5.532215	-4.628967	-1.87214
87	C	3.416977	-5.412519	0.347452
88	C	0.187892	-5.043838	1.636809
89	C	0.647785	-6.82552	-0.11847

90	C	-2.281106	-5.439343	0.060571
91	Cl	1.230891	1.46871	3.189642
92	C	0.544802	1.224084	1.639858
93	C	0.083762	0.062026	1.190425
94	F	0.467078	2.312761	0.854987
95	F	-0.414798	-0.077557	-0.030763
96	H	6.178159	-3.456537	5.4277
97	H	5.552683	-4.811699	4.518073
98	H	6.908236	-1.365266	3.666499
99	H	3.735968	-4.218959	3.188335
100	H	5.60128	0.300373	2.378893
101	H	2.427548	-2.55318	1.901334
102	H	3.358483	-0.278491	1.503361
103	H	8.235401	-3.417395	3.901346
104	H	-1.360132	7.236429	1.554226
105	H	-0.971191	4.724319	1.312909
106	H	-2.724595	4.498368	1.408358
107	H	-1.848734	5.190754	2.781773
108	H	-2.142239	7.335183	-0.684085
109	H	-4.255426	6.618055	0.812035
110	H	-5.305794	3.389644	6.158256
111	H	-5.917926	1.795284	6.536354
112	H	-3.728362	3.577825	4.278332
113	H	-6.252585	0.123286	4.585154
114	H	-2.400861	2.442973	2.516891
115	H	-4.941275	-0.996402	2.818823
116	H	-7.747405	2.328048	4.656536
117	H	-2.551506	-4.089863	4.930426
118	H	-1.838376	-4.967481	3.597985
119	H	-2.491473	-1.50781	2.317139
120	H	-1.536104	-2.254474	3.588877
121	H	-4.402456	-5.614921	3.050872
122	H	-9.158456	2.407395	-0.950617
123	H	-8.149236	1.416196	0.078003
124	H	-7.79388	4.048407	-0.018069
125	H	-6.850126	3.705736	-1.460986
126	H	-5.529443	3.883912	0.717223
127	H	-6.374517	2.476578	1.321717
128	H	-5.138315	1.078354	-0.139871
129	H	-1.868222	1.926503	-0.839846
130	H	-2.767783	0.891911	0.308553
131	H	-4.336738	4.319641	-1.221329
132	H	-2.613442	4.062412	-1.10826
133	H	-6.710328	0.91535	-2.019842
134	H	8.20644	5.7768	1.655604
135	H	9.122871	4.308864	1.88982

136	H	8.871162	4.393077	-1.820245
137	H	6.259252	3.984745	2.615213
138	H	4.043779	2.858428	2.57344
139	H	4.406377	2.119751	-1.654817
140	H	3.118988	1.965631	0.465211
141	H	9.721946	6.416246	-0.278206
142	H	-3.493657	-0.338617	-2.578281
143	H	3.398097	2.311683	-3.786186
144	H	4.750199	0.270421	-3.132078
145	H	5.418238	-3.353137	-0.141214
146	H	-0.92724	-5.54142	-1.594886
147	H	-1.160327	-2.911848	-0.079494
148	H	-1.384644	-3.289407	-4.368821
149	H	-1.706162	-4.735808	-3.405798
150	H	-0.077369	-4.041778	-3.452646
151	H	-4.955347	-3.03034	-2.645724
152	H	-3.750479	-3.765961	-3.690051
153	H	-3.937681	-4.325823	-2.027628
154	H	-2.667515	-1.233415	-0.327509
155	H	-3.21915	-2.872846	-0.051653
156	H	-6.393503	-1.635889	0.568823
157	H	-3.254686	-0.286992	-5.041999
158	H	-2.809952	-1.991029	-5.072925
159	H	-1.641839	2.638711	-3.486999
160	H	-1.982701	1.62792	-4.901165
161	H	-2.890612	1.434662	-3.399682
162	H	1.452404	3.733926	-5.141212
163	H	-0.190388	3.662753	-4.514856
164	H	0.361927	2.434992	-5.652926
165	H	2.279838	3.772315	-2.601779
166	H	1.800943	2.45523	-1.547592
167	H	-0.255075	5.70981	-1.075637
168	H	2.736131	1.998392	-6.168925
169	H	1.939184	0.465058	-5.76377
170	H	5.728637	-0.962686	0.014684
171	H	6.403414	-0.142399	-1.397995
172	H	7.188087	-1.597873	-0.761485
173	H	5.48658	-2.740602	-3.948326
174	H	6.280159	-1.189291	-3.661653
175	H	7.046634	-2.67037	-3.108831
176	H	6.617761	-4.57515	-1.985921
177	H	5.313617	-5.534869	-1.309666
178	H	4.40056	-5.095564	0.693608
179	H	2.85203	-5.666671	1.243803
180	H	3.53534	-6.335737	-0.229005
181	H	-0.231316	-4.05022	1.822737

182	H	-0.470064	-5.78451	2.098767
183	H	1.153316	-5.102086	2.144897
184	H	-0.25019	-7.421749	0.066307
185	H	1.42724	-7.208242	0.541707
186	H	-3.143731	-5.362017	-0.602927
187	H	-2.502213	-4.83965	0.94177
188	F	0.143046	-1.079767	1.849656
189	H	5.089655	-4.741201	-2.86548
190	H	0.958878	-6.998972	-1.153572
191	H	-2.217194	-6.486419	0.371822
192	H	-4.416195	-1.498332	-4.516542
193	H	3.694095	0.622061	-5.606885
194	H	-7.932986	1.801494	-2.950537
195	H	-4.642065	-5.298157	4.763318
196	H	-3.521909	-6.564872	4.238191
197	H	-7.974625	2.964858	6.26876
198	H	-7.330593	4.009078	4.982551
199	H	-3.598152	8.188553	1.339604
200	H	-3.779677	6.838047	2.491401
201	H	7.629521	-4.893971	3.129647
202	H	8.006998	-4.878712	4.87056
203	H	10.673924	4.946776	0.006235
204	H	10.584957	6.201923	1.253179
205	H	6.85483	3.160816	-2.67845
206	H	1.395463	5.618334	-1.667655
207	H	-5.056228	-2.649112	1.098074
208	C	-8.690833	0.014965	-2.067712
209	O	-9.888757	0.119444	-2.31044
210	N	-8.102932	-1.150442	-1.675387
211	H	-7.090812	-1.171371	-1.632679
212	C	-8.880367	-2.363465	-1.553797
213	H	-8.387711	-3.046451	-0.855539
214	H	-9.874225	-2.1109	-1.179078
215	H	-9.008806	-2.879261	-2.51468

Cl1F3-conf2_Int, E_{opt}= -5017.7150992

		Coordinates (Angstroms)		

Center number	Atomic Symbol	X	Y	Z
1	C	7.598088	-4.295696	4.045093
2	C	6.54145	-3.33932	4.64472
3	C	5.694432	-2.547122	3.667477
4	C	5.970446	-1.201141	3.387891
5	C	4.563664	-3.125938	3.06887
6	C	5.137846	-0.448236	2.555114
7	C	3.72439	-2.380165	2.238582
8	C	4.007338	-1.036015	1.982754
9	C	-3.523069	7.104487	1.465946
10	C	-2.068206	6.720131	1.15131
11	O	-1.897665	6.51115	-0.275423
12	C	-1.642692	5.425328	1.830875
13	C	-7.310021	2.999127	5.404997
14	C	-5.865553	2.608995	5.7865
15	C	-5.084934	2.003225	4.638095
16	C	-5.313169	0.66385	4.282014
17	C	-4.128432	2.719236	3.908385
18	C	-4.613141	0.049283	3.248491
19	C	-3.402946	2.117145	2.875263
20	C	-3.649876	0.782269	2.551788
21	O	-2.925315	0.124178	1.580311
22	C	-3.891481	-5.564297	4.014358
23	C	-2.749465	-4.550034	3.911641
24	C	-3.081134	-3.363986	3.009232
25	O	-4.131299	-3.31617	2.356342
26	N	-2.136067	-2.399033	2.915199
27	C	-7.758643	1.211634	-2.044129
28	C	-8.1066	2.204693	-0.913651
29	C	-7.270971	3.499162	-0.874122
30	C	-6.048357	3.43634	0.048778
31	N	-5.042691	2.463056	-0.375612
32	C	-3.760985	2.706465	-0.650315
33	N	-3.231728	3.92484	-0.702218
34	N	-2.934306	1.640508	-0.841923
35	C	10.017434	5.683936	0.480244
36	C	8.683116	5.209576	1.081507
37	C	7.824503	4.349669	0.195212
38	C	8.104975	3.875165	-1.061003
39	C	6.521009	3.828074	0.538401
40	N	7.054009	3.09473	-1.523931
41	C	6.069387	3.042706	-0.559798
42	C	5.696858	3.941437	1.672054
43	C	4.836949	2.38004	-0.549835

44	C	4.468956	3.286377	1.688459
45	C	4.045642	2.512123	0.587537
46	Co	1.134734	-1.77541	-2.00605
47	C	-1.422469	-2.913264	-2.264381
48	C	-2.911219	-2.406387	-2.230085
49	C	-2.831948	-1.162661	-3.184037
50	C	-1.383354	-0.721328	-3.018694
51	C	-0.874509	0.568641	-3.392229
52	C	0.478484	0.846197	-3.298062
53	C	1.217362	2.164786	-3.675246
54	C	2.603672	1.576853	-4.064957
55	C	2.667762	0.374141	-3.162702
56	C	3.858728	-0.228594	-2.8101
57	C	4.003805	-1.422932	-2.117731
58	C	5.371931	-2.026952	-1.870588
59	C	5.003484	-3.378255	-1.177038
60	C	3.478705	-3.335533	-1.080729
61	C	2.729069	-4.318483	-0.462103
62	C	1.29598	-4.30543	-0.519162
63	C	0.333092	-5.334434	0.115494
64	C	-0.990245	-4.99478	-0.641176
65	C	-0.814706	-3.509384	-0.973125
66	C	-1.119458	-3.852044	-3.451591
67	N	-0.642906	-1.661559	-2.464154
68	N	1.429839	-0.076302	-2.854838
69	N	2.975968	-2.186725	-1.687446
70	N	0.64212	-3.328123	-1.110012
71	C	-3.94191	-3.441074	-2.666734
72	C	-3.251209	-1.950707	-0.774002
73	C	-4.616351	-1.33375	-0.564168
74	O	-4.963691	-0.282376	-1.158344
75	N	-5.448928	-1.961023	0.27516
76	C	-3.248175	-1.386386	-4.652636
77	C	-1.860836	1.584847	-3.94645
78	C	0.664535	3.051902	-4.805276
79	C	1.499328	3.028618	-2.392974
80	C	0.34359	3.794323	-1.789387
81	O	-0.726564	3.24217	-1.479408
82	N	0.521623	5.111082	-1.565023
83	C	2.72743	1.122836	-5.532571
84	C	6.235355	-1.145626	-0.947379
85	C	6.080918	-2.172441	-3.228142
86	C	5.532053	-4.631857	-1.89889
87	C	3.428824	-5.406767	0.331666
88	C	0.214712	-5.043976	1.628383
89	C	0.647787	-6.825498	-0.11847

90	C	-2.272378	-5.386	0.09221
91	Cl	1.306129	1.54928	3.044177
92	C	0.320256	1.338412	1.659985
93	C	0.175998	0.194242	0.995634
94	F	-0.366007	2.423198	1.255049
95	F	-0.609618	0.099321	-0.065265
96	H	7.048493	-2.641904	5.321983
97	H	5.867197	-3.933495	5.274731
98	H	6.838548	-0.73149	3.844592
99	H	4.326576	-4.167157	3.277596
100	H	5.36403	0.594964	2.353047
101	H	2.85263	-2.842104	1.787631
102	H	3.359838	-0.457757	1.331852
103	H	8.404778	-3.751299	3.546041
104	H	-1.395186	7.528164	1.47127
105	H	-0.658909	5.100987	1.481748
106	H	-2.357677	4.627983	1.618597
107	H	-1.595245	5.568008	2.914132
108	H	-2.110681	7.336888	-0.731369
109	H	-4.199296	6.533416	0.818653
110	H	-5.334728	3.47999	6.186506
111	H	-5.899803	1.874445	6.599653
112	H	-3.919778	3.754399	4.168462
113	H	-6.04998	0.087697	4.836628
114	H	-2.613419	2.65203	2.357212
115	H	-4.780886	-0.990356	2.986848
116	H	-2.634899	0.752271	0.890385
117	H	-7.714723	2.288554	4.675659
118	H	-2.447907	-4.174503	4.897277
119	H	-1.854781	-5.017713	3.486764
120	H	-2.317039	-1.568985	2.356425
121	H	-1.313507	-2.419994	3.497668
122	H	-4.425757	-5.616048	3.062634
123	H	-9.165805	2.449666	-1.028001
124	H	-8.025091	1.690751	0.055455
125	H	-7.895252	4.319344	-0.500066
126	H	-6.95191	3.784296	-1.884087
127	H	-5.583485	4.423048	0.11904
128	H	-6.355521	3.171475	1.068094
129	H	-5.308277	1.478323	-0.39729
130	H	-2.01737	1.839325	-1.241679
131	H	-3.419811	0.806708	-1.176544
132	H	-3.749229	4.765463	-0.518911
133	H	-2.234259	4.00026	-0.933612
134	H	-6.711054	0.911298	-2.022207
135	H	8.102155	6.087138	1.39905

136	H	8.890459	4.653665	2.007205
137	H	8.980037	4.031059	-1.675277
138	H	6.01933	4.530888	2.525831
139	H	3.827516	3.360889	2.561599
140	H	4.513735	1.760892	-1.378909
141	H	3.091817	1.993223	0.633052
142	H	9.86257	6.36306	-0.36389
143	H	-3.514834	-0.399758	-2.810823
144	H	3.407766	2.291482	-3.852164
145	H	4.764704	0.258936	-3.152773
146	H	5.415474	-3.37044	-0.160052
147	H	-0.947844	-5.558158	-1.578792
148	H	-1.118097	-2.898869	-0.117519
149	H	-1.312876	-3.356947	-4.40205
150	H	-1.709246	-4.77069	-3.413924
151	H	-0.058372	-4.109602	-3.429285
152	H	-4.948714	-3.007825	-2.665437
153	H	-3.757709	-3.816477	-3.67321
154	H	-3.959607	-4.298053	-1.99049
155	H	-2.516413	-1.210798	-0.454353
156	H	-3.162452	-2.798692	-0.096455
157	H	-6.332549	-1.507033	0.470619
158	H	-3.128667	-0.459736	-5.218635
159	H	-2.653766	-2.150781	-5.154593
160	H	-1.68246	2.580724	-3.54637
161	H	-1.819641	1.649778	-5.039184
162	H	-2.888541	1.327379	-3.687696
163	H	1.448571	3.744783	-5.129901
164	H	-0.189984	3.653717	-4.492426
165	H	0.360579	2.459867	-5.669573
166	H	2.308086	3.727959	-2.629319
167	H	1.881969	2.371768	-1.603614
168	H	-0.212207	5.640443	-1.095131
169	H	2.708076	1.966796	-6.22777
170	H	1.917866	0.434131	-5.794514
171	H	5.753193	-0.988099	0.018078
172	H	6.423315	-0.162962	-1.392666
173	H	7.205123	-1.624991	-0.771748
174	H	5.479303	-2.729688	-3.950839
175	H	6.281573	-1.187522	-3.658613
176	H	7.04678	-2.675702	-3.122219
177	H	6.6193	-4.584472	-2.005969
178	H	5.303505	-5.543066	-1.347798
179	H	4.41497	-5.082965	0.666133
180	H	2.872062	-5.654287	1.236678
181	H	3.5515	-6.33989	-0.2289

182	H	-0.190277	-4.044218	1.812668
183	H	-0.441239	-5.779256	2.104672
184	H	1.186127	-5.104592	2.124338
185	H	-0.245049	-7.422875	0.093248
186	H	1.445086	-7.201427	0.52456
187	H	-3.144988	-5.315711	-0.560265
188	H	-2.463501	-4.751149	0.957546
189	F	0.746839	-0.945399	1.325075
190	H	5.094819	-4.731524	-2.895896
191	H	0.935641	-7.006603	-1.158851
192	H	-2.224653	-6.421852	0.442738
193	H	-4.300931	-1.676588	-4.716729
194	H	3.673391	0.591831	-5.670048
195	H	-7.951112	1.695703	-3.008505
196	H	-4.61685	-5.278069	4.780156
197	H	-3.521878	-6.564754	4.238197
198	H	-7.974584	2.964846	6.26869
199	H	-7.358112	3.995149	4.954693
200	H	-3.723616	8.167803	1.307735
201	H	-3.779738	6.837987	2.491387
202	H	7.147417	-4.983756	3.322383
203	H	8.007073	-4.87856	4.87059
204	H	10.618628	4.836881	0.132889
205	H	10.58498	6.201897	1.253183
206	H	7.039329	2.603304	-2.401896
207	H	1.398947	5.56223	-1.770425
208	H	-5.075962	-2.623248	0.958795
209	C	-8.692192	0.026535	-1.92122
210	O	-9.910161	0.1484	-1.971362
211	N	-8.077999	-1.162443	-1.649893
212	H	-7.083773	-1.231925	-1.813939
213	C	-8.880391	-2.363474	-1.553798
214	H	-8.302377	-3.148119	-1.058571
215	H	-9.775313	-2.145643	-0.968673
216	H	-9.205308	-2.728013	-2.53619

Cl1F3-conf2_TS, E_{opt}=-5017.6654349

		Coordinates (Angstroms)		

Center number	Atomic Symbol	X	Y	Z
1	C	7.598241	-4.295696	4.045027
2	C	6.189036	-3.832279	4.474333
3	C	5.438797	-2.904906	3.536985
4	C	5.915443	-1.61376	3.259732
5	C	4.196509	-3.274949	3.001481
6	C	5.167379	-0.720364	2.49023
7	C	3.442342	-2.385752	2.230716
8	C	3.925237	-1.102722	1.973113
9	C	-3.523137	7.104544	1.466006
10	C	-2.107587	6.568441	1.189707
11	O	-1.811805	6.527463	-0.226059
12	C	-1.924785	5.144131	1.703598
13	C	-6.641517	2.909954	5.456655
14	C	-5.219653	2.405562	5.781395
15	C	-4.478172	1.8702	4.570352
16	C	-4.933729	0.690064	3.959753
17	C	-3.332335	2.475337	4.040833
18	C	-4.263901	0.121251	2.883893
19	C	-2.646639	1.923483	2.950358
20	C	-3.110358	0.735241	2.384941
21	O	-2.482083	0.114155	1.322862
22	C	-3.891505	-5.564295	4.014477
23	C	-2.752829	-4.539617	3.948899
24	C	-3.044857	-3.391835	2.985229
25	O	-4.102788	-3.336864	2.345774
26	N	-2.064517	-2.476836	2.814529
27	C	-7.758706	1.211689	-2.044213
28	C	-8.129866	2.182054	-0.90154
29	C	-7.198855	3.393405	-0.711054
30	C	-5.989685	3.137899	0.204299
31	N	-4.932026	2.300082	-0.358631
32	C	-3.755077	2.749392	-0.81231
33	N	-3.578819	4.040216	-1.137701
34	N	-2.731805	1.89558	-0.923863
35	C	10.017466	5.683939	0.480236
36	C	8.536335	5.620205	0.889318
37	C	7.680506	4.64344	0.129256
38	C	8.056724	3.731166	-0.82333
39	C	6.266703	4.437943	0.34234
40	N	6.963973	2.966336	-1.212754
41	C	5.851518	3.37177	-0.506808
42	C	5.31994	5.046825	1.182155
43	C	4.535499	2.895932	-0.519069

44	C	4.007397	4.585407	1.168394
45	C	3.621835	3.516914	0.331684
46	Co	1.116279	-1.699748	-1.873851
47	C	-1.428396	-2.913	-2.242449
48	C	-2.916777	-2.410927	-2.212783
49	C	-2.841359	-1.149315	-3.147736
50	C	-1.387411	-0.725524	-3.004754
51	C	-0.877617	0.570846	-3.401819
52	C	0.462206	0.867286	-3.282812
53	C	1.191194	2.194178	-3.641154
54	C	2.602655	1.62399	-3.97742
55	C	2.656001	0.418016	-3.079124
56	C	3.855442	-0.176372	-2.715727
57	C	4.009322	-1.390917	-2.058872
58	C	5.380402	-2.017958	-1.85822
59	C	5.010988	-3.368058	-1.158162
60	C	3.487932	-3.31864	-1.03999
61	C	2.736214	-4.311179	-0.44407
62	C	1.296501	-4.298114	-0.500495
63	C	0.335777	-5.337437	0.123767
64	C	-0.991411	-4.99342	-0.625385
65	C	-0.824077	-3.504913	-0.945456
66	C	-1.109869	-3.838309	-3.435432
67	N	-0.662547	-1.654672	-2.446286
68	N	1.422191	-0.035091	-2.797883
69	N	2.990045	-2.147541	-1.604852
70	N	0.632586	-3.32395	-1.064977
71	C	-3.941851	-3.441072	-2.666723
72	C	-3.272017	-1.973177	-0.764617
73	C	-4.663199	-1.402001	-0.587053
74	O	-5.052945	-0.397902	-1.218349
75	N	-5.473198	-2.045097	0.272131
76	C	-3.297106	-1.354462	-4.60588
77	C	-1.869462	1.541377	-4.01802
78	C	0.66448	3.051882	-4.805323
79	C	1.399032	3.035244	-2.333454
80	C	0.284617	3.941127	-1.838278
81	O	-0.919414	3.716829	-2.056883
82	N	0.6661	5.009651	-1.115443
83	C	2.792635	1.167554	-5.438032
84	C	6.265512	-1.145997	-0.946
85	C	6.081005	-2.172444	-3.228184
86	C	5.531436	-4.623009	-1.88322
87	C	3.429651	-5.423351	0.319425
88	C	0.231787	-5.042838	1.63684
89	C	0.64782	-6.825576	-0.118511

90	C	-2.27011	-5.396113	0.105887
91	Cl	1.785399	1.675946	2.895086
92	C	0.786702	1.180735	1.578948
93	C	0.243176	-0.043832	1.349579
94	F	0.408622	2.214447	0.780966
95	F	0.465475	-0.678875	-0.161165
96	H	6.282469	-3.324872	5.444026
97	H	5.572417	-4.719235	4.662195
98	H	6.869724	-1.297224	3.674041
99	H	3.803589	-4.266458	3.214772
100	H	5.545337	0.280946	2.300781
101	H	2.47429	-2.679319	1.842376
102	H	3.337879	-0.409912	1.379527
103	H	8.26377	-3.44838	3.853427
104	H	-1.372833	7.22146	1.68529
105	H	-0.955865	4.734442	1.401783
106	H	-2.707078	4.489394	1.313589
107	H	-1.978855	5.127715	2.79568
108	H	-1.899437	7.421501	-0.58328
109	H	-4.23717	6.609083	0.799548
110	H	-4.637552	3.200955	6.259431
111	H	-5.29052	1.596339	6.518975
112	H	-2.954379	3.389096	4.492883
113	H	-5.814422	0.191104	4.357172
114	H	-1.75795	2.400017	2.548577
115	H	-4.591719	-0.816247	2.450064
116	H	-1.45567	0.191053	1.33464
117	H	-7.104518	2.278484	4.69016
118	H	-2.524947	-4.119792	4.936443
119	H	-1.823513	-5.008912	3.607189
120	H	-2.227345	-1.685383	2.199857
121	H	-1.216618	-2.47887	3.360025
122	H	-4.403384	-5.611034	3.050478
123	H	-9.152044	2.522882	-1.088437
124	H	-8.172495	1.622458	0.043845
125	H	-7.771016	4.205113	-0.244743
126	H	-6.857327	3.773475	-1.683632
127	H	-5.534059	4.085281	0.50973
128	H	-6.323743	2.650493	1.127734
129	H	-5.067868	1.289461	-0.412482
130	H	-1.862137	2.240658	-1.322428
131	H	-2.707029	1.094994	-0.296249
132	H	-4.369747	4.627159	-1.342749
133	H	-2.642122	4.366077	-1.375516
134	H	-6.715144	0.904358	-1.990175
135	H	8.09539	6.623958	0.81441

136	H	8.474843	5.359903	1.955697
137	H	9.027486	3.55441	-1.263036
138	H	5.614186	5.860227	1.839268
139	H	3.270723	5.031635	1.830741
140	H	4.240778	2.056895	-1.143282
141	H	2.60112	3.155429	0.380551
142	H	10.158487	6.194826	-0.476617
143	H	-3.481092	-0.373166	-2.724329
144	H	3.389732	2.345214	-3.73237
145	H	4.756564	0.32459	-3.050182
146	H	5.427786	-3.360424	-0.142876
147	H	-0.948247	-5.548753	-1.568152
148	H	-1.129383	-2.890735	-0.095012
149	H	-1.323886	-3.34069	-4.380754
150	H	-1.682531	-4.767113	-3.399567
151	H	-0.044332	-4.081818	-3.424204
152	H	-4.945412	-3.003575	-2.677949
153	H	-3.74855	-3.820308	-3.670603
154	H	-3.972018	-4.296863	-1.988845
155	H	-2.569036	-1.208965	-0.436037
156	H	-3.170822	-2.818366	-0.084956
157	H	-6.376563	-1.624444	0.450707
158	H	-3.177881	-0.434502	-5.179929
159	H	-2.736786	-2.136024	-5.122667
160	H	-1.663308	2.562746	-3.720442
161	H	-1.856094	1.48975	-5.112652
162	H	-2.885735	1.324026	-3.691387
163	H	1.453702	3.742927	-5.120829
164	H	-0.198711	3.651068	-4.522088
165	H	0.390191	2.438132	-5.664647
166	H	2.300056	3.644887	-2.450058
167	H	1.610299	2.346922	-1.507328
168	H	-0.064906	5.606772	-0.727214
169	H	2.785319	2.013389	-6.1295
170	H	2.004349	0.468133	-5.734486
171	H	5.797732	-0.984822	0.025665
172	H	6.461443	-0.166868	-1.396596
173	H	7.230652	-1.637874	-0.783507
174	H	5.475945	-2.737197	-3.941882
175	H	6.274172	-1.188922	-3.665227
176	H	7.046363	-2.673216	-3.118857
177	H	6.616377	-4.571304	-2.006145
178	H	5.316582	-5.53219	-1.324853
179	H	4.417302	-5.114325	0.661077
180	H	2.873251	-5.683953	1.219656
181	H	3.540195	-6.343231	-0.264628

182	H	-0.138612	-4.030908	1.819481
183	H	-0.447861	-5.756557	2.110967
184	H	1.201284	-5.137515	2.13077
185	H	-0.248314	-7.418888	0.087859
186	H	1.440384	-7.209549	0.525175
187	H	-3.143776	-5.309679	-0.542809
188	H	-2.45613	-4.777849	0.983022
189	F	0.605143	-1.0414	2.19172
190	H	5.08007	-4.729643	-2.873178
191	H	0.936265	-7.004797	-1.15919
192	H	-2.223062	-6.439115	0.434048
193	H	-4.356672	-1.621326	-4.636226
194	H	3.752519	0.653892	-5.541104
195	H	-7.915681	1.716945	-3.003995
196	H	-4.637657	-5.290082	4.764143
197	H	-3.521894	-6.564892	4.238137
198	H	-7.306074	2.87535	6.320371
199	H	-6.624648	3.931918	5.065259
200	H	-3.601426	8.187403	1.334773
201	H	-3.779674	6.837936	2.491367
202	H	7.564519	-4.917609	3.145358
203	H	8.006942	-4.878564	4.870656
204	H	10.443521	4.678007	0.395217
205	H	10.585034	6.201925	1.253178
206	H	6.98787	2.225424	-1.893019
207	H	1.632862	5.153685	-0.857799
208	H	-5.082968	-2.686676	0.96447
209	C	-8.699266	0.02725	-1.97092
210	O	-9.913203	0.143611	-2.090568
211	N	-8.092157	-1.154247	-1.652998
212	H	-7.087315	-1.205321	-1.74909
213	C	-8.880361	-2.363488	-1.553785
214	H	-8.330664	-3.111611	-0.975835
215	H	-9.820636	-2.131722	-1.050555
216	H	-9.124587	-2.786959	-2.536418

F4_React, E_{opt}= -4656.7713893

		Coordinates (Angstroms)

Center number	Atomic Symbol	X	Y	Z
1	C	7.598068	-4.295635	4.045198
2	C	6.149809	-3.917888	4.423329
3	C	5.4183	-2.939682	3.522236
4	C	5.934164	-1.657241	3.274552
5	C	4.152709	-3.249477	3.002878
6	C	5.203118	-0.712943	2.552374
7	C	3.412833	-2.309047	2.279238
8	C	3.935403	-1.034858	2.056452
9	C	-3.523138	7.104517	1.466009
10	C	-2.138978	6.540822	1.106187
11	O	-1.980812	6.447757	-0.33656
12	C	-1.923444	5.138405	1.659953
13	C	-7.310118	2.999206	5.404992
14	C	-5.866665	2.565184	5.750226
15	C	-5.099402	1.939659	4.597526
16	C	-5.393539	0.625217	4.196997
17	C	-4.051576	2.585917	3.927351
18	C	-4.673639	-0.024572	3.199315
19	C	-3.313092	1.94544	2.927357
20	C	-3.591375	0.612742	2.543273
21	O	-2.868368	-0.022697	1.619915
22	C	-3.891407	-5.564284	4.014496
23	C	-2.771776	-4.521571	3.943023
24	C	-3.105897	-3.390961	2.967446
25	O	-4.117946	-3.47736	2.253577
26	N	-2.230946	-2.375388	2.857348
27	C	-7.758646	1.211686	-2.044181
28	C	-8.123243	2.07232	-0.813175
29	C	-7.206622	3.281017	-0.542275
30	C	-6.014614	2.984119	0.379548
31	N	-5.004395	2.085795	-0.17731
32	C	-3.762653	2.443562	-0.536558
33	N	-3.518938	3.701003	-0.955501
34	N	-2.781926	1.541332	-0.484264
35	C	10.017413	5.68394	0.480262
36	C	8.784739	5.010249	1.102253
37	C	7.882188	4.31821	0.119694
38	C	8.068241	4.185173	-1.232626
39	C	6.624571	3.67147	0.418433
40	N	7.004064	3.496117	-1.797441
41	C	6.10037	3.173768	-0.808429
42	C	5.885955	3.470058	1.598257
43	C	4.874831	2.505837	-0.878739

44	C	4.665601	2.801427	1.534142
45	C	4.166091	2.324874	0.304583
46	Co	1.120668	-1.742697	-1.941497
47	C	-1.460484	-2.891031	-2.208339
48	C	-2.94866	-2.39592	-2.155526
49	C	-2.875578	-1.082783	-3.026797
50	C	-1.419279	-0.672123	-2.902724
51	C	-0.896798	0.632699	-3.275961
52	C	0.452465	0.893366	-3.237091
53	C	1.2057	2.197232	-3.6439
54	C	2.588279	1.589981	-4.029779
55	C	2.648414	0.379451	-3.141321
56	C	3.841809	-0.240649	-2.813516
57	C	3.992866	-1.432098	-2.117205
58	C	5.359494	-2.036895	-1.872751
59	C	4.993415	-3.39237	-1.18101
60	C	3.473495	-3.339308	-1.046519
61	C	2.718758	-4.313975	-0.427576
62	C	1.278245	-4.295464	-0.479202
63	C	0.315875	-5.340523	0.131095
64	C	-1.005021	-5.001652	-0.632997
65	C	-0.85248	-3.505167	-0.925488
66	C	-1.147164	-3.785473	-3.426329
67	N	-0.691891	-1.628353	-2.396699
68	N	1.414754	-0.041731	-2.812398
69	N	2.97798	-2.185592	-1.65101
70	N	0.610302	-3.32062	-1.042584
71	C	-3.941924	-3.441111	-2.666758
72	C	-3.338439	-2.025805	-0.699657
73	C	-4.748181	-1.483074	-0.546171
74	O	-5.191996	-0.581004	-1.288889
75	N	-5.502189	-2.034335	0.411801
76	C	-3.363313	-1.203633	-4.482403
77	C	-1.900613	1.670685	-3.740394
78	C	0.664544	3.051903	-4.805351
79	C	1.477214	3.099888	-2.386527
80	C	0.298556	3.858967	-1.794443
81	O	-0.712264	3.272198	-1.387754
82	N	0.404676	5.199653	-1.699997
83	C	2.720078	1.128595	-5.496586
84	C	6.199719	-1.144043	-0.93691
85	C	6.080943	-2.172471	-3.228247
86	C	5.482671	-4.64662	-1.929914
87	C	3.411345	-5.419056	0.346205
88	C	0.185115	-5.054837	1.644756
89	C	0.647815	-6.825531	-0.118475

90	C	-2.286986	-5.438646	0.072345
91	F	1.515773	1.156363	2.682297
92	C	0.797048	1.088938	1.575024
93	C	0.197294	-0.012151	1.147862
94	F	0.740896	2.234411	0.904525
95	F	-0.460296	-0.060294	-0.004121
96	H	6.1676	-3.4836	5.432325
97	H	5.556592	-4.835935	4.508574
98	H	6.908023	-1.386896	3.674654
99	H	3.729496	-4.232774	3.193482
100	H	5.610445	0.279241	2.379047
101	H	2.425829	-2.562699	1.906218
102	H	3.367765	-0.295149	1.501932
103	H	8.227478	-3.41086	3.909071
104	H	-1.355933	7.207103	1.497879
105	H	-1.016488	4.684554	1.24862
106	H	-2.768072	4.4914	1.415613
107	H	-1.827435	5.176686	2.748848
108	H	-2.216524	7.304923	-0.716945
109	H	-4.273264	6.638138	0.817161
110	H	-5.308351	3.420334	6.149189
111	H	-5.910486	1.831138	6.564842
112	H	-3.785412	3.603118	4.21261
113	H	-6.197222	0.08985	4.700822
114	H	-2.473104	2.44586	2.453056
115	H	-4.907602	-1.050609	2.934527
116	H	-7.73778	2.315521	4.662509
117	H	-2.545171	-4.093948	4.92708
118	H	-1.836662	-4.973995	3.594239
119	H	-2.481492	-1.517411	2.30605
120	H	-1.491659	-2.290186	3.53918
121	H	-4.40561	-5.614844	3.052655
122	H	-9.15537	2.408672	-0.948871
123	H	-8.133259	1.430542	0.07953
124	H	-7.790222	4.06828	-0.048786
125	H	-6.849514	3.707402	-1.490132
126	H	-5.514085	3.913335	0.670144
127	H	-6.366894	2.52894	1.312442
128	H	-5.166998	1.084796	-0.144422
129	H	-1.872885	1.854587	-0.806245
130	H	-2.802537	0.845437	0.333326
131	H	-4.272361	4.324862	-1.187936
132	H	-2.559204	4.01493	-1.055897
133	H	-6.710415	0.914847	-2.02387
134	H	8.199665	5.763221	1.648625
135	H	9.111548	4.288691	1.864268

136	H	8.883355	4.530946	-1.851259
137	H	6.262139	3.837678	2.548708
138	H	4.082417	2.64056	2.435957
139	H	4.49032	2.133072	-1.819297
140	H	3.209813	1.812066	0.280067
141	H	9.724714	6.420761	-0.275268
142	H	-3.500034	-0.329438	-2.544968
143	H	3.399594	2.292982	-3.813699
144	H	4.745675	0.241878	-3.161923
145	H	5.428327	-3.400295	-0.174489
146	H	-0.937718	-5.537073	-1.586421
147	H	-1.168643	-2.911588	-0.063063
148	H	-1.392956	-3.270403	-4.355159
149	H	-1.702678	-4.724226	-3.399077
150	H	-0.078434	-4.019727	-3.446252
151	H	-4.96023	-3.043291	-2.633851
152	H	-3.751627	-3.748408	-3.695718
153	H	-3.922129	-4.335191	-2.041106
154	H	-2.675648	-1.257306	-0.30018
155	H	-3.246191	-2.893353	-0.047885
156	H	-6.393372	-1.595186	0.601978
157	H	-3.274587	-0.24887	-5.004077
158	H	-2.809073	-1.947675	-5.060074
159	H	-1.630011	2.663472	-3.396762
160	H	-1.991641	1.692917	-4.831789
161	H	-2.888326	1.466805	-3.328442
162	H	1.456138	3.724896	-5.150214
163	H	-0.180995	3.673085	-4.508088
164	H	0.34678	2.437306	-5.649129
165	H	2.273139	3.803324	-2.652134
166	H	1.866692	2.480622	-1.572453
167	H	-0.336986	5.708964	-1.212887
168	H	2.71263	1.973867	-6.188279
169	H	1.905495	0.449736	-5.76817
170	H	5.710377	-1.012311	0.028761
171	H	6.365569	-0.152359	-1.368148
172	H	7.176144	-1.609374	-0.765191
173	H	5.490028	-2.72605	-3.962466
174	H	6.292072	-1.185085	-3.646447
175	H	7.042326	-2.678566	-3.107583
176	H	6.567302	-4.615218	-2.059181
177	H	5.250765	-5.558995	-1.382872
178	H	4.405229	-5.112045	0.670738
179	H	2.860911	-5.661509	1.255083
180	H	3.508041	-6.346512	-0.227199
181	H	-0.234074	-4.062652	1.837053

182	H	-0.47176	-5.799003	2.102591
183	H	1.151356	-5.115734	2.151314
184	H	-0.249225	-7.42527	0.060302
185	H	1.425742	-7.208961	0.542923
186	H	-3.151209	-5.357441	-0.588698
187	H	-2.504616	-4.841614	0.956432
188	F	0.256686	-1.181751	1.76758
189	H	5.022977	-4.72706	-2.918543
190	H	0.962344	-6.99351	-1.153444
191	H	-2.223718	-6.48686	0.379857
192	H	-4.419496	-1.482448	-4.491012
193	H	3.663053	0.591293	-5.629796
194	H	-7.936475	1.797077	-2.952819
195	H	-4.639355	-5.295834	4.765137
196	H	-3.521891	-6.564871	4.238095
197	H	-7.974626	2.964893	6.268772
198	H	-7.340333	4.004847	4.973373
199	H	-3.581281	8.190675	1.34753
200	H	-3.779682	6.837962	2.491367
201	H	7.638	-4.889086	3.126899
202	H	8.0071	-4.878626	4.870555
203	H	10.675496	4.950252	0.003242
204	H	10.585029	6.201916	1.253187
205	H	6.908117	3.281624	-2.775681
206	H	1.245775	5.684527	-1.969109
207	H	-5.063205	-2.617439	1.134409
208	C	-8.692027	0.015745	-2.060128
209	O	-9.891493	0.121534	-2.295321
210	N	-8.102823	-1.15046	-1.67415
211	H	-7.090715	-1.170792	-1.629675
212	C	-8.880378	-2.363473	-1.553785
213	H	-8.377895	-3.053965	-0.870139
214	H	-9.868558	-2.114594	-1.161703
215	H	-9.023562	-2.868644	-2.518197

F4_Int, E_{opt}= -4657.3534664

1	C	7.598063	-4.29575	4.04505
2	C	6.18203	-3.85201	4.474493

3	C	5.449216	-2.89737	3.553277
4	C	5.934955	-1.59871	3.332337
5	C	4.230753	-3.25579	2.96037
6	C	5.22495	-0.68853	2.549257
7	C	3.512099	-2.34929	2.175137
8	C	4.009981	-1.06356	1.967013
9	C	-3.52307	7.10437	1.465918
10	C	-2.07355	6.712386	1.137835
11	O	-1.9109	6.536291	-0.29384
12	C	-1.66065	5.39585	1.782865
13	C	-7.31002	2.999139	5.405001
14	C	-5.86837	2.601731	5.788171
15	C	-5.09425	1.994667	4.637327
16	C	-5.28046	0.639596	4.319164
17	C	-4.1948	2.731068	3.857639
18	C	-4.5892	0.027134	3.278658
19	C	-3.48067	2.131636	2.815711
20	C	-3.6755	0.778353	2.53615
21	O	-2.95156	0.123852	1.564293
22	C	-3.89149	-5.56429	4.014358
23	C	-2.75069	-4.54976	3.907071
24	C	-3.08964	-3.36516	3.005852
25	O	-4.15658	-3.30849	2.379886
26	N	-2.13908	-2.41085	2.887265
27	C	-7.75865	1.211638	-2.04414
28	C	-8.13752	2.227941	-0.94519
29	C	-7.2972	3.518195	-0.90273
30	C	-6.07933	3.450014	0.026064
31	N	-5.06968	2.479379	-0.39529
32	C	-3.79341	2.729604	-0.6884
33	N	-3.27804	3.953464	-0.76835
34	N	-2.95912	1.668343	-0.86711
35	C	10.01743	5.683936	0.480241
36	C	8.748978	5.067688	1.086117
37	C	7.865244	4.356814	0.10114
38	C	8.076473	4.189396	-1.2441
39	C	6.606367	3.712613	0.394691
40	N	7.021592	3.488728	-1.81027
41	C	6.107434	3.178531	-0.82681
42	C	5.854269	3.530876	1.569082
43	C	4.896903	2.485122	-0.89843
44	C	4.648829	2.836576	1.50452
45	C	4.179055	2.318074	0.280263
46	Co	1.12832	-1.76539	-1.9808
47	C	-1.42406	-2.90856	-2.24915
48	C	-2.91458	-2.40583	-2.22375

49	C	-2.8342	-1.16357	-3.17928
50	C	-1.38793	-0.71844	-3.00773
51	C	-0.8802	0.569364	-3.38783
52	C	0.472071	0.847877	-3.29125
53	C	1.209681	2.165195	-3.67211
54	C	2.595983	1.575761	-4.0591
55	C	2.661893	0.374616	-3.15353
56	C	3.851569	-0.23512	-2.81012
57	C	3.997334	-1.42851	-2.11712
58	C	5.365579	-2.03515	-1.87466
59	C	4.996874	-3.39163	-1.19214
60	C	3.473628	-3.34248	-1.0828
61	C	2.724981	-4.32812	-0.46886
62	C	1.292274	-4.30557	-0.51294
63	C	0.332373	-5.33592	0.121128
64	C	-0.99425	-4.99202	-0.62798
65	C	-0.82024	-3.50498	-0.95525
66	C	-1.11138	-3.84383	-3.43671
67	N	-0.64839	-1.65326	-2.44249
68	N	1.423013	-0.07074	-2.84005
69	N	2.970034	-2.18759	-1.67709
70	N	0.636593	-3.31976	-1.08778
71	C	-3.9419	-3.44107	-2.66673
72	C	-3.25722	-1.94915	-0.76894
73	C	-4.61945	-1.32756	-0.55631
74	O	-4.96558	-0.27656	-1.1527
75	N	-5.44958	-1.94808	0.289979
76	C	-3.24414	-1.39084	-4.64927
77	C	-1.86376	1.580558	-3.95618
78	C	0.664535	3.051906	-4.80527
79	C	1.478418	3.030632	-2.38861
80	C	0.321326	3.813561	-1.80944
81	O	-0.75418	3.274492	-1.49321
82	N	0.504738	5.133367	-1.60766
83	C	2.719419	1.116144	-5.52526
84	C	6.225377	-1.15635	-0.946
85	C	6.080927	-2.17243	-3.22815
86	C	5.509899	-4.64043	-1.93402
87	C	3.424703	-5.43135	0.302796
88	C	0.22426	-5.04939	1.635439
89	C	0.647788	-6.82549	-0.11846
90	C	-2.27377	-5.38238	0.110202
91	F	1.753764	1.05968	2.408315
92	C	0.899027	1.111178	1.401909
93	C	0.207979	0.070349	0.959773
94	F	0.807969	2.314252	0.845001

95	F	-0.64527	0.172771	-0.05081
96	H	6.263506	-3.37463	5.460496
97	H	5.565725	-4.74605	4.626492
98	H	6.872949	-1.29582	3.791751
99	H	3.834518	-4.25472	3.126304
100	H	5.612591	0.31291	2.383897
101	H	2.573105	-2.64528	1.719803
102	H	3.471237	-0.36325	1.339836
103	H	8.253714	-3.4385	3.864662
104	H	-1.39061	7.504638	1.475789
105	H	-0.67859	5.072275	1.425612
106	H	-2.38459	4.614538	1.543322
107	H	-1.6155	5.506743	2.869832
108	H	-2.12025	7.374317	-0.72861
109	H	-4.20614	6.540848	0.819108
110	H	-5.33349	3.470253	6.1884
111	H	-5.9061	1.866799	6.600501
112	H	-4.02414	3.78136	4.082491
113	H	-5.97742	0.049173	4.90903
114	H	-2.74312	2.693139	2.250274
115	H	-4.72926	-1.02265	3.043691
116	H	-2.65697	0.750915	0.87622
117	H	-7.7167	2.291694	4.6738
118	H	-2.44742	-4.17128	4.891043
119	H	-1.85628	-5.01676	3.481117
120	H	-2.31078	-1.58946	2.314457
121	H	-1.28518	-2.45277	3.420946
122	H	-4.42909	-5.6174	3.064599
123	H	-9.19114	2.475811	-1.09822
124	H	-8.09025	1.733721	0.036276
125	H	-7.92037	4.340431	-0.53108
126	H	-6.97271	3.802122	-1.91141
127	H	-5.61425	4.435957	0.107076
128	H	-6.39198	3.178586	1.041894
129	H	-5.32604	1.492089	-0.39994
130	H	-2.04467	1.866608	-1.27296
131	H	-3.43924	0.822768	-1.18139
132	H	-3.80477	4.789869	-0.59148
133	H	-2.28065	4.036962	-0.99761
134	H	-6.71257	0.911565	-1.99038
135	H	8.166448	5.855861	1.584068
136	H	9.032065	4.368325	1.885453
137	H	8.903292	4.518406	-1.85636
138	H	6.213199	3.925949	2.515378
139	H	4.060126	2.680354	2.403496
140	H	4.534746	2.06584	-1.82777

141	H	3.245676	1.767802	0.252117
142	H	9.769414	6.409305	-0.30197
143	H	-3.52015	-0.40141	-2.80981
144	H	3.398057	2.294817	-3.85361
145	H	4.758335	0.247501	-3.15548
146	H	5.418688	-3.39879	-0.17918
147	H	-0.95716	-5.55303	-1.56721
148	H	-1.12738	-2.89648	-0.10038
149	H	-1.29809	-3.34592	-4.38702
150	H	-1.70096	-4.76291	-3.40626
151	H	-0.05035	-4.10085	-3.40787
152	H	-4.94912	-3.00869	-2.66916
153	H	-3.75223	-3.81498	-3.67268
154	H	-3.96174	-4.29927	-1.99176
155	H	-2.51987	-1.21112	-0.45049
156	H	-3.1701	-2.79731	-0.09127
157	H	-6.32942	-1.48867	0.489737
158	H	-3.1215	-0.46601	-5.21742
159	H	-2.64826	-2.15705	-5.14665
160	H	-1.68994	2.579561	-3.56172
161	H	-1.81381	1.638622	-5.04894
162	H	-2.89319	1.323383	-3.70404
163	H	1.451014	3.744786	-5.12416
164	H	-0.19197	3.654205	-4.49871
165	H	0.366093	2.461047	-5.6723
166	H	2.302146	3.717709	-2.60836
167	H	1.828391	2.373735	-1.5863
168	H	-0.22661	5.671392	-1.14418
169	H	2.70121	1.95706	-6.22422
170	H	1.909169	0.427192	-5.78442
171	H	5.747097	-1.02026	0.023617
172	H	6.395385	-0.16505	-1.37806
173	H	7.201361	-1.62759	-0.78268
174	H	5.483355	-2.7234	-3.95907
175	H	6.286882	-1.18484	-3.64996
176	H	7.045326	-2.67776	-3.11968
177	H	6.596359	-4.5999	-2.05059
178	H	5.279523	-5.55663	-1.39219
179	H	4.41662	-5.11977	0.629941
180	H	2.877214	-5.68493	1.211463
181	H	3.53183	-6.35808	-0.27139
182	H	-0.17456	-4.04801	1.822699
183	H	-0.43167	-5.7828	2.114572
184	H	1.19895	-5.11465	2.124542
185	H	-0.24698	-7.42326	0.084418
186	H	1.440088	-7.20596	0.527637

187	H	-3.15013	-5.30392	-0.53646
188	H	-2.45557	-4.75084	0.979821
189	F	0.261353	-1.14031	1.493857
190	H	5.062788	-4.72454	-2.9281
191	H	0.941615	-7.00186	-1.15797
192	H	-2.2287	-6.42025	0.455106
193	H	-4.29682	-1.68036	-4.71738
194	H	3.664651	0.583175	-5.65991
195	H	-7.92503	1.675517	-3.02342
196	H	-4.61498	-5.27727	4.781605
197	H	-3.52187	-6.56476	4.238194
198	H	-7.97459	2.964837	6.268688
199	H	-7.35383	3.996601	4.957463
200	H	-3.716	8.169861	1.31218
201	H	-3.77974	6.838077	2.49142
202	H	7.575273	-4.90959	3.139937
203	H	8.007083	-4.87851	4.870622
204	H	10.66084	4.915671	0.038929
205	H	10.58498	6.201895	1.253185
206	H	6.96686	3.200738	-2.77288
207	H	1.385472	5.576668	-1.81539
208	H	-5.07535	-2.60703	0.977243
209	C	-8.69327	0.026919	-1.92278
210	O	-9.91095	0.147971	-1.9823
211	N	-8.07991	-1.16092	-1.64561
212	H	-7.08412	-1.22902	-1.80036
213	C	-8.88039	-2.36348	-1.5538
214	H	-8.30611	-3.14477	-1.04905
215	H	-9.78221	-2.14541	-0.97947
216	H	-9.19344	-2.73263	-2.53834

F4_TS, E_{opt}= -4657.3014561

Center number	Atomic Symbol	Coordinates (Angstroms)		
		X	Y	Z

1	C	7.59823	-4.295613	4.045046
2	C	6.184729	-3.847176	4.474244
3	C	5.442583	-2.898592	3.553461
4	C	5.930588	-1.606741	3.301255
5	C	4.200644	-3.251272	3.006705
6	C	5.193134	-0.693455	2.546013
7	C	3.455521	-2.342425	2.250412
8	C	3.948306	-1.05753	2.022864
9	C	-3.523138	7.104518	1.466003
10	C	-2.108951	6.575193	1.171065
11	O	-1.842506	6.52576	-0.251884
12	C	-1.90312	5.157005	1.692114
13	C	-6.641543	2.909878	5.456647
14	C	-5.220993	2.404039	5.783984
15	C	-4.470417	1.870728	4.577327
16	C	-4.91547	0.688667	3.962151
17	C	-3.318883	2.47628	4.061208
18	C	-4.227838	0.115823	2.899382
19	C	-2.615423	1.920108	2.984334
20	C	-3.064458	0.726313	2.417183
21	O	-2.413782	0.098759	1.375269
22	C	-3.891351	-5.564282	4.014517
23	C	-2.748985	-4.541419	3.978875
24	C	-3.01739	-3.386587	3.017352
25	O	-4.06824	-3.317217	2.367464
26	N	-2.027613	-2.479558	2.863706
27	C	-7.758669	1.211686	-2.044216
28	C	-8.125655	2.156386	-0.878674
29	C	-7.197343	3.367083	-0.669489
30	C	-5.992419	3.100189	0.24751
31	N	-4.94501	2.249241	-0.314447
32	C	-3.763427	2.682382	-0.770982
33	N	-3.571788	3.971143	-1.097632
34	N	-2.752442	1.814776	-0.887924
35	C	10.017429	5.684014	0.48021
36	C	8.739256	5.080391	1.075811
37	C	7.854792	4.39039	0.07706
38	C	8.063039	4.257563	-1.272396
39	C	6.597422	3.73833	0.357242
40	N	7.007944	3.568927	-1.853326
41	C	6.095818	3.234926	-0.875913
42	C	5.8481	3.527064	1.528059
43	C	4.88603	2.541399	-0.959866
44	C	4.643644	2.832787	1.450597
45	C	4.170122	2.342016	0.216037
46	Co	1.09616	-1.695259	-1.85076

47	C	-1.442685	-2.900533	-2.215966
48	C	-2.932594	-2.40626	-2.190391
49	C	-2.860326	-1.133918	-3.112336
50	C	-1.406683	-0.708692	-2.96799
51	C	-0.89481	0.586091	-3.369381
52	C	0.447674	0.875857	-3.260264
53	C	1.191423	2.188462	-3.647538
54	C	2.579231	1.587147	-4.01983
55	C	2.637081	0.388353	-3.112357
56	C	3.831712	-0.229108	-2.784836
57	C	3.983679	-1.420943	-2.090324
58	C	5.353578	-2.038895	-1.873531
59	C	4.982537	-3.395884	-1.190779
60	C	3.465323	-3.321699	-1.027459
61	C	2.7164	-4.306812	-0.416926
62	C	1.277912	-4.292021	-0.471684
63	C	0.320711	-5.341706	0.138066
64	C	-1.005327	-4.99581	-0.61361
65	C	-0.842445	-3.502651	-0.920561
66	C	-1.113472	-3.814248	-3.41553
67	N	-0.681738	-1.638121	-2.408603
68	N	1.403203	-0.034197	-2.78469
69	N	2.965681	-2.155384	-1.599266
70	N	0.613465	-3.31499	-1.032506
71	C	-3.941921	-3.441109	-2.666775
72	C	-3.301305	-1.992209	-0.737978
73	C	-4.69834	-1.437752	-0.559417
74	O	-5.102041	-0.439834	-1.192457
75	N	-5.498558	-2.084791	0.305854
76	C	-3.318296	-1.327787	-4.571384
77	C	-1.884245	1.557805	-3.987742
78	C	0.664555	3.051833	-4.805285
79	C	1.437865	3.05795	-2.364345
80	C	0.301396	3.926575	-1.848948
81	O	-0.892904	3.597228	-1.940325
82	N	0.644293	5.083893	-1.253305
83	C	2.716107	1.113168	-5.481079
84	C	6.204973	-1.15983	-0.934321
85	C	6.08095	-2.172477	-3.228176
86	C	5.435433	-4.647645	-1.966262
87	C	3.415093	-5.419605	0.340648
88	C	0.208247	-5.066398	1.65385
89	C	0.647792	-6.825513	-0.118504
90	C	-2.285144	-5.411176	0.109195
91	F	1.852833	1.443955	2.492886
92	C	0.96987	1.161027	1.54125

93	C	0.28952	-0.005988	1.368041
94	F	0.693352	2.240381	0.79872
95	F	0.436795	-0.649942	-0.125299
96	H	6.269635	-3.36278	5.456537
97	H	5.568853	-4.739798	4.636304
98	H	6.887628	-1.306817	3.721652
99	H	3.801744	-4.244785	3.198292
100	H	5.579386	0.306099	2.366446
101	H	2.488757	-2.623307	1.84973
102	H	3.373435	-0.349972	1.436537
103	H	8.257522	-3.44167	3.861889
104	H	-1.369204	7.238206	1.645085
105	H	-0.935601	4.757231	1.372238
106	H	-2.686394	4.490917	1.323938
107	H	-1.933685	5.149713	2.785275
108	H	-1.968977	7.412422	-0.616018
109	H	-4.240644	6.608893	0.803026
110	H	-4.641132	3.198033	6.267181
111	H	-5.295246	1.593162	6.519452
112	H	-2.949178	3.392703	4.514969
113	H	-5.801389	0.189582	4.347816
114	H	-1.720099	2.397183	2.597373
115	H	-4.547492	-0.82441	2.464904
116	H	-1.379855	0.208141	1.381372
117	H	-7.104133	2.278339	4.689829
118	H	-2.542032	-4.12815	4.973805
119	H	-1.813654	-5.011651	3.655271
120	H	-2.171796	-1.685271	2.246928
121	H	-1.182474	-2.499135	3.413046
122	H	-4.384201	-5.604639	3.040428
123	H	-9.150178	2.49745	-1.052081
124	H	-8.159968	1.576385	0.05471
125	H	-7.771248	4.173432	-0.196093
126	H	-6.851956	3.758346	-1.636278
127	H	-5.526183	4.042483	0.552797
128	H	-6.332619	2.617141	1.170972
129	H	-5.09684	1.241242	-0.374732
130	H	-1.876367	2.151546	-1.281457
131	H	-2.725963	1.023448	-0.248004
132	H	-4.354598	4.571071	-1.295649
133	H	-2.630393	4.287209	-1.326614
134	H	-6.714125	0.905906	-2.002623
135	H	8.163368	5.871953	1.57601
136	H	9.007758	4.370656	1.870897
137	H	8.887647	4.602988	-1.878449
138	H	6.208325	3.900296	2.482642

139	H	4.056102	2.656743	2.345454
140	H	4.524467	2.150697	-1.901681
141	H	3.239308	1.786382	0.180249
142	H	9.781486	6.409842	-0.305459
143	H	-3.500832	-0.362754	-2.681082
144	H	3.384772	2.29965	-3.811544
145	H	4.733906	0.244846	-3.150578
146	H	5.43852	-3.424668	-0.193532
147	H	-0.956218	-5.54244	-1.561191
148	H	-1.155002	-2.89754	-0.066079
149	H	-1.32824	-3.310156	-4.357427
150	H	-1.67862	-4.747919	-3.3893
151	H	-0.045992	-4.048719	-3.402342
152	H	-4.951681	-3.017721	-2.674503
153	H	-3.739624	-3.800372	-3.675948
154	H	-3.962568	-4.309267	-2.00426
155	H	-2.607603	-1.226833	-0.390604
156	H	-3.195737	-2.845438	-0.069582
157	H	-6.403687	-1.670322	0.489183
158	H	-3.206218	-0.402168	-5.137456
159	H	-2.753775	-2.100952	-5.096404
160	H	-1.66702	2.581398	-3.705903
161	H	-1.883038	1.492744	-5.081784
162	H	-2.899263	1.353624	-3.648154
163	H	1.461506	3.727093	-5.135223
164	H	-0.184429	3.668663	-4.51281
165	H	0.362243	2.443641	-5.658965
166	H	2.306759	3.699501	-2.546463
167	H	1.716128	2.403366	-1.531032
168	H	-0.099474	5.654474	-0.848375
169	H	2.7086	1.95057	-6.182961
170	H	1.904109	0.4285	-5.746089
171	H	5.719085	-1.029531	0.033077
172	H	6.376817	-0.166768	-1.361093
173	H	7.178852	-1.633035	-0.767314
174	H	5.490927	-2.721672	-3.966189
175	H	6.293863	-1.183346	-3.642172
176	H	7.041567	-2.679923	-3.105437
177	H	6.516633	-4.631877	-2.125203
178	H	5.204094	-5.563214	-1.424233
179	H	4.41347	-5.117532	0.656479
180	H	2.875854	-5.66787	1.25519
181	H	3.504803	-6.34507	-0.237916
182	H	-0.167885	-4.058591	1.846631
183	H	-0.470203	-5.788855	2.11666
184	H	1.176394	-5.162355	2.15024

185	H	-0.246047	-7.428286	0.07024
186	H	1.435421	-7.209635	0.530949
187	H	-3.156915	-5.32331	-0.542093
188	H	-2.478347	-4.802389	0.991838
189	F	0.645777	-1.009168	2.216511
190	H	4.947141	-4.710208	-2.942646
191	H	0.950104	-6.989981	-1.15766
192	H	-2.233693	-6.457065	0.427587
193	H	-4.376278	-1.600746	-4.602488
194	H	3.660155	0.575357	-5.604434
195	H	-7.92317	1.736805	-2.991782
196	H	-4.651241	-5.295933	4.752283
197	H	-3.521938	-6.564868	4.238073
198	H	-7.30605	2.875414	6.320383
199	H	-6.62353	3.931857	5.065344
200	H	-3.606276	8.187258	1.33585
201	H	-3.779682	6.837961	2.491378
202	H	7.573191	-4.912068	3.141508
203	H	8.006957	-4.878614	4.870629
204	H	10.657368	4.909551	0.044776
205	H	10.585022	6.201844	1.25322
206	H	6.944013	3.320335	-2.826189
207	H	1.60871	5.348635	-1.125432
208	H	-5.097632	-2.718368	0.9986
209	C	-8.698577	0.024939	-1.989903
210	O	-9.91	0.139907	-2.135756
211	N	-8.095486	-1.152747	-1.654917
212	H	-7.087431	-1.19694	-1.71695
213	C	-8.880372	-2.363479	-1.553779
214	H	-8.340681	-3.100203	-0.952097
215	H	-9.833081	-2.128481	-1.075829
216	H	-9.100229	-2.80418	-2.534638

F3H1_React, E_{opt}= -4557.5484686

Center number	Atomic Symbol	Coordinates (Angstroms)		
		X	Y	Z
1	C	7.598166	-4.2957	4.045091

2	C	6.146103	-3.907832	4.400501
3	C	5.440258	-2.908167	3.498465
4	C	5.972363	-1.626817	3.278659
5	C	4.178468	-3.193488	2.954592
6	C	5.259147	-0.661324	2.565631
7	C	3.457057	-2.23154	2.238999
8	C	3.993549	-0.957422	2.047052
9	C	-3.523113	7.104585	1.466049
10	C	-2.092398	6.596067	1.212606
11	O	-1.78898	6.501538	-0.200375
12	C	-1.880523	5.198438	1.782224
13	C	-7.033183	2.960346	5.427913
14	C	-5.586999	2.522176	5.75072
15	C	-4.850215	1.914579	4.569493
16	C	-5.18913	0.622083	4.134883
17	C	-3.809291	2.559028	3.888063
18	C	-4.517881	-0.008047	3.093549
19	C	-3.117302	1.937434	2.842841
20	C	-3.441041	0.625527	2.423145
21	O	-2.768921	0.000002	1.457482
22	C	-3.891382	-5.564332	4.014296
23	C	-2.772747	-4.521203	3.901148
24	C	-3.114667	-3.40787	2.90566
25	O	-4.141852	-3.490986	2.215652
26	N	-2.221339	-2.410204	2.750928
27	C	-7.758612	1.211608	-2.044182
28	C	-8.088674	2.040645	-0.781384
29	C	-7.177078	3.252709	-0.505891
30	C	-5.962151	2.959386	0.388205
31	N	-4.92192	2.127806	-0.216038
32	C	-3.711031	2.563411	-0.589764
33	N	-3.542038	3.854136	-0.942485
34	N	-2.683307	1.710806	-0.588728
35	C	10.01745	5.684137	0.480252
36	C	8.555281	5.544133	0.941629
37	C	7.698483	4.573273	0.173637
38	C	8.066933	3.699431	-0.816777
39	C	6.292831	4.334906	0.412325
40	N	6.979293	2.924213	-1.202576
41	C	5.874894	3.290793	-0.462605
42	C	5.352681	4.902863	1.287647
43	C	4.563299	2.800497	-0.467853
44	C	4.044346	4.428948	1.2791
45	C	3.654084	3.383912	0.414908
46	Co	1.137785	-1.759314	-1.982728
47	C	-1.444521	-2.918493	-2.248869

48	C	-2.92669	-2.40774	-2.196948
49	C	-2.852242	-1.128706	-3.109834
50	C	-1.396614	-0.719263	-2.992388
51	C	-0.882347	0.580653	-3.383885
52	C	0.459617	0.864383	-3.29696
53	C	1.198847	2.188133	-3.647843
54	C	2.603921	1.613122	-4.004619
55	C	2.660755	0.396267	-3.125139
56	C	3.859718	-0.198143	-2.75789
57	C	4.013521	-1.407088	-2.092908
58	C	5.381754	-2.017712	-1.859229
59	C	5.017315	-3.365798	-1.150092
60	C	3.492537	-3.335475	-1.056528
61	C	2.735536	-4.316002	-0.447567
62	C	1.294346	-4.302311	-0.505817
63	C	0.32853	-5.337933	0.117087
64	C	-0.991241	-5.004843	-0.649153
65	C	-0.835526	-3.513745	-0.959235
66	C	-1.142582	-3.838189	-3.450643
67	N	-0.671111	-1.663219	-2.461168
68	N	1.427705	-0.057718	-2.84647
69	N	2.997886	-2.180892	-1.659728
70	N	0.627586	-3.334058	-1.080768
71	C	-3.94202	-3.441189	-2.666721
72	C	-3.293886	-1.98857	-0.749579
73	C	-4.701068	-1.441664	-0.59695
74	O	-5.132614	-0.521864	-1.324804
75	N	-5.470426	-2.013483	0.337749
76	C	-3.339001	-1.29717	-4.562288
77	C	-1.889106	1.576528	-3.925954
78	C	0.664572	3.051988	-4.805467
79	C	1.429713	3.027339	-2.340976
80	C	0.300366	3.892741	-1.798379
81	O	-0.894294	3.594679	-1.941163
82	N	0.668926	4.998921	-1.125727
83	C	2.779891	1.168824	-5.471666
84	C	6.249321	-1.130276	-0.94536
85	C	6.080906	-2.172453	-3.228056
86	C	5.566504	-4.622324	-1.85064
87	C	3.424776	-5.419298	0.332314
88	C	0.202278	-5.036904	1.627722
89	C	0.647775	-6.82549	-0.118464
90	C	-2.274893	-5.431276	0.058589
91	H	0.739107	1.49401	2.678614
92	C	0.413591	1.351337	1.659344
93	C	0.082275	0.181968	1.134204

94	F	0.295207	2.471406	0.90098
95	F	-0.311661	0.025205	-0.124071
96	H	6.148029	-3.485216	5.414481
97	H	5.540754	-4.81971	4.462616
98	H	6.943187	-1.372576	3.696169
99	H	3.741322	-4.174958	3.121185
100	H	5.683142	0.32877	2.420778
101	H	2.472225	-2.468539	1.848161
102	H	3.433313	-0.197819	1.507671
103	H	8.231379	-3.412806	3.912287
104	H	-1.375163	7.288264	1.680521
105	H	-0.906661	4.793296	1.491615
106	H	-2.653108	4.517935	1.41937
107	H	-1.93013	5.222049	2.874634
108	H	-1.928692	7.369589	-0.601962
109	H	-4.217597	6.581649	0.8003
110	H	-5.020568	3.371153	6.151303
111	H	-5.618299	1.775864	6.554842
112	H	-3.515375	3.562367	4.194051
113	H	-5.992436	0.090111	4.642927
114	H	-2.290378	2.440628	2.349553
115	H	-4.790463	-1.015914	2.799056
116	H	-7.468664	2.283359	4.683885
117	H	-2.531661	-4.072104	4.872225
118	H	-1.842665	-4.981338	3.54944
119	H	-2.453279	-1.566334	2.180173
120	H	-1.470781	-2.31516	3.418726
121	H	-4.43168	-5.622824	3.06693
122	H	-9.127282	2.371236	-0.87675
123	H	-8.065897	1.376604	0.094522
124	H	-7.760819	4.023016	0.013424
125	H	-6.84565	3.702718	-1.45223
126	H	-5.493302	3.893647	0.712123
127	H	-6.284382	2.449463	1.303414
128	H	-5.039714	1.119027	-0.223318
129	H	-1.805966	2.062308	-0.954541
130	H	-2.669953	0.970963	0.177012
131	H	-4.33613	4.391082	-1.250341
132	H	-2.610195	4.181005	-1.188277
133	H	-6.709521	0.916249	-2.05841
134	H	8.07695	6.533256	0.929719
135	H	8.547143	5.235952	1.996934
136	H	9.029411	3.555714	-1.285862
137	H	5.646828	5.700036	1.964266
138	H	3.312846	4.855452	1.95979
139	H	4.267553	1.983582	-1.12042

140	H	2.629732	3.025898	0.454462
141	H	10.100753	6.233673	-0.461676
142	H	-3.477232	-0.358877	-2.654402
143	H	3.398162	2.325155	-3.757833
144	H	4.761493	0.310013	-3.077094
145	H	5.416382	-3.339662	-0.128363
146	H	-0.924509	-5.551039	-1.596202
147	H	-1.147898	-2.90795	-0.103909
148	H	-1.379386	-3.338178	-4.389598
149	H	-1.71082	-4.768744	-3.406768
150	H	-0.076977	-4.086541	-3.464793
151	H	-4.948688	-3.01325	-2.660099
152	H	-3.75574	-3.79958	-3.679837
153	H	-3.957284	-4.308233	-2.003675
154	H	-2.625602	-1.208177	-0.385191
155	H	-3.193222	-2.835819	-0.072536
156	H	-6.3635	-1.576396	0.524698
157	H	-3.238566	-0.363722	-5.117833
158	H	-2.792642	-2.067381	-5.111629
159	H	-1.651312	2.583406	-3.606468
160	H	-1.932103	1.5494	-5.02038
161	H	-2.888652	1.363686	-3.548677
162	H	1.452383	3.740959	-5.127458
163	H	-0.193099	3.652839	-4.507533
164	H	0.373637	2.441633	-5.66177
165	H	2.307491	3.662845	-2.492603
166	H	1.694974	2.342831	-1.527244
167	H	-0.069152	5.572478	-0.711998
168	H	2.772064	2.022834	-6.15235
169	H	1.984412	0.478838	-5.770578
170	H	5.768818	-0.962641	0.019654
171	H	6.445969	-0.154411	-1.401365
172	H	7.214696	-1.615402	-0.766891
173	H	5.478558	-2.743237	-3.93935
174	H	6.271288	-1.190556	-3.668954
175	H	7.048116	-2.669566	-3.118089
176	H	6.652503	-4.55556	-1.95396
177	H	5.354348	-5.527335	-1.284324
178	H	4.407082	-5.10291	0.682437
179	H	2.857305	-5.67824	1.225804
180	H	3.54637	-6.340291	-0.247138
181	H	-0.218267	-4.043673	1.809808
182	H	-0.452623	-5.776888	2.095034
183	H	1.169909	-5.091696	2.132445
184	H	-0.250756	-7.417808	0.076402
185	H	1.43064	-7.208028	0.537896

186	H	-3.136658	-5.358835	-0.606466
187	H	-2.493602	-4.821061	0.932512
188	F	0.177816	-0.974518	1.771264
189	H	5.135795	-4.746689	-2.847559
190	H	0.950384	-7.007516	-1.15467
191	H	-2.213939	-6.474857	0.381785
192	H	-4.398029	-1.564883	-4.564168
193	H	3.736234	0.651615	-5.589242
194	H	-7.964489	1.81673	-2.93343
195	H	-4.618872	-5.285887	4.781197
196	H	-3.521916	-6.564793	4.238239
197	H	-7.697587	2.92606	6.291341
198	H	-7.063418	3.967825	5.000171
199	H	-3.6272	8.183792	1.323333
200	H	-3.779689	6.837884	2.49132
201	H	7.646196	-4.888996	3.126941
202	H	8.007122	-4.878579	4.870568
203	H	10.483185	4.701566	0.345671
204	H	10.584988	6.2017	1.253208
205	H	6.99396	2.230568	-1.930995
206	H	1.637661	5.190348	-0.914409
207	H	-5.047401	-2.615041	1.053875
208	C	-8.690152	0.015283	-2.055868
209	O	-9.893003	0.119481	-2.27148
210	N	-8.096812	-1.153981	-1.680855
211	H	-7.084868	-1.182147	-1.666941
212	C	-8.880398	-2.36345	-1.553799
213	H	-8.358133	-3.069826	-0.902192
214	H	-9.850754	-2.113959	-1.11997
215	H	-9.064467	-2.848671	-2.521317

F3H1_Int, E_{opt}= -4558.1302759

Center number	Atomic Symbol	Coordinates (Angstroms)		
		X	Y	Z
1	C	7.598158	-4.295704	4.045092
2	C	6.191447	-3.816333	4.472569
3	C	5.459777	-2.86947	3.537839

4	C	5.927836	-1.563047	3.321571
5	C	4.250894	-3.241243	2.931533
6	C	5.207906	-0.659676	2.537503
7	C	3.523687	-2.342915	2.143944
8	C	4.000695	-1.046625	1.944874
9	C	-3.523133	7.104515	1.466034
10	C	-2.069276	6.685522	1.190241
11	O	-1.832738	6.514102	-0.227223
12	C	-1.72598	5.351422	1.842436
13	C	-7.033186	2.960396	5.42792
14	C	-5.587596	2.56802	5.798479
15	C	-4.831248	1.97272	4.630082
16	C	-5.11246	0.6542	4.236908
17	C	-3.867087	2.681607	3.904581
18	C	-4.461588	0.054089	3.164864
19	C	-3.1882	2.090912	2.833898
20	C	-3.493348	0.780289	2.46829
21	O	-2.824695	0.135887	1.450256
22	C	-3.891369	-5.564352	4.014254
23	C	-2.746449	-4.555685	3.879664
24	C	-3.079417	-3.380748	2.961758
25	O	-4.143327	-3.324419	2.332244
26	N	-2.117517	-2.437048	2.828481
27	C	-7.758685	1.211675	-2.044225
28	C	-8.139566	2.219178	-0.936788
29	C	-7.266113	3.484122	-0.835339
30	C	-6.069868	3.355919	0.118467
31	N	-5.028198	2.439063	-0.341399
32	C	-3.793034	2.763181	-0.726578
33	N	-3.353302	4.016298	-0.813794
34	N	-2.916528	1.751534	-0.980909
35	C	10.017474	5.684128	0.480247
36	C	8.537797	5.613161	0.894701
37	C	7.684576	4.61823	0.154784
38	C	8.06404	3.666068	-0.757438
39	C	6.266241	4.434997	0.354085
40	N	6.969479	2.898979	-1.133918
41	C	5.851762	3.341315	-0.460626
42	C	5.313128	5.088345	1.152089
43	C	4.530317	2.880279	-0.479235
44	C	3.994978	4.643926	1.130453
45	C	3.610148	3.548031	0.328699
46	Co	1.15955	-1.798914	-2.058849
47	C	-1.402822	-2.945864	-2.31314
48	C	-2.885455	-2.421953	-2.271517
49	C	-2.807445	-1.208554	-3.259229

50	C	-1.355843	-0.775038	-3.114473
51	C	-0.855355	0.512811	-3.499378
52	C	0.487031	0.814091	-3.358188
53	C	1.200985	2.158606	-3.67231
54	C	2.619666	1.617514	-4.013165
55	C	2.682289	0.403858	-3.123519
56	C	3.876477	-0.171993	-2.728706
57	C	4.027934	-1.392617	-2.082608
58	C	5.398722	-2.005327	-1.855647
59	C	5.034758	-3.35078	-1.147304
60	C	3.506027	-3.340897	-1.105433
61	C	2.753978	-4.336064	-0.508223
62	C	1.320935	-4.321647	-0.566377
63	C	0.354087	-5.330759	0.093073
64	C	-0.971568	-4.9961	-0.65897
65	C	-0.788398	-3.5199	-1.017415
66	C	-1.119553	-3.90953	-3.484322
67	N	-0.616267	-1.703542	-2.541903
68	N	1.444994	-0.090415	-2.884498
69	N	3.002383	-2.185623	-1.699382
70	N	0.668441	-3.349127	-1.166716
71	C	-3.941933	-3.44108	-2.666743
72	C	-3.19679	-1.923958	-0.824475
73	C	-4.557613	-1.301598	-0.61334
74	O	-4.902753	-0.25259	-1.212357
75	N	-5.391502	-1.923504	0.229292
76	C	-3.242804	-1.470715	-4.716376
77	C	-1.842424	1.488869	-4.120793
78	C	0.664517	3.051862	-4.805402
79	C	1.401956	2.965412	-2.342019
80	C	0.273494	3.823829	-1.814068
81	O	-0.927272	3.513699	-1.930598
82	N	0.63089	4.957825	-1.182664
83	C	2.812695	1.18559	-5.479963
84	C	6.295346	-1.130556	-0.960719
85	C	6.080935	-2.172455	-3.228079
86	C	5.642849	-4.599502	-1.811545
87	C	3.448538	-5.434111	0.275962
88	C	0.254356	-5.012639	1.601592
89	C	0.647768	-6.825499	-0.118454
90	C	-2.251659	-5.364113	0.088604
91	H	1.513463	1.13249	2.212052
92	C	0.80778	1.170827	1.395292
93	C	0.255493	0.110345	0.82248
94	F	0.490207	2.402712	0.921442
95	F	-0.587692	0.19919	-0.195871

96	H	6.28717	-3.318302	5.446718
97	H	5.56152	-4.696252	4.64829
98	H	6.85763	-1.246581	3.788299
99	H	3.866299	-4.24578	3.090035
100	H	5.587395	0.347119	2.383527
101	H	2.59595	-2.654239	1.676551
102	H	3.450823	-0.352878	1.314314
103	H	8.268517	-3.453102	3.849022
104	H	-1.387201	7.457738	1.574651
105	H	-0.735007	5.001369	1.53917
106	H	-2.456413	4.593673	1.552078
107	H	-1.739651	5.447981	2.931848
108	H	-1.95525	7.366188	-0.667288
109	H	-4.200017	6.544833	0.810554
110	H	-5.05119	3.43514	6.199108
111	H	-5.614683	1.825951	6.605132
112	H	-3.621064	3.701258	4.191233
113	H	-5.856504	0.085131	4.789092
114	H	-2.399316	2.62023	2.308029
115	H	-4.673308	-0.969073	2.87321
116	H	-2.523454	0.77739	0.779975
117	H	-7.442334	2.253319	4.697763
118	H	-2.430502	-4.165366	4.855032
119	H	-1.859226	-5.034751	3.45139
120	H	-2.274028	-1.625944	2.237649
121	H	-1.271098	-2.474906	3.374261
122	H	-4.445102	-5.620594	3.07386
123	H	-9.181239	2.500452	-1.11255
124	H	-8.134558	1.704075	0.03474
125	H	-7.878705	4.30777	-0.449088
126	H	-6.913109	3.795367	-1.826414
127	H	-5.621841	4.336447	0.301665
128	H	-6.407405	2.996695	1.098003
129	H	-5.242752	1.441965	-0.377386
130	H	-2.050629	2.014668	-1.452875
131	H	-3.369499	0.889086	-1.287618
132	H	-3.929305	4.809895	-0.597823
133	H	-2.398188	4.175858	-1.156619
134	H	-6.712275	0.91159	-1.985812
135	H	8.088175	6.611951	0.806093
136	H	8.482672	5.369554	1.965516
137	H	9.038642	3.462418	-1.176562
138	H	5.605889	5.924999	1.780156
139	H	3.251465	5.132697	1.754742
140	H	4.235136	2.026097	-1.083273
141	H	2.579756	3.212755	0.35873

142	H	10.153738	6.1967	-0.476209
143	H	-3.478229	-0.428728	-2.897769
144	H	3.393742	2.351956	-3.762391
145	H	4.780786	0.350165	-3.02319
146	H	5.399204	-3.305192	-0.112376
147	H	-0.939147	-5.576484	-1.586476
148	H	-1.079288	-2.89184	-0.171339
149	H	-1.313451	-3.430177	-4.442472
150	H	-1.720256	-4.820003	-3.42526
151	H	-0.061497	-4.179719	-3.464485
152	H	-4.935851	-2.979567	-2.683435
153	H	-3.769918	-3.86102	-3.657822
154	H	-3.986382	-4.27101	-1.958292
155	H	-2.454561	-1.177696	-0.542367
156	H	-3.099819	-2.753652	-0.125506
157	H	-6.279044	-1.473643	0.416614
158	H	-3.110336	-0.566693	-5.314139
159	H	-2.670386	-2.263689	-5.198772
160	H	-1.679565	2.505236	-3.777276
161	H	-1.783363	1.487072	-5.215005
162	H	-2.871444	1.23565	-3.862675
163	H	1.443166	3.768084	-5.090841
164	H	-0.212352	3.625721	-4.50862
165	H	0.407879	2.466258	-5.689337
166	H	2.287641	3.598662	-2.448283
167	H	1.639274	2.254983	-1.54206
168	H	-0.098775	5.545957	-0.782546
169	H	2.793637	2.038838	-6.163558
170	H	2.034219	0.477619	-5.781782
171	H	5.832278	-0.946153	0.009161
172	H	6.499492	-0.161285	-1.429325
173	H	7.258486	-1.625182	-0.791799
174	H	5.464867	-2.747057	-3.92463
175	H	6.258683	-1.19328	-3.681773
176	H	7.052181	-2.667312	-3.13596
177	H	6.729371	-4.500156	-1.888099
178	H	5.445084	-5.503801	-1.238711
179	H	4.418443	-5.104331	0.64876
180	H	2.871551	-5.716221	1.156983
181	H	3.601105	-6.349016	-0.307298
182	H	-0.137448	-4.005527	1.770304
183	H	-0.404389	-5.731366	2.098653
184	H	1.230731	-5.074994	2.087948
185	H	-0.24975	-7.409036	0.11133
186	H	1.446778	-7.203536	0.521216
187	H	-3.12719	-5.296834	-0.560018

188	H	-2.430001	-4.711295	0.942396
189	F	0.442776	-1.134682	1.22555
190	H	5.24281	-4.752823	-2.817119
191	H	0.923027	-7.027553	-1.158439
192	H	-2.211549	-6.393764	0.457843
193	H	-4.301826	-1.740605	-4.760882
194	H	3.778987	0.685225	-5.588573
195	H	-7.920098	1.681974	-3.021053
196	H	-4.599675	-5.268683	4.792252
197	H	-3.52192	-6.564784	4.23827
198	H	-7.697588	2.92601	6.291331
199	H	-7.081092	3.957529	4.979788
200	H	-3.697536	8.172167	1.308383
201	H	-3.77968	6.837947	2.491351
202	H	7.557762	-4.918835	3.146817
203	H	8.007122	-4.878572	4.870564
204	H	10.448453	4.680533	0.392856
205	H	10.58496	6.201706	1.2532
206	H	6.994824	2.128488	-1.780902
207	H	1.602368	5.172403	-1.00686
208	H	-5.02689	-2.592288	0.911756
209	C	-8.691608	0.025699	-1.928291
210	O	-9.909159	0.14327	-1.988911
211	N	-8.076504	-1.161997	-1.645144
212	H	-7.084743	-1.235327	-1.823784
213	C	-8.880361	-2.36347	-1.553775
214	H	-8.302235	-3.150744	-1.063004
215	H	-9.773772	-2.147374	-0.965821
216	H	-9.207715	-2.722733	-2.537256

F3H1_TS, E_{opt}= -4558.069389

Center number	Atomic Symbol	Coordinates (Angstroms)		
		X	Y	Z
1	C	7.5982	-4.295665	4.04506
2	C	6.161323	-3.882993	4.432779
3	C	5.430077	-2.915614	3.517935
4	C	5.933203	-1.628461	3.269654

5	C	4.173316	-3.240854	2.985563
6	C	5.195596	-0.695525	2.537202
7	C	3.428487	-2.312294	2.251996
8	C	3.935893	-1.031353	2.029216
9	C	-3.523105	7.104493	1.466017
10	C	-2.115707	6.550864	1.184209
11	O	-1.822452	6.517661	-0.231898
12	C	-1.95439	5.120104	1.688281
13	C	-7.033086	2.960528	5.427778
14	C	-5.609419	2.448522	5.747193
15	C	-4.825528	1.930225	4.548812
16	C	-5.233314	0.745646	3.911894
17	C	-3.649995	2.535274	4.083686
18	C	-4.49096	0.173339	2.883182
19	C	-2.892651	1.980759	3.041513
20	C	-3.309085	0.786493	2.451031
21	O	-2.617818	0.162383	1.428347
22	C	-3.891427	-5.564282	4.014368
23	C	-2.760196	-4.534507	3.925602
24	C	-3.09134	-3.378738	2.983225
25	O	-4.165767	-3.334633	2.369986
26	N	-2.132195	-2.444447	2.802508
27	C	-7.758655	1.211676	-2.044205
28	C	-8.127246	2.177061	-0.896337
29	C	-7.201171	3.392736	-0.709013
30	C	-5.989321	3.142722	0.203646
31	N	-4.936152	2.297937	-0.357446
32	C	-3.755527	2.738859	-0.81001
33	N	-3.575767	4.024296	-1.152162
34	N	-2.73346	1.880364	-0.901754
35	C	10.017392	5.683924	0.480208
36	C	8.536329	5.623395	0.894776
37	C	7.674837	4.632819	0.158243
38	C	8.044459	3.695045	-0.771904
39	C	6.260102	4.438831	0.378912
40	N	6.947702	2.925013	-1.139154
41	C	5.838125	3.352962	-0.441871
42	C	5.316211	5.075086	1.201369
43	C	4.519428	2.883554	-0.441379
44	C	4.000559	4.62169	1.198135
45	C	3.607435	3.532137	0.391675
46	Co	1.101826	-1.691433	-1.854111
47	C	-1.431863	-2.916688	-2.243743
48	C	-2.9178	-2.409667	-2.211838
49	C	-2.84272	-1.150063	-3.149213
50	C	-1.388294	-0.728828	-3.009053

51	C	-0.879931	0.570829	-3.400592
52	C	0.457518	0.871565	-3.274757
53	C	1.188333	2.196814	-3.636155
54	C	2.59916	1.623012	-3.968686
55	C	2.650047	0.419597	-3.06659
56	C	3.850252	-0.177842	-2.707413
57	C	4.00271	-1.392948	-2.05276
58	C	5.372788	-2.026933	-1.859672
59	C	4.999406	-3.381038	-1.170491
60	C	3.479328	-3.316284	-1.02791
61	C	2.728781	-4.307128	-0.428931
62	C	1.289454	-4.295587	-0.490291
63	C	0.329947	-5.337941	0.129103
64	C	-0.998531	-4.992954	-0.618635
65	C	-0.830049	-3.504672	-0.942635
66	C	-1.116326	-3.848441	-3.432421
67	N	-0.664608	-1.661735	-2.457798
68	N	1.416222	-0.029213	-2.782665
69	N	2.982327	-2.142262	-1.586938
70	N	0.625687	-3.324324	-1.058595
71	C	-3.941927	-3.441102	-2.666749
72	C	-3.268613	-1.963741	-0.762226
73	C	-4.663064	-1.398274	-0.589934
74	O	-5.054551	-0.397062	-1.224458
75	N	-5.474964	-2.047772	0.262448
76	C	-3.299739	-1.355812	-4.606738
77	C	-1.872778	1.541299	-4.014517
78	C	0.664666	3.05186	-4.805233
79	C	1.396057	3.041102	-2.330533
80	C	0.277848	3.94301	-1.838852
81	O	-0.924665	3.708941	-2.054757
82	N	0.653815	5.019402	-1.124845
83	C	2.790558	1.161354	-5.427537
84	C	6.253121	-1.164544	-0.933231
85	C	6.080935	-2.172474	-3.2282
86	C	5.484448	-4.63445	-1.922807
87	C	3.423537	-5.42267	0.328205
88	C	0.225628	-5.046086	1.642767
89	C	0.647775	-6.825535	-0.118493
90	C	-2.276309	-5.389679	0.11779
91	H	1.495091	1.433899	2.334677
92	C	0.80173	1.166384	1.544223
93	C	0.160452	-0.01068	1.35307
94	F	0.46262	2.220379	0.733188
95	F	0.360307	-0.712761	-0.196882
96	H	6.200606	-3.426382	5.431259

97	H	5.554748	-4.789229	4.546655
98	H	6.900156	-1.34478	3.677963
99	H	3.760613	-4.229374	3.173504
100	H	5.598366	0.299957	2.36856
101	H	2.449358	-2.572064	1.866366
102	H	3.353708	-0.307217	1.46627
103	H	8.242272	-3.424425	3.890163
104	H	-1.371246	7.189632	1.684084
105	H	-0.999231	4.689674	1.371265
106	H	-2.755825	4.484225	1.306363
107	H	-1.995152	5.100798	2.781153
108	H	-1.908156	7.414241	-0.583225
109	H	-4.245437	6.620773	0.799939
110	H	-5.036197	3.235045	6.250434
111	H	-5.68791	1.625915	6.469214
112	H	-3.300619	3.451283	4.554489
113	H	-6.131979	0.23915	4.256321
114	H	-1.978382	2.45725	2.701661
115	H	-4.784572	-0.77094	2.439697
116	H	-1.599054	0.268784	1.444843
117	H	-7.498077	2.333403	4.658245
118	H	-2.504191	-4.123633	4.909943
119	H	-1.839867	-4.994927	3.550149
120	H	-2.327828	-1.640908	2.212052
121	H	-1.263678	-2.444296	3.314706
122	H	-4.418039	-5.616765	3.058554
123	H	-9.152336	2.513033	-1.076007
124	H	-8.160736	1.614528	0.047679
125	H	-7.775422	4.202735	-0.24227
126	H	-6.863582	3.773068	-1.682886
127	H	-5.530458	4.091668	0.499211
128	H	-6.319755	2.662954	1.132426
129	H	-5.073858	1.287419	-0.399318
130	H	-1.855349	2.216527	-1.28878
131	H	-2.723988	1.091448	-0.260465
132	H	-4.365636	4.608846	-1.367844
133	H	-2.636984	4.344548	-1.391301
134	H	-6.714746	0.904938	-1.994398
135	H	8.093823	6.625206	0.804999
136	H	8.480022	5.381803	1.965855
137	H	9.013379	3.502938	-1.209305
138	H	5.614081	5.905294	1.835449
139	H	3.266259	5.096634	1.843321
140	H	4.219938	2.032498	-1.047324
141	H	2.581489	3.182755	0.439743
142	H	10.15747	6.193323	-0.477409

143	H	-3.480561	-0.372541	-2.725646
144	H	3.387331	2.343402	-3.724717
145	H	4.751053	0.319924	-3.047522
146	H	5.436365	-3.394315	-0.163994
147	H	-0.959357	-5.551561	-1.559578
148	H	-1.133717	-2.885838	-0.095607
149	H	-1.332213	-3.355883	-4.380038
150	H	-1.688462	-4.777249	-3.390726
151	H	-0.050636	-4.091745	-3.421539
152	H	-4.94664	-3.006327	-2.676485
153	H	-3.747412	-3.817215	-3.671376
154	H	-3.969614	-4.298481	-1.990689
155	H	-2.561857	-1.198042	-0.439229
156	H	-3.165671	-2.806295	-0.079548
157	H	-6.381163	-1.630182	0.434487
158	H	-3.180715	-0.435882	-5.180959
159	H	-2.73961	-2.137519	-5.123661
160	H	-1.664656	2.562038	-3.716328
161	H	-1.862686	1.489676	-5.10922
162	H	-2.888321	1.324603	-3.684899
163	H	1.455373	3.740915	-5.121183
164	H	-0.198504	3.652596	-4.524902
165	H	0.390945	2.435447	-5.662842
166	H	2.295024	3.653435	-2.450488
167	H	1.605985	2.357428	-1.500278
168	H	-0.080644	5.610484	-0.733825
169	H	2.785474	2.005059	-6.121582
170	H	2.001433	0.462408	-5.722917
171	H	5.779447	-1.014115	0.037815
172	H	6.449647	-0.180372	-1.372306
173	H	7.218065	-1.656884	-0.771139
174	H	5.478109	-2.729272	-3.949719
175	H	6.280101	-1.185924	-3.655726
176	H	7.043863	-2.677332	-3.116392
177	H	6.567523	-4.601169	-2.06608
178	H	5.262794	-5.547032	-1.372435
179	H	4.416739	-5.119231	0.658742
180	H	2.874607	-5.679816	1.234141
181	H	3.52305	-6.343598	-0.256095
182	H	-0.151867	-4.036542	1.826407
183	H	-0.447552	-5.765782	2.116886
184	H	1.196311	-5.133341	2.1357
185	H	-0.248356	-7.421488	0.080468
186	H	1.437599	-7.209975	0.528172
187	H	-3.152027	-5.301165	-0.527859
188	H	-2.456324	-4.768234	0.994142

189	F	0.527376	-1.016204	2.202326
190	H	5.011972	-4.718997	-2.905101
191	H	0.941901	-6.998952	-1.158517
192	H	-2.232041	-6.431996	0.448629
193	H	-4.359358	-1.622467	-4.636279
194	H	3.749749	0.645866	-5.527924
195	H	-7.919433	1.720763	-3.001332
196	H	-4.628082	-5.286474	4.772271
197	H	-3.521877	-6.564869	4.238188
198	H	-7.697689	2.92591	6.291476
199	H	-7.012546	3.98387	5.040185
200	H	-3.587378	8.188815	1.338897
201	H	-3.779696	6.837972	2.491356
202	H	7.614974	-4.899687	3.13284
203	H	8.007005	-4.878619	4.870629
204	H	10.442512	4.6775	0.39673
205	H	10.584935	6.201966	1.253171
206	H	6.966196	2.169616	-1.803458
207	H	1.619054	5.165589	-0.863478
208	H	-5.085441	-2.680472	0.965251
209	C	-8.698922	0.026948	-1.972124
210	O	-9.913049	0.143181	-2.090612
211	N	-8.091676	-1.154807	-1.655772
212	H	-7.086882	-1.205793	-1.752623
213	C	-8.880371	-2.363463	-1.55378
214	H	-8.328665	-3.112245	-0.978607
215	H	-9.818356	-2.131363	-1.046409
216	H	-9.129176	-2.786264	-2.535557

F3H1-conf1_React, E_{opt}= -4557.5475476

Center number	Atomic Symbol	Coordinates (Angstroms)		
		X	Y	Z
1	C	7.59818	-4.295671	4.0451
2	C	6.14315	-3.952082	4.424096
3	C	5.388188	-2.994084	3.523916
4	C	5.882773	-1.708424	3.254336
5	C	4.119849	-3.328032	3.028241

6	C	5.124516	-0.781538	2.538012
7	C	3.354006	-2.405197	2.310769
8	C	3.851937	-1.12449	2.069972
9	C	-3.523087	7.104667	1.466067
10	C	-2.153482	6.532938	1.071318
11	O	-2.08232	6.355358	-0.373927
12	C	-1.888519	5.171751	1.699406
13	C	-7.033186	2.960404	5.427967
14	C	-5.58024	2.573913	5.780224
15	C	-4.835992	1.922051	4.631745
16	C	-5.15934	0.605675	4.263253
17	C	-3.813289	2.554348	3.914244
18	C	-4.491261	-0.060133	3.243027
19	C	-3.123199	1.895873	2.891845
20	C	-3.435361	0.564075	2.537209
21	O	-2.759718	-0.090347	1.591439
22	C	-3.891402	-5.564377	4.01433
23	C	-2.757636	-4.533234	3.96585
24	C	-3.047419	-3.407391	2.973175
25	O	-4.069677	-3.459023	2.270788
26	N	-2.129673	-2.432362	2.83859
27	C	-7.758695	1.21163	-2.044219
28	C	-8.024242	2.027716	-0.759439
29	C	-7.104731	3.246571	-0.536441
30	C	-5.879714	2.970618	0.346968
31	N	-4.888744	2.065705	-0.233768
32	C	-3.619303	2.380062	-0.527469
33	N	-3.2892	3.634364	-0.904626
34	N	-2.685571	1.428333	-0.493884
35	C	10.01749	5.68431	0.48036
36	C	8.751865	5.050407	1.075708
37	C	7.875998	4.359864	0.068043
38	C	8.086667	4.26857	-1.284451
39	C	6.6362	3.664363	0.330671
40	N	7.053159	3.564325	-1.884325
41	C	6.15163	3.178458	-0.916854
42	C	5.885047	3.40256	1.491115
43	C	4.965576	2.448557	-1.022643
44	C	4.701515	2.674112	1.391218
45	C	4.251531	2.193582	0.14385
46	Co	1.118299	-1.745967	-1.942185
47	C	-1.465056	-2.885452	-2.205938
48	C	-2.954338	-2.393115	-2.147424
49	C	-2.881507	-1.06377	-2.997208
50	C	-1.424534	-0.654681	-2.867849
51	C	-0.892608	0.653101	-3.225364

52	C	0.461381	0.891144	-3.228821
53	C	1.219159	2.18493	-3.659921
54	C	2.588029	1.563955	-4.070286
55	C	2.651124	0.356009	-3.178067
56	C	3.840687	-0.267654	-2.852438
57	C	3.987324	-1.447095	-2.133397
58	C	5.352882	-2.04576	-1.877611
59	C	4.987476	-3.403456	-1.191588
60	C	3.468986	-3.345952	-1.04996
61	C	2.714602	-4.319037	-0.430355
62	C	1.274107	-4.2979	-0.480676
63	C	0.31236	-5.342243	0.12921
64	C	-1.007582	-5.004352	-0.637483
65	C	-0.855844	-3.506535	-0.925781
66	C	-1.153313	-3.773004	-3.429179
67	N	-0.696071	-1.622415	-2.387914
68	N	1.418933	-0.053773	-2.827951
69	N	2.973274	-2.19353	-1.658161
70	N	0.606734	-3.323224	-1.044295
71	C	-3.941927	-3.441082	-2.666754
72	C	-3.348238	-2.04984	-0.688074
73	C	-4.751787	-1.494952	-0.521599
74	O	-5.201642	-0.592386	-1.262997
75	N	-5.494488	-2.028918	0.453319
76	C	-3.360988	-1.158306	-4.456832
77	C	-1.883424	1.732147	-3.620504
78	C	0.664525	3.051834	-4.805382
79	C	1.529817	3.092025	-2.414212
80	C	0.35673	3.763873	-1.71989
81	O	-0.368252	3.14068	-0.926976
82	N	0.140115	5.070197	-1.97642
83	C	2.687882	1.101174	-5.538586
84	C	6.179665	-1.150589	-0.932018
85	C	6.080939	-2.172499	-3.228105
86	C	5.464741	-4.655098	-1.952794
87	C	3.407066	-5.425603	0.34093
88	C	0.18095	-5.058512	1.642862
89	C	0.647763	-6.825485	-0.118453
90	C	-2.289312	-5.447214	0.064596
91	F	1.66874	1.359179	2.456302
92	C	0.876514	1.191605	1.379606
93	C	0.325087	0.012366	1.128011
94	H	0.705736	2.052491	0.747203
95	F	-0.39405	-0.209494	0.02928
96	H	6.152051	-3.517463	5.433217
97	H	5.57276	-4.884418	4.510996

98	H	6.859543	-1.420466	3.634449
99	H	3.715114	-4.315873	3.235391
100	H	5.515329	0.214284	2.349928
101	H	2.365272	-2.676566	1.957332
102	H	3.254799	-0.393973	1.536554
103	H	8.212886	-3.399038	3.921472
104	H	-1.355848	7.228054	1.370129
105	H	-1.01298	4.695273	1.248414
106	H	-2.74442	4.505084	1.572102
107	H	-1.707849	5.285921	2.771922
108	H	-2.35041	7.188032	-0.7861
109	H	-4.289365	6.648161	0.828136
110	H	-5.033061	3.454774	6.135854
111	H	-5.597869	1.870309	6.622118
112	H	-3.529057	3.572374	4.177569
113	H	-5.948192	0.086595	4.80585
114	H	-2.293801	2.377163	2.38101
115	H	-4.747233	-1.084393	2.993862
116	H	-7.44298	2.251043	4.700181
117	H	-2.558668	-4.096831	4.952201
118	H	-1.815772	-4.999171	3.654212
119	H	-2.359321	-1.572736	2.282734
120	H	-1.382523	-2.364485	3.514061
121	H	-4.39155	-5.608059	3.045012
122	H	-9.068924	2.351035	-0.794145
123	H	-7.947449	1.358428	0.109379
124	H	-7.67477	4.03735	-0.033194
125	H	-6.780671	3.661375	-1.50021
126	H	-5.375285	3.907573	0.603891
127	H	-6.196443	2.533134	1.301791
128	H	-5.096991	1.07001	-0.261494
129	H	-1.757204	1.726471	-0.770025
130	H	-2.721238	0.732949	0.343484
131	H	-3.981239	4.361051	-0.971964
132	H	-2.322347	3.917848	-0.809283
133	H	-6.712455	0.910647	-2.108759
134	H	8.16519	5.824588	1.589952
135	H	9.036539	4.333465	1.858338
136	H	8.900683	4.654041	-1.880481
137	H	6.225892	3.766023	2.456425
138	H	4.108787	2.464176	2.276034
139	H	4.620427	2.07928	-1.9793
140	H	3.341585	1.60223	0.093589
141	H	9.763125	6.417203	-0.292936
142	H	-3.50699	-0.319473	-2.504753
143	H	3.408307	2.26067	-3.868064

144	H	4.745748	0.202868	-3.212227
145	H	5.42891	-3.420042	-0.188207
146	H	-0.93661	-5.537379	-1.592197
147	H	-1.170513	-2.916281	-0.06065
148	H	-1.403212	-3.253689	-4.354683
149	H	-1.706583	-4.713136	-3.404874
150	H	-0.084174	-4.004948	-3.453087
151	H	-4.964696	-3.056583	-2.614833
152	H	-3.760774	-3.727118	-3.703775
153	H	-3.902009	-4.34559	-2.057676
154	H	-2.679322	-1.29987	-0.266979
155	H	-3.267233	-2.931708	-0.054571
156	H	-6.379556	-1.581676	0.652345
157	H	-3.264834	-0.193523	-4.960482
158	H	-2.804796	-1.893465	-5.0441
159	H	-1.619227	2.696156	-3.188113
160	H	-1.951372	1.85603	-4.706386
161	H	-2.880607	1.499889	-3.251381
162	H	1.456866	3.719352	-5.159557
163	H	-0.171742	3.675925	-4.488514
164	H	0.329536	2.446279	-5.64925
165	H	2.247899	3.852282	-2.740103
166	H	2.03018	2.495273	-1.647584
167	H	-0.56809	5.568346	-1.432025
168	H	2.675985	1.946268	-6.230614
169	H	1.862218	0.429783	-5.794929
170	H	5.680419	-1.023664	0.029479
171	H	6.343018	-0.156984	-1.358974
172	H	7.156596	-1.611932	-0.752523
173	H	5.492826	-2.717637	-3.970851
174	H	6.301264	-1.183349	-3.636934
175	H	7.0396	-2.68384	-3.106398
176	H	6.549094	-4.631274	-2.085778
177	H	5.227027	-5.570016	-1.412412
178	H	4.403847	-5.122165	0.659375
179	H	2.861327	-5.663379	1.253733
180	H	3.496022	-6.354168	-0.231836
181	H	-0.233094	-4.064992	1.836512
182	H	-0.480712	-5.800156	2.098089
183	H	1.146458	-5.126607	2.149894
184	H	-0.248862	-7.427218	0.056035
185	H	1.422701	-7.207511	0.547095
186	H	-3.152644	-5.369488	-0.598107
187	H	-2.510585	-4.852123	0.949112
188	F	0.464312	-1.088287	1.851283
189	H	5.000737	-4.7234	-2.940403

190	H	0.967693	-6.99407	-1.151736
191	H	-2.221982	-6.49552	0.370903
192	H	-4.418023	-1.433885	-4.477294
193	H	3.623224	0.555123	-5.688777
194	H	-7.996831	1.830942	-2.915339
195	H	-4.646924	-5.296329	4.757047
196	H	-3.521898	-6.564744	4.2382
197	H	-7.69759	2.926006	6.291294
198	H	-7.091156	3.95584	4.975819
199	H	-3.578742	8.191118	1.350414
200	H	-3.779728	6.837799	2.491312
201	H	7.653391	-4.881407	3.122705
202	H	8.007096	-4.878591	4.870553
203	H	10.665663	4.926318	0.028376
204	H	10.584953	6.201558	1.253093
205	H	6.993941	3.346693	-2.864766
206	H	0.767425	5.598154	-2.562293
207	H	-5.048314	-2.611645	1.171209
208	C	-8.698252	0.02059	-2.044199
209	O	-9.90194	0.134745	-2.252367
210	N	-8.104794	-1.150503	-1.681767
211	H	-7.092935	-1.168831	-1.631219
212	C	-8.880355	-2.363438	-1.553781
213	H	-8.388221	-3.042642	-0.851283
214	H	-9.87499	-2.109913	-1.18174
215	H	-9.006118	-2.884451	-2.512209

F3H1-conf1_Int, E_{opt}= -4558.1278356

Center number	Atomic Symbol	Coordinates (Angstroms)		
		X	Y	Z
1	C	7.59816	-4.295673	4.045088
2	C	6.168141	-3.900422	4.472312
3	C	5.39687	-2.976471	3.553279
4	C	5.857819	-1.678755	3.283522
5	C	4.160797	-3.362922	3.017448
6	C	5.103325	-0.794487	2.512297
7	C	3.398884	-2.482276	2.245126

8	C	3.869094	-1.194594	1.990837
9	C	-3.523142	7.104525	1.466048
10	C	-2.075078	6.734523	1.114864
11	O	-1.960619	6.50652	-0.317648
12	C	-1.608348	5.459804	1.804493
13	C	-7.033209	2.960398	5.427954
14	C	-5.575156	2.645301	5.821122
15	C	-4.813975	2.000226	4.686133
16	C	-5.062431	0.651543	4.383767
17	C	-3.898223	2.696196	3.891044
18	C	-4.421374	0.007437	3.333345
19	C	-3.229704	2.062382	2.839161
20	C	-3.49433	0.719453	2.569816
21	O	-2.827756	0.031988	1.583444
22	C	-3.891373	-5.564365	4.014236
23	C	-2.736499	-4.561442	3.941605
24	C	-3.024063	-3.375642	3.025735
25	O	-4.073969	-3.296821	2.373242
26	N	-2.046586	-2.447467	2.926134
27	C	-7.758687	1.211667	-2.044222
28	C	-8.045898	2.23117	-0.922172
29	C	-7.211413	3.526774	-0.973991
30	C	-5.953953	3.511921	-0.097911
31	N	-4.980809	2.495391	-0.49954
32	C	-3.671637	2.676896	-0.671807
33	N	-3.079007	3.867302	-0.685479
34	N	-2.882374	1.578651	-0.814607
35	C	10.017501	5.68432	0.480361
36	C	8.717558	5.111187	1.059462
37	C	7.851267	4.409902	0.052156
38	C	8.068523	4.299195	-1.298133
39	C	6.615323	3.712219	0.320571
40	N	7.036828	3.587608	-1.891974
41	C	6.138642	3.203106	-0.919924
42	C	5.869183	3.456753	1.48515
43	C	4.966669	2.450113	-1.016425
44	C	4.700536	2.70465	1.394986
45	C	4.260733	2.197161	0.154481
46	Co	1.128428	-1.776601	-2.001788
47	C	-1.428232	-2.907047	-2.257697
48	C	-2.91793	-2.403339	-2.221927
49	C	-2.839368	-1.147383	-3.161483
50	C	-1.392452	-0.704196	-2.986647
51	C	-0.87913	0.588711	-3.346272
52	C	0.479897	0.848864	-3.284288
53	C	1.222134	2.157397	-3.685016

54	C	2.596892	1.556313	-4.094326
55	C	2.665591	0.35574	-3.188688
56	C	3.851831	-0.256347	-2.845394
57	C	3.992609	-1.441426	-2.136126
58	C	5.359793	-2.0403	-1.879137
59	C	4.992663	-3.397908	-1.19976
60	C	3.469993	-3.349483	-1.092092
61	C	2.72207	-4.333579	-0.477632
62	C	1.289419	-4.310343	-0.523662
63	C	0.329131	-5.336667	0.114268
64	C	-0.996445	-4.995795	-0.638511
65	C	-0.821423	-3.508664	-0.966801
66	C	-1.123318	-3.839927	-3.44886
67	N	-0.64943	-1.653363	-2.450086
68	N	1.427274	-0.0808	-2.859716
69	N	2.965744	-2.198647	-1.695894
70	N	0.635134	-3.328062	-1.10476
71	C	-3.941941	-3.441079	-2.666752
72	C	-3.260205	-1.965696	-0.762571
73	C	-4.617409	-1.333468	-0.542035
74	O	-4.963577	-0.284853	-1.143905
75	N	-5.440635	-1.941517	0.319068
76	C	-3.24391	-1.352921	-4.635626
77	C	-1.860263	1.63715	-3.848299
78	C	0.664531	3.051859	-4.805392
79	C	1.529561	3.031919	-2.412076
80	C	0.370873	3.743858	-1.756179
81	O	-0.465066	3.133585	-1.055736
82	N	0.278225	5.078176	-1.920619
83	C	2.691859	1.097187	-5.562519
84	C	6.204128	-1.156045	-0.941752
85	C	6.080931	-2.172497	-3.228097
86	C	5.504089	-4.645225	-1.94552
87	C	3.421058	-5.434758	0.296957
88	C	0.219091	-5.046728	1.627581
89	C	0.647766	-6.825494	-0.118448
90	C	-2.27575	-5.389333	0.098448
91	F	1.853397	1.243752	2.200248
92	C	0.929162	1.209595	1.222284
93	C	0.308831	0.079819	0.910669
94	H	0.730652	2.131582	0.693916
95	F	-0.609772	0.037383	-0.05119
96	H	6.233472	-3.419299	5.458001
97	H	5.585398	-4.816192	4.627877
98	H	6.809441	-1.353769	3.697334
99	H	3.784923	-4.362821	3.221621

100	H	5.471521	0.207777	2.312989
101	H	2.444651	-2.797969	1.838258
102	H	3.284795	-0.510076	1.388999
103	H	8.234064	-3.420187	3.883813
104	H	-1.399317	7.555661	1.390589
105	H	-0.64952	5.126015	1.397777
106	H	-2.337682	4.657402	1.673237
107	H	-1.486731	5.639955	2.876144
108	H	-2.223907	7.319136	-0.771154
109	H	-4.20453	6.531133	0.825588
110	H	-5.066701	3.554914	6.158859
111	H	-5.573887	1.958149	6.675274
112	H	-3.682693	3.739916	4.107546
113	H	-5.775899	0.097347	4.988879
114	H	-2.4778	2.586382	2.255971
115	H	-4.611263	-1.034504	3.099541
116	H	-2.520753	0.649061	0.893965
117	H	-7.407447	2.210499	4.722794
118	H	-2.463231	-4.183318	4.93452
119	H	-1.83298	-5.041541	3.549826
120	H	-2.184025	-1.625208	2.344508
121	H	-1.211621	-2.505376	3.487829
122	H	-4.406898	-5.608158	3.05198
123	H	-9.110601	2.471421	-0.98243
124	H	-7.907446	1.744319	0.054519
125	H	-7.821048	4.365132	-0.617061
126	H	-6.930909	3.762251	-2.007773
127	H	-5.476605	4.494448	-0.129602
128	H	-6.223748	3.328053	0.950057
129	H	-5.285343	1.520927	-0.514993
130	H	-1.943615	1.725408	-1.179423
131	H	-3.386392	0.753207	-1.141251
132	H	-3.551632	4.741357	-0.536583
133	H	-2.056421	3.876454	-0.770684
134	H	-6.713276	0.90181	-2.062324
135	H	8.141704	5.920244	1.531285
136	H	8.957937	4.413082	1.873377
137	H	8.885436	4.675475	-1.896227
138	H	6.207046	3.836584	2.445394
139	H	4.117878	2.48949	2.284912
140	H	4.629531	2.047436	-1.962067
141	H	3.372485	1.575655	0.110168
142	H	9.808929	6.404457	-0.318356
143	H	-3.529166	-0.39326	-2.784236
144	H	3.407243	2.269194	-3.900619
145	H	4.760249	0.218527	-3.195498

146	H	5.415244	-3.408217	-0.187336
147	H	-0.955858	-5.55729	-1.577524
148	H	-1.123459	-2.900551	-0.109619
149	H	-1.319536	-3.342022	-4.397418
150	H	-1.710373	-4.760406	-3.414041
151	H	-0.06149	-4.094492	-3.428598
152	H	-4.953412	-3.018738	-2.648949
153	H	-3.763431	-3.798699	-3.680931
154	H	-3.943198	-4.30867	-2.004093
155	H	-2.516078	-1.242301	-0.43062
156	H	-3.181867	-2.822937	-0.095704
157	H	-6.31767	-1.478546	0.522247
158	H	-3.129453	-0.415401	-5.185775
159	H	-2.637895	-2.102915	-5.14547
160	H	-1.681634	2.614796	-3.396215
161	H	-1.817835	1.767568	-4.934633
162	H	-2.889267	1.370284	-3.606307
163	H	1.452648	3.73656	-5.138182
164	H	-0.180188	3.66255	-4.483129
165	H	0.346123	2.465784	-5.668728
166	H	2.292324	3.766746	-2.690261
167	H	1.973062	2.390506	-1.645692
168	H	-0.421782	5.603432	-1.39801
169	H	2.66891	1.938618	-6.260856
170	H	1.872537	0.414285	-5.808715
171	H	5.71132	-1.019381	0.021038
172	H	6.375603	-0.165264	-1.373716
173	H	7.179207	-1.623841	-0.763755
174	H	5.486592	-2.720072	-3.964162
175	H	6.2923	-1.184251	-3.645301
176	H	7.044317	-2.679672	-3.11752
177	H	6.590948	-4.607589	-2.059239
178	H	5.269079	-5.562647	-1.40768
179	H	4.414205	-5.124105	0.620502
180	H	2.875933	-5.681408	1.208696
181	H	3.523481	-6.364091	-0.273799
182	H	-0.175257	-4.043935	1.813871
183	H	-0.441867	-5.776334	2.105622
184	H	1.192627	-5.117678	2.118187
185	H	-0.24668	-7.424353	0.082842
186	H	1.437617	-7.202178	0.532677
187	H	-3.151461	-5.317817	-0.549964
188	H	-2.463017	-4.756795	0.966205
189	F	0.468618	-1.096406	1.49483
190	H	5.05908	-4.723969	-2.941005
191	H	0.946299	-7.006039	-1.155954

192	H	-2.225859	-6.426208	0.445681
193	H	-4.293337	-1.652515	-4.71176
194	H	3.630848	0.557468	-5.7127
195	H	-7.985966	1.679958	-3.009054
196	H	-4.628587	-5.280662	4.769237
197	H	-3.521918	-6.564768	4.238283
198	H	-7.697572	2.926011	6.291303
199	H	-7.12348	3.938109	4.945053
200	H	-3.734838	8.16646	1.311178
201	H	-3.779675	6.837931	2.491337
202	H	7.600019	-4.899527	3.133075
203	H	8.007119	-4.878596	4.870569
204	H	10.647888	4.891741	0.064072
205	H	10.584949	6.201553	1.253092
206	H	7.00541	3.311829	-2.859193
207	H	0.978808	5.58526	-2.438236
208	H	-5.060103	-2.59879	1.003701
209	C	-8.695602	0.035352	-1.872472
210	O	-9.913338	0.170758	-1.861879
211	N	-8.079595	-1.161792	-1.650417
212	H	-7.088171	-1.2294	-1.83124
213	C	-8.880362	-2.36347	-1.553776
214	H	-8.303113	-3.148184	-1.057649
215	H	-9.775302	-2.144605	-0.96935
216	H	-9.203519	-2.728594	-2.536625

F3H1-conf1_TS, E_{opt}= -4558.0645262

Center number	Atomic Symbol	Coordinates (Angstroms)		
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1	C	7.598245	-4.295632	4.045026
2	C	6.139357	-3.957188	4.414506
3	C	5.389058	-2.98299	3.525989
4	C	5.884992	-1.693996	3.276228
5	C	4.118111	-3.305663	3.028463
6	C	5.123488	-0.75232	2.581192
7	C	3.349738	-2.368278	2.333125
8	C	3.846969	-1.08301	2.115965

9	C	-3.523192	7.103378	1.465152
10	C	-2.159556	6.520547	1.06677
11	O	-2.069586	6.401449	-0.381213
12	C	-1.926233	5.130314	1.645251
13	C	-7.033072	2.960403	5.427809
14	C	-5.60226	2.480969	5.765059
15	C	-4.803044	1.932243	4.590885
16	C	-5.196398	0.727384	3.982612
17	C	-3.618282	2.523636	4.132662
18	C	-4.429326	0.120826	2.992576
19	C	-2.834162	1.934307	3.129917
20	C	-3.233328	0.717125	2.573775
21	O	-2.513725	0.048662	1.603816
22	C	-3.891419	-5.564317	4.014384
23	C	-2.74343	-4.549302	4.029415
24	C	-2.984514	-3.370839	3.092485
25	O	-4.018068	-3.28623	2.415247
26	N	-1.995431	-2.45929	2.993024
27	C	-7.758655	1.211601	-2.044161
28	C	-8.081123	2.138489	-0.852428
29	C	-7.157396	3.361031	-0.685924
30	C	-5.945223	3.120608	0.225374
31	N	-4.94212	2.196901	-0.303273
32	C	-3.699486	2.520152	-0.6779
33	N	-3.357471	3.771048	-1.02086
34	N	-2.770995	1.556187	-0.746487
35	C	10.01706	5.683793	0.480077
36	C	8.733414	5.086616	1.072074
37	C	7.864817	4.371025	0.076707
38	C	8.08815	4.221833	-1.268555
39	C	6.615494	3.701117	0.355361
40	N	7.049739	3.507809	-1.849181
41	C	6.135849	3.169167	-0.874091
42	C	5.856742	3.493455	1.521261
43	C	4.943419	2.445742	-0.958244
44	C	4.668363	2.771995	1.443215
45	C	4.21984	2.246589	0.21323
46	Co	1.076274	-1.675992	-1.803152
47	C	-1.457306	-2.880294	-2.186012
48	C	-2.95118	-2.396461	-2.153444
49	C	-2.882227	-1.089077	-3.032187
50	C	-1.427304	-0.664935	-2.882736
51	C	-0.903196	0.636296	-3.259811
52	C	0.44907	0.893409	-3.21383
53	C	1.203618	2.18677	-3.650335
54	C	2.574155	1.561558	-4.044924

55	C	2.630763	0.367701	-3.130996
56	C	3.820667	-0.262129	-2.818991
57	C	3.966846	-1.441465	-2.10132
58	C	5.333871	-2.062379	-1.882705
59	C	4.956388	-3.427241	-1.223288
60	C	3.447443	-3.324709	-1.012664
61	C	2.700424	-4.303456	-0.393206
62	C	1.261848	-4.285285	-0.445005
63	C	0.30761	-5.344274	0.151421
64	C	-1.01905	-4.993637	-0.59941
65	C	-0.858034	-3.495791	-0.893587
66	C	-1.122696	-3.77685	-3.396827
67	N	-0.700081	-1.614697	-2.365319
68	N	1.398888	-0.029507	-2.766695
69	N	2.949664	-2.155144	-1.581149
70	N	0.596942	-3.306179	-1.000053
71	C	-3.9417	-3.440935	-2.666832
72	C	-3.333	-2.029877	-0.689705
73	C	-4.738025	-1.494683	-0.503373
74	O	-5.163423	-0.505155	-1.136471
75	N	-5.521804	-2.151748	0.368299
76	C	-3.332617	-1.227307	-4.498701
77	C	-1.886272	1.683962	-3.751956
78	C	0.664376	3.051997	-4.804842
79	C	1.500831	3.088955	-2.397144
80	C	0.330193	3.819101	-1.767292
81	O	-0.540407	3.223297	-1.107186
82	N	0.270031	5.157301	-1.919029
83	C	2.678302	1.081386	-5.506485
84	C	6.161257	-1.195046	-0.910239
85	C	6.080964	-2.172436	-3.22816
86	C	5.339128	-4.668659	-2.052183
87	C	3.401729	-5.424033	0.34947
88	C	0.189518	-5.083415	1.669879
89	C	0.647751	-6.825481	-0.118491
90	C	-2.297384	-5.418085	0.122091
91	F	1.822026	1.561804	2.384158
92	C	0.861196	1.289513	1.463299
93	C	0.221168	0.09294	1.395651
94	H	0.633913	2.108648	0.791798
95	F	0.387312	-0.66553	-0.148019
96	H	6.13936	-3.541152	5.431734
97	H	5.568311	-4.891163	4.47984
98	H	6.864708	-1.414323	3.655698
99	H	3.714066	-4.297659	3.217938
100	H	5.515184	0.24637	2.4088

101	H	2.358534	-2.623597	1.975681
102	H	3.239271	-0.344638	1.60755
103	H	8.209603	-3.39582	3.927238
104	H	-1.354537	7.186862	1.408306
105	H	-1.056638	4.655114	1.182375
106	H	-2.792125	4.482696	1.487424
107	H	-1.752239	5.203678	2.72258
108	H	-2.288602	7.263298	-0.76155
109	H	-4.295308	6.658495	0.825824
110	H	-5.042576	3.292259	6.24363
111	H	-5.670492	1.682721	6.514991
112	H	-3.278456	3.45343	4.582992
113	H	-6.103004	0.231967	4.322633
114	H	-1.903521	2.392879	2.809545
115	H	-4.707807	-0.839169	2.572459
116	H	-1.497976	0.25457	1.547464
117	H	-7.483419	2.310991	4.668105
118	H	-2.555914	-4.163082	5.03888
119	H	-1.804296	-5.018675	3.714717
120	H	-2.139034	-1.638497	2.408445
121	H	-1.164044	-2.49765	3.562068
122	H	-4.351861	-5.593347	3.024312
123	H	-9.117338	2.46676	-0.97315
124	H	-8.06159	1.549263	0.075693
125	H	-7.727238	4.179525	-0.229028
126	H	-6.819339	3.724565	-1.665129
127	H	-5.446404	4.066316	0.460559
128	H	-6.28058	2.712119	1.186545
129	H	-5.162289	1.200468	-0.345155
130	H	-1.823026	1.830612	-0.992408
131	H	-2.849487	0.798349	-0.071888
132	H	-4.029994	4.517229	-1.054308
133	H	-2.370001	4.012755	-1.0134
134	H	-6.71342	0.903055	-2.043212
135	H	8.148708	5.885169	1.550481
136	H	8.995545	4.39345	1.883693
137	H	8.912913	4.57176	-1.871851
138	H	6.197847	3.89076	2.473232
139	H	4.070248	2.59988	2.331529
140	H	4.601763	2.032671	-1.898305
141	H	3.304998	1.662592	0.183996
142	H	9.788171	6.406209	-0.310797
143	H	-3.524101	-0.335989	-2.573958
144	H	3.393759	2.261955	-3.849695
145	H	4.722763	0.192203	-3.208156
146	H	5.448599	-3.498345	-0.246232

147	H	-0.969246	-5.532236	-1.551776
148	H	-1.17038	-2.896468	-0.035232
149	H	-1.350284	-3.265094	-4.332039
150	H	-1.674653	-4.718385	-3.377874
151	H	-0.05209	-3.997331	-3.391367
152	H	-4.962051	-3.044236	-2.649319
153	H	-3.737072	-3.750313	-3.691943
154	H	-3.930119	-4.335278	-2.040321
155	H	-2.642178	-1.274742	-0.311143
156	H	-3.22491	-2.900564	-0.045852
157	H	-6.428118	-1.744612	0.561359
158	H	-3.218516	-0.277538	-5.025746
159	H	-2.761341	-1.976848	-5.050715
160	H	-1.6667	2.665897	-3.333875
161	H	-1.884837	1.777226	-4.843172
162	H	-2.90295	1.443978	-3.445634
163	H	1.461792	3.71934	-5.148586
164	H	-0.176604	3.678742	-4.50461
165	H	0.344082	2.444244	-5.652508
166	H	2.268064	3.814342	-2.687716
167	H	1.933567	2.469673	-1.606383
168	H	-0.447635	5.672469	-1.40733
169	H	2.671443	1.916139	-6.211672
170	H	1.852628	0.407045	-5.755186
171	H	5.655439	-1.08989	0.050953
172	H	6.330017	-0.191358	-1.311991
173	H	7.135853	-1.663844	-0.735467
174	H	5.498324	-2.699093	-3.987883
175	H	6.314258	-1.177307	-3.615787
176	H	7.033953	-2.693075	-3.099902
177	H	6.414317	-4.688065	-2.246734
178	H	5.089151	-5.591288	-1.530172
179	H	4.408849	-5.132177	0.646126
180	H	2.877243	-5.667126	1.273882
181	H	3.470913	-6.349192	-0.232098
182	H	-0.201313	-4.082772	1.872963
183	H	-0.479577	-5.819409	2.125119
184	H	1.158598	-5.170427	2.166177
185	H	-0.245694	-7.434538	0.051084
186	H	1.427752	-7.211466	0.538609
187	H	-3.170737	-5.329177	-0.527263
188	H	-2.492172	-4.816987	1.01044
189	F	0.621403	-0.875141	2.266386
190	H	4.817119	-4.681244	-3.013207
191	H	0.96498	-6.976234	-1.155245
192	H	-2.241295	-6.466285	0.43206

193	H	-4.389644	-1.501933	-4.546728
194	H	3.61243	0.530084	-5.644088
195	H	-7.951879	1.754422	-2.976045
196	H	-4.67317	-5.304718	4.731995
197	H	-3.521894	-6.564822	4.238168
198	H	-7.697685	2.926011	6.291429
199	H	-7.031459	3.977481	5.023593
200	H	-3.566694	8.19075	1.352419
201	H	-3.779369	6.83881	2.491932
202	H	7.661771	-4.877308	3.120578
203	H	8.006941	-4.878633	4.870638
204	H	10.655643	4.904134	0.051747
205	H	10.585082	6.20197	1.25327
206	H	7.006782	3.23128	-2.815527
207	H	1.002941	5.665407	-2.388165
208	H	-5.101694	-2.767053	1.066197
209	C	-8.702112	0.027162	-1.988102
210	O	-9.913037	0.147729	-2.137032
211	N	-8.100789	-1.150714	-1.658043
212	H	-7.090198	-1.18441	-1.676425
213	C	-8.880356	-2.363385	-1.553808
214	H	-8.357741	-3.083351	-0.917382
215	H	-9.850381	-2.12368	-1.114398
216	H	-9.06316	-2.82922	-2.530779

F3H1-conf2_React, E_{opt}= -4557.5388426

Center number	Atomic Symbol	Coordinates (Angstroms)		
		X	Y	Z
1	C	7.598087	-4.295747	4.04503
2	C	6.308347	-3.629164	4.57159
3	C	5.573602	-2.708609	3.619448
4	C	6.024637	-1.398132	3.393544
5	C	4.384227	-3.113502	2.99767
6	C	5.299797	-0.512661	2.594453
7	C	3.650912	-2.234332	2.195756
8	C	4.104664	-0.928578	1.998249
9	C	-3.522982	7.104244	1.46592

10	C	-2.165802	6.499951	1.076064
11	O	-2.053433	6.394453	-0.371628
12	C	-1.973428	5.096867	1.635577
13	C	-7.033216	2.960377	5.427877
14	C	-5.582951	2.559236	5.775081
15	C	-4.840682	1.921634	4.616877
16	C	-5.187114	0.621963	4.211889
17	C	-3.79671	2.549059	3.925826
18	C	-4.524967	-0.030396	3.17931
19	C	-3.11149	1.903862	2.891756
20	C	-3.451504	0.590322	2.494949
21	O	-2.789052	-0.047415	1.530512
22	C	-3.891376	-5.564302	4.014309
23	C	-2.765326	-4.530395	3.915103
24	C	-3.107128	-3.403316	2.938345
25	O	-4.157233	-3.461157	2.279067
26	N	-2.202263	-2.422518	2.77239
27	C	-7.758711	1.211655	-2.044222
28	C	-8.151182	2.115659	-0.85393
29	C	-7.221746	3.314727	-0.586991
30	C	-6.043858	3.008222	0.35071
31	N	-5.033674	2.102649	-0.190599
32	C	-3.796862	2.451206	-0.573059
33	N	-3.539995	3.705486	-0.999074
34	N	-2.829302	1.535948	-0.559226
35	C	10.017749	5.683551	0.480363
36	C	8.762321	5.06783	1.120783
37	C	7.858691	4.302328	0.194551
38	C	8.065506	3.985739	-1.123339
39	C	6.570531	3.747273	0.542814
40	N	6.986228	3.263444	-1.6178
41	C	6.047563	3.111619	-0.61868
42	C	5.807338	3.731469	1.723185
43	C	4.794448	2.489087	-0.627699
44	C	4.560958	3.111757	1.720307
45	C	4.05649	2.499896	0.553342
46	Co	1.1311	-1.76505	-1.975551
47	C	-1.448207	-2.901877	-2.23275
48	C	-2.935068	-2.400069	-2.170741
49	C	-2.86076	-1.089397	-3.044042
50	C	-1.404265	-0.679879	-2.923313
51	C	-0.889388	0.629119	-3.286637
52	C	0.457892	0.894345	-3.238545
53	C	1.208458	2.198757	-3.642014
54	C	2.594391	1.593498	-4.019456
55	C	2.649929	0.380501	-3.134823

56	C	3.846715	-0.230718	-2.803472
57	C	3.998979	-1.430361	-2.121268
58	C	5.369435	-2.020309	-1.867014
59	C	5.014961	-3.36663	-1.151637
60	C	3.490705	-3.34659	-1.063468
61	C	2.73426	-4.318205	-0.44124
62	C	1.294267	-4.300283	-0.501817
63	C	0.327552	-5.336869	0.1199
64	C	-0.994014	-4.996981	-0.641981
65	C	-0.834054	-3.504868	-0.949831
66	C	-1.150449	-3.806443	-3.446141
67	N	-0.672117	-1.643854	-2.430782
68	N	1.415823	-0.03987	-2.804632
69	N	2.986056	-2.206435	-1.686841
70	N	0.62862	-3.329708	-1.079772
71	C	-3.941857	-3.44103	-2.666769
72	C	-3.315719	-2.019596	-0.716185
73	C	-4.724538	-1.471735	-0.561311
74	O	-5.171048	-0.572351	-1.306719
75	N	-5.478197	-2.018237	0.399353
76	C	-3.346056	-1.211267	-4.500678
77	C	-1.892737	1.674217	-3.737
78	C	0.664464	3.051881	-4.805276
79	C	1.476479	3.102142	-2.384211
80	C	0.288682	3.834683	-1.776607
81	O	-0.659254	3.222726	-1.270095
82	N	0.318105	5.183689	-1.788235
83	C	2.736265	1.135643	-5.486762
84	C	6.217876	-1.108765	-0.956977
85	C	6.080923	-2.172384	-3.228095
86	C	5.574287	-4.624475	-1.841276
87	C	3.425723	-5.405993	0.357969
88	C	0.201017	-5.050292	1.633905
89	C	0.647783	-6.825472	-0.118463
90	C	-2.274919	-5.41704	0.075545
91	F	0.14915	1.741831	1.300256
92	C	0.104497	0.440082	1.069386
93	C	0.599317	-0.469589	1.894541
94	F	-0.434947	0.155422	-0.104358
95	F	0.508492	-1.782476	1.56155
96	H	6.565163	-3.056936	5.472019
97	H	5.621115	-4.416898	4.902399
98	H	6.940573	-1.062659	3.874484
99	H	4.011628	-4.120617	3.170342
100	H	5.651142	0.504068	2.440956
101	H	2.720441	-2.557513	1.74469

102	H	3.535801	-0.236605	1.384902
103	H	8.339114	-3.553365	3.733339
104	H	-1.35328	7.147149	1.437836
105	H	-1.107898	4.605194	1.181723
106	H	-2.851134	4.47473	1.449334
107	H	-1.818662	5.144105	2.717244
108	H	-2.295949	7.250209	-0.751121
109	H	-4.298462	6.669452	0.824455
110	H	-5.031189	3.431119	6.14567
111	H	-5.606597	1.842532	6.605968
112	H	-3.49433	3.55413	4.217734
113	H	-5.990367	0.104591	4.734842
114	H	-2.268323	2.379856	2.399319
115	H	-4.799939	-1.043095	2.90448
116	H	-7.44979	2.25972	4.695549
117	H	-2.521177	-4.095381	4.891857
118	H	-1.838886	-4.99247	3.556527
119	H	-2.433392	-1.564421	2.215566
120	H	-1.413096	-2.373372	3.398766
121	H	-4.423737	-5.618835	3.062558
122	H	-9.17167	2.465304	-1.035391
123	H	-8.204235	1.50321	0.057711
124	H	-7.800231	4.114188	-0.107144
125	H	-6.847703	3.726005	-1.534765
126	H	-5.540566	3.933146	0.651807
127	H	-6.415049	2.556454	1.277722
128	H	-5.204497	1.102073	-0.170345
129	H	-1.919313	1.852632	-0.873253
130	H	-2.829404	0.829373	0.247008
131	H	-4.27804	4.380422	-1.101952
132	H	-2.579583	4.029019	-1.021103
133	H	-6.712894	0.911881	-1.990421
134	H	8.177421	5.866879	1.597412
135	H	9.069511	4.406823	1.943887
136	H	8.905209	4.21752	-1.762387
137	H	6.185371	4.203545	2.625474
138	H	3.963871	3.099151	2.627182
139	H	4.412151	2.00239	-1.517367
140	H	3.073466	2.035246	0.580401
141	H	9.755772	6.401955	-0.302919
142	H	-3.484876	-0.33432	-2.564527
143	H	3.406987	2.291902	-3.791754
144	H	4.748939	0.262914	-3.140871
145	H	5.410493	-3.328815	-0.129079
146	H	-0.935896	-5.542689	-1.590144
147	H	-1.132161	-2.898394	-0.09107

148	H	-1.401314	-3.297155	-4.376829
149	H	-1.712936	-4.740671	-3.406336
150	H	-0.083421	-4.043535	-3.472738
151	H	-4.955917	-3.033097	-2.630111
152	H	-3.76171	-3.758341	-3.694699
153	H	-3.925319	-4.330169	-2.034209
154	H	-2.645915	-1.254007	-0.324129
155	H	-3.224304	-2.883789	-0.05966
156	H	-6.371918	-1.58143	0.583151
157	H	-3.251591	-0.257697	-5.024209
158	H	-2.794119	-1.959237	-5.075191
159	H	-1.634339	2.658959	-3.358348
160	H	-1.96584	1.73181	-4.8283
161	H	-2.885673	1.453199	-3.348511
162	H	1.453638	3.728365	-5.149054
163	H	-0.183637	3.669	-4.50756
164	H	0.351171	2.435405	-5.649283
165	H	2.255044	3.821998	-2.657784
166	H	1.88761	2.484295	-1.579339
167	H	-0.423012	5.68574	-1.293583
168	H	2.730488	1.984403	-6.174002
169	H	1.92439	0.456412	-5.76477
170	H	5.731515	-0.941343	0.004125
171	H	6.400043	-0.133542	-1.418489
172	H	7.188258	-1.581222	-0.772063
173	H	5.490508	-2.747956	-3.945782
174	H	6.275772	-1.190965	-3.667358
175	H	7.048417	-2.665599	-3.104571
176	H	6.660983	-4.555057	-1.935593
177	H	5.359358	-5.526505	-1.270863
178	H	4.40193	-5.074344	0.712332
179	H	2.852605	-5.660261	1.249098
180	H	3.562641	-6.332269	-0.209413
181	H	-0.219811	-4.061833	1.827226
182	H	-0.448806	-5.800887	2.09141
183	H	1.168954	-5.106336	2.137477
184	H	-0.249204	-7.417891	0.083284
185	H	1.435653	-7.205367	0.533482
186	H	-3.143176	-5.335399	-0.579985
187	H	-2.478216	-4.808032	0.95443
188	H	1.074333	-0.250121	2.838661
189	H	5.151883	-4.755857	-2.840888
190	H	0.943574	-7.008306	-1.156603
191	H	-2.218639	-6.462226	0.394497
192	H	-4.40354	-1.485336	-4.510088
193	H	3.681139	0.60115	-5.617201

194	H	-7.90992	1.767185	-2.976446
195	H	-4.623883	-5.285939	4.776299
196	H	-3.521912	-6.564778	4.238242
197	H	-7.697565	2.926015	6.291347
198	H	-7.082674	3.959316	4.982374
199	H	-3.548994	8.19268	1.356812
200	H	-3.779735	6.838127	2.49144
201	H	7.395509	-4.960277	3.199485
202	H	8.007153	-4.87855	4.870604
203	H	10.661458	4.91481	0.04019
204	H	10.584622	6.202195	1.25306
205	H	6.891832	2.944645	-2.567268
206	H	1.117627	5.691917	-2.130718
207	H	-5.044337	-2.601946	1.125935
208	C	-8.693387	0.017185	-2.04596
209	O	-9.894977	0.123499	-2.269851
210	N	-8.101693	-1.150033	-1.66636
211	H	-7.089013	-1.174357	-1.638209
212	C	-8.88035	-2.363459	-1.553782
213	H	-8.368599	-3.065649	-0.889382
214	H	-9.861656	-2.119256	-1.14208
215	H	-9.039768	-2.851858	-2.52419

F3H1-conf2_Int, E_{opt}= -4558.1303317

Center number	Atomic Symbol	Coordinates (Angstroms)		
		X	Y	Z
1	C	7.598146	-4.295729	4.045079
2	C	6.175651	-3.843943	4.442832
3	C	5.501314	-2.830853	3.536611
4	C	6.004802	-1.524785	3.424112
5	C	4.318082	-3.139791	2.851481
6	C	5.337244	-0.557629	2.672439
7	C	3.645328	-2.176765	2.091706
8	C	4.149779	-0.878625	2.006145
9	C	-3.523101	7.104504	1.466054
10	C	-2.071191	6.688829	1.182568
11	O	-1.83399	6.554197	-0.239292

12	C	-1.724971	5.33917	1.798441
13	C	-7.033214	2.960374	5.42788
14	C	-5.593274	2.544554	5.792491
15	C	-4.837759	1.958921	4.618165
16	C	-5.145331	0.657195	4.189312
17	C	-3.839177	2.655543	3.929256
18	C	-4.48102	0.060424	3.124358
19	C	-3.147968	2.069597	2.863265
20	C	-3.473487	0.772636	2.468679
21	O	-2.794212	0.113633	1.46876
22	C	-3.891372	-5.564309	4.014301
23	C	-2.745915	-4.553713	3.908818
24	C	-3.056978	-3.376456	2.986273
25	O	-4.083742	-3.337581	2.297535
26	N	-2.107337	-2.413536	2.912131
27	C	-7.7587	1.211665	-2.044217
28	C	-8.192236	2.268409	-1.004405
29	C	-7.315571	3.530546	-0.906735
30	C	-6.144316	3.415684	0.080446
31	N	-5.08833	2.497713	-0.338537
32	C	-3.862834	2.823848	-0.762825
33	N	-3.44631	4.085784	-0.893299
34	N	-2.993504	1.82706	-1.025311
35	C	10.017476	5.684151	0.480223
36	C	8.528931	5.660999	0.868751
37	C	7.663069	4.667948	0.140598
38	C	8.033063	3.680445	-0.737197
39	C	6.23711	4.526645	0.319645
40	N	6.925358	2.931032	-1.111592
41	C	5.808479	3.420886	-0.471057
42	C	5.287869	5.225661	1.082671
43	C	4.476141	2.99306	-0.500129
44	C	3.959431	4.813832	1.051581
45	C	3.561431	3.705973	0.274293
46	Co	1.155367	-1.797529	-2.04694
47	C	-1.407496	-2.941741	-2.303997
48	C	-2.889772	-2.418353	-2.265973
49	C	-2.810473	-1.203625	-3.25224
50	C	-1.357108	-0.774067	-3.111242
51	C	-0.85497	0.511606	-3.503777
52	C	0.484955	0.814999	-3.357234
53	C	1.196112	2.16244	-3.666734
54	C	2.621153	1.626374	-3.993406
55	C	2.680156	0.407022	-3.11002
56	C	3.873771	-0.172954	-2.715337
57	C	4.023876	-1.398982	-2.078922

58	C	5.392853	-2.019239	-1.857035
59	C	5.023921	-3.37283	-1.166592
60	C	3.49714	-3.34683	-1.101516
61	C	2.744205	-4.335872	-0.494917
62	C	1.31111	-4.317441	-0.547148
63	C	0.344044	-5.333666	0.100414
64	C	-0.976407	-5.001184	-0.661317
65	C	-0.79919	-3.520885	-1.007269
66	C	-1.119141	-3.901341	-3.477619
67	N	-0.621258	-1.69876	-2.530363
68	N	1.442956	-0.089979	-2.880608
69	N	2.996353	-2.189056	-1.69294
70	N	0.658962	-3.341732	-1.14386
71	C	-3.941918	-3.441089	-2.666762
72	C	-3.199926	-1.917301	-0.81969
73	C	-4.563037	-1.300454	-0.606878
74	O	-4.908219	-0.243418	-1.187957
75	N	-5.400987	-1.943489	0.218373
76	C	-3.25175	-1.462599	-4.707979
77	C	-1.841662	1.487437	-4.124791
78	C	0.664519	3.051848	-4.805363
79	C	1.380601	2.981328	-2.341496
80	C	0.244843	3.844398	-1.827944
81	O	-0.953947	3.528848	-1.934137
82	N	0.599937	4.990741	-1.215596
83	C	2.832143	1.202407	-5.460194
84	C	6.286787	-1.158207	-0.945448
85	C	6.08095	-2.172436	-3.228122
86	C	5.595814	-4.620371	-1.865126
87	C	3.438172	-5.439156	0.282588
88	C	0.2291	-5.02831	1.610544
89	C	0.647767	-6.825489	-0.118446
90	C	-2.261325	-5.385719	0.069708
91	F	1.843173	1.547438	1.906803
92	C	0.768261	1.44913	1.147588
93	C	0.473519	0.398231	0.396086
94	F	0.007797	2.545293	1.204632
95	F	-0.673828	0.445632	-0.326895
96	H	6.227667	-3.408928	5.449851
97	H	5.5344	-4.72872	4.532244
98	H	6.918245	-1.258844	3.950814
99	H	3.908907	-4.143886	2.927386
100	H	5.737854	0.450294	2.606485
101	H	2.738338	-2.447548	1.559683
102	H	3.632194	-0.120552	1.427223
103	H	8.256329	-3.437784	3.874976

104	H	-1.38918	7.450907	1.587223
105	H	-0.719033	5.021077	1.5127
106	H	-2.431356	4.577297	1.461277
107	H	-1.770324	5.397612	2.889714
108	H	-1.960418	7.416598	-0.657557
109	H	-4.202311	6.546765	0.811591
110	H	-5.04742	3.398193	6.209014
111	H	-5.630789	1.789729	6.587033
112	H	-3.573688	3.662513	4.242137
113	H	-5.915965	0.09621	4.71257
114	H	-2.333432	2.590974	2.370055
115	H	-4.704038	-0.954266	2.811849
116	H	-2.302355	0.734643	0.908265
117	H	-7.452834	2.267351	4.690169
118	H	-2.450052	-4.169106	4.892664
119	H	-1.84998	-5.030611	3.496082
120	H	-2.255826	-1.595149	2.327328
121	H	-1.313441	-2.424014	3.533243
122	H	-4.426373	-5.615213	3.062815
123	H	-9.220355	2.548094	-1.248893
124	H	-8.246167	1.796673	-0.012645
125	H	-7.935347	4.364059	-0.55412
126	H	-6.934685	3.819665	-1.894315
127	H	-5.70402	4.399599	0.266282
128	H	-6.508507	3.067336	1.05423
129	H	-5.289667	1.499375	-0.339034
130	H	-2.110715	2.070901	-1.469747
131	H	-3.418859	0.923751	-1.229828
132	H	-4.015629	4.872693	-0.640481
133	H	-2.4878	4.251464	-1.212466
134	H	-6.716924	0.915533	-1.924424
135	H	8.106355	6.66783	0.745634
136	H	8.448562	5.447548	1.944364
137	H	9.009077	3.439862	-1.132775
138	H	5.59116	6.071798	1.692663
139	H	3.218898	5.336357	1.651161
140	H	4.171607	2.12874	-1.084974
141	H	2.522674	3.397242	0.299522
142	H	10.184943	6.173142	-0.483569
143	H	-3.474241	-0.420211	-2.88463
144	H	3.390419	2.361555	-3.730621
145	H	4.779051	0.349948	-3.005385
146	H	5.407891	-3.35241	-0.138353
147	H	-0.930337	-5.57353	-1.593188
148	H	-1.105339	-2.903713	-0.15761
149	H	-1.31209	-3.418316	-4.43421

150	H	-1.718339	-4.813009	-3.423522
151	H	-0.060639	-4.170017	-3.456886
152	H	-4.93755	-2.983517	-2.683933
153	H	-3.765837	-3.857575	-3.658476
154	H	-3.984688	-4.272296	-1.95996
155	H	-2.459435	-1.166229	-0.541619
156	H	-3.101005	-2.744228	-0.117788
157	H	-6.289607	-1.498922	0.412632
158	H	-3.118748	-0.559258	-5.306434
159	H	-2.684293	-2.257937	-5.192644
160	H	-1.676309	2.503576	-3.782543
161	H	-1.78451	1.483442	-5.21911
162	H	-2.869782	1.237826	-3.861628
163	H	1.44248	3.770815	-5.085651
164	H	-0.216572	3.622533	-4.515541
165	H	0.416321	2.4647	-5.69073
166	H	2.263069	3.618384	-2.449797
167	H	1.623081	2.277526	-1.536486
168	H	-0.131843	5.583683	-0.825016
169	H	2.817188	2.059234	-6.139246
170	H	2.059881	0.493151	-5.77458
171	H	5.825122	-0.995033	0.029229
172	H	6.48526	-0.178492	-1.394771
173	H	7.252471	-1.65178	-0.788956
174	H	5.466075	-2.734721	-3.935474
175	H	6.266829	-1.188491	-3.668214
176	H	7.049041	-2.67334	-3.136161
177	H	6.683472	-4.546425	-1.951808
178	H	5.38084	-5.531642	-1.309309
179	H	4.423924	-5.125928	0.62658
180	H	2.87648	-5.702419	1.179773
181	H	3.559629	-6.361637	-0.29586
182	H	-0.184592	-4.031185	1.788893
183	H	-0.420056	-5.763554	2.096056
184	H	1.203519	-5.077488	2.102619
185	H	-0.251055	-7.414211	0.091649
186	H	1.436798	-7.205355	0.532456
187	H	-3.128914	-5.323991	-0.589909
188	H	-2.460448	-4.741688	0.925825
189	H	1.072422	-0.492046	0.248239
190	H	5.180861	-4.74184	-2.868973
191	H	0.941047	-7.018674	-1.155182
192	H	-2.214336	-6.417634	0.431685
193	H	-4.311784	-1.728825	-4.748136
194	H	3.80157	0.70652	-5.560981
195	H	-7.87101	1.637304	-3.048517

196	H	-4.6146	-5.276296	4.781427
197	H	-3.521917	-6.564775	4.238254
198	H	-7.697563	2.926016	6.291341
199	H	-7.067762	3.962882	4.990587
200	H	-3.697454	8.172674	1.31061
201	H	-3.77971	6.837935	2.491348
202	H	7.591258	-4.905557	3.136897
203	H	8.007107	-4.878563	4.87058
204	H	10.422891	4.667904	0.421031
205	H	10.584964	6.201684	1.253215
206	H	6.942235	2.139484	-1.732994
207	H	1.570803	5.228424	-1.069308
208	H	-5.035236	-2.622055	0.888058
209	C	-8.693028	0.027047	-1.922733
210	O	-9.910569	0.142354	-1.988798
211	N	-8.078102	-1.160412	-1.637061
212	H	-7.085723	-1.23211	-1.812746
213	C	-8.880355	-2.363466	-1.553781
214	H	-8.302161	-3.152245	-1.065457
215	H	-9.77561	-2.151816	-0.966976
216	H	-9.204906	-2.718514	-2.539775

F3H1-conf2_TS, E_{opt}= -4558.0736529

Center number	Atomic Symbol	Coordinates (Angstroms)		
		X	Y	Z
1	C	7.5982	-4.295665	4.04506
2	C	6.166652	-3.865123	4.438633
3	C	5.461728	-2.861436	3.542044
4	C	5.972324	-1.563686	3.371821
5	C	4.232237	-3.166323	2.939883
6	C	5.27114	-0.601232	2.643888
7	C	3.523461	-2.207919	2.206411
8	C	4.039115	-0.919499	2.062049
9	C	-3.523105	7.104493	1.466017
10	C	-2.123458	6.541705	1.165839
11	O	-1.856695	6.507293	-0.256727
12	C	-1.958298	5.111774	1.671138

13	C	-7.033086	2.960528	5.427778
14	C	-5.623518	2.414068	5.752471
15	C	-4.80762	1.923473	4.562417
16	C	-5.208941	0.773087	3.860725
17	C	-3.592193	2.509954	4.183986
18	C	-4.417652	0.207938	2.86358
19	C	-2.785168	1.963809	3.175212
20	C	-3.19031	0.793539	2.531892
21	O	-2.440219	0.156837	1.556275
22	C	-3.891427	-5.564282	4.014368
23	C	-2.75501	-4.53933	4.005525
24	C	-3.021683	-3.364516	3.068177
25	O	-4.032533	-3.309382	2.359112
26	N	-2.05598	-2.420252	3.004123
27	C	-7.758655	1.211676	-2.044205
28	C	-8.091284	2.131448	-0.848853
29	C	-7.164331	3.346018	-0.650318
30	C	-5.953488	3.088763	0.261372
31	N	-4.905613	2.234638	-0.296264
32	C	-3.727912	2.670344	-0.762743
33	N	-3.555846	3.948452	-1.131768
34	N	-2.701332	1.815304	-0.844337
35	C	10.017392	5.683924	0.480208
36	C	8.732763	5.095006	1.08006
37	C	7.852228	4.378573	0.095836
38	C	8.081769	4.184305	-1.242684
39	C	6.578976	3.758371	0.380686
40	N	7.027444	3.485318	-1.812204
41	C	6.091386	3.209887	-0.838941
42	C	5.80237	3.611526	1.544183
43	C	4.870019	2.535683	-0.917614
44	C	4.584687	2.939612	1.471468
45	C	4.125016	2.404533	0.249854
46	Co	1.080404	-1.670151	-1.787314
47	C	-1.449939	-2.882297	-2.173664
48	C	-2.944292	-2.40019	-2.163498
49	C	-2.869544	-1.12182	-3.078611
50	C	-1.417922	-0.690345	-2.9223
51	C	-0.904474	0.600561	-3.33561
52	C	0.438143	0.891226	-3.229821
53	C	1.1852	2.197972	-3.636241
54	C	2.572456	1.588332	-3.998144
55	C	2.625051	0.393202	-3.084566
56	C	3.819423	-0.236542	-2.772319
57	C	3.970939	-1.430025	-2.082319
58	C	5.338376	-2.061637	-1.87976

59	C	4.95749	-3.42955	-1.226971
60	C	3.448636	-3.325025	-1.013541
61	C	2.701497	-4.305266	-0.391787
62	C	1.263187	-4.283078	-0.432758
63	C	0.308865	-5.346027	0.159198
64	C	-1.016115	-4.992606	-0.591074
65	C	-0.857704	-3.493696	-0.877947
66	C	-1.099637	-3.780806	-3.379297
67	N	-0.696481	-1.612772	-2.347333
68	N	1.391783	-0.016619	-2.743105
69	N	2.953101	-2.151314	-1.569312
70	N	0.597897	-3.297116	-0.977466
71	C	-3.941927	-3.441102	-2.666749
72	C	-3.326911	-1.994831	-0.708423
73	C	-4.731758	-1.460379	-0.540276
74	O	-5.13819	-0.457686	-1.162318
75	N	-5.538071	-2.132101	0.302506
76	C	-3.32013	-1.311135	-4.540738
77	C	-1.892188	1.563982	-3.968801
78	C	0.664666	3.05186	-4.805233
79	C	1.427123	3.087985	-2.368259
80	C	0.289295	3.970247	-1.878168
81	O	-0.906882	3.666091	-2.018039
82	N	0.635907	5.110824	-1.253837
83	C	2.714738	1.106365	-5.456341
84	C	6.182658	-1.206458	-0.911272
85	C	6.080935	-2.172474	-3.2282
86	C	5.336455	-4.668611	-2.061086
87	C	3.40449	-5.42978	0.344714
88	C	0.186773	-5.098058	1.679438
89	C	0.647775	-6.825535	-0.118493
90	C	-2.297397	-5.421192	0.12186
91	F	1.439682	2.308945	2.094007
92	C	0.646832	1.545698	1.323364
93	C	0.431728	0.235123	1.409572
94	F	0.017001	2.362942	0.450807
95	F	0.51156	-0.649457	-0.073107
96	H	6.211979	-3.431808	5.447039
97	H	5.541993	-4.761696	4.529447
98	H	6.920278	-1.299736	3.834706
99	H	3.815877	-4.163147	3.063284
100	H	5.676287	0.400032	2.525029
101	H	2.57665	-2.463566	1.740159
102	H	3.495715	-0.173576	1.491188
103	H	8.248505	-3.430153	3.883892
104	H	-1.367188	7.178143	1.650438

105	H	-1.018124	4.673102	1.324093
106	H	-2.776274	4.478764	1.320464
107	H	-1.962598	5.099459	2.764894
108	H	-1.975655	7.3995	-0.609772
109	H	-4.254614	6.630196	0.802807
110	H	-5.05147	3.172808	6.297989
111	H	-5.730845	1.567902	6.443143
112	H	-3.247242	3.401472	4.702342
113	H	-6.140135	0.281777	4.133262
114	H	-1.839963	2.428657	2.911103
115	H	-4.706034	-0.716241	2.37495
116	H	-1.445793	0.319411	1.60095
117	H	-7.507643	2.354094	4.647468
118	H	-2.548022	-4.151983	5.010797
119	H	-1.819823	-5.000263	3.66774
120	H	-2.18865	-1.609002	2.405424
121	H	-1.264407	-2.437571	3.62787
122	H	-4.365701	-5.600404	3.030773
123	H	-9.123528	2.467996	-0.980751
124	H	-8.088711	1.532861	0.073312
125	H	-7.736826	4.153806	-0.177741
126	H	-6.826643	3.733648	-1.621278
127	H	-5.487249	4.035119	0.553897
128	H	-6.286427	2.613582	1.191612
129	H	-5.055607	1.22622	-0.343004
130	H	-1.827215	2.151667	-1.240624
131	H	-2.661779	1.052568	-0.174057
132	H	-4.349392	4.526512	-1.351534
133	H	-2.618013	4.271667	-1.372501
134	H	-6.713118	0.905232	-2.034491
135	H	8.154458	5.899451	1.556253
136	H	8.996061	4.406277	1.895281
137	H	8.921812	4.492238	-1.847847
138	H	6.15028	4.020447	2.488738
139	H	3.971991	2.819895	2.359137
140	H	4.520012	2.10935	-1.84837
141	H	3.181908	1.868354	0.220558
142	H	9.789604	6.403067	-0.313714
143	H	-3.514446	-0.354043	-2.647299
144	H	3.379936	2.298735	-3.789418
145	H	4.720323	0.228255	-3.153412
146	H	5.451293	-3.506402	-0.250929
147	H	-0.963181	-5.526111	-1.545796
148	H	-1.180407	-2.897659	-0.020789
149	H	-1.307561	-3.267597	-4.317816
150	H	-1.657048	-4.719459	-3.370431

151	H	-0.030378	-4.00596	-3.356306
152	H	-4.954563	-3.025137	-2.684213
153	H	-3.719861	-3.786615	-3.676119
154	H	-3.964417	-4.316351	-2.013454
155	H	-2.643503	-1.220366	-0.353072
156	H	-3.214463	-2.848529	-0.042295
157	H	-6.450939	-1.731617	0.478473
158	H	-3.209833	-0.383176	-5.102578
159	H	-2.749706	-2.079561	-5.066408
160	H	-1.668485	2.59092	-3.706872
161	H	-1.895517	1.477573	-5.061373
162	H	-2.906904	1.37038	-3.621116
163	H	1.465205	3.721925	-5.136979
164	H	-0.184123	3.672432	-4.521601
165	H	0.365185	2.43672	-5.655123
166	H	2.298174	3.724938	-2.557204
167	H	1.69887	2.447463	-1.522216
168	H	-0.110169	5.675002	-0.84432
169	H	2.713027	1.939968	-6.162715
170	H	1.901566	0.423053	-5.721417
171	H	5.690417	-1.104038	0.056622
172	H	6.352032	-0.201064	-1.309429
173	H	7.157538	-1.68033	-0.751992
174	H	5.495385	-2.699235	-3.985537
175	H	6.309462	-1.176608	-3.617404
176	H	7.034954	-2.692011	-3.103458
177	H	6.411115	-4.689692	-2.258557
178	H	5.085812	-5.592786	-1.541949
179	H	4.414644	-5.141889	0.636113
180	H	2.883066	-5.677201	1.270479
181	H	3.470866	-6.353996	-0.2388
182	H	-0.221456	-4.105351	1.890225
183	H	-0.471375	-5.847293	2.129489
184	H	1.157097	-5.174173	2.175656
185	H	-0.246375	-7.435136	0.045579
186	H	1.426464	-7.216836	0.537295
187	H	-3.167326	-5.322699	-0.530611
188	H	-2.495486	-4.827282	1.013953
189	H	1.015937	-0.324198	2.12897
190	H	4.811907	-4.676946	-3.020673
191	H	0.96705	-6.969991	-1.155449
192	H	-2.245812	-6.472446	0.422359
193	H	-4.376739	-1.588508	-4.578436
194	H	3.65742	0.564663	-5.572973
195	H	-7.948489	1.756627	-2.975478
196	H	-4.665757	-5.304074	4.740191

197	H	-3.521877	-6.564869	4.238188
198	H	-7.697689	2.92591	6.291476
199	H	-6.98875	3.989116	5.056736
200	H	-3.5799	8.189834	1.342912
201	H	-3.779696	6.837972	2.491356
202	H	7.604356	-4.901517	3.134096
203	H	8.007005	-4.878619	4.870629
204	H	10.653542	4.900017	0.056089
205	H	10.584935	6.201966	1.253171
206	H	6.971782	3.204605	-2.776781
207	H	1.600373	5.338774	-1.067761
208	H	-5.137987	-2.767815	0.991412
209	C	-8.700228	0.026111	-1.989111
210	O	-9.91226	0.143358	-2.128244
211	N	-8.096884	-1.152827	-1.659947
212	H	-7.088577	-1.194218	-1.718716
213	C	-8.880371	-2.363463	-1.55378
214	H	-8.338959	-3.098891	-0.951861
215	H	-9.832418	-2.128216	-1.074455
216	H	-9.102066	-2.807283	-2.532851

F2H2_React, E_{opt}= -4458.3308104

Center number	Atomic Symbol	Coordinates (Angstroms)		
		X	Y	Z
1	C	7.598252	-4.295565	4.045049
2	C	6.142457	-3.947067	4.422123
3	C	5.380312	-2.994909	3.518959
4	C	5.876508	-1.715254	3.223655
5	C	4.100739	-3.326564	3.049232
6	C	5.113077	-0.794476	2.50388
7	C	3.32994	-2.410819	2.326924
8	C	3.833443	-1.138211	2.054681
9	C	-3.523069	7.104766	1.466039
10	C	-2.148642	6.540691	1.076972
11	O	-2.058317	6.386852	-0.369012
12	C	-1.888795	5.169221	1.685663
13	C	-7.033084	2.960508	5.427729

14	C	-5.584146	2.5576	5.778348
15	C	-4.852627	1.900901	4.625336
16	C	-5.198766	0.591992	4.25116
17	C	-3.828223	2.523969	3.902713
18	C	-4.553076	-0.073188	3.216977
19	C	-3.159817	1.865234	2.866268
20	C	-3.499493	0.543116	2.500565
21	O	-2.85515	-0.115546	1.538165
22	C	-3.891407	-5.564254	4.014602
23	C	-2.763452	-4.527882	3.969204
24	C	-3.058167	-3.410505	2.96892
25	O	-4.050907	-3.500035	2.227629
26	N	-2.175998	-2.402412	2.867396
27	C	-7.758687	1.211626	-2.044187
28	C	-8.035066	2.026822	-0.761207
29	C	-7.11431	3.242126	-0.526777
30	C	-5.888604	2.956555	0.352723
31	N	-4.894192	2.062361	-0.23867
32	C	-3.630645	2.393875	-0.541726
33	N	-3.324802	3.653188	-0.918302
34	N	-2.682893	1.45482	-0.509425
35	C	10.017477	5.683915	0.480277
36	C	8.777231	5.021138	1.09533
37	C	7.885064	4.321948	0.109177
38	C	8.06928	4.194284	-1.243886
39	C	6.640188	3.65455	0.410454
40	N	7.01336	3.489869	-1.806254
41	C	6.120322	3.147663	-0.813796
42	C	5.912858	3.438649	1.593718
43	C	4.91132	2.449009	-0.876593
44	C	4.709089	2.741264	1.536144
45	C	4.213827	2.247541	0.310838
46	Co	1.119402	-1.73743	-1.921817
47	C	-1.463444	-2.872729	-2.200519
48	C	-2.954471	-2.391138	-2.146548
49	C	-2.882095	-1.057158	-2.991247
50	C	-1.428415	-0.636513	-2.843061
51	C	-0.895369	0.667824	-3.210625
52	C	0.459534	0.9068	-3.211308
53	C	1.219237	2.198148	-3.649851
54	C	2.588856	1.573597	-4.052566
55	C	2.650958	0.369685	-3.15518
56	C	3.840514	-0.254697	-2.833084
57	C	3.988481	-1.441521	-2.124577
58	C	5.355127	-2.041911	-1.875069
59	C	4.990119	-3.397698	-1.183604

60	C	3.470302	-3.345052	-1.048149
61	C	2.715328	-4.31786	-0.429945
62	C	1.273478	-4.301455	-0.487044
63	C	0.308926	-5.34446	0.12081
64	C	-1.005725	-5.013714	-0.658166
65	C	-0.856313	-3.515167	-0.934552
66	C	-1.137088	-3.730674	-3.440843
67	N	-0.701441	-1.596854	-2.342309
68	N	1.417351	-0.036972	-2.800807
69	N	2.97477	-2.191159	-1.6561
70	N	0.606714	-3.332138	-1.056335
71	C	-3.941949	-3.441211	-2.666793
72	C	-3.347465	-2.053131	-0.687808
73	C	-4.753572	-1.506236	-0.519646
74	O	-5.202513	-0.59486	-1.250595
75	N	-5.503183	-2.064112	0.436889
76	C	-3.353571	-1.148801	-4.453772
77	C	-1.88702	1.7372	-3.629187
78	C	0.664611	3.051855	-4.805338
79	C	1.527673	3.119897	-2.414316
80	C	0.355806	3.825214	-1.749208
81	O	-0.45066	3.206531	-1.040437
82	N	0.234809	5.1562	-1.937172
83	C	2.693898	1.103295	-5.518328
84	C	6.187942	-1.145606	-0.936129
85	C	6.080978	-2.172495	-3.228146
86	C	5.47794	-4.651217	-1.934388
87	C	3.405973	-5.420441	0.348398
88	C	0.164924	-5.054396	1.632335
89	C	0.647821	-6.825525	-0.118495
90	C	-2.290357	-5.467633	0.031357
91	H	0.353057	1.837795	1.030066
92	C	0.471712	0.814156	0.710132
93	C	0.18441	-0.180669	1.530674
94	H	0.759229	0.603867	-0.309853
95	F	0.270831	-1.473648	1.189153
96	H	6.15293	-3.505881	5.428442
97	H	5.572327	-4.878788	4.516761
98	H	6.860188	-1.426581	3.584917
99	H	3.692349	-4.308254	3.277615
100	H	5.50655	0.196791	2.297254
101	H	2.334617	-2.680577	1.990031
102	H	3.233692	-0.417929	1.506098
103	H	8.215309	-3.400697	3.919917
104	H	-1.356096	7.231902	1.398359
105	H	-1.009351	4.701989	1.23256

106	H	-2.742382	4.503753	1.538617
107	H	-1.717311	5.265923	2.761484
108	H	-2.333126	7.222083	-0.771467
109	H	-4.284559	6.643239	0.826114
110	H	-5.026575	3.431594	6.134795
111	H	-5.608479	1.852249	6.618654
112	H	-3.527335	3.535936	4.170954
113	H	-5.988808	0.079884	4.798603
114	H	-2.332471	2.342387	2.348403
115	H	-4.824464	-1.091789	2.960665
116	H	-7.449466	2.260272	4.695061
117	H	-2.569049	-4.087717	4.954597
118	H	-1.818372	-4.988948	3.659735
119	H	-2.413012	-1.565287	2.28498
120	H	-1.454359	-2.296011	3.564606
121	H	-4.388523	-5.608298	3.043876
122	H	-9.078394	2.353424	-0.805829
123	H	-7.968492	1.356177	0.10736
124	H	-7.683722	4.028058	-0.015123
125	H	-6.79139	3.666596	-1.486877
126	H	-5.386734	3.891259	0.622919
127	H	-6.204484	2.504786	1.301008
128	H	-5.086656	1.063418	-0.252725
129	H	-1.755317	1.768781	-0.772358
130	H	-2.732649	0.747442	0.313108
131	H	-4.03459	4.359998	-1.005591
132	H	-2.357527	3.950644	-0.868371
133	H	-6.711667	0.91159	-2.100182
134	H	8.188515	5.780512	1.628918
135	H	9.093019	4.304741	1.866742
136	H	8.877112	4.552993	-1.86473
137	H	6.286436	3.814513	2.541961
138	H	4.141455	2.568224	2.445574
139	H	4.533886	2.063062	-1.814851
140	H	3.276096	1.697945	0.295977
141	H	9.735071	6.419213	-0.280752
142	H	-3.515812	-0.318404	-2.500005
143	H	3.409899	2.269912	-3.851029
144	H	4.744951	0.21832	-3.192341
145	H	5.426134	-3.406119	-0.17762
146	H	-0.922262	-5.542009	-1.614993
147	H	-1.166648	-2.934913	-0.062588
148	H	-1.387106	-3.19696	-4.357774
149	H	-1.684028	-4.674982	-3.435457
150	H	-0.066505	-3.955685	-3.464367
151	H	-4.964041	-3.054589	-2.619573

152	H	-3.756817	-3.730207	-3.702512
153	H	-3.905175	-4.343048	-2.054213
154	H	-2.682159	-1.299354	-0.26824
155	H	-3.260031	-2.934587	-0.054959
156	H	-6.388659	-1.619943	0.640955
157	H	-3.26375	-0.181813	-4.953792
158	H	-2.791334	-1.877864	-5.042479
159	H	-1.628363	2.706972	-3.208704
160	H	-1.947055	1.842218	-4.717464
161	H	-2.885401	1.506087	-3.262672
162	H	1.455959	3.717156	-5.165583
163	H	-0.171652	3.679312	-4.495114
164	H	0.330528	2.437255	-5.642926
165	H	2.265923	3.85967	-2.74181
166	H	2.00534	2.528288	-1.628254
167	H	-0.482109	5.657003	-1.407546
168	H	2.683428	1.944899	-6.214546
169	H	1.86949	0.430113	-5.774063
170	H	5.691925	-1.013034	0.025908
171	H	6.354446	-0.154488	-1.367939
172	H	7.163901	-1.609279	-0.75743
173	H	5.492612	-2.722891	-3.966856
174	H	6.296256	-1.184164	-3.641684
175	H	7.041207	-2.680514	-3.105802
176	H	6.562947	-4.622458	-2.061459
177	H	5.241965	-5.564097	-1.389785
178	H	4.400824	-5.114328	0.670399
179	H	2.855873	-5.654731	1.259589
180	H	3.498593	-6.351252	-0.220185
181	H	-0.242767	-4.057527	1.820795
182	H	-0.504617	-5.791288	2.083553
183	H	1.126056	-5.127667	2.147109
184	H	-0.247353	-7.429557	0.055122
185	H	1.421163	-7.203555	0.551398
186	H	-3.147489	-5.396184	-0.639794
187	H	-2.526823	-4.876997	0.915172
188	F	-0.194326	-0.078454	2.788696
189	H	5.019911	-4.727957	-2.924132
190	H	0.972262	-6.998416	-1.149718
191	H	-2.217491	-6.516154	0.335619
192	H	-4.408453	-1.432173	-4.478292
193	H	3.630041	0.557128	-5.662775
194	H	-7.990507	1.830691	-2.917189
195	H	-4.649806	-5.299734	4.755911
196	H	-3.521898	-6.56485	4.238006
197	H	-7.697768	2.925945	6.291606

198	H	-7.081503	3.959719	4.982932
199	H	-3.583693	8.190934	1.349385
200	H	-3.779733	6.837734	2.491327
201	H	7.652213	-4.88279	3.123506
202	H	8.006898	-4.878635	4.870533
203	H	10.67329	4.943576	0.010356
204	H	10.584992	6.201942	1.25318
205	H	6.924225	3.263975	-2.782532
206	H	0.927718	5.681361	-2.44634
207	H	-5.055297	-2.65124	1.14994
208	C	-8.69667	0.019163	-2.050862
209	O	-9.899045	0.130873	-2.268083
210	N	-8.10438	-1.150756	-1.68296
211	H	-7.092904	-1.167536	-1.623909
212	C	-8.880359	-2.363434	-1.5538
213	H	-8.383871	-3.044751	-0.8565
214	H	-9.872804	-2.111007	-1.17497
215	H	-9.012347	-2.881604	-2.512918

F2H2_Int, E_{opt}= -4458.9142138

Center number	Atomic Symbol	Coordinates (Angstroms)		
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1	C	7.598252	-4.295565	4.045049
2	C	6.263663	-3.701864	4.551293
3	C	5.480749	-2.820869	3.597844
4	C	5.893007	-1.507364	3.322663
5	C	4.27839	-3.265745	3.029806
6	C	5.122278	-0.661445	2.522713
7	C	3.500345	-2.425816	2.228178
8	C	3.919349	-1.119132	1.974692
9	C	-3.523069	7.104766	1.466039
10	C	-2.0685	6.740545	1.13141
11	O	-1.930964	6.500549	-0.295704
12	C	-1.601132	5.473695	1.835499
13	C	-7.033084	2.960508	5.427729
14	C	-5.578101	2.620077	5.811232
15	C	-4.828939	1.978771	4.665574

16	C	-5.102892	0.639835	4.342209
17	C	-3.897538	2.667184	3.882438
18	C	-4.471584	-0.000737	3.284309
19	C	-3.239995	2.037089	2.820847
20	C	-3.530096	0.704179	2.530992
21	O	-2.88801	0.013065	1.532067
22	C	-3.891407	-5.564254	4.014602
23	C	-2.739353	-4.55964	3.946771
24	C	-3.039088	-3.367299	3.045109
25	O	-4.070899	-3.315759	2.359774
26	N	-2.096495	-2.404082	2.991287
27	C	-7.758687	1.211626	-2.044187
28	C	-8.054756	2.173455	-0.872847
29	C	-7.217386	3.468144	-0.841465
30	C	-5.970015	3.399683	0.047423
31	N	-4.979298	2.425887	-0.411218
32	C	-3.685159	2.653211	-0.635143
33	N	-3.134729	3.864656	-0.636191
34	N	-2.867845	1.585097	-0.838698
35	C	10.017477	5.683915	0.480277
36	C	8.713414	5.153413	1.099445
37	C	7.839283	4.327044	0.197287
38	C	8.077892	3.934562	-1.095105
39	C	6.557659	3.764467	0.556918
40	N	7.01946	3.168301	-1.565738
41	C	6.072986	3.042786	-0.570103
42	C	5.779284	3.793055	1.727298
43	C	4.84818	2.365961	-0.554674
44	C	4.561175	3.120517	1.750049
45	C	4.100094	2.414282	0.618469
46	Co	1.138884	-1.780901	-2.021287
47	C	-1.419069	-2.910686	-2.271611
48	C	-2.907694	-2.406803	-2.231774
49	C	-2.830636	-1.153097	-3.174697
50	C	-1.383962	-0.707	-2.999735
51	C	-0.871261	0.584979	-3.364316
52	C	0.486991	0.850519	-3.296855
53	C	1.227501	2.164823	-3.682572
54	C	2.609206	1.57259	-4.082562
55	C	2.675343	0.369421	-3.181353
56	C	3.8632	-0.231377	-2.822673
57	C	4.005124	-1.424933	-2.125876
58	C	5.373993	-2.021177	-1.870434
59	C	5.009778	-3.368975	-1.16805
60	C	3.483362	-3.341118	-1.092962
61	C	2.73413	-4.325967	-0.480162

62	C	1.299815	-4.312306	-0.537094
63	C	0.335674	-5.334351	0.106672
64	C	-0.98847	-4.996823	-0.650154
65	C	-0.809303	-3.513087	-0.985133
66	C	-1.115853	-3.840075	-3.465484
67	N	-0.640446	-1.654232	-2.458836
68	N	1.436415	-0.079864	-2.871673
69	N	2.978657	-2.193383	-1.704914
70	N	0.646971	-3.338033	-1.129951
71	C	-3.941949	-3.441211	-2.666793
72	C	-3.237804	-1.961261	-0.773292
73	C	-4.600708	-1.341979	-0.551535
74	O	-4.959691	-0.302448	-1.160847
75	N	-5.416858	-1.956383	0.311416
76	C	-3.2368	-1.36318	-4.647865
77	C	-1.855827	1.622596	-3.88235
78	C	0.664611	3.051855	-4.805338
79	C	1.527305	3.039166	-2.40874
80	C	0.372022	3.77385	-1.770422
81	O	-0.565166	3.169059	-1.213796
82	N	0.398805	5.120854	-1.789574
83	C	2.721103	1.120664	-5.551688
84	C	6.23336	-1.129059	-0.954488
85	C	6.080978	-2.172495	-3.228146
86	C	5.569965	-4.622076	-1.866118
87	C	3.431168	-5.415291	0.31318
88	C	0.220606	-5.035774	1.618647
89	C	0.647821	-6.825525	-0.118495
90	C	-2.268885	-5.383832	0.088712
91	H	0.092317	1.653053	0.720231
92	C	0.307999	0.609751	0.539915
93	C	0.181594	-0.276623	1.512575
94	H	0.629831	0.27219	-0.438258
95	F	0.407676	-1.582971	1.394773
96	H	6.476399	-3.120048	5.457734
97	H	5.619723	-4.529544	4.872293
98	H	6.817388	-1.137936	3.76106
99	H	3.936929	-4.27746	3.236908
100	H	5.449082	0.356388	2.32781
101	H	2.569747	-2.783956	1.80379
102	H	3.320143	-0.465971	1.347435
103	H	8.31312	-3.514333	3.769845
104	H	-1.402591	7.568402	1.411967
105	H	-0.624936	5.157461	1.456755
106	H	-2.312744	4.659424	1.681087
107	H	-1.513136	5.655143	2.910261

108	H	-2.19314	7.305978	-0.762434
109	H	-4.197202	6.526617	0.822373
110	H	-5.055833	3.518081	6.158771
111	H	-5.583325	1.922029	6.656535
112	H	-3.661672	3.703198	4.114035
113	H	-5.826624	0.089499	4.938558
114	H	-2.482701	2.560457	2.244299
115	H	-4.675607	-1.037524	3.040062
116	H	-2.46628	0.631208	0.907782
117	H	-7.420026	2.224711	4.714488
118	H	-2.456884	-4.194097	4.941594
119	H	-1.839123	-5.032634	3.538648
120	H	-2.242379	-1.58817	2.404851
121	H	-1.270103	-2.434965	3.566878
122	H	-4.404042	-5.606858	3.05089
123	H	-9.118039	2.42082	-0.93211
124	H	-7.928016	1.634245	0.077464
125	H	-7.831065	4.284883	-0.443375
126	H	-6.925461	3.762864	-1.856797
127	H	-5.503272	4.38633	0.103301
128	H	-6.250341	3.133959	1.074534
129	H	-5.259162	1.444581	-0.459133
130	H	-1.947278	1.774038	-1.231095
131	H	-3.35763	0.752327	-1.162656
132	H	-3.644886	4.713693	-0.472336
133	H	-2.124623	3.923664	-0.802477
134	H	-6.711449	0.907887	-2.069865
135	H	8.128284	6.00261	1.480402
136	H	8.960835	4.555105	1.988097
137	H	8.926761	4.14128	-1.730798
138	H	6.129416	4.330678	2.60405
139	H	3.957739	3.130023	2.653203
140	H	4.49859	1.798093	-1.409427
141	H	3.150276	1.887014	0.666018
142	H	9.820178	6.38028	-0.340863
143	H	-3.52128	-0.399	-2.798717
144	H	3.416506	2.284822	-3.874206
145	H	4.771104	0.25684	-3.1585
146	H	5.403407	-3.343741	-0.143833
147	H	-0.948518	-5.563448	-1.586321
148	H	-1.10051	-2.900889	-0.12788
149	H	-1.31189	-3.340775	-4.413168
150	H	-1.70533	-4.759064	-3.430821
151	H	-0.054855	-4.097886	-3.446593
152	H	-4.948656	-3.00817	-2.654703
153	H	-3.765041	-3.810002	-3.677309

154	H	-3.951888	-4.301633	-1.995179
155	H	-2.494788	-1.227814	-0.454293
156	H	-3.146579	-2.811301	-0.099193
157	H	-6.297959	-1.500957	0.514212
158	H	-3.124207	-0.427691	-5.20146
159	H	-2.632343	-2.114922	-5.156731
160	H	-1.680325	2.605332	-3.443448
161	H	-1.810638	1.734183	-4.970652
162	H	-2.884033	1.35496	-3.638286
163	H	1.446252	3.743768	-5.13812
164	H	-0.185443	3.656012	-4.485027
165	H	0.352558	2.461193	-5.667803
166	H	2.306356	3.759594	-2.678931
167	H	1.951352	2.39646	-1.630461
168	H	-0.315336	5.643176	-1.282957
169	H	2.69811	1.965804	-6.245364
170	H	1.908772	0.433554	-5.809057
171	H	5.745204	-0.959749	0.005793
172	H	6.425496	-0.152376	-1.410546
173	H	7.201055	-1.608027	-0.766916
174	H	5.48086	-2.73831	-3.94556
175	H	6.276467	-1.189981	-3.666061
176	H	7.049441	-2.670369	-3.120832
177	H	6.657549	-4.555337	-1.958727
178	H	5.351396	-5.530364	-1.306652
179	H	4.410647	-5.087779	0.661767
180	H	2.866544	-5.670737	1.210423
181	H	3.563198	-6.343379	-0.25367
182	H	-0.171559	-4.032008	1.801784
183	H	-0.44166	-5.764006	2.096762
184	H	1.192144	-5.105769	2.113374
185	H	-0.24571	-7.420407	0.097274
186	H	1.444536	-7.199421	0.526234
187	H	-3.143774	-5.314948	-0.560919
188	H	-2.456804	-4.746805	0.953233
189	F	-0.227608	-0.015562	2.752312
190	H	5.149035	-4.74268	-2.867809
191	H	0.934678	-7.014379	-1.157879
192	H	-2.220942	-6.418906	0.441469
193	H	-4.286288	-1.663103	-4.720073
194	H	3.665342	0.588704	-5.696687
195	H	-7.98613	1.7238	-2.985808
196	H	-4.630902	-5.28292	4.76847
197	H	-3.521898	-6.56485	4.238006
198	H	-7.697768	2.925945	6.291606
199	H	-7.110488	3.944986	4.956372

200	H	-3.738948	8.165237	1.307929
201	H	-3.779733	6.837734	2.491327
202	H	7.448393	-4.949417	3.180602
203	H	8.006898	-4.878635	4.870533
204	H	10.635996	4.866281	0.095221
205	H	10.584992	6.201942	1.25318
206	H	6.979452	2.728385	-2.469785
207	H	1.179606	5.619769	-2.185807
208	H	-5.030441	-2.608572	0.998838
209	C	-8.692735	0.027168	-1.921838
210	O	-9.910574	0.151619	-1.971883
211	N	-8.079315	-1.161836	-1.652111
212	H	-7.081929	-1.228257	-1.797451
213	C	-8.880359	-2.363434	-1.5538
214	H	-8.307473	-3.142444	-1.043883
215	H	-9.782889	-2.141438	-0.982071
216	H	-9.191939	-2.738168	-2.536715

F2H2_TS, E_{opt}= -4458.8376908

Center number	Atomic Symbol	Coordinates (Angstroms)		
		X	Y	Z
1	C	7.598196	-4.295632	4.045064
2	C	6.16325	-3.894678	4.447942
3	C	5.418651	-2.938471	3.536136
4	C	5.926927	-1.660255	3.256806
5	C	4.151141	-3.266934	3.033298
6	C	5.186788	-0.738259	2.514623
7	C	3.402714	-2.348886	2.291446
8	C	3.91949	-1.079186	2.030508
9	C	-3.523137	7.10451	1.466017
10	C	-2.109812	6.574044	1.174626
11	O	-1.841202	6.519376	-0.248412
12	C	-1.903659	5.158337	1.701508
13	C	-7.033091	2.960529	5.42778
14	C	-5.622497	2.414577	5.74742
15	C	-4.820642	1.92285	4.549197
16	C	-5.256103	0.795594	3.831368

17	C	-3.592026	2.483251	4.174049
18	C	-4.490431	0.234714	2.813309
19	C	-2.809702	1.939885	3.144231
20	C	-3.254964	0.799394	2.47323
21	O	-2.54466	0.180108	1.461556
22	C	-3.891417	-5.564279	4.014389
23	C	-2.766287	-4.529054	3.938035
24	C	-3.102959	-3.373263	2.998073
25	O	-4.180856	-3.331827	2.389209
26	N	-2.147894	-2.438734	2.81894
27	C	-7.758677	1.211667	-2.044208
28	C	-8.123396	2.135373	-0.860433
29	C	-7.209061	3.355292	-0.642695
30	C	-5.997056	3.092427	0.265089
31	N	-4.956038	2.241172	-0.307807
32	C	-3.775143	2.668427	-0.76923
33	N	-3.579194	3.954901	-1.102267
34	N	-2.769883	1.793145	-0.886257
35	C	10.017436	5.683954	0.480221
36	C	8.742566	5.077759	1.083754
37	C	7.842553	4.391357	0.09536
38	C	8.022948	4.271418	-1.259116
39	C	6.593885	3.729619	0.39529
40	N	6.958519	3.583222	-1.824481
41	C	6.06859	3.234244	-0.83137
42	C	5.871115	3.501097	1.579406
43	C	4.861379	2.532495	-0.89643
44	C	4.670589	2.798014	1.519612
45	C	4.170729	2.314117	0.292092
46	Co	1.088231	-1.674556	-1.818677
47	C	-1.441154	-2.900702	-2.213274
48	C	-2.930942	-2.405206	-2.188897
49	C	-2.858186	-1.134275	-3.113703
50	C	-1.404782	-0.70876	-2.969693
51	C	-0.89398	0.589068	-3.368099
52	C	0.446968	0.881571	-3.254459
53	C	1.192774	2.191254	-3.646911
54	C	2.578565	1.585863	-4.020018
55	C	2.635761	0.391159	-3.107025
56	C	3.831095	-0.224623	-2.779922
57	C	3.984288	-1.414422	-2.081795
58	C	5.355221	-2.033234	-1.871126
59	C	4.985716	-3.386738	-1.179869
60	C	3.468388	-3.312479	-1.012848
61	C	2.719644	-4.29922	-0.406201
62	C	1.280357	-4.287834	-0.466836

63	C	0.323578	-5.339195	0.142155
64	C	-1.006268	-4.988706	-0.601949
65	C	-0.84054	-3.496415	-0.913867
66	C	-1.112039	-3.819181	-3.409133
67	N	-0.681311	-1.639325	-2.414357
68	N	1.402035	-0.025548	-2.77435
69	N	2.968158	-2.144091	-1.581142
70	N	0.614634	-3.313137	-1.027063
71	C	-3.941918	-3.44111	-2.666753
72	C	-3.299123	-1.982934	-0.736271
73	C	-4.702491	-1.439333	-0.569756
74	O	-5.107996	-0.44629	-1.209445
75	N	-5.505518	-2.095365	0.285481
76	C	-3.314914	-1.329393	-4.572836
77	C	-1.884617	1.559706	-3.98607
78	C	0.664531	3.05186	-4.805306
79	C	1.446331	3.065736	-2.368631
80	C	0.304254	3.922661	-1.843487
81	O	-0.88547	3.57411	-1.91413
82	N	0.63907	5.088117	-1.257736
83	C	2.711467	1.106081	-5.47961
84	C	6.209137	-1.149517	-0.938327
85	C	6.080974	-2.172477	-3.228179
86	C	5.439951	-4.64148	-1.949521
87	C	3.41752	-5.413359	0.349648
88	C	0.218147	-5.059314	1.658138
89	C	0.64779	-6.825527	-0.118501
90	C	-2.283287	-5.393408	0.131751
91	H	1.566705	1.045465	2.732517
92	C	0.938012	1.02565	1.844135
93	C	0.228592	-0.068018	1.491854
94	H	0.838272	1.933097	1.259101
95	F	0.435687	-0.701099	-0.182094
96	H	6.210422	-3.435142	5.444955
97	H	5.566227	-4.806102	4.571461
98	H	6.902217	-1.376403	3.644824
99	H	3.735688	-4.24868	3.249748
100	H	5.590164	0.250737	2.314847
101	H	2.414036	-2.605069	1.928999
102	H	3.336156	-0.367347	1.456464
103	H	8.238054	-3.421615	3.890286
104	H	-1.370478	7.238705	1.646944
105	H	-0.925777	4.771647	1.398672
106	H	-2.673767	4.484122	1.320077
107	H	-1.952262	5.150987	2.793892
108	H	-1.973588	7.403608	-0.61639

109	H	-4.240758	6.608559	0.803364
110	H	-5.04607	3.173865	6.287582
111	H	-5.725966	1.569454	6.440067
112	H	-3.217622	3.351505	4.711468
113	H	-6.194682	0.319446	4.105395
114	H	-1.84711	2.372453	2.888865
115	H	-4.8076	-0.671904	2.311571
116	H	-1.522152	0.273939	1.498286
117	H	-7.509697	2.355632	4.64772
118	H	-2.520039	-4.119431	4.925284
119	H	-1.840245	-4.98148	3.567134
120	H	-2.333019	-1.639726	2.219881
121	H	-1.266121	-2.449597	3.307404
122	H	-4.410338	-5.616372	3.054343
123	H	-9.153875	2.465901	-1.018447
124	H	-8.140336	1.54082	0.064247
125	H	-7.789988	4.149955	-0.158275
126	H	-6.873233	3.760486	-1.607151
127	H	-5.528466	4.035427	0.564668
128	H	-6.325886	2.608842	1.192419
129	H	-5.10862	1.233359	-0.362818
130	H	-1.89342	2.121039	-1.284563
131	H	-2.744591	1.020475	-0.223569
132	H	-4.359864	4.56043	-1.291472
133	H	-2.636117	4.270802	-1.322154
134	H	-6.713194	0.908119	-2.013092
135	H	8.173568	5.868236	1.593587
136	H	9.018554	4.364958	1.873502
137	H	8.83337	4.6261	-1.878926
138	H	6.249518	3.865907	2.530202
139	H	4.110925	2.60625	2.430489
140	H	4.484441	2.146216	-1.834542
141	H	3.243212	1.747956	0.278554
142	H	9.776233	6.410767	-0.30294
143	H	-3.498174	-0.362283	-2.68318
144	H	3.386623	2.296645	-3.81511
145	H	4.732183	0.247885	-3.150301
146	H	5.442668	-3.409336	-0.182851
147	H	-0.967362	-5.538924	-1.547893
148	H	-1.145421	-2.883156	-0.062708
149	H	-1.329026	-3.320001	-4.353196
150	H	-1.6755	-4.753602	-3.377985
151	H	-0.044242	-4.052748	-3.396064
152	H	-4.95095	-3.016638	-2.677253
153	H	-3.736258	-3.800983	-3.674931
154	H	-3.964409	-4.308385	-2.003055

155	H	-2.605501	-1.213029	-0.393864
156	H	-3.190755	-2.833001	-0.064127
157	H	-6.417183	-1.68864	0.454828
158	H	-3.202106	-0.404422	-5.139939
159	H	-2.750509	-2.103582	-5.096564
160	H	-1.667805	2.583748	-3.705081
161	H	-1.885009	1.494326	-5.080074
162	H	-2.899051	1.355037	-3.645105
163	H	1.461559	3.724711	-5.139868
164	H	-0.182683	3.671222	-4.51246
165	H	0.358356	2.44178	-5.656218
166	H	2.304018	3.717978	-2.56609
167	H	1.751454	2.412763	-1.541613
168	H	-0.108077	5.656911	-0.85529
169	H	2.704302	1.941135	-6.184224
170	H	1.897324	0.422485	-5.740753
171	H	5.723861	-1.012731	0.028349
172	H	6.382873	-0.159556	-1.371421
173	H	7.181981	-1.624028	-0.769297
174	H	5.490892	-2.726806	-3.962318
175	H	6.291335	-1.184885	-3.647005
176	H	7.042334	-2.677642	-3.103283
177	H	6.520614	-4.622797	-2.111539
178	H	5.21374	-5.554838	-1.401927
179	H	4.414057	-5.110167	0.669995
180	H	2.876127	-5.664435	1.261814
181	H	3.509731	-6.336812	-0.231731
182	H	-0.149393	-4.047419	1.848611
183	H	-0.461782	-5.777444	2.124938
184	H	1.186955	-5.158532	2.15217
185	H	-0.246286	-7.42613	0.075676
186	H	1.439131	-7.210666	0.52573
187	H	-3.160058	-5.298497	-0.511825
188	H	-2.462184	-4.781371	1.015067
189	F	0.42629	-1.205562	2.235799
190	H	4.949327	-4.711071	-2.924295
191	H	0.942732	-6.988676	-1.159941
192	H	-2.238124	-6.439103	0.451635
193	H	-4.372978	-1.601665	-4.604092
194	H	3.654116	0.565937	-5.603132
195	H	-7.931549	1.75144	-2.981886
196	H	-4.636362	-5.29202	4.766296
197	H	-3.521881	-6.564873	4.238174
198	H	-7.69769	2.92591	6.291475
199	H	-6.98966	3.989535	5.057358
200	H	-3.60583	8.187181	1.335168

201	H	-3.779684	6.837968	2.491364
202	H	7.611975	-4.898198	3.131954
203	H	8.006995	-4.878608	4.870623
204	H	10.656754	4.911286	0.040956
205	H	10.585015	6.201896	1.25321
206	H	6.879085	3.338016	-2.797051
207	H	1.599938	5.37888	-1.164351
208	H	-5.104653	-2.713288	0.996932
209	C	-8.696961	0.022951	-2.000028
210	O	-9.907806	0.136554	-2.151682
211	N	-8.094481	-1.153921	-1.661596
212	H	-7.08625	-1.19757	-1.72227
213	C	-8.880368	-2.363464	-1.553783
214	H	-8.337764	-3.099826	-0.954313
215	H	-9.829975	-2.126835	-1.070403
216	H	-9.10689	-2.805512	-2.532497

F2H2-iso1_React, E_{opt}= -4458.323435

Center number	Atomic Symbol	Coordinates (Angstroms)		
		X	Y	Z
1	C	7.600129	-4.290799	4.048817
2	C	6.211703	-3.774525	4.475498
3	C	5.582707	-2.723055	3.584038
4	C	6.164513	-1.450224	3.463515
5	C	4.366167	-2.957501	2.929935
6	C	5.535351	-0.437059	2.74145
7	C	3.726251	-1.946846	2.205173
8	C	4.305683	-0.680292	2.121169
9	C	-3.520574	7.106927	1.467509
10	C	-2.08585	6.641249	1.171005
11	O	-1.847123	6.552829	-0.258391
12	C	-1.797426	5.257222	1.737549
13	C	-7.022032	2.93809	5.434343
14	C	-5.58859	2.504416	5.800182
15	C	-4.845595	1.891724	4.629258

16	C	-5.19516	0.602661	4.195848
17	C	-3.812955	2.539963	3.941109
18	C	-4.538184	-0.024033	3.144422
19	C	-3.13368	1.920001	2.887416
20	C	-3.467771	0.61248	2.47333
21	O	-2.800539	-0.022174	1.502622
22	C	-3.901334	-5.54671	4.028227
23	C	-2.781418	-4.51088	3.920834
24	C	-3.129776	-3.397972	2.934173
25	O	-4.165552	-3.472783	2.254533
26	N	-2.2413	-2.401015	2.782689
27	C	-7.755665	1.211419	-2.043646
28	C	-8.08661	2.048347	-0.786816
29	C	-7.174177	3.261497	-0.519511
30	C	-5.959231	2.968199	0.373493
31	N	-4.931901	2.118101	-0.226873
32	C	-3.701385	2.521719	-0.571659
33	N	-3.481385	3.808649	-0.905681
34	N	-2.702732	1.633977	-0.571239
35	C	10.004313	5.683793	0.482875
36	C	8.767652	5.045633	1.131872
37	C	7.862722	4.296705	0.19428
38	C	8.087659	3.964133	-1.117015
39	C	6.553802	3.781647	0.520146
40	N	6.999614	3.267142	-1.627385
41	C	6.035867	3.153368	-0.647545
42	C	5.767912	3.800987	1.685272
43	C	4.761554	2.57785	-0.679731
44	C	4.500108	3.228255	1.659053
45	C	3.998332	2.629884	0.484117
46	Co	1.114954	-1.716242	-1.895669
47	C	-1.456657	-2.880438	-2.197039
48	C	-2.947904	-2.394761	-2.161557
49	C	-2.870743	-1.086671	-3.04147
50	C	-1.418522	-0.66202	-2.89418
51	C	-0.895574	0.639604	-3.280398
52	C	0.453249	0.908247	-3.227114
53	C	1.210483	2.209604	-3.644604
54	C	2.590658	1.592542	-4.026844
55	C	2.646643	0.39001	-3.1272
56	C	3.842109	-0.226962	-2.792071
57	C	3.994043	-1.422434	-2.102554
58	C	5.360781	-2.034789	-1.869133
59	C	4.99023	-3.387707	-1.172316
60	C	3.471097	-3.323482	-1.019512
61	C	2.71568	-4.29146	-0.389886

62	C	1.273347	-4.284912	-0.454319
63	C	0.310632	-5.343645	0.138451
64	C	-1.005197	-5.004336	-0.635393
65	C	-0.855058	-3.506156	-0.917928
66	C	-1.122759	-3.758299	-3.42171
67	N	-0.697356	-1.606606	-2.356519
68	N	1.413739	-0.026884	-2.796764
69	N	2.977829	-2.169555	-1.626327
70	N	0.606572	-3.315643	-1.025242
71	C	-3.942166	-3.441227	-2.668206
72	C	-3.342586	-2.019789	-0.71036
73	C	-4.747673	-1.468194	-0.557605
74	O	-5.173639	-0.541821	-1.281484
75	N	-5.518799	-2.035402	0.37724
76	C	-3.340478	-1.222375	-4.50203
77	C	-1.89884	1.655457	-3.792711
78	C	0.664831	3.05092	-4.809865
79	C	1.503052	3.136352	-2.412899
80	C	0.348838	3.930684	-1.818109
81	O	-0.761566	3.424185	-1.609958
82	N	0.590614	5.215777	-1.491903
83	C	2.719963	1.118666	-5.48958
84	C	6.210408	-1.146934	-0.935819
85	C	6.08146	-2.173185	-3.228388
86	C	5.455293	-4.646366	-1.92879
87	C	3.40933	-5.390589	0.392053
88	C	0.162053	-5.083971	1.654636
89	C	0.647748	-6.826234	-0.119116
90	C	-2.291574	-5.446633	0.058715
91	F	0.445087	-1.822703	1.594654
92	C	0.149011	-0.593143	1.102877
93	C	1.08304	0.061731	0.421338
94	H	-0.868597	-0.245722	1.31284
95	F	0.786904	1.298315	-0.057498
96	H	6.301307	-3.347549	5.483092
97	H	5.526861	-4.625224	4.571242
98	H	7.106198	-1.245676	3.967292
99	H	3.89558	-3.93368	3.018547
100	H	5.984656	0.549715	2.670083
101	H	2.768507	-2.147743	1.738219
102	H	3.818202	0.126083	1.584385
103	H	8.291433	-3.464605	3.854477
104	H	-1.371861	7.360253	1.600544
105	H	-0.81911	4.898553	1.403268
106	H	-2.556714	4.541878	1.414873
107	H	-1.79638	5.28898	2.830795

108	H	-2.069573	7.406737	-0.653338
109	H	-4.211372	6.574466	0.804384
110	H	-5.028663	3.355937	6.204031
111	H	-5.637233	1.762804	6.607582
112	H	-3.514731	3.541349	4.248378
113	H	-5.995081	0.072596	4.710641
114	H	-2.305106	2.420414	2.393503
115	H	-4.813172	-1.030291	2.847171
116	H	-7.449291	2.23624	4.709134
117	H	-2.538619	-4.06479	4.892656
118	H	-1.852879	-4.970379	3.565482
119	H	-2.487627	-1.559174	2.219053
120	H	-1.457769	-2.330134	3.414263
121	H	-4.457699	-5.589152	3.09008
122	H	-9.124905	2.378698	-0.885793
123	H	-8.065319	1.390293	0.093701
124	H	-7.755801	4.035151	-0.002847
125	H	-6.84218	3.705536	-1.468259
126	H	-5.479469	3.900826	0.685712
127	H	-6.284135	2.472774	1.296061
128	H	-5.078757	1.112133	-0.251858
129	H	-1.798599	1.970051	-0.888713
130	H	-2.72532	0.907998	0.202681
131	H	-4.247796	4.398394	-1.183586
132	H	-2.529206	4.115576	-1.086398
133	H	-6.707678	0.913364	-2.056925
134	H	8.181206	5.828386	1.63248
135	H	9.091842	4.369281	1.935778
136	H	8.945034	4.168841	-1.741561
137	H	6.144457	4.266208	2.591747
138	H	3.882709	3.243079	2.552161
139	H	4.380881	2.097617	-1.573866
140	H	2.995685	2.213099	0.487899
141	H	9.721066	6.411624	-0.284112
142	H	-3.507188	-0.33384	-2.574301
143	H	3.406574	2.29186	-3.815158
144	H	4.745154	0.255989	-3.14361
145	H	5.439177	-3.399736	-0.172089
146	H	-0.930609	-5.535202	-1.590921
147	H	-1.174264	-2.923109	-0.050351
148	H	-1.358933	-3.23597	-4.348607
149	H	-1.674579	-4.699765	-3.410207
150	H	-0.052972	-3.98761	-3.430214
151	H	-4.958034	-3.035681	-2.653145
152	H	-3.742	-3.765006	-3.690399
153	H	-3.936579	-4.325862	-2.029276

154	H	-2.679802	-1.251001	-0.313099
155	H	-3.252212	-2.885706	-0.055891
156	H	-6.409569	-1.593425	0.563876
157	H	-3.252979	-0.271478	-5.030164
158	H	-2.775454	-1.966629	-5.068144
159	H	-1.646025	2.6566	-3.461749
160	H	-1.958916	1.651226	-4.886516
161	H	-2.894955	1.446779	-3.404172
162	H	1.453999	3.721995	-5.164217
163	H	-0.179989	3.672545	-4.512305
164	H	0.343686	2.430516	-5.648256
165	H	2.297719	3.828816	-2.709921
166	H	1.90465	2.533305	-1.595551
167	H	-0.149157	5.741772	-1.021229
168	H	2.714678	1.959053	-6.187254
169	H	1.902316	0.441171	-5.755294
170	H	5.727538	-1.009395	0.032718
171	H	6.383734	-0.157206	-1.369629
172	H	7.184661	-1.618001	-0.767795
173	H	5.487825	-2.725848	-3.961004
174	H	6.292875	-1.185921	-3.646808
175	H	7.041484	-2.681618	-3.107362
176	H	6.538305	-4.627568	-2.073376
177	H	5.219968	-5.556381	-1.378887
178	H	4.405349	-5.082149	0.710014
179	H	2.85893	-5.626138	1.30338
180	H	3.50536	-6.32343	-0.172782
181	H	-0.270306	-4.102415	1.857203
182	H	-0.485135	-5.848495	2.092704
183	H	1.124915	-5.13792	2.167941
184	H	-0.247472	-7.42935	0.057203
185	H	1.426912	-7.20902	0.541709
186	H	-3.151059	-5.36426	-0.608235
187	H	-2.515056	-4.853673	0.94424
188	H	2.098758	-0.260618	0.232256
189	H	4.980627	-4.720428	-2.91091
190	H	0.963792	-6.989843	-1.154484
191	H	-2.229229	-6.495834	0.362844
192	H	-4.394437	-1.509246	-4.521093
193	H	3.660775	0.576807	-5.62001
194	H	-7.958661	1.8128	-2.936192
195	H	-4.612754	-5.292657	4.817873
196	H	-3.511805	-6.574481	4.225576
197	H	-7.705394	2.943853	6.285508
198	H	-7.035592	3.928387	4.967039
199	H	-3.653997	8.183768	1.328309

200	H	-3.782736	6.836166	2.491325
201	H	7.545098	-4.90735	3.146521
202	H	8.00547	-4.882678	4.867488
203	H	10.647037	4.928962	0.017543
204	H	10.592822	6.200754	1.252923
205	H	6.910598	2.952876	-2.578939
206	H	1.501523	5.631033	-1.606837
207	H	-5.094755	-2.634889	1.095287
208	C	-8.691621	0.01725	-2.053837
209	O	-9.8939	0.126658	-2.270385
210	N	-8.101069	-1.153338	-1.680621
211	H	-7.089258	-1.178544	-1.654576
212	C	-8.88199	-2.364289	-1.553782
213	H	-8.370034	-3.060971	-0.883688
214	H	-9.861494	-2.113067	-1.142094
215	H	-9.044998	-2.862318	-2.518605

F2H2-iso1_Int, E_{opt}= -4458.9016422

Center number	Atomic Symbol	Coordinates (Angstroms)		
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1	C	7.59828	-4.295605	4.045072
2	C	6.162729	-3.87004	4.425741
3	C	5.465473	-2.873282	3.515705
4	C	5.982102	-1.580041	3.333247
5	C	4.237022	-3.179091	2.911784
6	C	5.286652	-0.624151	2.591085
7	C	3.53312	-2.227188	2.165885
8	C	4.05533	-0.942628	2.007881
9	C	-3.523063	7.104838	1.466045
10	C	-2.048444	6.731191	1.229975
11	O	-1.772889	6.502729	-0.171331
12	C	-1.667269	5.442003	1.949024
13	C	-7.033161	2.96049	5.427832
14	C	-5.579962	2.59477	5.795339
15	C	-4.83383	1.985438	4.628185
16	C	-5.117974	0.660286	4.259102
17	C	-3.888202	2.687497	3.873562

18	C	-4.487787	0.047003	3.183643
19	C	-3.231543	2.085213	2.794509
20	C	-3.538158	0.767683	2.455583
21	O	-2.902367	0.108348	1.429291
22	C	-3.891398	-5.564302	4.014595
23	C	-2.757331	-4.550079	3.862439
24	C	-3.122706	-3.376349	2.956183
25	O	-4.213865	-3.314396	2.372045
26	N	-2.167338	-2.436042	2.791377
27	C	-7.758689	1.211617	-2.044174
28	C	-8.046792	2.137788	-0.840376
29	C	-7.196416	3.421844	-0.746064
30	C	-5.961236	3.307417	0.15819
31	N	-4.92937	2.407866	-0.355718
32	C	-3.684016	2.735214	-0.696566
33	N	-3.223383	3.984136	-0.706873
34	N	-2.806718	1.729049	-0.984292
35	C	10.017472	5.684004	0.480214
36	C	8.546747	5.558336	0.924555
37	C	7.700993	4.558231	0.179174
38	C	8.085269	3.661561	-0.785664
39	C	6.297973	4.302886	0.421096
40	N	7.01124	2.863954	-1.157242
41	C	5.901796	3.22505	-0.423839
42	C	5.342549	4.87334	1.278833
43	C	4.604508	2.697811	-0.411668
44	C	4.047122	4.363311	1.288377
45	C	3.682976	3.280334	0.458571
46	Co	1.154663	-1.790513	-2.036832
47	C	-1.401965	-2.936025	-2.291401
48	C	-2.886907	-2.419557	-2.264007
49	C	-2.805011	-1.204658	-3.251115
50	C	-1.358993	-0.759248	-3.085853
51	C	-0.857623	0.529534	-3.470773
52	C	0.489145	0.827901	-3.342252
53	C	1.216045	2.160591	-3.681915
54	C	2.617876	1.593159	-4.050309
55	C	2.683035	0.39005	-3.147332
56	C	3.874444	-0.192269	-2.759189
57	C	4.02145	-1.404268	-2.094778
58	C	5.391581	-2.011953	-1.859401
59	C	5.028333	-3.358351	-1.152861
60	C	3.50069	-3.339432	-1.092802
61	C	2.750399	-4.329022	-0.487205
62	C	1.316156	-4.315313	-0.54468
63	C	0.351204	-5.332303	0.106046

64	C	-0.97644	-4.99195	-0.641742
65	C	-0.793735	-3.513778	-0.993283
66	C	-1.101443	-3.890821	-3.465547
67	N	-0.620365	-1.685672	-2.504922
68	N	1.444309	-0.086661	-2.881012
69	N	2.996324	-2.187654	-1.695849
70	N	0.663183	-3.340831	-1.137836
71	C	-3.941961	-3.441167	-2.666767
72	C	-3.202174	-1.920667	-0.81967
73	C	-4.558487	-1.28592	-0.614908
74	O	-4.896173	-0.246937	-1.235477
75	N	-5.391707	-1.886713	0.242299
76	C	-3.21814	-1.46906	-4.714484
77	C	-1.843714	1.500286	-4.102991
78	C	0.664602	3.051917	-4.80531
79	C	1.469288	2.992828	-2.381393
80	C	0.345244	3.820271	-1.793745
81	O	-0.859704	3.568701	-1.985289
82	N	0.717637	4.866893	-1.03368
83	C	2.769948	1.142514	-5.515916
84	C	6.273734	-1.132177	-0.954623
85	C	6.080952	-2.172481	-3.228167
86	C	5.613544	-4.609032	-1.83429
87	C	3.446307	-5.428589	0.293244
88	C	0.253866	-5.02708	1.617041
89	C	0.647797	-6.825529	-0.11852
90	C	-2.25549	-5.358081	0.108574
91	F	0.309852	1.97498	0.450054
92	C	0.325109	0.614129	0.404897
93	C	0.231976	-0.053691	1.54693
94	H	0.431435	0.176708	-0.58413
95	F	0.222654	-1.408214	1.546664
96	H	6.196814	-3.431355	5.432322
97	H	5.539836	-4.767964	4.514373
98	H	6.928575	-1.313223	3.797297
99	H	3.815776	-4.17271	3.044235
100	H	5.700044	0.373281	2.467085
101	H	2.580438	-2.482243	1.713745
102	H	3.515217	-0.197231	1.430682
103	H	8.246411	-3.427408	3.889113
104	H	-1.404203	7.545052	1.592947
105	H	-0.648344	5.136037	1.69523
106	H	-2.346846	4.637495	1.659893
107	H	-1.726434	5.578014	3.032534
108	H	-1.881387	7.335226	-0.650415
109	H	-4.168946	6.513551	0.807467

110	H	-5.04997	3.476423	6.171897
111	H	-5.589358	1.868726	6.616776
112	H	-3.64184	3.712819	4.138787
113	H	-5.8486	0.096582	4.834127
114	H	-2.464917	2.616334	2.23734
115	H	-4.700584	-0.981284	2.912487
116	H	-2.467983	0.740552	0.828933
117	H	-7.433809	2.240061	4.706176
118	H	-2.427777	-4.156901	4.831984
119	H	-1.873856	-5.020856	3.418289
120	H	-2.327334	-1.639595	2.18259
121	H	-1.265753	-2.515095	3.233754
122	H	-4.45793	-5.625806	3.082038
123	H	-9.106738	2.400429	-0.894891
124	H	-7.932039	1.561467	0.089202
125	H	-7.810711	4.225151	-0.322179
126	H	-6.888621	3.758442	-1.743667
127	H	-5.522756	4.295025	0.323121
128	H	-6.250274	2.939798	1.150356
129	H	-5.155663	1.413965	-0.439071
130	H	-1.955753	2.012939	-1.470165
131	H	-3.25972	0.879326	-1.318063
132	H	-3.807823	4.776561	-0.509712
133	H	-2.282601	4.153281	-1.084982
134	H	-6.710855	0.9094	-2.08064
135	H	8.068792	6.546607	0.873518
136	H	8.523922	5.285421	1.989335
137	H	9.051929	3.518264	-1.24626
138	H	5.616235	5.696753	1.932505
139	H	3.308656	4.787027	1.963978
140	H	4.329766	1.854418	-1.039409
141	H	2.673939	2.880817	0.518943
142	H	10.116879	6.222207	-0.466649
143	H	-3.489211	-0.432379	-2.899359
144	H	3.408341	2.317777	-3.823949
145	H	4.780274	0.31774	-3.068952
146	H	5.407771	-3.322268	-0.123251
147	H	-0.948428	-5.568835	-1.571596
148	H	-1.085514	-2.889308	-0.14485
149	H	-1.281426	-3.40497	-4.422937
150	H	-1.704374	-4.80062	-3.420719
151	H	-0.044316	-4.162986	-3.433299
152	H	-4.935362	-2.979063	-2.688368
153	H	-3.762604	-3.8591	-3.657382
154	H	-3.989085	-4.271673	-1.959154
155	H	-2.453788	-1.17879	-0.534067

156	H	-3.111745	-2.750627	-0.120075
157	H	-6.277206	-1.429406	0.421559
158	H	-3.087198	-0.562673	-5.308932
159	H	-2.633053	-2.255993	-5.191362
160	H	-1.678627	2.520332	-3.77188
161	H	-1.780493	1.485051	-5.196669
162	H	-2.873552	1.249433	-3.845768
163	H	1.446013	3.755931	-5.113132
164	H	-0.196302	3.638582	-4.489136
165	H	0.379944	2.467215	-5.68135
166	H	2.311935	3.66712	-2.566875
167	H	1.805178	2.318904	-1.586916
168	H	-0.005034	5.458995	-0.628834
169	H	2.745868	1.989137	-6.207414
170	H	1.974426	0.442843	-5.791634
171	H	5.801096	-0.95556	0.012378
172	H	6.47213	-0.158944	-1.417252
173	H	7.239638	-1.619273	-0.780161
174	H	5.469499	-2.743743	-3.931401
175	H	6.264221	-1.191988	-3.6764
176	H	7.051315	-2.668473	-3.132072
177	H	6.700257	-4.523308	-1.922029
178	H	5.409831	-5.515249	-1.266456
179	H	4.422853	-5.104515	0.653191
180	H	2.876925	-5.703322	1.181454
181	H	3.585717	-6.346182	-0.289009
182	H	-0.129922	-4.018858	1.791676
183	H	-0.407752	-5.747118	2.107963
184	H	1.229778	-5.097202	2.102728
185	H	-0.248494	-7.411765	0.109141
186	H	1.44887	-7.206218	0.516923
187	H	-3.133625	-5.2819	-0.535628
188	H	-2.424415	-4.708301	0.96661
189	H	0.14809	0.392314	2.531165
190	H	5.200401	-4.748481	-2.836657
191	H	0.920698	-7.019394	-1.160718
192	H	-2.219586	-6.389762	0.472354
193	H	-4.274037	-1.748231	-4.772088
194	H	3.726377	0.627694	-5.642544
195	H	-7.993662	1.748676	-2.969534
196	H	-4.592935	-5.268121	4.798682
197	H	-3.521913	-6.564867	4.238032
198	H	-7.697638	2.925952	6.29143
199	H	-7.097685	3.951629	4.968486
200	H	-3.732348	8.163509	1.293407
201	H	-3.779741	6.837673	2.49129

202	H	7.613222	-4.899367	3.132818
203	H	8.006929	-4.878624	4.870609
204	H	10.475485	4.695874	0.360407
205	H	10.584973	6.201845	1.253205
206	H	7.04973	2.114187	-1.827404
207	H	1.688795	5.026054	-0.802701
208	H	-5.032574	-2.548059	0.937576
209	C	-8.690658	0.024747	-1.936298
210	O	-9.908336	0.144861	-1.98803
211	N	-8.075373	-1.164061	-1.659155
212	H	-7.083859	-1.238389	-1.838083
213	C	-8.880363	-2.363439	-1.5538
214	H	-8.302732	-3.146402	-1.055545
215	H	-9.772547	-2.139636	-0.966895
216	H	-9.20958	-2.732689	-2.532966

F2H2-iso1_TS, E_{opt}= -4458.8322899

Center number	Atomic Symbol	Coordinates (Angstroms)		
		X	Y	Z
1	C	7.598196	-4.295632	4.045064
2	C	6.173562	-3.832956	4.427229
3	C	5.507688	-2.804023	3.526974
4	C	6.046473	-1.513772	3.388528
5	C	4.287186	-3.074554	2.890261
6	C	5.379835	-0.526394	2.661056
7	C	3.612751	-2.090722	2.156947
8	C	4.154808	-0.80922	2.046322
9	C	-3.523137	7.10451	1.466017
10	C	-2.113287	6.532916	1.222268
11	O	-1.801381	6.41967	-0.186015
12	C	-1.955974	5.130937	1.803549
13	C	-7.033091	2.960529	5.42778
14	C	-5.632136	2.393316	5.748414
15	C	-4.816429	1.922097	4.551434
16	C	-5.245366	0.81473	3.799162
17	C	-3.574969	2.478798	4.216058
18	C	-4.456955	0.262976	2.792762

19	C	-2.766966	1.944014	3.202137
20	C	-3.204095	0.816429	2.504509
21	O	-2.453944	0.178065	1.531737
22	C	-3.891417	-5.564279	4.014389
23	C	-2.758194	-4.535012	3.991333
24	C	-3.032502	-3.365876	3.046829
25	O	-4.032214	-3.334954	2.320167
26	N	-2.083491	-2.404307	2.994933
27	C	-7.758677	1.211667	-2.044208
28	C	-8.083352	2.105972	-0.825593
29	C	-7.180477	3.338582	-0.619895
30	C	-5.961753	3.095379	0.284723
31	N	-4.923242	2.241646	-0.288035
32	C	-3.75734	2.663985	-0.786296
33	N	-3.542459	3.946752	-1.106829
34	N	-2.77148	1.770807	-0.963137
35	C	10.017436	5.683954	0.480221
36	C	8.584062	5.428846	0.980344
37	C	7.767783	4.4551	0.1732
38	C	8.165539	3.699465	-0.900414
39	C	6.396348	4.080887	0.439879
40	N	7.128445	2.87589	-1.317906
41	C	6.031748	3.081685	-0.507464
42	C	5.4458	4.486199	1.393929
43	C	4.770358	2.476108	-0.508756
44	C	4.185233	3.895345	1.391482
45	C	3.854251	2.893979	0.453722
46	Co	1.089594	-1.67058	-1.800248
47	C	-1.4433	-2.891071	-2.18818
48	C	-2.936127	-2.403339	-2.17831
49	C	-2.859802	-1.136619	-3.108665
50	C	-1.40696	-0.70762	-2.95917
51	C	-0.891305	0.578637	-3.382816
52	C	0.451428	0.868069	-3.271789
53	C	1.196259	2.179315	-3.655332
54	C	2.587726	1.58089	-4.021418
55	C	2.637661	0.381357	-3.114737
56	C	3.833875	-0.23162	-2.775141
57	C	3.983468	-1.423307	-2.079485
58	C	5.351169	-2.053202	-1.873482
59	C	4.971795	-3.415803	-1.206635
60	C	3.459378	-3.319722	-1.011159
61	C	2.709684	-4.298217	-0.389704
62	C	1.270105	-4.279782	-0.43643
63	C	0.313934	-5.343496	0.155156
64	C	-1.008599	-4.992099	-0.599199

65	C	-0.851173	-3.493984	-0.888677
66	C	-1.096766	-3.799443	-3.387539
67	N	-0.688292	-1.625223	-2.374704
68	N	1.405096	-0.039957	-2.79015
69	N	2.965252	-2.149224	-1.575882
70	N	0.604732	-3.296682	-0.984659
71	C	-3.941918	-3.44111	-2.666753
72	C	-3.31367	-1.985951	-0.725579
73	C	-4.717133	-1.446318	-0.558935
74	O	-5.11959	-0.442889	-1.182827
75	N	-5.524462	-2.115554	0.284208
76	C	-3.311619	-1.342163	-4.568542
77	C	-1.874434	1.537223	-4.032976
78	C	0.664531	3.05186	-4.805306
79	C	1.450335	3.02975	-2.366239
80	C	0.315421	3.862299	-1.793295
81	O	-0.885742	3.630987	-2.02784
82	N	0.678085	4.874192	-0.986932
83	C	2.73501	1.115941	-5.483712
84	C	6.20901	-1.196912	-0.91919
85	C	6.080974	-2.172477	-3.228179
86	C	5.377927	-4.661454	-2.017217
87	C	3.408897	-5.420093	0.354409
88	C	0.187276	-5.090648	1.674292
89	C	0.64779	-6.825527	-0.118501
90	C	-2.291982	-5.421822	0.10891
91	F	0.835518	2.941364	1.624575
92	C	0.230515	1.877194	0.953723
93	C	0.427761	0.598362	1.241074
94	H	-0.427265	2.275445	0.190696
95	F	0.586563	-0.611207	-0.188066
96	H	6.217654	-3.406252	5.438312
97	H	5.524133	-4.713003	4.504613
98	H	6.987376	-1.274467	3.878456
99	H	3.847262	-4.063941	2.989749
100	H	5.807914	0.468502	2.573129
101	H	2.668821	-2.317945	1.670678
102	H	3.637989	-0.040322	1.479024
103	H	8.262106	-3.441885	3.875069
104	H	-1.373518	7.195783	1.696674
105	H	-0.964783	4.716781	1.596792
106	H	-2.706511	4.454481	1.38636
107	H	-2.089583	5.155491	2.888759
108	H	-1.830033	7.302154	-0.579634
109	H	-4.241136	6.616119	0.798525
110	H	-5.055561	3.134005	6.313614

111	H	-5.751915	1.533242	6.419866
112	H	-3.207268	3.334986	4.776629
113	H	-6.196448	0.344493	4.038086
114	H	-1.796212	2.377643	2.980524
115	H	-4.768172	-0.631193	2.264674
116	H	-1.481613	0.423758	1.534305
117	H	-7.515995	2.369079	4.641213
118	H	-2.544292	-4.141894	4.992777
119	H	-1.824353	-4.995177	3.648784
120	H	-2.216265	-1.594396	2.393362
121	H	-1.302204	-2.406334	3.632022
122	H	-4.374443	-5.60342	3.035091
123	H	-9.124472	2.42319	-0.933207
124	H	-8.052295	1.490505	0.084948
125	H	-7.767204	4.130774	-0.138806
126	H	-6.852189	3.740634	-1.587889
127	H	-5.496137	4.043934	0.571294
128	H	-6.280437	2.620522	1.219936
129	H	-5.091198	1.236825	-0.359756
130	H	-1.91868	2.083802	-1.417235
131	H	-2.72619	0.982777	-0.325693
132	H	-4.308252	4.590151	-1.212094
133	H	-2.595332	4.252479	-1.334354
134	H	-6.712111	0.90861	-2.053584
135	H	8.047692	6.385796	1.045923
136	H	8.631197	5.056123	2.013596
137	H	9.117318	3.675938	-1.410826
138	H	5.698414	5.246586	2.127783
139	H	3.442472	4.183394	2.129783
140	H	4.515884	1.692392	-1.214378
141	H	2.875969	2.429832	0.506011
142	H	10.031114	6.278681	-0.437857
143	H	-3.502348	-0.3617	-2.686284
144	H	3.393711	2.287515	-3.794917
145	H	4.734428	0.241451	-3.148212
146	H	5.450957	-3.473151	-0.221901
147	H	-0.951434	-5.52692	-1.552807
148	H	-1.175488	-2.895488	-0.033922
149	H	-1.300589	-3.291575	-4.329744
150	H	-1.658769	-4.7352	-3.37331
151	H	-0.028573	-4.030195	-3.361726
152	H	-4.950342	-3.015538	-2.692068
153	H	-3.72257	-3.804245	-3.670604
154	H	-3.974773	-4.306097	-2.000555
155	H	-2.624124	-1.215843	-0.372486
156	H	-3.205122	-2.837295	-0.055668

157	H	-6.438822	-1.716561	0.456302
158	H	-3.194117	-0.422853	-5.143013
159	H	-2.747418	-2.121925	-5.083885
160	H	-1.661946	2.566643	-3.769701
161	H	-1.856382	1.451027	-5.125265
162	H	-2.894405	1.337315	-3.705935
163	H	1.463709	3.724528	-5.135135
164	H	-0.177038	3.671228	-4.499942
165	H	0.356326	2.450135	-5.661603
166	H	2.289687	3.706142	-2.561333
167	H	1.785046	2.366805	-1.559634
168	H	-0.062384	5.434334	-0.563864
169	H	2.729782	1.959198	-6.178416
170	H	1.925215	0.432644	-5.759231
171	H	5.722295	-1.069538	0.048239
172	H	6.397558	-0.201368	-1.334754
173	H	7.177288	-1.68213	-0.755075
174	H	5.488317	-2.704748	-3.976098
175	H	6.303519	-1.179401	-3.627757
176	H	7.036525	-2.690773	-3.110441
177	H	6.455614	-4.670462	-2.198736
178	H	5.133016	-5.581531	-1.488498
179	H	4.416242	-5.130225	0.653463
180	H	2.880008	-5.66684	1.275987
181	H	3.48097	-6.344973	-0.227419
182	H	-0.221743	-4.097469	1.881436
183	H	-0.472173	-5.83924	2.12325
184	H	1.155447	-5.165059	2.174714
185	H	-0.247858	-7.431186	0.050553
186	H	1.427367	-7.216907	0.536406
187	H	-3.158537	-5.328104	-0.548602
188	H	-2.497091	-4.826252	0.997906
189	H	1.138555	0.263862	1.988758
190	H	4.867842	-4.690937	-2.984048
191	H	0.963606	-6.974716	-1.155884
192	H	-2.239176	-6.472105	0.41234
193	H	-4.37023	-1.611908	-4.602549
194	H	3.68099	0.581369	-5.606985
195	H	-7.963739	1.773214	-2.962047
196	H	-4.659838	-5.302284	4.745988
197	H	-3.521881	-6.564873	4.238174
198	H	-7.69769	2.92591	6.291475
199	H	-6.974953	3.992476	5.067932
200	H	-3.580714	8.187428	1.329682
201	H	-3.779684	6.837968	2.491364
202	H	7.595144	-4.907385	3.137793

203	H	8.006995	-4.878608	4.870623
204	H	10.536416	4.739539	0.282217
205	H	10.585015	6.201896	1.25321
206	H	7.181372	2.215739	-2.075242
207	H	1.632507	4.97589	-0.671569
208	H	-5.124675	-2.751565	0.974467
209	C	-8.697132	0.023434	-1.995385
210	O	-9.909055	0.138484	-2.135194
211	N	-8.093403	-1.154973	-1.66332
212	H	-7.086131	-1.200229	-1.733519
213	C	-8.880368	-2.363464	-1.553783
214	H	-8.333449	-3.103997	-0.963332
215	H	-9.824913	-2.127652	-1.060168
216	H	-9.117204	-2.800467	-2.532282

F2H2-iso2_React, E_{opt}= -4458.3241717

Center number	Atomic Symbol	Coordinates (Angstroms)		
		X	Y	Z
1	C	7.598288	-4.295554	4.04512
2	C	6.207086	-3.772828	4.461933
3	C	5.628579	-2.675878	3.591453
4	C	6.190622	-1.389306	3.611618
5	C	4.495264	-2.887305	2.795565
6	C	5.625024	-0.343248	2.883375
7	C	3.925926	-1.842485	2.060931
8	C	4.480623	-0.562026	2.110308
9	C	-3.523069	7.104746	1.466061
10	C	-2.107603	6.559456	1.205946
11	O	-1.80197	6.4805	-0.206795
12	C	-1.937865	5.146713	1.752913
13	C	-7.033117	2.960487	5.427797
14	C	-5.603901	2.474273	5.748168
15	C	-4.869781	1.897939	4.55119
16	C	-5.245189	0.640891	4.04908
17	C	-3.791656	2.536974	3.925298
18	C	-4.565548	0.034227	2.998724
19	C	-3.093307	1.938476	2.871858

20	C	-3.447267	0.657253	2.39429
21	O	-2.753405	0.036001	1.43498
22	C	-3.891391	-5.564239	4.014389
23	C	-2.779309	-4.510219	3.929309
24	C	-3.095645	-3.406136	2.914199
25	O	-4.065731	-3.521725	2.152393
26	N	-2.227094	-2.374839	2.820631
27	C	-7.758766	1.211619	-2.044223
28	C	-8.111245	2.070838	-0.80825
29	C	-7.211207	3.294558	-0.551641
30	C	-5.989101	3.020256	0.337762
31	N	-4.951931	2.182189	-0.263133
32	C	-3.747999	2.613757	-0.659334
33	N	-3.567237	3.897135	-1.020929
34	N	-2.725526	1.751193	-0.677793
35	C	10.017502	5.683991	0.480176
36	C	8.536846	5.607752	0.898938
37	C	7.686485	4.598035	0.17361
38	C	8.067107	3.674091	-0.765611
39	C	6.280387	4.362204	0.41678
40	N	6.987002	2.873528	-1.117516
41	C	5.877313	3.26523	-0.398528
42	C	5.329412	4.965678	1.256241
43	C	4.576769	2.747706	-0.367639
44	C	4.029699	4.467486	1.278976
45	C	3.658119	3.362305	0.483749
46	Co	1.119567	-1.739641	-1.943132
47	C	-1.455194	-2.898104	-2.217928
48	C	-2.941388	-2.400968	-2.180718
49	C	-2.867068	-1.115861	-3.086358
50	C	-1.414022	-0.696902	-2.953895
51	C	-0.895334	0.593935	-3.362911
52	C	0.450998	0.868929	-3.287513
53	C	1.195561	2.187222	-3.647237
54	C	2.597239	1.604963	-4.005664
55	C	2.64868	0.38474	-3.131741
56	C	3.846987	-0.218597	-2.775777
57	C	3.998179	-1.421414	-2.103235
58	C	5.363752	-2.038755	-1.868982
59	C	4.990832	-3.396588	-1.184636
60	C	3.471136	-3.336397	-1.047797
61	C	2.715031	-4.305218	-0.415589
62	C	1.27483	-4.290642	-0.468882
63	C	0.310332	-5.343777	0.127784
64	C	-0.998448	-5.014032	-0.657561
65	C	-0.857248	-3.514518	-0.933768

66	C	-1.128959	-3.792525	-3.433268
67	N	-0.692584	-1.63172	-2.397898
68	N	1.414972	-0.058638	-2.843369
69	N	2.979672	-2.184234	-1.653014
70	N	0.605815	-3.314998	-1.032062
71	C	-3.941986	-3.44115	-2.666767
72	C	-3.323006	-1.999125	-0.733197
73	C	-4.730327	-1.456727	-0.581247
74	O	-5.14799	-0.505303	-1.273693
75	N	-5.516534	-2.069834	0.315648
76	C	-3.345156	-1.281439	-4.54225
77	C	-1.895573	1.584844	-3.926242
78	C	0.664582	3.051875	-4.805289
79	C	1.431002	3.029786	-2.344638
80	C	0.303291	3.897216	-1.800171
81	O	-0.896104	3.650495	-2.000706
82	N	0.681759	4.956104	-1.060367
83	C	2.773781	1.167762	-5.474792
84	C	6.223447	-1.166749	-0.932116
85	C	6.080961	-2.172511	-3.228184
86	C	5.46949	-4.650022	-1.94174
87	C	3.407139	-5.411408	0.358787
88	C	0.154516	-5.06802	1.640201
89	C	0.647795	-6.825508	-0.118501
90	C	-2.289002	-5.478118	0.012902
91	H	-0.86954	0.465354	1.164356
92	C	0.097078	0.800545	0.770173
93	C	1.146759	0.022606	0.537413
94	F	0.162561	2.129215	0.498702
95	F	2.3073	0.515495	0.031071
96	H	6.280396	-3.385265	5.486109
97	H	5.504593	-4.613213	4.508007
98	H	7.068628	-1.204177	4.225967
99	H	4.039128	-3.873686	2.772031
100	H	6.066614	0.648517	2.92293
101	H	3.052639	-2.037741	1.446968
102	H	4.037766	0.253262	1.549515
103	H	8.287125	-3.467897	3.847263
104	H	-1.372339	7.221894	1.688662
105	H	-0.978491	4.716469	1.449516
106	H	-2.732695	4.494875	1.385726
107	H	-1.979927	5.154051	2.845682
108	H	-1.902384	7.361575	-0.591538
109	H	-4.234694	6.606212	0.79962
110	H	-5.0212	3.291465	6.189063
111	H	-5.662796	1.696695	6.520573

112	H	-3.470733	3.513037	4.286306
113	H	-6.078894	0.115765	4.513043
114	H	-2.232377	2.431645	2.428767
115	H	-4.855075	-0.952058	2.651889
116	H	-7.486549	2.312474	4.668875
117	H	-2.570895	-4.054955	4.904975
118	H	-1.834982	-4.963682	3.607825
119	H	-2.455839	-1.547735	2.23108
120	H	-1.560639	-2.233524	3.566004
121	H	-4.412378	-5.619795	3.055944
122	H	-9.150013	2.392577	-0.927835
123	H	-8.097636	1.430459	0.08539
124	H	-7.799492	4.064248	-0.036603
125	H	-6.889231	3.737566	-1.50442
126	H	-5.520618	3.960482	0.645386
127	H	-6.30242	2.521135	1.26207
128	H	-5.070681	1.172064	-0.258836
129	H	-1.844049	2.106713	-1.032009
130	H	-2.719669	1.017809	0.077395
131	H	-4.357727	4.460918	-1.286035
132	H	-2.630748	4.224239	-1.252149
133	H	-6.710594	0.914808	-2.036897
134	H	8.081211	6.60305	0.802636
135	H	8.485939	5.373915	1.971905
136	H	9.033829	3.511098	-1.219392
137	H	5.609987	5.805044	1.886139
138	H	3.29374	4.916081	1.940527
139	H	4.29511	1.882869	-0.961028
140	H	2.651688	2.96304	0.560269
141	H	10.15032	6.199109	-0.475374
142	H	-3.4975	-0.349358	-2.631684
143	H	3.395054	2.310696	-3.752811
144	H	4.748868	0.284651	-3.102207
145	H	5.426554	-3.412172	-0.178712
146	H	-0.902978	-5.538628	-1.614423
147	H	-1.194508	-2.934493	-0.069736
148	H	-1.357728	-3.277009	-4.365628
149	H	-1.688986	-4.728928	-3.414443
150	H	-0.06101	-4.03112	-3.438576
151	H	-4.952984	-3.022777	-2.670186
152	H	-3.740749	-3.793269	-3.679129
153	H	-3.95826	-4.310478	-2.007278
154	H	-2.663678	-1.218243	-0.352462
155	H	-3.223994	-2.85381	-0.065912
156	H	-6.410533	-1.638327	0.511311
157	H	-3.251056	-0.34489	-5.092979

158	H	-2.789572	-2.043784	-5.092963
159	H	-1.658039	2.594963	-3.616886
160	H	-1.925793	1.545098	-5.020677
161	H	-2.899157	1.377498	-3.556886
162	H	1.456226	3.736805	-5.126487
163	H	-0.189342	3.657252	-4.506747
164	H	0.371549	2.443803	-5.662667
165	H	2.300186	3.673912	-2.511321
166	H	1.711511	2.353857	-1.530077
167	H	-0.053616	5.53939	-0.657998
168	H	2.771257	2.025335	-6.150951
169	H	1.975714	0.483	-5.778969
170	H	5.744136	-1.022101	0.036553
171	H	6.40925	-0.179555	-1.367639
172	H	7.193502	-1.647249	-0.767562
173	H	5.485254	-2.72134	-3.961879
174	H	6.290219	-1.184073	-3.645312
175	H	7.041125	-2.68212	-3.112658
176	H	6.554454	-4.627813	-2.070262
177	H	5.228456	-5.563921	-1.400841
178	H	4.411614	-5.116426	0.661611
179	H	2.866549	-5.641439	1.277824
180	H	3.485874	-6.345214	-0.207002
181	H	-0.28165	-4.08357	1.834245
182	H	-0.499978	-5.823319	2.083337
183	H	1.114273	-5.120852	2.159834
184	H	-0.249477	-7.428123	0.047641
185	H	1.417411	-7.208664	0.553195
186	H	-3.13563	-5.410321	-0.671703
187	H	-2.543494	-4.892207	0.894264
188	H	1.168936	-1.036973	0.757176
189	H	5.009656	-4.719826	-2.93101
190	H	0.97731	-6.992851	-1.148946
191	H	-2.214145	-6.527062	0.314942
192	H	-4.401569	-1.559268	-4.550972
193	H	3.727942	0.646833	-5.593641
194	H	-7.948663	1.796855	-2.950501
195	H	-4.6352	-5.296591	4.769538
196	H	-3.521903	-6.564869	4.238189
197	H	-7.697679	2.92592	6.291465
198	H	-7.03263	3.977691	5.022827
199	H	-3.59826	8.187414	1.331723
200	H	-3.779762	6.837744	2.491329
201	H	7.548274	-4.917821	3.146639
202	H	8.006937	-4.878677	4.870581
203	H	10.450528	4.681479	0.390321

204	H	10.585027	6.201884	1.253239
205	H	7.0186	2.120095	-1.783163
206	H	1.647727	5.103485	-0.803084
207	H	-5.096453	-2.689586	1.013718
208	C	-8.692482	0.017118	-2.04572
209	O	-9.896885	0.123288	-2.250529
210	N	-8.097731	-1.153757	-1.678758
211	H	-7.086327	-1.182307	-1.672115
212	C	-8.880342	-2.363437	-1.553794
213	H	-8.352114	-3.075343	-0.913073
214	H	-9.846481	-2.117301	-1.108752
215	H	-9.073936	-2.840993	-2.523198

F2H2-iso2_Int, E_{opt}= -4458.8991161

Center number	Atomic Symbol	Coordinates (Angstroms)		
		X	Y	Z
1	C	7.598294	-4.295551	4.045107
2	C	6.1829	-3.823587	4.443943
3	C	5.566694	-2.749881	3.568597
4	C	6.100131	-1.451143	3.561247
5	C	4.423439	-2.996806	2.797011
6	C	5.501114	-0.430788	2.823348
7	C	3.819279	-1.978583	2.05211
8	C	4.351643	-0.688111	2.069454
9	C	-3.523067	7.104814	1.466056
10	C	-2.063623	6.6938	1.203344
11	O	-1.820208	6.490517	-0.20856
12	C	-1.711747	5.378718	1.889159
13	C	-7.033134	2.960482	5.42778
14	C	-5.590501	2.560544	5.79987
15	C	-4.848514	1.961906	4.625869
16	C	-5.130974	0.640164	4.246214
17	C	-3.91158	2.678337	3.872484
18	C	-4.506785	0.043196	3.156616
19	C	-3.25938	2.090686	2.784431
20	C	-3.566829	0.77726	2.43237
21	O	-2.914499	0.134006	1.398168

22	C	-3.891422	-5.564265	4.014496
23	C	-2.756926	-4.546382	3.876706
24	C	-3.097445	-3.384936	2.944707
25	O	-4.137447	-3.361351	2.275307
26	N	-2.158686	-2.413084	2.837901
27	C	-7.758693	1.211597	-2.044174
28	C	-8.080859	2.194693	-0.896867
29	C	-7.2333	3.481424	-0.846639
30	C	-6.006931	3.399092	0.071049
31	N	-4.983077	2.461427	-0.38944
32	C	-3.730678	2.75697	-0.734546
33	N	-3.242834	3.989746	-0.783045
34	N	-2.874654	1.714927	-0.987953
35	C	10.017438	5.683988	0.480191
36	C	8.519388	5.709319	0.830641
37	C	7.660327	4.691167	0.130018
38	C	8.038212	3.708659	-0.750179
39	C	6.244133	4.507793	0.347609
40	N	6.944993	2.924234	-1.092032
41	C	5.831923	3.379818	-0.41915
42	C	5.291909	5.177678	1.133178
43	C	4.521986	2.890168	-0.388366
44	C	3.98239	4.7066	1.156398
45	C	3.603598	3.56693	0.414659
46	Co	1.137616	-1.782067	-2.016396
47	C	-1.412576	-2.92982	-2.28569
48	C	-2.896697	-2.414869	-2.265275
49	C	-2.813035	-1.203428	-3.255394
50	C	-1.367994	-0.755817	-3.083564
51	C	-0.864003	0.528806	-3.475655
52	C	0.482616	0.822907	-3.344502
53	C	1.211046	2.155555	-3.681653
54	C	2.613209	1.588048	-4.04774
55	C	2.6731	0.378883	-3.15228
56	C	3.865316	-0.20644	-2.767722
57	C	4.00964	-1.414257	-2.098762
58	C	5.37636	-2.033367	-1.866545
59	C	5.003233	-3.391992	-1.189553
60	C	3.479838	-3.341283	-1.08567
61	C	2.728478	-4.322679	-0.464441
62	C	1.295892	-4.306201	-0.517539
63	C	0.331501	-5.336842	0.110805
64	C	-0.984711	-5.005079	-0.659191
65	C	-0.815898	-3.519035	-0.987059
66	C	-1.101453	-3.878044	-3.463321
67	N	-0.63491	-1.678148	-2.490077

68	N	1.434864	-0.09235	-2.882776
69	N	2.980405	-2.187598	-1.684536
70	N	0.641697	-3.326862	-1.107041
71	C	-3.941973	-3.441169	-2.666763
72	C	-3.220224	-1.912311	-0.824114
73	C	-4.581019	-1.285726	-0.62677
74	O	-4.916945	-0.235377	-1.228399
75	N	-5.424256	-1.909662	0.204638
76	C	-3.221649	-1.471598	-4.719321
77	C	-1.846621	1.502555	-4.108959
78	C	0.6646	3.051897	-4.805323
79	C	1.456813	2.979786	-2.3725
80	C	0.331926	3.8155	-1.807568
81	O	-0.869292	3.504669	-1.932579
82	N	0.687981	4.930419	-1.144689
83	C	2.771145	1.147169	-5.515736
84	C	6.253066	-1.173557	-0.936448
85	C	6.080935	-2.172488	-3.228162
86	C	5.516464	-4.63991	-1.932945
87	C	3.424809	-5.428634	0.307572
88	C	0.203006	-5.048949	1.622687
89	C	0.647799	-6.825522	-0.1185
90	C	-2.273722	-5.407517	0.054693
91	H	-0.699381	0.652054	1.407683
92	C	0.195466	1.047409	0.948875
93	C	1.180817	0.279426	0.506125
94	F	0.221008	2.402771	0.85017
95	F	2.298414	0.800947	-0.037668
96	H	6.230844	-3.434512	5.469569
97	H	5.513352	-4.690936	4.482815
98	H	6.984518	-1.236632	4.156795
99	H	3.993032	-3.994738	2.790528
100	H	5.925046	0.569543	2.835425
101	H	2.945258	-2.203825	1.448254
102	H	3.893959	0.105263	1.48932
103	H	8.264494	-3.444298	3.872122
104	H	-1.389613	7.480547	1.571661
105	H	-0.723065	5.021176	1.588644
106	H	-2.441953	4.611492	1.624576
107	H	-1.718498	5.50433	2.975596
108	H	-1.925162	7.335501	-0.666424
109	H	-4.192731	6.537299	0.809624
110	H	-5.047509	3.424665	6.197769
111	H	-5.620354	1.818674	6.606379
112	H	-3.666467	3.700523	4.150204
113	H	-5.856158	0.068407	4.82009

114	H	-2.489897	2.623968	2.234584
115	H	-4.720744	-0.980649	2.868861
116	H	-2.711554	0.765952	0.675333
117	H	-7.444934	2.257207	4.695655
118	H	-2.448167	-4.144451	4.849596
119	H	-1.863658	-5.019353	3.455305
120	H	-2.335259	-1.606779	2.24699
121	H	-1.355279	-2.398917	3.44672
122	H	-4.446918	-5.623808	3.075183
123	H	-9.139126	2.451133	-0.992982
124	H	-7.991637	1.666586	0.063817
125	H	-7.850026	4.30093	-0.458867
126	H	-6.916801	3.778759	-1.853888
127	H	-5.556879	4.388293	0.186772
128	H	-6.307967	3.085569	1.077954
129	H	-5.224108	1.469984	-0.438021
130	H	-1.996643	1.973268	-1.443243
131	H	-3.351173	0.894152	-1.367056
132	H	-3.787193	4.804436	-0.561729
133	H	-2.28063	4.119381	-1.131365
134	H	-6.71125	0.909339	-2.044326
135	H	8.119201	6.716465	0.648182
136	H	8.406723	5.549113	1.912401
137	H	9.011414	3.494307	-1.167212
138	H	5.579846	6.043113	1.723492
139	H	3.243979	5.20292	1.780887
140	H	4.23716	2.002086	-0.944618
141	H	2.591638	3.18699	0.502496
142	H	10.22304	6.143806	-0.490659
143	H	-3.499688	-0.430417	-2.90926
144	H	3.40408	2.309431	-3.813088
145	H	4.771317	0.302592	-3.078455
146	H	5.421582	-3.403046	-0.17537
147	H	-0.92579	-5.566764	-1.596704
148	H	-1.138381	-2.915512	-0.133379
149	H	-1.272775	-3.385242	-4.418599
150	H	-1.702487	-4.789654	-3.430858
151	H	-0.043793	-4.147478	-3.423742
152	H	-4.938531	-2.985744	-2.697884
153	H	-3.755709	-3.864495	-3.653804
154	H	-3.990423	-4.267067	-1.954299
155	H	-2.474502	-1.16775	-0.539881
156	H	-3.132198	-2.739144	-0.120461
157	H	-6.319188	-1.468242	0.376474
158	H	-3.088743	-0.566344	-5.315135
159	H	-2.634761	-2.259179	-5.192734

160	H	-1.690564	2.519212	-3.761952
161	H	-1.771849	1.502956	-5.202012
162	H	-2.878752	1.244473	-3.867823
163	H	1.44611	3.760075	-5.102811
164	H	-0.20068	3.634597	-4.49231
165	H	0.387087	2.471973	-5.686864
166	H	2.316614	3.637342	-2.534253
167	H	1.755894	2.293246	-1.575861
168	H	-0.041657	5.501776	-0.723142
169	H	2.753123	1.998415	-6.201795
170	H	1.974954	0.451638	-5.8001
171	H	5.78249	-1.026924	0.036433
172	H	6.439987	-0.185976	-1.373126
173	H	7.22387	-1.656806	-0.779465
174	H	5.473172	-2.720302	-3.952626
175	H	6.282993	-1.185028	-3.652454
176	H	7.044096	-2.681945	-3.129004
177	H	6.603611	-4.602521	-2.044148
178	H	5.280432	-5.556764	-1.394483
179	H	4.425343	-5.129369	0.619748
180	H	2.8825	-5.673024	1.222609
181	H	3.516423	-6.359445	-0.262781
182	H	-0.217921	-4.056079	1.807293
183	H	-0.444793	-5.793798	2.095263
184	H	1.174121	-5.09698	2.121031
185	H	-0.249561	-7.421903	0.075719
186	H	1.431383	-7.205223	0.539065
187	H	-3.133974	-5.353011	-0.615096
188	H	-2.490776	-4.770237	0.91167
189	H	1.157982	-0.802671	0.515426
190	H	5.073729	-4.720405	-2.929099
191	H	0.954659	-7.007408	-1.15331
192	H	-2.219557	-6.440452	0.41261
193	H	-4.277242	-1.751816	-4.779961
194	H	3.726652	0.630321	-5.641373
195	H	-7.970526	1.703507	-3.000375
196	H	-4.600962	-5.272604	4.79315
197	H	-3.52188	-6.56488	4.23809
198	H	-7.697652	2.925947	6.291473
199	H	-7.076209	3.95924	4.982966
200	H	-3.705542	8.170272	1.303951
201	H	-3.779742	6.837701	2.491308
202	H	7.583472	-4.905379	3.13709
203	H	8.006923	-4.878674	4.870579
204	H	10.393681	4.655189	0.454847
205	H	10.58501	6.201864	1.253232

206	H	6.97428	2.115176	-1.689893
207	H	1.658801	5.124664	-0.939531
208	H	-5.069276	-2.590965	0.878407
209	C	-8.691064	0.027785	-1.906162
210	O	-9.909255	0.151245	-1.920675
211	N	-8.072324	-1.165674	-1.655608
212	H	-7.089463	-1.243133	-1.876346
213	C	-8.880359	-2.36342	-1.553803
214	H	-8.291742	-3.160591	-1.092305
215	H	-9.752718	-2.149673	-0.934328
216	H	-9.241921	-2.707606	-2.530608

F2H2-iso2_TS, E_{opt}= -4458.8368308

Center number	Atomic Symbol	Coordinates (Angstroms)		
		X	Y	Z
1	C	7.598196	-4.295632	4.045064
2	C	6.162274	-3.882598	4.439952
3	C	5.4416	-2.895545	3.538928
4	C	5.953132	-1.604635	3.328049
5	C	4.198066	-3.209351	2.97112
6	C	5.241466	-0.658022	2.589881
7	C	3.478874	-2.266585	2.228366
8	C	3.997765	-0.985435	2.038756
9	C	-3.523137	7.10451	1.466017
10	C	-2.136729	6.518747	1.152721
11	O	-1.877158	6.500259	-0.271223
12	C	-2.00198	5.079375	1.639295
13	C	-7.033091	2.960529	5.42778
14	C	-5.622171	2.413322	5.748012
15	C	-4.808522	1.920613	4.556393
16	C	-5.196363	0.754582	3.873149
17	C	-3.606359	2.521653	4.157995
18	C	-4.404382	0.188983	2.876377
19	C	-2.799263	1.975511	3.149182
20	C	-3.189168	0.789462	2.525139
21	O	-2.437724	0.153199	1.551062
22	C	-3.891417	-5.564279	4.014389

23	C	-2.754592	-4.538962	4.004336
24	C	-3.019966	-3.36569	3.063943
25	O	-4.03289	-3.311949	2.357431
26	N	-2.05248	-2.42395	2.994287
27	C	-7.758677	1.211667	-2.044208
28	C	-8.093679	2.13809	-0.855048
29	C	-7.16169	3.348782	-0.658968
30	C	-5.948861	3.086822	0.248775
31	N	-4.903151	2.231307	-0.310865
32	C	-3.725084	2.667636	-0.778375
33	N	-3.561193	3.942353	-1.160667
34	N	-2.694353	1.816478	-0.847624
35	C	10.017436	5.683954	0.480221
36	C	8.747819	5.067344	1.08333
37	C	7.849444	4.387409	0.089355
38	C	8.038152	4.269915	-1.264355
39	C	6.594452	3.733751	0.379024
40	N	6.974466	3.588756	-1.838359
41	C	6.074204	3.246206	-0.853173
42	C	5.860155	3.508882	1.556668
43	C	4.860125	2.558745	-0.930369
44	C	4.652036	2.8195	1.484318
45	C	4.156661	2.345771	0.251113
46	Co	1.078778	-1.663534	-1.780072
47	C	-1.44846	-2.88643	-2.17929
48	C	-2.941903	-2.401669	-2.167151
49	C	-2.867296	-1.12563	-3.08503
50	C	-1.415687	-0.695379	-2.931446
51	C	-0.903859	0.596718	-3.343985
52	C	0.436824	0.890301	-3.233885
53	C	1.181119	2.19972	-3.633046
54	C	2.572117	1.595476	-3.989074
55	C	2.625032	0.39932	-3.076933
56	C	3.821192	-0.22642	-2.760321
57	C	3.974146	-1.422145	-2.075355
58	C	5.342593	-2.054293	-1.876657
59	C	4.96339	-3.419109	-1.215001
60	C	3.452566	-3.320299	-1.009951
61	C	2.704796	-4.302972	-0.392449
62	C	1.266157	-4.282562	-0.435862
63	C	0.311227	-5.345143	0.157274
64	C	-1.014226	-4.991887	-0.59214
65	C	-0.855729	-3.493601	-0.881687
66	C	-1.100679	-3.789128	-3.382404
67	N	-0.694721	-1.618576	-2.359525
68	N	1.391976	-0.014981	-2.743978

69	N	2.956807	-2.146001	-1.563736
70	N	0.599713	-3.298223	-0.980897
71	C	-3.941918	-3.44111	-2.666753
72	C	-3.322145	-1.99172	-0.712986
73	C	-4.728542	-1.460278	-0.547104
74	O	-5.139365	-0.465319	-1.178523
75	N	-5.531	-2.126693	0.303186
76	C	-3.319545	-1.316738	-4.546343
77	C	-1.892892	1.558794	-3.976057
78	C	0.664531	3.05186	-4.805306
79	C	1.408382	3.083194	-2.356504
80	C	0.271947	3.979753	-1.891008
81	O	-0.923924	3.690838	-2.05852
82	N	0.619138	5.11775	-1.261598
83	C	2.723831	1.114402	-5.44682
84	C	6.19354	-1.194443	-0.918422
85	C	6.080974	-2.172477	-3.228179
86	C	5.357897	-4.662386	-2.035245
87	C	3.406389	-5.426491	0.346676
88	C	0.190207	-5.094276	1.677196
89	C	0.64779	-6.825527	-0.118501
90	C	-2.295269	-5.41868	0.122177
91	H	1.306233	2.220009	1.949922
92	C	0.664059	1.599786	1.326345
93	C	0.42785	0.291574	1.398069
94	F	-0.036577	2.404542	0.438721
95	F	0.495047	-0.661869	-0.115422
96	H	6.204353	-3.442757	5.445816
97	H	5.55073	-4.787376	4.538188
98	H	6.911346	-1.33317	3.764503
99	H	3.780686	-4.201178	3.127634
100	H	5.650413	0.336826	2.436826
101	H	2.52101	-2.527971	1.789059
102	H	3.446021	-0.256331	1.454266
103	H	8.241658	-3.42416	3.889747
104	H	-1.365559	7.133459	1.642262
105	H	-1.094103	4.606843	1.253448
106	H	-2.854111	4.477975	1.31585
107	H	-1.968453	5.058356	2.732536
108	H	-1.99485	7.39712	-0.612742
109	H	-4.26695	6.647647	0.804247
110	H	-5.048157	3.172643	6.29078
111	H	-5.727851	1.568294	6.440253
112	H	-3.27209	3.426427	4.660444
113	H	-6.117356	0.252033	4.159888
114	H	-1.865326	2.45017	2.866126

115	H	-4.683681	-0.745353	2.401995
116	H	-1.441884	0.336471	1.573938
117	H	-7.508257	2.354433	4.647451
118	H	-2.549005	-4.149232	5.008979
119	H	-1.818915	-5.000831	3.669136
120	H	-2.183153	-1.611592	2.395428
121	H	-1.260429	-2.441166	3.617491
122	H	-4.366603	-5.600342	3.031248
123	H	-9.123861	2.478531	-0.993057
124	H	-8.097586	1.544371	0.070253
125	H	-7.72978	4.158498	-0.184253
126	H	-6.825995	3.735443	-1.631056
127	H	-5.480028	4.032076	0.540869
128	H	-6.281018	2.61193	1.179453
129	H	-5.04971	1.222548	-0.348616
130	H	-1.812744	2.162335	-1.217889
131	H	-2.651892	1.067851	-0.160974
132	H	-4.358976	4.512747	-1.384906
133	H	-2.625218	4.269251	-1.405069
134	H	-6.713545	0.904482	-2.029825
135	H	8.176454	5.850868	1.601081
136	H	9.029202	4.348899	1.866108
137	H	8.8545	4.621879	-1.877827
138	H	6.234035	3.867779	2.511505
139	H	4.084255	2.632434	2.391702
140	H	4.4851	2.1819	-1.872602
141	H	3.218365	1.799042	0.221512
142	H	9.769524	6.413016	-0.298707
143	H	-3.510099	-0.356189	-2.653726
144	H	3.375913	2.308877	-3.777131
145	H	4.721982	0.243864	-3.134748
146	H	5.449875	-3.484299	-0.234262
147	H	-0.962364	-5.526989	-1.545996
148	H	-1.176999	-2.894069	-0.026659
149	H	-1.310068	-3.278977	-4.322273
150	H	-1.658557	-4.727413	-3.369672
151	H	-0.031466	-4.015262	-3.360438
152	H	-4.953262	-3.022281	-2.68532
153	H	-3.721272	-3.790621	-3.675131
154	H	-3.966944	-4.314232	-2.010664
155	H	-2.640036	-1.214576	-0.360824
156	H	-3.208101	-2.842911	-0.043868
157	H	-6.443991	-1.726413	0.478783
158	H	-3.208297	-0.389987	-5.109965
159	H	-2.751166	-2.087298	-5.071194
160	H	-1.670769	2.585001	-3.710816

161	H	-1.895394	1.473845	-5.068785
162	H	-2.907064	1.362912	-3.628447
163	H	1.464272	3.725894	-5.131068
164	H	-0.188983	3.667973	-4.527399
165	H	0.374398	2.435235	-5.6574
166	H	2.293556	3.707062	-2.522218
167	H	1.642748	2.436786	-1.503374
168	H	-0.129295	5.682926	-0.857904
169	H	2.723234	1.948518	-6.152496
170	H	1.914093	0.428985	-5.716864
171	H	5.706197	-1.083654	0.050659
172	H	6.364258	-0.192716	-1.325
173	H	7.167589	-1.670311	-0.760086
174	H	5.49436	-2.706666	-3.979539
175	H	6.303806	-1.178343	-3.62492
176	H	7.036982	-2.687921	-3.102844
177	H	6.433856	-4.674896	-2.226235
178	H	5.1135	-5.584255	-1.509353
179	H	4.414105	-5.136361	0.644074
180	H	2.88095	-5.674767	1.269757
181	H	3.477148	-6.350273	-0.237008
182	H	-0.217892	-4.101248	1.886359
183	H	-0.468182	-5.842542	2.128367
184	H	1.160462	-5.170139	2.17355
185	H	-0.246645	-7.433546	0.049468
186	H	1.42809	-7.216382	0.535689
187	H	-3.165293	-5.320392	-0.530132
188	H	-2.492628	-4.823266	1.013273
189	H	1.015646	-0.2914	2.098936
190	H	4.839321	-4.684559	-2.997803
191	H	0.963785	-6.972486	-1.156123
192	H	-2.244278	-6.469559	0.424061
193	H	-4.376638	-1.592303	-4.581981
194	H	3.668797	0.57564	-5.558777
195	H	-7.944673	1.752297	-2.978861
196	H	-4.66505	-5.303685	4.740795
197	H	-3.521881	-6.564873	4.238174
198	H	-7.69769	2.92591	6.291475
199	H	-6.989295	3.989198	5.056778
200	H	-3.561308	8.191566	1.349469
201	H	-3.779684	6.837968	2.491364
202	H	7.611567	-4.898351	3.132155
203	H	8.006995	-4.878608	4.870623
204	H	10.660114	4.917363	0.035192
205	H	10.585015	6.201896	1.25321
206	H	6.892281	3.359425	-2.814608

207	H	1.581772	5.331713	-1.051893
208	H	-5.128273	-2.757976	0.994865
209	C	-8.700947	0.026705	-1.987505
210	O	-9.913113	0.144273	-2.126059
211	N	-8.097475	-1.152488	-1.659776
212	H	-7.088862	-1.192846	-1.716621
213	C	-8.880368	-2.363464	-1.553783
214	H	-8.340009	-3.097757	-0.949489
215	H	-9.833717	-2.128085	-1.077103
216	H	-9.099394	-2.808877	-2.532779

F1H3_React, E_{opt}= -4359.1008772

Center number	Atomic Symbol	Coordinates (Angstroms)		
		X	Y	Z
1	C	7.598334	-4.295434	4.045127
2	C	6.200483	-3.804954	4.469805
3	C	5.555004	-2.763602	3.580116
4	C	6.123449	-1.485701	3.452716
5	C	4.33754	-3.014224	2.934344
6	C	5.479333	-0.481874	2.730903
7	C	3.682399	-2.013228	2.210838
8	C	4.24857	-0.74137	2.120233
9	C	-3.522996	7.104452	1.465992
10	C	-2.155785	6.524001	1.07614
11	O	-2.073141	6.357632	-0.369132
12	C	-1.911054	5.153993	1.693112
13	C	-7.033359	2.959964	5.427818
14	C	-5.578435	2.584044	5.779467
15	C	-4.846311	1.930909	4.625614
16	C	-5.19817	0.624939	4.24711
17	C	-3.82307	2.555724	3.903796
18	C	-4.557868	-0.03731	3.208245
19	C	-3.158485	1.898965	2.863084
20	C	-3.504091	0.580848	2.497603
21	O	-2.860631	-0.080641	1.529885

22	C	-3.891573	-5.56431	4.015087
23	C	-2.770804	-4.528146	3.931736
24	C	-3.124488	-3.399947	2.965896
25	O	-4.194923	-3.43855	2.337681
26	N	-2.215925	-2.428815	2.777728
27	C	-7.75866	1.211728	-2.044228
28	C	-8.066249	2.071935	-0.798933
29	C	-7.149576	3.292836	-0.58702
30	C	-5.919072	3.020556	0.288344
31	N	-4.931726	2.113318	-0.297068
32	C	-3.661518	2.424586	-0.585764
33	N	-3.317725	3.677838	-0.94851
34	N	-2.732715	1.465943	-0.558128
35	C	10.017491	5.683955	0.480232
36	C	8.774656	5.036489	1.104464
37	C	7.894129	4.293519	0.141214
38	C	8.127428	4.022711	-1.182762
39	C	6.605256	3.721904	0.448126
40	N	7.061007	3.313483	-1.720223
41	C	6.107374	3.122805	-0.743013
42	C	5.824004	3.666362	1.614953
43	C	4.858923	2.495769	-0.793303
44	C	4.580965	3.042773	1.570564
45	C	4.102426	2.467863	0.374851
46	Co	1.111162	-1.715828	-1.89051
47	C	-1.463461	-2.872278	-2.178312
48	C	-2.957628	-2.392317	-2.14284
49	C	-2.88211	-1.067308	-3.000422
50	C	-1.42973	-0.645728	-2.850793
51	C	-0.894988	0.655062	-3.227972
52	C	0.461008	0.886741	-3.24084
53	C	1.221345	2.179459	-3.668364
54	C	2.589546	1.558767	-4.080181
55	C	2.649403	0.353423	-3.184676
56	C	3.840695	-0.259667	-2.842182
57	C	3.987451	-1.441763	-2.126022
58	C	5.353523	-2.043643	-1.875991
59	C	4.985618	-3.399302	-1.18575
60	C	3.467794	-3.332932	-1.029914
61	C	2.713622	-4.297816	-0.396743
62	C	1.270632	-4.285284	-0.451847
63	C	0.31187	-5.342895	0.14549
64	C	-1.011095	-4.997754	-0.615026
65	C	-0.858583	-3.499087	-0.898897
66	C	-1.127733	-3.746778	-3.405053
67	N	-0.705425	-1.598445	-2.335318

68	N	1.41727	-0.058122	-2.841161
69	N	2.973336	-2.182536	-1.642914
70	N	0.602626	-3.311338	-1.012021
71	C	-3.94199	-3.441147	-2.666832
72	C	-3.362732	-2.043516	-0.689283
73	C	-4.765066	-1.484139	-0.528397
74	O	-5.20187	-0.571551	-1.265523
75	N	-5.520724	-2.02686	0.43117
76	C	-3.341475	-1.170004	-4.466248
77	C	-1.88351	1.728929	-3.64365
78	C	0.66454	3.05192	-4.805321
79	C	1.534672	3.077745	-2.418424
80	C	0.359859	3.733719	-1.712877
81	O	-0.388523	3.086122	-0.96501
82	N	0.170923	5.054688	-1.917822
83	C	2.687864	1.09714	-5.54839
84	C	6.185134	-1.150666	-0.931689
85	C	6.081007	-2.17249	-3.228188
86	C	5.444692	-4.654854	-1.951194
87	C	3.409755	-5.399407	0.379532
88	C	0.181063	-5.080813	1.662871
89	C	0.647802	-6.825555	-0.118504
90	C	-2.291521	-5.432322	0.095573
91	H	0.780572	0.997032	-0.179367
92	C	1.063916	0.060495	0.286841
93	C	0.160302	-0.568414	1.031941
94	H	2.077921	-0.305163	0.178516
95	F	0.455581	-1.750911	1.639205
96	H	6.281377	-3.378921	5.478641
97	H	5.531226	-4.668229	4.56327
98	H	7.066148	-1.270555	3.950153
99	H	3.8791	-3.995576	3.029612
100	H	5.918448	0.508775	2.650452
101	H	2.724445	-2.21936	1.748226
102	H	3.744165	0.053742	1.584236
103	H	8.276507	-3.458022	3.853227
104	H	-1.354035	7.208202	1.388822
105	H	-1.037301	4.670374	1.246116
106	H	-2.774291	4.499679	1.55207
107	H	-1.739048	5.255697	2.768314
108	H	-2.314128	7.200018	-0.778291
109	H	-4.290727	6.652486	0.826716
110	H	-5.032928	3.469147	6.126732
111	H	-5.588004	1.884172	6.624602
112	H	-3.51999	3.566108	4.174723
113	H	-5.988307	0.112968	4.794068

114	H	-2.331841	2.375165	2.343389
115	H	-4.831861	-1.053885	2.948108
116	H	-7.440944	2.245743	4.703925
117	H	-2.536568	-4.096107	4.912228
118	H	-1.838451	-4.980116	3.576533
119	H	-2.475026	-1.577643	2.232684
120	H	-1.393039	-2.387282	3.35974
121	H	-4.415723	-5.618049	3.058919
122	H	-9.107388	2.397273	-0.88229
123	H	-8.022711	1.434928	0.096155
124	H	-7.718478	4.082786	-0.08101
125	H	-6.833391	3.706172	-1.554015
126	H	-5.41257	3.957837	0.539875
127	H	-6.230068	2.584649	1.245691
128	H	-5.139549	1.116206	-0.321326
129	H	-1.79627	1.767765	-0.805711
130	H	-2.803796	0.76719	0.255777
131	H	-4.000979	4.41261	-1.016309
132	H	-2.348016	3.949913	-0.851447
133	H	-6.71278	0.907412	-2.070378
134	H	8.175532	5.811849	1.601901
135	H	9.087187	4.353351	1.907136
136	H	8.976402	4.280594	-1.798866
137	H	6.18587	4.111146	2.537523
138	H	3.967612	2.994659	2.465219
139	H	4.495144	2.034329	-1.703637
140	H	3.125649	1.993212	0.370697
141	H	9.742043	6.410003	-0.291582
142	H	-3.518725	-0.323745	-2.520868
143	H	3.41109	2.254246	-3.876509
144	H	4.745318	0.214713	-3.199779
145	H	5.436767	-3.418843	-0.186609
146	H	-0.949496	-5.529638	-1.571039
147	H	-1.173323	-2.914123	-0.030439
148	H	-1.364413	-3.221747	-4.330622
149	H	-1.678407	-4.688752	-3.396655
150	H	-0.057815	-3.975171	-3.413853
151	H	-4.962702	-3.049186	-2.633659
152	H	-3.746353	-3.736673	-3.69858
153	H	-3.917204	-4.341105	-2.04965
154	H	-2.695559	-1.29016	-0.273257
155	H	-3.28111	-2.921968	-0.050604
156	H	-6.405684	-1.575662	0.622947
157	H	-3.229038	-0.20944	-4.974268
158	H	-2.782662	-1.91293	-5.040462
159	H	-1.634452	2.693751	-3.204269

160	H	-1.927672	1.854124	-4.730482
161	H	-2.886968	1.48994	-3.297011
162	H	1.455925	3.72118	-5.158624
163	H	-0.16947	3.675875	-4.482074
164	H	0.326607	2.45184	-5.652062
165	H	2.248004	3.843582	-2.74134
166	H	2.042461	2.476866	-1.660742
167	H	-0.540947	5.546436	-1.373112
168	H	2.673805	1.943273	-6.239168
169	H	1.861834	0.425821	-5.803554
170	H	5.687349	-1.017038	0.029896
171	H	6.357705	-0.15951	-1.362253
172	H	7.159284	-1.617038	-0.750454
173	H	5.49193	-2.718396	-3.969464
174	H	6.300951	-1.183795	-3.638148
175	H	7.039487	-2.684212	-3.106074
176	H	6.52739	-4.639753	-2.098285
177	H	5.206499	-5.566745	-1.40568
178	H	4.40678	-5.09179	0.694909
179	H	2.863612	-5.637734	1.292261
180	H	3.503862	-6.330328	-0.188892
181	H	-0.227545	-4.088845	1.864565
182	H	-0.477676	-5.830639	2.108773
183	H	1.146916	-5.153706	2.167603
184	H	-0.247007	-7.4288	0.059955
185	H	1.429466	-7.210389	0.537794
186	H	-3.160395	-5.343523	-0.558605
187	H	-2.49519	-4.836838	0.984318
188	H	-0.863443	-0.251475	1.239409
189	H	4.966966	-4.722005	-2.932476
190	H	0.959145	-6.98549	-1.155852
191	H	-2.232117	-6.48181	0.399448
192	H	-4.400503	-1.43652	-4.503182
193	H	3.623196	0.551681	-5.700953
194	H	-7.968464	1.802475	-2.942529
195	H	-4.633263	-5.292572	4.770505
196	H	-3.521781	-6.564736	4.237584
197	H	-7.69745	2.92637	6.291417
198	H	-7.097092	3.952972	4.971112
199	H	-3.571711	8.191271	1.350878
200	H	-3.77979	6.838004	2.491397
201	H	7.557682	-4.915412	3.144529
202	H	8.006907	-4.878776	4.870581
203	H	10.672046	4.933574	0.024246
204	H	10.584975	6.201901	1.253198
205	H	6.989024	3.016845	-2.678819

206	H	0.824012	5.59487	-2.46284
207	H	-5.08608	-2.608369	1.159727
208	C	-8.700512	0.022821	-2.034569
209	O	-9.90576	0.140443	-2.231112
210	N	-8.105833	-1.150204	-1.680537
211	H	-7.093687	-1.171171	-1.645869
212	C	-8.880383	-2.363525	-1.55377
213	H	-8.393046	-3.039779	-0.845124
214	H	-9.877503	-2.108588	-1.189836
215	H	-8.998751	-2.888106	-2.511185

F1H3_Int, E_{opt}= -4359.6759042

Center number	Atomic Symbol	Coordinates (Angstroms)		
		X	Y	Z
1	C	7.598234	-4.295571	4.045034
2	C	6.192022	-3.83023	4.485528
3	C	5.452025	-2.876915	3.567287
4	C	5.925111	-1.5723	3.351274
5	C	4.23256	-3.241367	2.979634
6	C	5.199372	-0.660832	2.583165
7	C	3.49774	-2.334982	2.20913
8	C	3.980954	-1.041224	2.010341
9	C	-3.523007	7.104436	1.465968
10	C	-2.124008	6.604526	1.086184
11	O	-2.005982	6.513793	-0.359393
12	C	-1.82955	5.217573	1.643513
13	C	-7.033347	2.959966	5.427811
14	C	-5.609877	2.491734	5.793313
15	C	-4.831442	1.928324	4.621339
16	C	-5.171716	0.66402	4.110768
17	C	-3.752145	2.597525	4.035412
18	C	-4.450029	0.068705	3.082581
19	C	-3.010032	2.017092	3.000191
20	C	-3.353063	0.745485	2.542268
21	O	-2.623367	0.078715	1.58354
22	C	-3.891566	-5.564343	4.014726
23	C	-2.740947	-4.560331	3.954681

24	C	-3.029876	-3.367416	3.04907
25	O	-4.069638	-3.282602	2.381766
26	N	-2.058742	-2.432795	2.982758
27	C	-7.75866	1.211712	-2.044218
28	C	-8.152146	2.360937	-1.09661
29	C	-7.253942	3.610728	-1.16658
30	C	-6.076621	3.602212	-0.181276
31	N	-5.072607	2.573219	-0.444667
32	C	-3.79023	2.76617	-0.778439
33	N	-3.276359	3.975109	-1.049773
34	N	-2.980083	1.703361	-0.85415
35	C	10.017405	5.684003	0.480201
36	C	8.72825	5.123531	1.103116
37	C	7.84765	4.32264	0.18476
38	C	8.074988	3.975366	-1.122603
39	C	6.569525	3.747467	0.535807
40	N	7.013003	3.22558	-1.610396
41	C	6.074679	3.066722	-0.612371
42	C	5.800491	3.735546	1.712581
43	C	4.847817	2.394451	-0.612396
44	C	4.580421	3.065792	1.72021
45	C	4.108542	2.403468	0.566941
46	Co	1.122974	-1.763506	-1.978257
47	C	-1.4358	-2.887833	-2.222841
48	C	-2.930332	-2.397362	-2.186754
49	C	-2.858815	-1.119256	-3.100069
50	C	-1.407731	-0.679482	-2.937016
51	C	-0.887992	0.606008	-3.317101
52	C	0.473571	0.862129	-3.2682
53	C	1.214418	2.173535	-3.667317
54	C	2.600743	1.582143	-4.052291
55	C	2.661477	0.375039	-3.155836
56	C	3.849997	-0.23688	-2.813672
57	C	3.992937	-1.4335	-2.124215
58	C	5.360861	-2.038077	-1.876731
59	C	4.991701	-3.395428	-1.19699
60	C	3.468906	-3.344908	-1.086113
61	C	2.721717	-4.326414	-0.46634
62	C	1.287883	-4.305036	-0.510022
63	C	0.32931	-5.337005	0.123825
64	C	-0.99812	-4.991072	-0.624266
65	C	-0.824267	-3.501484	-0.94087
66	C	-1.122845	-3.79881	-3.429066
67	N	-0.662621	-1.627155	-2.397886
68	N	1.422389	-0.06733	-2.840468
69	N	2.965302	-2.192791	-1.688143

70	N	0.632132	-3.321019	-1.084513
71	C	-3.941939	-3.441116	-2.666761
72	C	-3.27957	-1.987487	-0.719116
73	C	-4.649023	-1.383682	-0.497116
74	O	-4.974855	-0.286196	-1.006235
75	N	-5.510721	-2.082642	0.25931
76	C	-3.283207	-1.302267	-4.571623
77	C	-1.866105	1.645041	-3.842824
78	C	0.664539	3.051909	-4.805322
79	C	1.493221	3.059646	-2.396425
80	C	0.323771	3.799648	-1.785759
81	O	-0.593274	3.209895	-1.181871
82	N	0.315582	5.144532	-1.887605
83	C	2.731404	1.134157	-5.521405
84	C	6.212415	-1.159699	-0.939826
85	C	6.080992	-2.17248	-3.228181
86	C	5.500094	-4.643716	-1.942962
87	C	3.422341	-5.426335	0.309021
88	C	0.220514	-5.054554	1.638872
89	C	0.647797	-6.825539	-0.118491
90	C	-2.275949	-5.391027	0.112104
91	H	0.534341	1.700225	0.521853
92	C	0.349195	0.63369	0.495767
93	C	0.536481	-0.102849	1.584315
94	H	0.075135	0.173194	-0.448431
95	F	0.299422	-1.438211	1.586938
96	H	6.291258	-3.344161	5.465561
97	H	5.568704	-4.716279	4.654695
98	H	6.863949	-1.264823	3.806086
99	H	3.845724	-4.244717	3.14132
100	H	5.57494	0.346839	2.426191
101	H	2.5544	-2.633057	1.765306
102	H	3.426225	-0.339108	1.396601
103	H	8.263221	-3.448134	3.852477
104	H	-1.362194	7.306043	1.454597
105	H	-0.926419	4.798578	1.190787
106	H	-2.65722	4.53169	1.448997
107	H	-1.685191	5.277415	2.726198
108	H	-2.204815	7.38451	-0.730502
109	H	-4.263935	6.612864	0.823593
110	H	-5.051657	3.313835	6.254767
111	H	-5.681892	1.70879	6.55815
112	H	-3.463211	3.577649	4.40739
113	H	-6.004879	0.122232	4.55206
114	H	-2.150246	2.530967	2.579632
115	H	-4.682283	-0.929203	2.725723

116	H	-1.844267	0.593156	1.313139
117	H	-7.471385	2.29622	4.674014
118	H	-2.470524	-4.190929	4.951458
119	H	-1.836005	-5.035332	3.559451
120	H	-2.163673	-1.629182	2.370102
121	H	-1.194683	-2.535617	3.490362
122	H	-4.400082	-5.605771	3.048479
123	H	-9.186347	2.627388	-1.329302
124	H	-8.173101	1.987754	-0.062588
125	H	-7.851607	4.496652	-0.920079
126	H	-6.875401	3.760921	-2.185183
127	H	-5.589072	4.580685	-0.1792
128	H	-6.449318	3.446985	0.83868
129	H	-5.342682	1.60041	-0.321768
130	H	-2.026509	1.826923	-1.177868
131	H	-3.427501	0.791511	-0.891505
132	H	-3.745724	4.829169	-0.803145
133	H	-2.256231	4.018301	-1.094202
134	H	-6.718404	0.910288	-1.915861
135	H	8.142169	5.955249	1.519439
136	H	8.993677	4.498814	1.968064
137	H	8.918016	4.204939	-1.758132
138	H	6.15822	4.241151	2.605137
139	H	3.983754	3.04604	2.627863
140	H	4.488652	1.859671	-1.483995
141	H	3.156851	1.87915	0.598425
142	H	9.7994	6.389309	-0.328097
143	H	-3.534566	-0.36587	-2.693905
144	H	3.404713	2.295238	-3.83319
145	H	4.757301	0.247008	-3.157518
146	H	5.415771	-3.40609	-0.185122
147	H	-0.959292	-5.544903	-1.568124
148	H	-1.118792	-2.898407	-0.077488
149	H	-1.329426	-3.287777	-4.368715
150	H	-1.699599	-4.726132	-3.406107
151	H	-0.058299	-4.041857	-3.416932
152	H	-4.959434	-3.03524	-2.640045
153	H	-3.751975	-3.767816	-3.688771
154	H	-3.930827	-4.325566	-2.026687
155	H	-2.554496	-1.247699	-0.373464
156	H	-3.185837	-2.851454	-0.063489
157	H	-6.395677	-1.637091	0.46868
158	H	-3.175263	-0.360078	-5.113731
159	H	-2.686091	-2.048479	-5.09798
160	H	-1.678274	2.632535	-3.421432
161	H	-1.826216	1.738275	-4.933221

162	H	-2.893636	1.393658	-3.582139
163	H	1.450993	3.73992	-5.134729
164	H	-0.188754	3.65931	-4.500545
165	H	0.361423	2.453731	-5.665847
166	H	2.274405	3.780022	-2.660987
167	H	1.906157	2.422133	-1.608516
168	H	-0.402797	5.675053	-1.395961
169	H	2.716316	1.980017	-6.21435
170	H	1.923317	0.446334	-5.790009
171	H	5.7202	-1.014902	0.022302
172	H	6.396767	-0.171826	-1.374439
173	H	7.183292	-1.635252	-0.759859
174	H	5.486503	-2.723864	-3.961125
175	H	6.286594	-1.184312	-3.648943
176	H	7.045895	-2.67642	-3.11727
177	H	6.586671	-4.607324	-2.059947
178	H	5.265797	-5.560341	-1.403416
179	H	4.415816	-5.114714	0.631487
180	H	2.876675	-5.67412	1.220296
181	H	3.526597	-6.356077	-0.260756
182	H	-0.164249	-4.049293	1.829173
183	H	-0.44549	-5.782286	2.112577
184	H	1.19326	-5.134666	2.129703
185	H	-0.245345	-7.425594	0.084553
186	H	1.441335	-7.204826	0.526782
187	H	-3.153613	-5.308653	-0.532577
188	H	-2.458823	-4.769518	0.98889
189	H	0.881083	0.243444	2.554018
190	H	5.051973	-4.723267	-2.936963
191	H	0.940983	-7.000491	-1.158495
192	H	-2.227355	-6.432526	0.445401
193	H	-4.334042	-1.599204	-4.636736
194	H	3.67781	0.6032	-5.65544
195	H	-7.873885	1.554301	-3.080445
196	H	-4.635657	-5.285668	4.764977
197	H	-3.521776	-6.564732	4.237877
198	H	-7.697454	2.926369	6.291427
199	H	-7.035008	3.972689	5.013454
200	H	-3.630131	8.186065	1.340902
201	H	-3.779775	6.838016	2.491417
202	H	7.558447	-4.915257	3.144485
203	H	8.006965	-4.878658	4.870648
204	H	10.644909	4.882852	0.075666
205	H	10.585047	6.201857	1.253229
206	H	6.963535	2.820219	-2.530035
207	H	1.075839	5.639187	-2.326573

208	H	-5.148237	-2.771971	0.916787
209	C	-8.701186	0.044958	-1.835114
210	O	-9.918231	0.181647	-1.802962
211	N	-8.084333	-1.156955	-1.631138
212	H	-7.096356	-1.222056	-1.830325
213	C	-8.88038	-2.363507	-1.553775
214	H	-8.299156	-3.154204	-1.071787
215	H	-9.774945	-2.157419	-0.964176
216	H	-9.203971	-2.713976	-2.541937

F1H3_TS, E_{opt}= -4359.5948278

1	C	7.598189	-4.29567	4.045006
2	C	6.162085	-3.88885	4.445402
3	C	5.421147	-2.91977	3.541405
4	C	5.920374	-1.63013	3.297231
5	C	4.165754	-3.24836	3.00896
6	C	5.184956	-0.69871	2.562168
7	C	3.423623	-2.32132	2.269716
8	C	3.929844	-1.04047	2.046386
9	C	-3.52314	7.104446	1.466023
10	C	-2.10853	6.586773	1.15749
11	O	-1.87565	6.512038	-0.27212
12	C	-1.86916	5.184843	1.70579
13	C	-7.03306	2.960599	5.427803
14	C	-5.61986	2.40986	5.737902
15	C	-4.80248	1.916607	4.546109
16	C	-5.19454	0.759355	3.849747
17	C	-3.58163	2.496113	4.170968
18	C	-4.39129	0.184413	2.866861
19	C	-2.76109	1.939427	3.177521
20	C	-3.15544	0.760466	2.538333
21	O	-2.39722	0.107443	1.586112
22	C	-3.89144	-5.5643	4.014386
23	C	-2.75363	-4.53914	4.031142
24	C	-2.99409	-3.36674	3.082861
25	O	-3.9894	-3.31835	2.350029
26	N	-2.02803	-2.42455	3.035223
27	C	-7.75865	1.211674	-2.04428

28	C	-8.0895	2.07858	-0.80844
29	C	-7.1708	3.291814	-0.56295
30	C	-5.96667	3.009583	0.349617
31	N	-4.93027	2.152785	-0.22366
32	C	-3.74356	2.573308	-0.67658
33	N	-3.54125	3.856955	-1.01217
34	N	-2.74121	1.691444	-0.79021
35	C	10.01741	5.68396	0.480209
36	C	8.749135	5.064965	1.083751
37	C	7.859156	4.368537	0.093553
38	C	8.05684	4.230943	-1.25684
39	C	6.604943	3.714079	0.385883
40	N	6.998715	3.536948	-1.8267
41	C	6.094206	3.204666	-0.84108
42	C	5.864386	3.505558	1.562657
43	C	4.883395	2.510174	-0.91382
44	C	4.659554	2.810413	1.4946
45	C	4.173288	2.313287	0.266763
46	Co	1.064299	-1.64337	-1.74062
47	C	-1.45467	-2.87895	-2.16512
48	C	-2.94999	-2.39979	-2.1523
49	C	-2.87789	-1.1103	-3.0541
50	C	-1.4254	-0.68024	-2.90149
51	C	-0.9095	0.614427	-3.30932
52	C	0.434239	0.899105	-3.21546
53	C	1.187682	2.198358	-3.63594
54	C	2.566262	1.576797	-4.00909
55	C	2.618889	0.387795	-3.08818
56	C	3.81349	-0.24211	-2.7786
57	C	3.965157	-1.42907	-2.07833
58	C	5.332811	-2.06237	-1.88084
59	C	4.950711	-3.42886	-1.22565
60	C	3.444268	-3.31684	-0.99656
61	C	2.697207	-4.29684	-0.37594
62	C	1.258226	-4.27566	-0.41828
63	C	0.304995	-5.34639	0.163699
64	C	-1.01861	-4.99187	-0.58856
65	C	-0.86493	-3.49	-0.8666
66	C	-1.1011	-3.77424	-3.37199
67	N	-0.70444	-1.60923	-2.34197
68	N	1.386926	-0.01026	-2.73401
69	N	2.949142	-2.14025	-1.54719
70	N	0.589904	-3.28886	-0.95575
71	C	-3.94192	-3.4411	-2.66671
72	C	-3.33973	-2.01612	-0.69457
73	C	-4.75208	-1.50216	-0.52614

74	O	-5.18077	-0.51833	-1.1647
75	N	-5.53967	-2.16957	0.336048
76	C	-3.33157	-1.28292	-4.51689
77	C	-1.89859	1.595221	-3.91334
78	C	0.664545	3.051872	-4.80533
79	C	1.452415	3.092916	-2.37357
80	C	0.30543	3.939849	-1.84948
81	O	-0.85948	3.513427	-1.7817
82	N	0.602093	5.177076	-1.41064
83	C	2.690833	1.085156	-5.46566
84	C	6.174956	-1.20483	-0.91201
85	C	6.080957	-2.17247	-3.22816
86	C	5.311951	-4.66811	-2.06727
87	C	3.39948	-5.42293	0.358013
88	C	0.177441	-5.10346	1.684573
89	C	0.647791	-6.82552	-0.1185
90	C	-2.3005	-5.43092	0.117066
91	H	1.254674	2.457524	1.710495
92	C	0.545134	1.845271	1.143582
93	C	0.339817	0.538306	1.38243
94	H	-0.02494	2.381668	0.385275
95	F	0.456925	-0.63972	-0.15142
96	H	6.208833	-3.4377	5.446123
97	H	5.560034	-4.79816	4.559765
98	H	6.886894	-1.34602	3.706523
99	H	3.756629	-4.23898	3.193547
100	H	5.583908	0.296481	2.386591
101	H	2.454117	-2.59234	1.863508
102	H	3.356187	-0.32345	1.467543
103	H	8.23936	-3.42265	3.889388
104	H	-1.36764	7.270932	1.598228
105	H	-0.90153	4.79655	1.374513
106	H	-2.64688	4.49471	1.371305
107	H	-1.87513	5.202968	2.799297
108	H	-2.05255	7.383319	-0.65224
109	H	-4.2407	6.602731	0.807188
110	H	-5.04367	3.16683	6.281832
111	H	-5.72435	1.563517	6.428916
112	H	-3.23886	3.390951	4.685894
113	H	-6.12804	0.268711	4.116997
114	H	-1.8054	2.388723	2.926227
115	H	-4.67748	-0.74441	2.385764
116	H	-1.37393	0.330599	1.550243
117	H	-7.51194	2.358568	4.646303
118	H	-2.57285	-4.14858	5.040148
119	H	-1.81006	-5.0025	3.720368

120	H	-2.14653	-1.60792	2.435583
121	H	-1.25436	-2.43891	3.681441
122	H	-4.34919	-5.59466	3.022921
123	H	-9.12489	2.413136	-0.92024
124	H	-8.07708	1.440458	0.086691
125	H	-7.75061	4.082144	-0.07032
126	H	-6.82633	3.710835	-1.51836
127	H	-5.49052	3.945219	0.660006
128	H	-6.30499	2.52189	1.271271
129	H	-5.09763	1.14881	-0.2939
130	H	-1.86262	2.019104	-1.18396
131	H	-2.69666	0.937148	-0.10642
132	H	-4.31853	4.468719	-1.19503
133	H	-2.59446	4.171465	-1.22186
134	H	-6.71127	0.910253	-2.05211
135	H	8.170666	5.849511	1.591967
136	H	9.032116	4.35596	1.874542
137	H	8.875513	4.576752	-1.87078
138	H	6.231011	3.882011	2.513574
139	H	4.084188	2.638053	2.399782
140	H	4.515101	2.117735	-1.8528
141	H	3.234353	1.767223	0.244234
142	H	9.768481	6.411753	-0.29955
143	H	-3.51986	-0.34705	-2.61085
144	H	3.381341	2.281566	-3.81137
145	H	4.713096	0.217145	-3.16876
146	H	5.454105	-3.51104	-0.25511
147	H	-0.96005	-5.5193	-1.54633
148	H	-1.19208	-2.89572	-0.01043
149	H	-1.3145	-3.26103	-4.30953
150	H	-1.65277	-4.71608	-3.36359
151	H	-0.03037	-3.99395	-3.35096
152	H	-4.95626	-3.02969	-2.6803
153	H	-3.7185	-3.77604	-3.67954
154	H	-3.95931	-4.32236	-2.02157
155	H	-2.66553	-1.24209	-0.32092
156	H	-3.22232	-2.87669	-0.03893
157	H	-6.45329	-1.77507	0.520391
158	H	-3.22338	-0.34852	-5.06897
159	H	-2.76154	-2.04539	-5.05199
160	H	-1.67024	2.615992	-3.62755
161	H	-1.91137	1.537809	-5.00762
162	H	-2.9111	1.396794	-3.56253
163	H	1.465548	3.717338	-5.14427
164	H	-0.18037	3.678797	-4.51932
165	H	0.35395	2.436243	-5.65078

166	H	2.304798	3.745483	-2.59062
167	H	1.757769	2.454055	-1.53605
168	H	-0.1434	5.723203	-0.97489
169	H	2.689857	1.914804	-6.17663
170	H	1.869624	0.407303	-5.7197
171	H	5.681437	-1.10235	0.054907
172	H	6.343648	-0.1998	-1.31097
173	H	7.149595	-1.67855	-0.75123
174	H	5.498607	-2.69816	-3.98871
175	H	6.313179	-1.17644	-3.61446
176	H	7.03323	-2.69361	-3.09808
177	H	6.384334	-4.69652	-2.27551
178	H	5.059439	-5.59183	-1.54829
179	H	4.410388	-5.1364	0.647625
180	H	2.879831	-5.66725	1.285357
181	H	3.461543	-6.34711	-0.22577
182	H	-0.23069	-4.11143	1.898277
183	H	-0.4837	-5.8538	2.128134
184	H	1.145499	-5.18329	2.184647
185	H	-0.24652	-7.43599	0.040211
186	H	1.424201	-7.2174	0.539564
187	H	-3.16797	-5.33427	-0.53895
188	H	-2.50646	-4.84316	1.01129
189	H	0.970797	0.00287	2.09285
190	H	4.777903	-4.66921	-3.02175
191	H	0.971112	-6.96624	-1.15475
192	H	-2.24396	-6.48366	0.411308
193	H	-4.38795	-1.56062	-4.55446
194	H	3.627948	0.535212	-5.58799
195	H	-7.95761	1.792717	-2.95105
196	H	-4.67724	-5.30969	4.729698
197	H	-3.52188	-6.56488	4.238188
198	H	-7.69772	2.925852	6.291456
199	H	-6.98885	3.990394	5.059491
200	H	-3.61703	8.186179	1.333764
201	H	-3.77969	6.838016	2.491383
202	H	7.611204	-4.89865	3.132214
203	H	8.007016	-4.87857	4.870659
204	H	10.66131	4.917875	0.03586
205	H	10.58502	6.20188	1.253216
206	H	6.927521	3.285385	-2.79825
207	H	1.54752	5.526493	-1.4102
208	H	-5.12116	-2.78776	1.030914
209	C	-8.69636	0.020313	-2.02913
210	O	-9.90204	0.131722	-2.22052
211	N	-8.10062	-1.15121	-1.66578

212	H	-7.08953	-1.1816	-1.66783
213	C	-8.88037	-2.36346	-1.55375
214	H	-8.35658	-3.07881	-0.91312
215	H	-9.85081	-2.12237	-1.11558
216	H	-9.06392	-2.83623	-2.52725